



BASE24 Banknet Interface Manual

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Version Record

05/2011, Version 44

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PREFACE

This manual describes the Banknet Interface process, which is used to establish a communications link between a BASE24 network and the Banknet Interchange. This manual provides basic information for installing, configuring, and operating the Banknet Interface process.

Reading Prerequisites

While no firm prerequisites exist for reading this manual, it is helpful if readers are somewhat knowledgeable about BASE24 and BASE24 files maintenance.

Document Location

This document is located on the SW60BNET subvolume in a file called BNETMNWD. It may be copied as needed by ACI customers who have purchased ACI's Banknet Interface.

BASE24 Compatibility

The information contained in this manual is applicable to BASE24 Release 6.0 and above.

NET24 –XPNET Compatibility

The information in this manual is applicable to XPNET 3.0 and above. For previous releases, refer to the appropriate NET24 - XPNET manuals.

Document Control

Version 1, 06/01	Banknet Interface Release 6.0
Version 2, 07/01	Banknet Release 01.2 Mandates and Removal of PCS/PLF Support
Version 3, 01/02	Triple/Single DES Enhancement and Format 2 ILF Support
Version 4, 02/02	Banknet Release 02.1 Mandates
Version 5, 06/02	Transaction Security Services
Version 6, 07/02	Banknet Release 02.2 Mandates
Version 7, 10/02	Discretionary Data Suppression Enhancement
Version 8, 06/03	Banknet Release 03.2 Mandates
Version 9, 07/04	Banknet Release 04.2 Mandates
Version 10, 08/04	Auto signon start clarification
Version 11, 01/05	Banknet Interface CVD/CVD2 Processing Enhancement and Banknet Release 05.1 Mandates
Version 12, 08/05	Switch Common Data Token Enhancement, Issuer Suppression of Discretionary Data Enhancement, and Banknet Release 5.2 Mandates
Version 13, 02/06	Banknet Release 06.1 Mandates
Version 14, 02/06	Added LCONF param USE-OWNER-NAME-FOR-DE43
Version 15, 04/06	Forward Declined Advices Enhancement
Version 16, 05/06	Installation Steps corrections
Version 17, 07/06	Banknet Release 06.2 Mandates
Version 18, 11/06	SPDH & RTAU Enhancements for Visa and MasterCard
Version 19, 02/07	Banknet MCC104 Accounts Management File Enhancement Banknet Release 07.1 Mandates
Version 20, 03/07	Banknet Release 06.2 Healthcare Eligibility Transaction Support
Version 21, 07/07	Acquiring Institution ID Enhancement Banknet Cardholder Billing Currency per BIN Enhancement Banknet Release 07.2 Mandates
Version 22, 01/08	Banknet Release 08.1 Mandates and Banknet Global Authorization Enhancements
Version 23, 03/08	Added LCONF parameters EKM-PIN-BLOCK-FORMAT, EKM-ACQ-PEK-INDEX and EKM-ISS-PEK-INDEX
Version 24, 08/08	Banknet Release 08.2 Mandates
Version 25, 11/08	Warmboot clarification
Version 26, 12/08	UKDM Globalization and Maestro Best In Class for BASE24 Banknet Interface
Version 27, 02/09	Banknet Release 09.1 Mandates
Version 28, 03/09	Correct ICFE GMT Offset descriptions
Version 29, 03/09	Update the param USE-OWNER-NAME-FOR-DE43 for ATM processing.
Version 30, 04/09	Modified BNRLF MCC103 screen
Version 31, 04/09	ICFE Logon Type clarification.
Version 32, 05/09	Identifying MasterCard network ids for Maestro and Prepaid cards
Version 33, 05/09	Banknet Release 09.1 Partial Approval Edit Changes
Version 34, 09/09	Banknet Release 09.2 Mandates

Version 35, 10/09 Banknet Currency Exponent Table(BNETTBL)
 Version 36, 01/10 TCP/IP Line Configuration Modification
 Version 37, 02/10 Banknet Release 10.1 Mandates
 Clarification of the use of timestamps in field 7.
 Version 38, 04/10 Installation Steps Modification
 Version 39, 04/10 Banknet Release 10.1 Banknet Release 10.1 Release Item LAC 111
 Allow PIN change on non-EMV transactions
 Version 40, 05/10 Update of Switch Token description
 Version 41, 08/10 Banknet Release 10.2 Mandates
 Version 42, 09/10 Banknet UK Domestic Maestro Technical Alignment
 Version 43, 04/11 Banknet Release 11.1 Mandates
 Version 44, 05/11 Banknet Release 10.1: AFD Completion Advice

Document History

30JUN2001 D00/1693
Symptom: Release 6.0 Enhancements.
Problem: None.
Fix: Created Release 6.0 manual for Banknet Interface.
Dependency: Restore BNETMAN.
Reference: WO #000101-1 (General Release 6.0)

31JUL2001 D00/1693
Symptom: Banknet Release 01.2 Mandates.
Problem: None.
Fix: 1. Modified the INTRODUCTION section to document the
the scope of the Banknet Release 01.2 Mandates.
Modified Subsections:
CHIP-BASED DEBIT AND CREDIT PROCESSING
New Subsections:
U.S. DEFERRED BILLING INDICATOR
PAYMENT TRANSACTION AUTHORIZATION
2. Modified the TOKEN DOCUMENTATION section to
document the addition of DE 3 (Processing Code) and
SE 78 (Visa Deferred Billing Indicator) of DE 48
(Additional Data) within the Switch Token.
3. Modified the SUBFIELD DOCUMENTATION section to
include a description of the new SE 78 of DE 48.
Dependency: Apply changes to BNETDDLs and BNETLIBS and remake.
Replace BNETMAN and BNETUPDT.
Reference: (retro of SW5P569 from WO #010602-01)

31JUL2001 D00/1693
Symptom: Removal of PCS/PLF support from Banknet Release 6.0
Interface.
Problem: None.
Fix: Removed references to PCS/PLF from this manual.
Dependency: Apply fix to BNETG and BNETS and re-make. Replace
BNETLOGM and BNETMAN.
Reference: Case #337091

08JAN2002 dtw/577
Symptom: Triple/Single DES Enhancement.
Problem: None.
Fix: Modified the following sections in order to support
the Triple/Single DES Enhancement:
PIN Encryption
LCONF Assigns and Params
Initialization
Section added:
KEY6 Settings
Dependency: Apply changes to BNETDDLs, BNETG, BNETLIBS, BNETM,
BNETS, HISWUTLS, SECUTILS and run MAKE.
Replace BNETLOGM, BNETMAN and BNETUPDT.
Reference: WO #000211-01 (Triple DES)

08JAN2002 jds/1802
 Symptom: Release 6.0 Format 2 File Support
 Problem: None
 Fix: Added the following section for support of format 2
 ILF files.
 Format 2 File Support
 Dependency: Apply fixes to BNETG, BNETLIBS, BNETS and RP09BNTS.
 Replace files BNETLOGM, BNETMAN and BNETUPDT. Apply
 corresponding fixes to BAUTILS, HISWUTLS and OKLIBTS.
 Run MAKE.
 Reference: WO #000202-01

08FEB2002 D00/1693
 Symptom: Banknet Release 02.1 Mandates.
 Problem: None.
 Fix: 1. Added the following new sections to document the
 scope of the Banknet Release 02.1 Mandates:
 UNIVERSAL CARD AUTHENTICATION FIELD SUPPORT
 CHIP CARD AUTHORIZATION SUPPORT
 NEW VISA AUTHORIZATION CHARACTERISTIC INDICATOR
 VALUES SUPPORT
 INTERNATIONAL ADDRESS VERIFICATION SUPPORT
 2. Modified the SUBFIELD DOCUMENTATION section to
 include a description of the new SE 43 of DE 48. Also
 noted in this section are the changes to SE 42 of DE 48
 and DE 120.
 Dependency: New files: ILB0102M, ILB0102R, and ILB0102S. Apply
 fixes to BNETDDL, BNETG, BNETLIBS, BNETS, BACOUTLS,
 COBNAMES, DDL PSTKN, PSTKN CVS, and PSTKNID and re-MAKE.
 Replace BNETLOGM, BNETMAN, and BNETUPDT.
 Reference: WO #011112-02, #011112-03, and #011112-04

30JUN2002 tmh/173
 Symptom: Transaction Security Services Enhancement
 Problem: None.
 Fix: Added support for Transaction Security Services.
 Sections modified: Pin Encryption
 LCONF Assigns and Params
 KEYF Settings
 KEY6 Settings
 Dependency: Apply fixes to BNETS, BNETG, HISWUTLS and SECUTLS and
 run MAKE. Replace BNETLOGM, BNETMAN and BNETUPDT.
 Reference: WO #000211-01.

11JUN2002 kjw/571
 Symptom: Problems configuring BNPATh server.
 Problem: BNPATh server contains misspelled param.
 SET SERVER PARAM EXT, should be ETX.
 Fix: Corrected spelling of param ETX.
 Sections modified: INSTALLATION STEPS, #7
 Dependency: Replace BNETMAN.
 Reference: Case #348904.

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13JUL2002 gg/724
Symptom: Banknet Release 02.2 Mandates
Problem: None
Fix: Add documentation for a redefine of subelement 43 and new subelements 44 and 45 of DE 48 to the SUBFIELD DOCUMENTATION section.
Dependency: New files ILB0202M, ILB0202R, and ILB0202S. Apply fixes to BNETDDL, BNETG and BNETLIBS. Apply fix to DDLPSTKN. Run MAKE. Replace files BNETMAN and BNETUPDT. Run ILF conversion ILB0202.
Reference: WO #020424-02

18OCT2002 gg/724
Symptom: Discretionary Data Suppression
Problem: None
Fix: Added documentation for the Discretionary Data Suppression enhancement.
Dependency: Replace BNETUPDT and BNETMAN. Apply fixes to BAUTILS, HISWUTLS, BNETG, BNETLIBS, and BNETS. Run MAKE.
Reference: WO #020625-02

08JAN2003 RAF/100
Symptom: RACAL users are unable to complete a Key Exchange.
Problem: MasterCard is using the ANSI X9.17 encryption method for Key Exchanges. The interface is only coded to use the Variant encryption method.
Fix: Added documentation for the new Lconf param RACAL-ANSI-METHOD.
Dependency: Replace BNETMAN. Apply fixes to BNETG and BNETS. Run MAKE.
Reference: Case #357289.

19JUN2003 D00/1693
Symptom: Banknet Release 03.2 Mandates.
Problem: None.
Fix: Added documentation for Banknet Release 03.2 Mandates.
Modified sections: LCONF ASSIGNS AND PARAMS
BNIIF SETTINGS
TOKEN DOCUMENTATION
Dependency: BA10SC04 subvolume. New files: BII0203M, BII0203R, BII0203S, ILB0203M, ILB0203R, and ILB0203S. Apply fixes to BNETDDL, BNETG, BNETLIBS, BNETS, and SCRNBIIIF and re-MAKE. Replace BNETLOGM, BNETMAN, and BNETUPDT.
Reference: WO #030323-01

26JUN2003 D00/1693
Symptom: CVV2 response codes are always logged to the PS50 token.
Problem: The 6.0 token POS Data1 token also has a field meant specifically for these response codes. Customers should have the option of which token to log the responses to.
Fix: Added documentation for the new LCONF parameter CVV2-RESULT-OPTION.
Dependency: Apply fixes BNETG, BNETLIBS, and BNETS and re-MAKE. Replace BNETLOGM and BNETMAN.
Reference: WO #030323-06

11AUG2003 D00/1693
Symptom: Inconsistency in the length of field MCI-ID.
Problem: The size of this field was increased by one to 6 bytes as a result of the 03.2. This document incorrectly states that customers may configure institution ids up to 11 bytes in length. If configured field for more than 6 bytes, other data may be overwritten and the effects may be undesirable.
Fix: Increased the size of screen field DESTINATION MCI ID to allow entries that are up to 6 bytes long.
Modified section: SC-BNADM-MESSAGE
Dependency: Apply fixes BNETDDLs, BNETLIBS, and BNETS and re-MAKE. Replace BNETMAN and SCRNBADM.
Reference: Case #366959, 366962, and 366629.

28AUG2003 D00/1693
Symptoms: Banknet Release 03.2 Mandates.
Problem: None.
Fix: RLF conversion for Banknet Release 03.2 Mandates.
Modified section BASE24 CRT OPERATIONS to indicate that the MCI ID/Customer Identifier can be up to 6 bytes long.
Dependency: BA10SC04 subvolume. New files: BRL0203M, BRL0203R, BRL0203S. Apply fix BNETDDLs and re-MAKE. Replace SCRNBRLF and BNETMAN.
Reference: Case #367529

23FEB2004 engelk
Symptom: Descriptions of the MCI ID in its various functions were not clear.
Problem: The MCI ID field must be all numeric. The descriptions of this field did not indicate that it needed to be right justified and zero filled to the left.
Fix: Modified BNIIIF MCI ID and BNRLF DESTINATION MCI ID descriptions. Modified the LOGOFFG, LOGONG, and LOGONS command descriptions so that the format of the group id is correctly described.
Dependency: Apply fix to BNETS. Run MAKE. Replace BNETLOGM and BNETMAN.
Reference: Case #374259

22JUN2004 frederr
Symptom: There is no option on the ICFE Screen 12 Protocol field for the TCP/IP protocol.
Problem: The TCP/IP protocol behaves the exact same way as the SNA protocol to the interface, so no new value was added for TCP/IP. The documentation should reflect this.
Fix: Modified the Protocol field definition for ICFE screen 12 to tell customers that when using the TCP/IP protocol to set this field to a 3 (SNA LU0).
Dependency: Replace BNETMAN.
Reference: Case #378756.

PREFACE

05AUG2004 hayest
Symptom: Banknet Release 04.2 Mandates.
Problem: None.
Fix: Added documentation for Banknet Release 04.2 Mandates.
Modified sections: LCONF ASSIGNS AND PARAMS
TOKEN DOCUMENTATION
SUBFIELD DOCUMENTATION
Dependency: BA10SC04 subvolume. New files: ILB0404M, ILB0404R,
and ILB0404S. Run ILF conversion ILB0404. Apply fixes
to BNETDDL, BNITEMVS, BNETG, BNETLIBS, BNETS, and
DDL PSTKN and re-MAKE. Replace BNETLOGM, BNETMAN, and
BNETUPDT.
Reference: WO #040505-02

27AUG2004 engelk
Symptom: Mastercard interface previously logged on is marked
logged off after a warmboot.
Problem: Logon information is not saved on warmboot. This is
because the BNIIF is re-read on warmboot and too much
overhead would be involved in comparing before and
after images. Instead the interface will initiate
logons anew after the warmboot if auto signon start is
set to "Y" on screen 3 of the ICFE. However, nothing
was stated in the documentation to alert customers to
this fact.
Fix: Added information to the AUTO SIGNON START description
to alert customers to the fact that either:
1) AUTO SIGNON START must be set to "Y" to
automatically initiate logons after a warmboot
or
2) Customers must manually initiate logons after a
warmboot if AUTO SIGNON START is set to "N".
Modified section: ICFE Settings
Dependency: Replace BNETMAN.
Reference: Case #381435

26JAN2005 hayest
Symptom: Banknet CVD/CVD2 Processing Enhancement
and Release 05.1 Mandates
Problem: None.
Fix: Removed support for LCONF parameter
CVV2-RESULT-OPTION. Added three new screen
fields to the MCC102 Accounts File Screen. Added screen
descriptions for the new screen fields.
Modified sections: LCONF Assigns and Params
MCC102 Accounts File Screen
Dependency: Apply fixes to BNETDDL, BNITEMVS, BNETG,
BNETLIBS, BNETS, RQBNRLFS, SCRNBRLF, and SVBNRLFS and
run MAKE. Replace BNETMAN and BNETUPDT.
Reference: WO #040826-01 and #040918-04

26APR2005 owallad
Symptom: BNRLF screen fields CARD SEQUENCE NUMBER and CARD EXPIRATION DATE are documented in this manual as required fields.
Problem: Not all customers participate in the MasterCard program requiring these fields.
Fix: Modified the fields descriptions to indicate that they are not required.
Modified section: MCC102 Accounts File Screen
Dependency: Apply fix to BNETLIBS and RQBNRLFS and run MAKE. Replace BNETMAN and BNETUPDT.
Reference: Case #393071

11AUG2005 hayest
Symptom: Switch Common Data Token Enhancement, Issuer Suppression of Discretionary Data Enhancement, and Banknet Release 05.2 Mandates
Problem: None.
Fix: Added support for purchase with cash back transactions to section Supported Transaction Types. Added 0400/0410 as supported message types in section Supported Message Types. Added LCONF parameters SWI-ISS-AVS-DATA-SPPRS, SWI-ISS-CVD2-DATA-SPPRS, and SWI-ISS-DISC-DATA-SPPRS to section LCONF Assigns and Params. Added support for 0400 messages to section Outbound Request Timer. Added support for the Switch Common Data token to section Token Documentation. Added a diagram to sections Message Flows From Banknet to BASE24 and Message Flows From BASE24 To Banknet, depicting the 0400/0410 Acquirer Reversal Request/ Response Messages.
Dependency: BA10SC04 subvolume. New files: ILB0605M, ILB0605R, and ILB0605S. Apply fixes to BATKNID, BATKNCVS, COBTKN, DDLBATKN, BNETDDL, BNETG, BNETLIBS, and BNETS and run MAKE. Run ILF conversion ILB0605. Replace BNETLOGM, BNETMAN, and BNETUPDT.
Reference: WO #041014-01, #050305-03, and #050426-01

13FEB2006 hayest
Symptom: Banknet Release 06.1 Mandates
Problem: None
Fix: Modified section TOKEN DOCUMENTATION in order to document the two new fields added to the Switch token. Modified section SUBFIELD DOCUMENTATION in order to document the new subfields added to data element 48. Replaced section MCC105 AVS Transaction File Screen with MCC105 Recurring Payment Cancellation Service File Screen.
Dependency: BA10SC04 subvolume. New files: BRL0206M, BRL0206S, BRL0206R, ILB0206M, ILB0206R and ILB0206S. Apply fixes to BNETDDL, BNETEMVS, BNETG, BNETLIBS, BNETS, RQBNRLFS, SCRNBRFL and SVBNRLFS. Run MAKE. MAKE the ILF conversion using ILB0206M. Run the ILF conversion using ILB0206R. MAKE the BNRLF conversion using BRL0206M. Run the BNRLF conversion using BRL0206R. Replace BNETMAN and BNETUPDT.
Reference: WO #051110-01.

PREFACE

17FEB2006 frederr
Symptom: DE-43, Card Acceptor Name/Locataion, is being sent to MasterCard with an invalid value in 0200 request messages.
Problem: The interface uses PSTM.TERM^NAME^LOC to populate DE-43. This field is populated by Device Handlers using the PTD's term-name-loc field. Some customers want to use PSTM.TERM^OWNER^NAME to populate DE-43 instead. This field is populated by Device Handlers using the PTD's term-owner-name field.
Fix: Added LCONF param USE-OWNER-NAME-FOR-DE43 to the LCONF Assigns and Params section of this document.
Dependency: Apply fix to BNETG, BNETS and BNETLIBS. Replace BNETMAN. Run MAKE.
Reference: Case #405686.

13APR2006 saderc
Symptom: Banknet ISO Forward Declined Advices Enhancement
Problem: None
Fix: Added documentation for Banknet ISO Forward Declined Advices Enhancement. Added LCONF parameter FRWD-DCLN-ADVC to section LCONF Assigns and Params.
Dependency: Apply fixes to BNETG, BNETLIBS and BNETS. Run MAKE. Replace BNETMAN and BNETUPDT.
Reference: WO #040826-02.

07MAY2006 engelk
Symptom: 1) The Pathway server setup documented in this file referenced setting the server volume to \<sys>.\$<volume>.LNETEXEC. The EXEC subvolume is no longer used.
 2) Some steps in setting up the Banknet pathway server to talk to the Banknet Interface process were missing and/or unclear.
Problem: 1) The EXEC and CNTL subvolumes were combined into the CNTL subvolume. This should now reference the CNTL subvolume.
 2) The Banknet pathway server device declaration was missing from the pathway server setup instructions.
Fix: 1) Modified the Installation Steps to reference the CNTL subvolume when setting the server volume for the SERVER-BNPATHxxx. It now references \<sys>.\$<volume>.<lnet>CNTL
 Modified system, volume, and subvolume references to be surrounded by <> throughout the Installation Steps to indicate that these are to be modified to fit the system settings where the code is being installed.
 2) Added the Banknet pathway server device declaration to the pathway server setup instructions within the Installation Steps. Added additional information to the Installation Steps regarding SERVER-BNPATH and its associated device, line, and station.
Dependency: Replace BNETMAN.
Reference: Case #411988

26JUL2006 hayest
Symptom: Banknet Release 06.2 Mandates.
Problem: None
Fix: 1. Modified sections Network Management Messages and Banknet Interface Process Functions in order to support the Network Connection Status Network Management Request/0800 message.
2. Modified section Network Control Commands in order to support the NETWK CONN STAT command.
3. Modified sections Acquirer Reversal to Banknet and Store-and-Forward Messages to Banknet in order to depict the SAF processing of the Acquirer Reversal Request/0400 messages.
4. Added a Network Connection Status diagram to section Network Management Message Flows in order to depict the Network Connection Status network management message.
5. Added the Healthcare Token to section Token Documentation.
Dependency: Apply fixes to BNETG, BNETLIBS and BNETS. Run MAKE. Replace BNETLOGM and BNETMAN.
Reference: WO #060321-01.

03NOV2006 hayest
Symptom: SPDH & RTAU Enhancements for Visa and MasterCard
Problem: None
Fix: Added the Healthcare/Transit Token to Appendix A, Token Documentation.
Dependency: Apply fixes to BNETG, BNETLIBS, COBTKN, DDL PSTKN, PSTKN CVS and PSTKNID. Run MAKE. Replace BNETMAN.
Reference: WO #050909-01.

12DEC2006 singhM
Symptom: Banknet MCC104 Accounts Management File Enhancement.
Problem: None.
Fix: Added documentation for Banknet MCC104 Accounts Management File Enhancement.
Added section: MCC104 Accounts File Screen
Dependency: BA10SC04 subvolume. Apply fix to BNETDDL S, BNETLIBS, BNETS, BRL0207M, BRL0207R, BRL0207S, RQBNRLFS, SCRNBRLF, SVBNRLFS, and Run Make. Make the BNRLF conversion using BRL0207M. Run the BNRLF conversion using BRL0207R. Replace BNETMAN and BNETUPDT.
Reference: WO #061010-02.

PREFACE

13FEB2007 HAYEST
Symptom: Banknet Release 07.1 Mandates
Problem: None.
Fix: Added documentation for the ACTVT STA and DEACTVT STA network control commands. Added documentation for Banknet MCC106 PayPass Mapping File, MCC107 MasterCard Enhanced Value Platform Account File and MCC108 MasterCard Product Graduation Account File.
Modified sections: Network Control Commands
Network Management Messages
LCONF Assigns and Params
Switch Token (B0, B1)
Subfield Documentation
Added sections: MCC106 PayPass Mapping File Screen
MCC107 MasterCard Enhanced Value Platform Account File Screen
MCC108 Mastercard Product Graduation Account File Screen
Dependency: BA10SC04 subvolume. New files: BRL0207M, BRL0207R, BRL0207S, ILB0207M, ILB0207R and ILB0207S. Apply fixes to BNETDDLs, BNETG, BNETLIBS, BNETS, RQBNRLFS, SCRNBRLF and SVBNRLFS. Run Make. Make the BNRLF conversion using BRL0207M. Run the BNRLF conversion using BRL0207R. Make the ILF conversion using ILB0207M. Run the ILF conversion using ILB0207R. Replace BNETLOGM, BNETMAN and BNETUPDT.
Reference: WO #061120-01.

30MAR2007 SADERC
Symptom: Banknet Release 06.2 Healthcare Eligibility Transaction Support
Problem: None.
Fix: 1. Added support for healthcare eligibility transactions to section Supported Transaction Types.
2. Added the Healthcare/Service Token to Appendix A, Token Documentation.
3. Removed Healthcare Token from Appendix A, Token Documentation.
Dependency: BA10SC04 subvolume. New files: ILB0307M, ILB0307R and ILB0307S. Apply fixes to BNETDDLs, BNETG and BNETLIBS. Run Make. Make the ILF conversion using ILB0307M. Run the ILF conversion using ILB0307R. Replace BNETLOGM, BNETMAN and BNETUPDT.
Reference: WO #070212-03

01AUG2007 SaderC
Symptom: Banknet Cardholder Billing Currency per BIN Enhancement
Problem: None
Fix: Added support for new LCONF parameter SPPT-MULTI-CRNCY (Support Multi-Currency).
Modified section: LCONF ASSIGNS AND PARAMS
Dependency: Apply fixes to BNETG, BNETLIBS and BNETS. Run Make. Replace BNETUPDT. New interface documentation file BNETMNWD (Microsoft Word) supersedes BNETMAN.
Reference: WO #070425-01

01AUG2007 SaderC
Symptom: Banknet Acquiring Institution ID Enhancement
Problem: None.
Fix: Modified the interface to support different MasterCard assigned customer IDs to be used for the same institution, based on transaction currency.
Dependency: BA10SC04 subvolume. New files: BII0807M, BII0807R and BII0807S. Apply fixes to BNETDDL, BNETLIBS and BNETS. Run Make. Make the BNIIF conversion using BII0807M. Run the BNIIF conversion using BII0807R. Replace BNETUPDT. New interface documentation file BNETMNWD (Microsoft Word) supersedes BNETMAN.
Reference: WO #060925-03.

01AUG2007 swiftD
Symptom: Banknet Release 07.2 Mandates
Problem: None
Fix: 1. Added code to store and retrieve new global variable ntwk^assign^tcc^g.
2. Added support for new values that may be received in DE 48 SE 71 to the existing on-behalf processing.
3. Added processing for MCC109 specific data in a 0312 file request response.
Modified sections: LCONF ASSIGNS AND PARAMS
BASE24 CRT OPERATIONS
TOKEN DOCUMENTATION
Dependency: New files: ILB0807M, ILB0807R, and ILB0807S. Apply fixes to BNETEMVS, BNETDDL, BNETG, BNETLIBS, BNETS, COBTKN, DDLPSTKN, PSTKNVS, PSTKNID, RQBNRLFS, and SCRNBRLF. Run Make. Make the ILF conversion using ILB0807M. Run the ILF conversion using ILB0807R. Replace BNETLOGM and BNETUPDT. New interface documentation file BNETMNWD (Microsoft Word) supersedes BNETMAN.
Reference: WO #070419-01

31JAN2008 bartond
Symptom: Banknet Global Authorization Migration
Problem: None
Fix: Added and modified code to support Global Authorization Migration for European customers
Modified sections: INTRODUCTION
INSTALLATION AND CONFIGURATION
STARTUP AND CONTROL
TRANSACTION HANDLING
CUTOVER AND REPORTING
BASE24 CRT OPERATIONS
Added sections: FUNCTIONAL OVERVIEW
INTERFACE FUNCTIONS
IPF REFRESH
Dependency: New subvolume: SW60ABNT.
New files: ILB1107M, ILB1107R, ILB1107S, STF1107M, STF1107R and STF1107S.
Apply fixes to BNETD, BNETDDL, BNETEMVM, BNETEMVS, BNETG, BNETLIBS, BNETM, BNETS, DDLFSTF, OMFXS, RQBNETS, RQSTFS, SCRNBNET, SCRNSTF, SWBNETM, SWIFM and SWIMM.
Run Make. Make the ILF conversion using ILB1107M.
Run the ILF conversion using ILB1107R. Make the STF conversion using STF1107M. Run the STF conversion using STF1107R.
Replace BNETLOGM and BNETMNWD.
Reference: WO #070202-01

13FEB2008 BensonE
Symptom: Add ICF/ICFE flags to control ATM and NCD processing.
Problem: There was no control over ATM and NCD processing at the Banknet level.
Fix: Modified section: ICFE Settings
Dependency: Apply fixes to BNETS, BNETDDL, RQBNETS and SCRNBNET.
Run Make. Replace BNETMNWD.
Reference: Case #455664.

15JAN2008 hayest
Symptom: Banknet Release 08.1 Mandates
Problem: None
Fix: Added support for new screen fields on the MCC108 MasterCard Product Graduation Account File screen. Added support for new fields in the Banknet Switch Data token. Added support for new subfields within DE 48.
Modified sections: BASE24 CRT OPERATIONS
TOKEN DOCUMENTATION
SUBFIELD DOCUMENTATION
Dependency: New files: ILB0108M, ILB0108R, and ILB0108S. Apply fixes to BNETG, BNETDDL, BNETLIBS, BNETS, RQBNRLFS, SCRNBRLF and SVBNRLFS. Run Make. Make the ILF conversion using ILB0108M. Run the ILF conversion using ILB0108R. Replace BNETMNWD and BNETUPDT.
Reference: WO #071016-01

09APR2008 bensone
Symptom: Customer could not certify with MasterCard.
Problem: MasterCard wants the PIN Block Format Code of DE 53 to be (10=ISO Format 0) in the Authorization Request. MasterCard wants the PIN Key Index of DE 53 to be 0003 in the issuer GSI logon.
Fix: Added support for LCONF parameters EKM-PIN-BLOCK-FORMAT, EKM-ACQ-PEK-INDEX and EKM-ISS-PEK-INDEX.
Modified sections: LCONF ASSIGNS AND PARAMS.
Dependency: Apply fix to BNETATMS, BNETLIBS, BNETG and BNETS.
Run Make. Replace BNETMNWD and BNETLOGM.
Reference: Case #459869

11AUG2008 bensone
Symptom: Banknet Release 08.2 Mandates
Problem: None
Fix: Added new LCONF parameters PROCESS-MONEYSSEND-TXN and PRVT-LABEL-USES-DE48SE32. Added PROCESSING OPTION on page 3 of the ICFE. Added ALTERNATE HOST on page 12 of the ICFE. Added ALT ID to the STF screen.
Modified sections: LCONF ASSIGNS AND PARAMS
ICFE SCREEN 3 SETTINGS
ICFE SCREEN 12 SETTINGS
STF SCREEN SETTINGS
Section 6
Appendix A, B and C
Dependency: Apply fixes to BATKNCVS, BATKNID, BNETATMS, BNETDDL, BNETEMVS, BNETG, BNETLIBS, BNETS, COBTKN, DDLBATKN, DDLFSTF, DDLPSTKN, RQBNETS, SCRNBNET, AND SCRNSTF.
Run Make. Replace BNETLOGM, BNETMNWD and BNETUPDT.
Reference: WO #080401-02

17NOV2008 engelk
Symptom: Mastercard interface previously logged on is marked logged off after a warmboot.
Problem: Logon information is not saved on warmboot. This is because the BNIIF is re-read on warmboot and too much overhead would be involved in comparing before and after images. Instead the interface will initiate logons anew after the warmboot if auto signon start is set to "Y" on screen 3 of the ICFE. However, nothing was stated in the documentation description of the warmboot command to alert customers to this fact.
Fix: Added information to the Warmboot description to alert customers to the fact that either:
1) AUTO SIGNON START must be set to "Y" to automatically initiate logons after a warmboot
or
2) Customers must manually initiate logons after a warmboot if AUTO SIGNON START is set to "N".
Modified section: Network Control Commands
Dependency: Replace BNETMNWD.
Reference: Case #479953

PREFACE

10DEC2008 SaderC
Symptom: UKDM Globalization and Maestro Best In Class for BASE24 Banknet Interface
Problem: None
Fix: Added new LCONF parameters ACTVT-INTERVAL-TIMER and SAF-0120-MESSAGES.
Modified section: LCONF ASSIGNS AND PARAMS
Dependency: Apply fixes to BNETATMS, BNETEMVS, BNETG, BNETLIBS and BNETS. Run Make. Replace BNETLOGM, BNETUPDT, and BNETMNWD.
Reference: WO #080905-01

16FEB2009 bensone
Symptom: Banknet Release 09.1 Mandates
Problem: None
Fix: Added info on new F1 E-Commerce Additional Data Token.
Modified sections:
Section 6
Appendix A and B
Dependency: Replace BNETMNWD.
Reference: WO #081007-04

02MAR2009 bensone
Symptom: Description for ICFE GMT Offset not consistent.
Problem: None
Fix: Modified ICFE GMT Offset description to match Visa manual.
Modified sections:
Section 3
ICFE Screen 12
Dependency: Replace BNETMNWD.
Reference: Case #466635

19MAR2009 frederr
Symptom: DE-42 (Card Acceptor ID code) is being sent with an incorrect value in ATM 0200 request messages to MasterCard.
Problem: The interface uses the STM.TERM-OWNER-FIID to populate DE-42. Customers would like to use STM.TERM-OWNER-NAME in some cases.
Fix: Modified the documentation for the LCONF param USE-OWNER-NAME-FOR-DE43. This param will now also control how DE-42 is populated. If the param is set to 'Y' then STM.TERM-OWNER-FIID will be used. If the param is set to 'N' then STM.TERM-OWNER-NAME will be used.
Modified section: LCONF Assigns and Params.
Dependency: Apply fix to BNETATMS. Run Make. Replace BNETMNWD.
Reference: Case #488286.

30MAR2009 bensone
Symptom: Contents of DE 61 won't necessarily match the contents of the Point of Service Data Token.
Problem: After moving the contents of the C4 token to DE 61, the Banknet Interface can modify the data in DE 61.
Fix: The contents of the condition codes in DE 61 will be included in the generic switch token.
Dependency: Apply fixes to BNETDDLs and BNETLIBS. Run Make. Replace BNETMNWD.
Reference: Case #489863

01APR2009 engelk
 Symptom: MCC103 Regional Indicator/Purge Date was incorrectly populated for ADDs and UPDATES.
 Problem: User was required to know region codes and enter them in the right order with no intervening spaces.
 Fix: Modified the MCC103 screen to more closely match the MCC104 screen with its regions labeled next to Y/N indicators. Modified the BANKNET Request Log File (BNRLF) MCC103 screens in this document to reflect the new screen layout.
 Dependency: Apply fixes to BNETDDLs, BNETLIBS, SCRNBRLF, and RQBNRLFS. Run Make. Replace BNETMNWD.
 Reference: Case #489863

09APR2009 bensone
 Symptom: When the LOGON TYPE on ICFE Screen12 is "G", but the customer performs a manual "LOGONP" DE 70 can be set incorrectly. DE 2 formatted incorrectly for prefixes.
 Problem: The code assumed prefixes were not allowed when LOGON TYPE was "G". Release 9.1 removed code to calculate prefix length.
 Fix: Modified the code allow prefix logons when LOGON TYPE set to "G". Added code to calculate the length of the prefix.
 Modified sections: ICFE SCREEN 12 SETTINGS
 NETWORK CONTROL COMMANDS
 Dependency: Apply fix to BNETLIBS and run Make. Replace BNETMNWD.
 Reference: Case #493997

PREFACE

06MAY2009 engelk
Symptom: Non-Maestro fuel transactions were converted to preauth requests.
Problem: MasterCard only allows preauth/preauth completion message sets for Maestro fuel transactions. However, MasterCard does not edit transactions to ensure that only Maestro fuel transactions have DE-61, subfield 7, POS Transaction Status, set to "4" (preauthorized request).
Fix: Added new LCONF PARAMS, MAESTRO-NETWORK-IDxx (where xx are numbers), to allow issuers to define up to 50 Mastercard assigned Maestro network ids. Added code to read and sort the ids in those LCONF PARAMS, as well as, code to search those ids for the network id in the external message to determine if the transaction is for Maestro.
 Altered a previous fix designed to identify prepaid transactions to use the same approach for MasterCard prepaid network ids. Added new LCONF PARAMS, PREPAID-NETWORK-IDxx (where xx are numbers), to allow customers to identify up to 50 MasterCard assigned Prepaid network ids. Added code to read and sort the ids in those LCONF PARAMS, as well as, code to search those ids for the network id in the external message to determine whether the transaction was for a prepaid card.
 Modified Section: LCONF ASSIGNS AND PARAMS
 IMPORTANT NOTE: Unless the Maestro and Prepaid LCONF PARAMS are added, the interface will NOT correctly identify these transactions.
Dependency: Apply fixes to BNETG, BNETS, and BNETLIBS. Run Make. Replace BNETMNWD and BNETLOGM.
Reference: Case #495806

27MAY2009 ArumugP
Symptom: Banknet Release 09.1 Partial Approval Edit Changes
Problem: None.
Fix: Added new LCONF parameter GREATER-AMT-APPROVAL-ALWD and modified the interface to support partial authorization for greater amount
 Modified Section: LCONF PARAMS
Dependency: Apply fixes to BNETS, BNETLIBS and BNETG. Run Make. Replace BNETMNWD and BNETUPDT.
Reference: WO #090409-02

16SEP2009 PrasadS
 Symptom: Banknet Release 09.2 Mandate.
 Problem: None.
 Fix: 1. Modified the interface to map the value within
 DE 28 to the Surcharge Data token or vice versa.
 2. Modified the interface to support the new value "2"
 in DE 48 SE 42 subfield 1 position 2 and DE 48 SE
 42 subfield 1 position 1-3 values of "222" and "922"
 for the online checkout service.
 3. Modified the interface to remove support for
 currency conversion.
 4. Modified the interface to process all entries within
 DE 48 SE 71.
 5. Modified the interface to include the contents of
 DE 48 SE 33, DE 48 SE 42, DE 48 SE 80 and
 DE 48 SE 83 in the generic switch token.
 6. Modified the interface to include DE 28 in the
 generic switch token.
 Modified sections: LCONF Assigns and Params
 Currency Conversion
 On-Behalf Processing by Banknet
 E-Commerce Transactions
 Appendix A – Token Documentation
 Dependency: Apply fixes to BNETATMS, BNETDDL, BNETMVS, BNETG,
 BNETLIBS, BNETS, DDLBATKN and DDLSTKN. Run Make.
 New files: ILB0909M, ILB0909R, ILB0909S depend on
 BA10SC04 subvolume. Make the ILF conversion using
 ILB0909M. Run the ILF conversion using ILB0909R.
 Replace BNETLOGM, BNETMNWD and BNETUPDT.
 Reference: WO #090331-04

23OCT2009 bensone
 Symptom: Amount fields are not represented correctly by this
 interface under certain currency codes.
 Problem: The interface uses an ISO table to determine decimal
 positions for each of the currencies of the world.
 MasterCard uses an internal table which does not match
 the ISO table in many currency decimal positions.
 The error in decimal placement can occur in acquirer
 and issuer traffic flows.
 Fix: Modified the logic used to format amount fields to
 invoke a new HISWUTL proc that will make the necessary
 corrections to amount fields that are impacted by the
 decimal position inconsistency.
 Modified sections: INSTALLATION AND CONFIGURATION
 INTERFACE FUNCTIONS
 Dependency: New file BNETTBL.
 Apply fixes to ABNTMM, BNETATMS, BNETATMM, BNETG,
 BNETLIBS, BNETM, BNETMM and HISWUTLS. Run Make.
 Replace BNETMNWD
 Reference: Case #929517.

10FEB2010 bensone
Symptom: Question about ICFE GMT parameter for Banknet.
Problem: The use of GMT OFFSET field on Screen 12 of the ICF(E) varies by
 interface.
Fix: The Banknet Interface does not use the GMT OFFSET field in the
 ICF(E) when setting DE 7 in messages sent to Banknet. Additional
 information regarding the setting of DE 7 and the system
 configuration necessary has been provided in this document.
 Modified section: INSTALLATION STEPS
 Added new section: HP NONSTOP CONFIGURATION
Dependency: Replace BNETMNWD.
Reference: Case #1011783

01APR2010 saderc
 Symptom: The Banknet process abends after it is started.
 Problem: The call to NETINIT contains an output length that is greater than the MAXPROMSG setting for the Node.
 Fix: Updated the installation steps to state that the MAXPROMSG setting must be increased to a value equal to or greater than 16384.
 Modified section: INSTALLATION STEPS
 Dependency: Replace BNETMNWD and BNETUPDT.
 Reference: Case #1068665

31MAR2010 swift
 Symptom: Banknet Release 10.1 Release Item LAC 111
 Problem: None
 Fix: The interface has been modified to support PIN Change transactions, where EMV data is not present.
 Procs Modified: cmd^warmboot^retrieve^globals
 cmd^warmboot^store^globals
 init^paramproc
 Dependency: Apply fixes to BNETATMS, BNETG, and BNETS.
 Run Make. Replace BNETUPDT.
 Reference: WO#100305-01

06MAY2010 wielerk
 Symptom: Switch Token description is out of date.
 Problem: None
 Fix: Updated Switch Token description in Appendix A.
 Dependency: Replace BNETMNWD.
 Reference: Case #1072396.

20AUG2010 bensone
 Symptom: Banknet Release 10.2 Mandates.
 Problem: None
 Fix: Add LCONF parameters SW-BNET-CREATE-ILF and SW-BNET-TRAN-TIME-SET.
 Add new CUTOVER timer.
 Remove the LOGONS command and message handling descriptions.
 Add *MoneySend* funding transaction.
 Modify Preauthorization Processing.
 Modified sections: LCONF ASSIGNS AND PARAMS
 INITIALIZATION
 NETWORK CONTROL COMMANDS
 TIMERS
 MESSAGE HANDLING
 INTERFACE FUNCTIONS
 CUTOVER PROCESSING
 TOKEN DOCUMENTATION
 Dependency: Apply fixes to the ATM modules AUTHG and AUTHLIBS.
 Apply fixes to BNETATMS, BNETDDL, BNETG, BNETLIBS and BNETS. Run Make.
 New files: ILB0910M, ILB0910R, ILB0910S depend on BA10SC04 subvolume. Make the ILF conversion using ILB0910M. Run the ILF conversion using ILB0910R.
 Replace BNETLOGM, BNETMNWD and BNETUPDT.
 Reference: WO #100318-04

PREFACE

17SEP2010 bensone
Symptom: Banknet UK Domestic Maestro Technical Alignment
Problem: None
Fix: Banknet will align the UK Maestro program with the
 standard product in two phases. The first phase is a
 partial alignment. By the second phase, the UK Maestro
 program will be supported by a standard Banknet interface.
Modified sections: ICFE SCREEN 3 SETTINGS
 INTERFACE FUNCTIONS
Dependency: Replace BNETMNWD.
Reference: WO #PR001116

10FEB2011 bensone
Symptom: Banknet Release 11.1 Mandates
Problem: None
Fix: Added support for new fields in the Banknet Switch Data token.
 Added support for new subfields within DE 48.
Modified sections: TOKEN DOCUMENTATION
 SUBFIELD DOCUMENTATION
Dependency: Apply fixes to BNETATMS, BNETDDL, BNETEMVS, BNETG,
 BNETLIBS, and BNETS.
 Run Make. Replace BNETLOGM, BNETMNWD, and BNETUPDT.
Reference: WO #001164

10MAY2011 swift
Symptom: Banknet Release 10.1 AFD Completion Advice.
Problem: None
Fix: The Banknet Interface has been enhanced to provide full
 support of Release Item 502, AFD Completion Advice, as
 specified in the Authorization and Clearing Release 10.1
 Document and Authorization and Clearing Release 10.1
 Clarifications.
Modified sections: LCONF ASSIGNS AND PARAMS
 INTERCHANGE COMPLIANCE
Added new section: INTERCHANGE COMPLIANCE FOR AFD
 COMPLETION ADVICES
Dependency: Apply fixes to BNETG, BNETLIBS, BNETS, and
 PS60EXT.PTLFXS. Run Make.
 Replace BNETLOGM, BNETMNWD and BNETUPDT.
Reference: WO #002006

Section 1

BANKNET INTERFACE PROCESS

INTRODUCTION

This section introduces the Banknet Interchange and the Banknet Interface process.

THE BANKNET INTERCHANGE

The Banknet Interchange serves as a link between two networks. It allows a cardholder on one network to initiate a transaction at a terminal belonging to the other network.

Banknet connects an acquirer with an issuer. Both the acquirer and the issuer must conform to Banknet standards. These standards include sending and accepting the appropriate form of the Banknet external message and providing the correct message flows.

Authorization of transactions is performed using online authorization messages through Banknet. An authorization request is sent from the acquirer to the issuer, and the issuer returns an authorization response to the acquirer. The message standards are documented in MasterCard's ***Customer Interface Specification***.

Settlement of transactions is performed using the MasterCard Global Clearing Management System (GCMS).

THE BANKNET INTERFACE PROCESS FUNCTIONS

The Banknet Interface process is responsible for maintaining the relationship between BASE24 and Banknet. In order to maintain this relationship, the Banknet Interface process must perform a variety of functions. These functions include the following:

- ☐ Facilitating message exchange between BASE24 and Banknet.

When BASE24 receives an inbound transaction from Banknet, the Banknet ISO Interface process translates the message into the appropriate internal message and sends the message to a BASE24 Authorization process. When BASE24 receives a transaction and sends it to Banknet, the Banknet Interface process translates the message from the internal message format to the external message format and forwards the message to Banknet.

- ☐ Providing store-and-forward capabilities.

When the link between BASE24 and Banknet is down, the Banknet Interface process stores non-interactive messages intended for Banknet in its Store-and-Forward File (SAF). Non-interactive messages are messages not requiring an immediate response to complete a transaction. When the link is restored, the messages are retrieved from the SAF and sent to Banknet.

- ☐ Facilitating settlement between BASE24 and MasterCard.

The Banknet Interface process processes transactions based on the Banknet Interface's reporting day. Since Banknet's cutover rarely occurs at the same time as the BASE24 cutover, the reporting day and the transactions to be settled will differ between the two. The Banknet Interface process produces a set of interchange reports to reconcile these points of view. These reports allow the BASE24 participants to reconcile the settlement positions of Banknet and BASE24.

- ☐ Managing communications between BASE24 and Banknet.

The Banknet Interface process sends and receives network management messages to determine the status of the link between BASE24 and Banknet. Network management messages include logon, logoff and network connection status.

The Banknet Interface supports the message standards documented in MasterCard's ***Customer Interface Specification***, and can function on behalf of both issuers and acquirers.

An issuer processes transactions on behalf of the card-issuing financial institution. In the case of Banknet, this means that the Banknet Interface process receives transactions from Banknet that require authorization by a BASE24 Authorization process.

An acquirer accepts transaction requests that must be sent to another network for authorization. In the case of Banknet, this means that the Banknet Interface process forwards transactions initiated on the BASE24 network to Banknet for authorization.

SUPPORTED TRANSACTION TYPES

The Banknet Interface process provides support for BASE24-pos transactions. These include:

- ☐ Purchases
- ☐ Pre-Authorization Requests
- ☐ Cash Advances/Withdrawals
- ☐ Mail/Phone Orders
- ☐ Purchases with Cash Back
- ☐ Card Verifications

Any of the above transactions can be sent to Banknet for authorization.

Card verification transactions are forwarded to Banknet as “account status inquiry” requests or “test” recurring payments.

Pre-Authorization Completion messages may be mapped to completion advice messages before being sent to Banknet.

Merchandise return transactions (refunds) may be mapped to payment transaction messages before being sent to Banknet.

Adjustment transactions may be mapped to reversal messages before being sent to Banknet.

BASE24-pos transactions that cannot be sent to Banknet for authorization may be authorized “blind”, or routed to an alternate authorizing destination, (e.g., the POS Router/Authorization process).

SUPPORT FOR ATM TRANSACTIONS

The Banknet Interface process provides support for BASE24-atm transactions for European members only. These include:

- ☐ Cash Withdrawals
- ☐ Balance Inquiry
- ☐ PIN Changes
- ☐ EMV PIN Unblocks

Any of the above transactions can be sent to Banknet for authorization.

Non-currency dispense (NCD) transactions originating from an ATM (including mobile top-ups) are forwarded to Banknet as either normal purchase requests or cash withdrawal requests.

To support ATM processing, a BASE24-atm system is required, and the Banknet ATM extension module must be licensed and installed.

MESSAGE FORMATS

The Banknet Interface process routes messages using two different formats--internal or external.

The internal message format is used whenever information is routed between the Banknet Interface process and the rest of BASE24. For BASE24-atm, the internal message is called the BASE24-atm Standard Internal Message (STM). For more information on the STM, see the ***BASE24-atm Transaction Processing Manual***. For BASE24-pos, the internal message is called the BASE24-pos Standard Internal Message (PSTM). For more information on the PSTM, see the ***BASE24-pos Transaction Processing Manual***.

An internal Pathway message is used whenever information is routed between the BASE24 CRT operations screens and the Banknet Interface process. The following types of messages are generated from BASE24 CRT operations screens:

- ☐ Administrative messages
- ☐ File Requests (MCC102, MCC103, MCC104, MCC105, MCC106, MCC107, MCC108, MCC109)

These messages may also be generated using the automated hot card update facility from a Host system.

The external message format is used whenever information is routed between the Banknet Interface process and the interchange. A message sent in the external message format is called a Standard External Message (SEM). The external message format for authorization messages comprises a subset of the 128 data elements (DEs) defined in the ISO 8583 (1987) Message Standard. Some of these DEs, e.g. DE 48, are divided into sub-elements (SEs).

The EBCDIC character set is used for all non-binary message data passed between the Banknet Interface process and the interchange.

SOFTWARE DEPENDENCIES

The Banknet Interface is a Switch 6.0 product, and therefore operates in a Release 3.4/4.0, a Release 5.0, a Release 5.1, a Release 5.3, or a Release 6.0 BASE24 system.

The BASE24-pos product and/or the BASE24-atm product must be installed in order to process authorization messages. The BASE24-fhm product must be installed in order to send File Action request messages, if the messages are generated using the automated hot card update facility.

To support EMV processing, the BASE24 EMV product must be licensed and installed. For Banknet, EMV support is provided by a separately licensable module, which is built on top of the Switch 6.0 version of the Banknet Interface.

The BASE24 subvolume, PSxxADDR, contains the source file required to support address verification within the POS Router/Authorization process. This subvolume is required to allow address data to be verified during authorization processing.

To provide full support for the encryption of PIN blocks using Triple DES, the Transaction Security Services (TSS) must be installed and configured.

The BASE24 subvolume, SW60SWPM, contains the source file libraries required to support Transactional Interface Monitoring within an Interchange Interface. This subvolume is required for the Banknet Interface to support T.I.M. Transactional Interface Monitoring (T.I.M.) also requires ACI's NET24-OM (XN-NT100) and Compaq's Measure product.

SNA customers must license the 6100SNA Device Support - Primary LU type 0 XPNET product ID XN-p1213. The implementation of SNA LU0 in the Banknet Interface supports 5 stations, Send and Receive on the same station. BASE24 will always be the Primary LU. No protocol characters are added. The "End Bracket" parameter must be set in the Network Environment Source File.

LOG MESSAGE DOCUMENTATION

Log messages generated by the Banknet Interface process are documented on the SW60BNET subvolume in a file named BNETLOGM.

Section 2

FUNCTIONAL OVERVIEW

INTRODUCTION

This section introduces provides an overview of Banknet Interface process functionality. This section also identifies the BASE24 files used by the Banknet Interface.

DATA COMMUNICATION LINES

The Banknet Interface process supports bisynchronous multipoint (3270), bisynchronous point-to-point (2780/3780), X.25, TCP/IP and SNA data communications lines between BASE24 and Banknet. For more information about configuring the data communications lines used by the Banknet Interface process, see Section 3.

BASE24 FILES

The Banknet Interface process uses several files. These files can be divided into control files and transaction files and are identified below.

Control Files

Control files contain configuration information and parameters that determine how the Banknet Interface process performs certain functions. The control files used by the Banknet Interface process include:

- ☐ Acquirer Processing Code File (APCF)
- ☐ Banknet Institution Identification File (BNIIF)
- ☐ Enhanced Interchange Configuration File (ICFE)
- ☐ Interchange Configuration File (ICF)
- ☐ Issuer Processing Code File (IPCF)
- ☐ Key File (KEYF)

- ☐ Key6 File (KEY6)
- ☐ Logical Network Configuration File (LCONF)
- ☐ Station Name File (SNF)
- ☐ Switch Terminal File (STF)
- ☐ Token File (TKN)

For more information on these files and how they are used by the Banknet Interface process, see Section 3.

Transaction Files

Transaction files are used to store information about individual transactions processed through the Banknet Interface process. The transaction files used by the Banknet Interface process include:

- ☐ Interchange Log File (ILF)
- ☐ Store-and-Forward File (SAF)
- ☐ Banknet Request Log File (BNRLF)

For more information on these files and how they are used by the Banknet Interface process, see Section 5.

The ILF and SAF record definitions can be accessed using DDLDOC. For information on DDLDOC, see the ***BASE24 Files and Record Structures Reference Manual***. For more information on the BNRLF user interface see Section 8.

PROCESS CONTROL

The Banknet Interface connects a BASE24 system with Banknet. The communications between the networks are managed by ACI's XPNET software and application-level network management messages.

XPNET

Stations provide the means by which communications takes place between BASE24 and Banknet. All messages sent to Banknet are actually sent to the stations defined for the Banknet process. Likewise, all messages received from Banknet are actually received from stations.

The XPNET process handles all protocol-level functions between BASE24 and the stations, using the communications control software, provided with the HP NonStop (formally Tandem) hardware. The Banknet Interface process sends and receives messages through the XPNET process. It is the responsibility of the XPNET process to maintain the communications links with the stations according to the protocols under which the stations are defined.

Configuration of an XPNET system includes defining attributes for each XPNET node and all of its associated processes, links, lines, stations, and devices. All objects, including the XPNET node itself, can be added, changed, and deleted online without stopping the XPNET system. Configuration information relating to each XPNET node is held in the node's Network Environment File (NEF). There is one NEF for each XPNET node in the system. Each NEF is managed by the XPNET process in that XPNET node.

No two XPNET systems are configured exactly alike. In addition to being composed of different applications, processes, devices, and stations, there are a number of settings that depend entirely on how you want your system to work.

For more information on XPNET configuration for the Banknet Interface process, refer to Section 3.

XPNET entities, such as stations, can be managed by using network control commands. These commands can be issued from the Network Control Supervisor's (NCS) screen or from the NCPcom prompt. For complete information on the syntax of the commands, attributes, and directives used to manage an XPNET system, see the ***NET24-XPNET Network Control Command Reference Manual***.

For more information on XPNET command processing for the Banknet Interface process, refer to Section 4.

Network Management Messages

Network management messages are used to manage the operational status of the communications lines between BASE24 and Banknet. These messages are sent as 0800 messages and require 0810 messages in response. BASE24 supports the following types of network management messages.

- ☐ Logon and Logoff messages
- ☐ Network Connection Status messages
- ☐ PIN Encryption Key (PEK) Exchange messages
- ☐ Host Session Activation and Deactivation messages

For more information on network management message flows in the Banknet Interface process, refer to Section 5.

MESSAGE PROCESSING

The following paragraphs provide an overview of transaction flows for normal and link-down processing. They also identify the message set (different to the transaction set) supported by Banknet.

For more information on transaction message flows in the Banknet Interface process, refer to Section 5.

Normal Message Flow

When a message from Banknet is received by the Banknet Interface process, it is translated from the external message into the appropriate internal message format. The internal message is then sent to the appropriate BASE24 Authorization process. When the message has been processed, the authorizing entity sends an internal response to the Banknet Interface process. Using information from the internal message, the Banknet Interface process formats an external response message and sends it to Banknet.

When a message from BASE24 is received by the Banknet Interface process, it is translated from the internal message into the external message format. The external message is then sent to the interchange for authorization. When the message has been processed, Banknet sends an external response to the Banknet Interface process. Using information from the external message, the Banknet Interface process formats an internal response message and sends it to BASE24.

Link-Down Message Flow

If the link to Banknet is down and an alternate destination is available, the link-down situation is handled by using alternate routing to send the transaction to an alternate destination. If an alternate destination is not available, the Banknet Interface process responds to a link-down situation by denying all interactive transactions. All non-interactive messages (e.g., reversals) are logged to the Store-and-Forward File (SAF).

Supported Message Types

The Banknet Interface process supports the following internal and external message types:

Type	Description	Internal	External
0100	Authorization Request		X
0110	Authorization Request Response		X
0120	Authorization Advice		X
0130	Authorization Advice Acknowledgement		X
0180	Authorization Acknowledgement		X
0190	Authorization Negative Acknowledgement		X
0200	Financial Transaction Request	X	
0210	Financial Transaction Request Response	X	
0220	Financial Transaction Advice	X	
0310	Acquirer File Update Request Response		X
0302	Issuer File Update Request		X
0312	Issuer File Update Request Response		X
0400	Acquirer Reversal Request		X
0410	Acquirer Reversal Response		X
0420	Acquirer Reversal Advice	X	X
0430	Acquirer Reversal Advice Response		X
0620	Administrative Advice		X
0630	Administrative Advice Response		X

Type	Description	Internal	External
0800	Network Management Request		X
0810	Network Management Request Response		X
0820	Network Management Advice		X

TRANSACTION PROCESSING FUNCTIONS

The Banknet Interface connects a BASE24-atm or BASE24-pos network with MasterCard's Banknet authorization network. It allows a MasterCard cardholder to perform a transaction at a terminal belonging to a network not operated by the card issuer.

To facilitate transaction message exchange, the Banknet Interface supports a number of transaction processing functions. These functions allow the Banknet Interface to comply with the requirements documented in MasterCard's ***Customer Interface Specification***, and to ensure that the necessary transaction information is provided to BASE24.

For more information on these functions, refer to Section 6.

INTERCHANGE REPORTING

The Banknet Interface produces two reports: SETL01 and STAT09. The purpose of these reports is to facilitate settlement between BASE24 and Banknet.

For more information on these reports, refer to Section 7.

ADMINISTRATIVE PROCESSING FUNCTIONS

BASE24 supports several Banknet-specific operations that are performed from BASE24 CRT screens. These operations include:

- ☐ Creating and sending administrative messages to Banknet
- ☐ Perusing administrative messages sent to or received from Banknet
- ☐ File maintenance request messages

For more information on these functions, refer to Section 8.

File maintenance request messages can also be sent to Banknet by using the automated hot card update facility. The BASE24 automated hot card update function allows a Host system to initiate a Card Status Update message, which is then sent to as many as three interchanges, and may also update a cardholder file on BASE24.

For more information on administrative message flows (including automated hot card updates) in the Banknet Interface process, refer to Section 5.

INTERCHANGE PREFIX FILE (IPF) REFRESH

The BASE24 Banknet Interchange Prefix File (IPF) Refresh module, associated with the BASE24 Banknet Interchange Interface, creates a file containing all MasterCard BINS, allowing an acquiring BASE24 system to route transaction messages to Banknet.

The ATM Authorization process and the POS Router module use the SPROUTE file for routing transactions to the appropriate interchange interface. One of the files used to create SPROUTE records is the Interchange Prefix File (IPF), which contains card prefixes supported by individual authorization networks.

The Banknet IPF Refresh module runs as a stand-alone batch process. It creates a standard BASE24 Interchange Prefix File (IPF), using the MasterCard Optimized Current Parameters File as input.

For more information on the Banknet IPF Refresh module, see Section 9.

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Section 3

INSTALLATION AND CONFIGURATION

INTRODUCTION

This section lists the steps required to install the Banknet Interface process. In addition, it provides guidelines for configuring the required control files.

BANKNET INTERFACE PROCESS FILES

The Banknet Interface comprises the files on the SW60BNET subvolume and (optionally) the SW60IBNT subvolume (if EMV transactions are supported) and the SW60ABNT subvolume (if ATM transactions are supported).

SW60BNET SUBVOLUME

The following is a list of all files on the SW60BNET subvolume and a brief description of each file.

☐ BNET	-	EXECUTABLE OBJECT FOR BNET INTERFACE
☐ BNETCOB	-	DDL COBOL FILE SOURCED IN BY REPORTS AND REQUESTERS
☐ BNETD	-	DUMMY PROCEDURE FILE FOR BNET INTERFACE
☐ BNETDDL	-	MAKE FILE FOR BNET DDLS
☐ BNETDDL	-	DDL SOURCE FILE USED TO CREATE DDLCOB, DDLFUP
☐ BNETE	-	EXTERNALS FOR BNETS SOURCE
☐ BNETFM	-	MAKE CONFIGURATION FILE
☐ BNETFUP	-	DDL FUP FILE USED TO CREATE BNET FILES
☐ BNETG	-	GLOBALS SOURCED IN BY BNETS
☐ BNETLIBE	-	EXTERNALS FOR BNETLIBS SOURCE
☐ BNETLIBO	-	NON-EXECUTABLE OBJECT FOR BNET INTERFACE
☐ BNETLIBS	-	SOURCE FILE FOR BNET INTERFACE
☐ BNETLOGM	-	LOG MESSAGE DOCUMENTATION FILE
☐ BNETM	-	MAKE FILE FOR BNET INTERFACE
☐ BNETMAN	-	TGAL VERSION OF THIS MANUAL (NOW OBSOLETE)
☐ BNETMM	-	MACRO DEFINITION FILE FOR BNET INTERFACE
☐ BNETMNWD	-	SOURCE FILE OF THIS MANUAL
☐ BNETO	-	NON-EXECUTABLE OBJECT FOR BNET INTERFACE

☐	BNETTBL	-	BANKNET CURRENCY EXPONENT TABLE
☐	BNETS	-	SOURCE FILE FOR BNET INTERFACE
☐	BNETTAL	-	DDL TAL FILE SOURCED IN BY BNETS/BNETLIBS
☐	BNETUPDT*	-	ENHANCEMENT UPDATE DOCUMENTATION
☐	RP09BNT	-	EXECUTABLE OBJECT FOR BNET REPORT 9
☐	RP09BNTM	-	MAKE FILE FOR BNET REPORT 9
☐	RP09BNTS	-	SOURCE FILE FOR BNET REPORT 9
☐	RQBNADMM	-	MAKE FILE FOR BNET ADMIN REQUESTER
☐	RQBNADMO	-	PSEUDO FILE FOR RQBNADMS
☐	RQBNADMS	-	SOURCE FILE FOR BNET ADMIN REQUESTER
☐	RQBNETM	-	MAKE FILE FOR BNET ICF/ICFE SWITCH DEPENDENT SCREEN REQUESTER
☐	RQBNETO	-	PSEUDO FILE FOR RQBNETS
☐	RQBNETS	-	SOURCE FILE FOR BNET ICF/ICFE SWITCH DEPENDENT SCREEN REQUESTER
☐	RQBNIIFM	-	MAKE FILE FOR BNET IIF REQUESTER
☐	RQBNIIFO	-	PSEUDO FILE FOR RQBNIIFS
☐	RQBNIIFS	-	SOURCE FILE FOR BNET IIF REQUESTER
☐	RQBNRLFM	-	MAKE FILE FOR BNET RLF REQUESTER
☐	RQBNRLFO	-	PSEUDO FILE FOR RQBNRLFS
☐	RQBNRLFS	-	SOURCE FILE FOR BNET RLF REQUESTER
☐	SCRNBADM	-	SOURCE FILE FOR BADM SCREEN
☐	SCRNBIIF	-	SOURCE FILE FOR BIIF SCREEN
☐	SCRNBNET	-	SOURCE FILE FOR BNET SCREEN
☐	SCRNBRLF	-	SOURCE FILE FOR BRLF SCREEN
☐	SVBNIIF	-	OBJECT FILE FOR BNET IIF SERVER
☐	SVBNIIFM	-	MAKE FILE FOR BNET IIF SERVER
☐	SVBNIIFS	-	SOURCE FILE FOR BNET IIF SERVER
☐	SVBNRLF	-	OBJECT FILE FOR BNET RLF SERVER
☐	SVBNRLFM	-	MAKE FILE FOR BNET RLF SERVER
☐	SVBNRLFS	-	SOURCE FILE FOR BNET RLF SERVER
☐	SWBNETM	-	MAKE FILE FOR SUBVOLUME SW60BNET

* This file will only exist when enhancements or major fixes have been made to the Banknet Interface.

SW60IBNT SUBVOLUME

The following is a list of all files on the SW60IBNT subvolume and a brief description of each file.

- ☐ BNETEMVE - EXTERNALS FOR BNETEMVS SOURCE
- ☐ BNETEMVM - MAKE FILE FOR BNET EMV MODULE
- ☐ BNETEMVO - NON-EXECUTABLE OBJECT FOR BNET EMV MODULE
- ☐ BNETEMVS - SOURCE FILE FOR BNET EMV FUNCTIONS
- ☐ IBNTFM - MAKE CONFIGURATION FILE
- ☐ IBNTMAN - INSTALLATION GUIDE FOR BNET EMV SUBVOLUME
- ☐ IBNTMM - MACRO DEFINITION FILE FOR BNET EMV SUBVOLUME
- ☐ SWIBNTM - MAKE FILE FOR SUBVOLUME SW60IBNT

SW60ABNT SUBVOLUME

The following is a list of all files on the SW60ABNT subvolume and a brief description of each file.

- ☐ ABNTFM - MAKE CONFIGURATION FILE
- ☐ ABNTMM - MACRO DEFINITION FILE FOR BNET ATM SUBVOLUME
- ☐ BNETATME - EXTERNALS FOR BNETATMS SOURCE
- ☐ BNETATMM - MAKE FILE FOR BNET ATM MODULE
- ☐ BNETATMO - NON-EXECUTABLE OBJECT FOR BNET ATM MODULE
- ☐ BNETATMS - SOURCE FILE FOR BNET ATM FUNCTIONS
- ☐ SWABNTM - MAKE FILE FOR SUBVOLUME SW60ABNT

INSTALLATION STEPS

During the course of the installation, it may be necessary to refer to the following manuals:

- ☐ ACI MAKE User Guide
- ☐ BASE24-atm EMV Support Manual
- ☐ BASE24-atm Transaction Processing Manual
- ☐ BASE24-pos EMV Support Manual
- ☐ BASE24-pos Transaction Processing Manual
- ☐ BASE24 Base Files Maintenance Manual
- ☐ BASE24 Device Control Manual
- ☐ BASE24 Logical Network Configuration File Manual
- ☐ BASE24 Text Command Reference Manual
- ☐ BASE24 Tokens Manual
- ☐ BASE24 Transaction Security Manual
- ☐ NET24-XPNET Network Control Command Reference Manual
- ☐ NET24-XPNET Network Control Operations Guide
- ☐ Tandem's Communication Management Interface Operator Reference Manual
- ☐ Tandem's Subsystem Control Facility (SCF) Reference Manual
- ☐ Tandem's System Management Manual
- ☐ Tandem's Guardian Programmer's Guide

The installation steps are shown below.

Please note the change in step 7 concerning the setting of the server process. Originally, SERVER-BNPATH was configured to send messages to a device, but now it should be configured to send messages to a process. The configuration for a process uses #p instead of #d in the line SET SERVER PROCESS. If the server is configured to send messages to a device, then an escape character is inserted in front of the message. The interface no longer expects the escape character in front of the message.

1. Restore the software tape to the appropriate volume and subvolume by entering the following command at your TACL prompt:

```
RESTORE /OUT $S.#BNET/ $TAPE, *.*.*,LISTALL,TAPEDATE,
        OPEN,MYID,VOL <$your vol>
```

2. Configure a Tandem port and line via a network control facility. If a Tandem communication subsystem controller (CSS 6100) is used, the line should be configured using the EBCDIC character set qualifier.
3. Volume to \$<vol>.BA60MAKE and edit CUSTMACS
 - a. Set CLASSIC_ON to true.
 - b. Set SWITCH_ON to true.
 - c. Set SWI_RPT_ON to true if intending to use the Banknet Interface reports.
 - d. Set SDF_ON to true.
 - e. Set STF_ON to true.
 - f. Set BNET_ON to true.
 - g. Set BNET_ENHANCED_MODE_ON to true.
 - h. Set BNET_EMV_ON to true if required.
 - i. Set SW_PRE_60_LCONF_ON to false.

Set SWITCH_PERFMON_ON to true if required.

NOTE: It is recommended that you use B24C/B24M files so all necessary modules are rebuilt. If using B24C/B24M files, part bin step 4 may be skipped.

4. Volume to \$<vol>.SW60BNET and perform the following:
 - a. Edit the macro Make file BNETMM and ensure that the system, volume and subvolume references are correct.
 - b. Make the SW60BNET subvolume by entering the following command at the TACL prompt:

```
RUN $<vol>.<subvol>.makevXXX/OUT $S.#BNET/SWBNETM -FORCE
where: XXX is the Make version.
```

5. If using the optional BNET Currency Exponent Table(BNETTBL):
 - a. Volume to SW60BNET. Edit BNETTBL and add the currency exponents you need as defined in the MasterCard Quick Reference Booklet. Only add currencies that differ in their exponent and that you process. See the description of BNET Currency Exponent Table in Section 6 of this manual.
6. If part b in step 4 above was skipped in order to run the B24C/B24M files, perform the following:
 - b. Volume to \$<vol>.SW60SRC.
 - c. Edit configuration Make files SWICSEFM and SWIFM and ensure that the system, volume and subvolume references are correct.
 - d. Volume to \$<vol>.BA60SRC.
 - e. Edit B24C and ensure that the -force option is NOT included in the Make command unless this is a first time install.
 - f. Make all necessary modules on your BASE24 system by entering the following command at the TACL prompt:

/IN B24C/

7. Make will move the object files to the locations specified in the configuration Make file if the move_obj_subvol_on define is set to TRUE. Whoever runs the Make must have security to write to the subvolumes where Make attempts to place the objects.
8. Add the ICF/ICFE, BNPATH, BNIIF, and BNRLF servers to the PATHCONF file.

The following are server definition examples. Different values can be used for various fields. Use previous server definitions in the PATHCONF to assist in determining values.

```
RESET SERVER
SET SERVER CPUS          x:y
SET SERVER PROGRAM      \<sys>.$<vol>.xxxxOBJ.SVICFE
SET SERVER DELETEDELAY  10 MINS
SET SERVER PRI          145
SET SERVER NUMSTATIC     0
SET SERVER HOMETERM     \<sys>.$PLOG
SET SERVER TIMEOUT      60 SECS
ADD SERVER SERVER-ICFE
```

```
( This BNPATH server definition example is for XPNET 2.1 or above)
( For additional configuration information regarding              )
( SERVER-BNPATH for XPNET 2.1 or above, refer to the            )
( XPNET CTS/E PWS XN-PL320-01 Implementation Guide.          )
```

```

RESET SERVER
SET SERVER AUTORESTART      0
SET SERVER CPUS             x:y (must be valid CPUs, e.g., 0:1)
SET SERVER PRI              140
SET SERVER PROGRAM          $SYSACI.XPNET.PATH
SET SERVER PROCESS          $S1AP.#BNETxxx (ASSOCIATIVE ON)
SET SERVER LINKDEPTH        1
SET SERVER MAXLINKS         1
SET SERVER MAXSERVERS       1
SET SERVER NUMSTATIC        1
SET SERVER TMF              OFF
SET SERVER TIMEOUT          60 SECS
SET SERVER CREATEDELAY      30 SECS
SET SERVER DEBUG            OFF
SET SERVER DELETEDELAY      1 MINS
SET SERVER PARAM ETX        "yes"
SET SERVER PARAM GENERIC    "no"
SET SERVER PARAM MODE       "path"
SET SERVER HOMETERM         \<sys>.$PLOG
SET SERVER VOLUME           \<sys>.$<volume>.<lnet>CNTL
ADD SERVER SERVER-BNPATHxxx

```

Where \$S1AP is any valid PPD name. When SERVER-BNPATHxxx is started, a path process with this PPD name will be created which will pass messages from pathway to the Banknet Interface process.

xxx is the optional Banknet Interface process number. If used, the process number field on the pathway screen must contain the same value.

The ASSOCIATIVE ON param instructs PATHMON that the process may have been started external to PATHWAY and, if it has, the existing process is to be adopted by PATHWAY instead of trying to start up a process of its own. It is typical that the SERVER-BNPATH process is started external to PATHWAY using an OBEY file.

The Server Process PPD name (\$S1AP in the example above) and the port attribute in the Banknet Server Path line declaration in the Network Environment File (NEF) must be the same.

```

RESET SERVER
SET SERVER CPUS             x:y (must be valid CPUs, e.g., 0:1)
SET SERVER PROGRAM          $<vol>.SW600BJ.SVBNIIF
SET SERVER DELETEDELAY      10 MINS
SET SERVER PRI              145
SET SERVER NUMSTATIC        0
SET SERVER HOMETERM         \<sys>.$PLOG
SET SERVER TIMEOUT          60 SECS
SET SERVER VOLUME           \<sys>.$<volume>.<subvol>
ADD SERVER SERVER-BNIIF

```

```
RESET SERVER
SET SERVER CPUS           x:y
SET SERVER PROGRAM        $<vol> .SW600BJ.SVBNRLF
SET SERVER DELETEDELAY    10 MINS
SET SERVER PRI            145
SET SERVER NUMSTATIC      0
SET SERVER HOMETERM       \<sys>.$PLOG
SET SERVER TIMEOUT        60 SECS
SET SERVER VOLUME         \<sys>.$<volume>.<subvol>
ADD SERVER SERVER-BNRLF
```

Upon completion of the changes, a PATHCOLD should be performed, generating the new PATHCTL file. A REFRESH-COLD<tcp> will also be required.

The modifications noted above may be done online via PATHCOM as well.

9. Ensure that the following are defined in the Network Environment File (NEF). If the following are not defined, add the declarations via a network control facility.

- a. A Banknet process declaration
- b. At least one Banknet station declaration (Banknet can support 16 stations.)
- c. A Banknet device declaration
- d. At least one Banknet line declaration (Banknet can support 16 lines.)
- e. A Banknet Server Path device declaration (XPNET 2.1 or greater)

A Banknet Server Path line declaration (XPNET 2.1 or greater) which has as a PORT \$S1AP.#BNETxxx.P1 where:

- * \$S1AP.#BNETxxx matches the PPD of the SERVER-BNPATH process
- * xxx (optional) is the Banknet interface process number to receive the pathway requests
- * P1 is an additional qualifier used by XPNET for internal configuration purposes.

The PROTOCOL for this line must be CTS.

- f. A Banknet Server Path station declaration (XPNET 2.1 or greater) specifying:
 - * the Banknet interface process as the DESTINATION
 - * the Banknet Server Path device specified in e) as the DEVICENAME
 - * the Banknet Server Path line defined in f) as the LINENAME.

For details on the previous declarations, refer to the **NET24-XPNET Network Control Command Reference Manual**.

NOTE 1: If 3270 multipoint protocol is to be used, only one station and one line may be configured. Poll and select characters for the station must be "4040 4040" and "6161 4040", respectively. These may be entered in the Network Environment File using the ADDRESS declaration. An example of this declaration for an environment using 3270-multipoint is as follows:

```
ADDRESS " ":/ "
```

NOTE 2: The node attribute MAXPROMSG specifies the maximum I/O buffer size (in bytes) for application processes on the node. To support application messages of 4000 bytes (as required by MasterCard), it is recommended that this attribute is set to a value greater than or equal to 16512.

The following commands are used to add a process, line, device and station. They are sent to the XPNET 3.0 Network from a network control facility. These are only examples and many of the input values must be determined based on your own network. If necessary, refer to the **NET24-XPNET Network Control Operations Guide**.

```
RESET PROCESS
SET PROCESS BCPU 3
SET PROCESS PROGRAM $<vol>.xxxxOBJ.BNET
SET PROCESS LIBRARY $SYSACI.XPNET.SKELB
SET PROCESS PPD $BNET
SET PROCESS PRIORITY 135
SET PROCESS CPU 2
SET PROCESS STARTUP AUTOMATIC
SET PROCESS QAT 100
SET PROCESS QMT 150
SET PROCESS SAVEABEND OFF
SET PROCESS QMI 90
ADD PROCESS P1A^BNET, UNDER SYSNAME \CFS, UNDER NODE P1A^GATE
```

The following device, line and station samples are for non-SNA protocols:

```
RESET DEVICE
SET DEVICE RECVLEN 4128 (see Note)
SET DEVICE XMITLEN 4128 (see Note)
SET DEVICE TYPE 29
ADD DEVICE D1A^BNET, UNDER SYSNAME \CFS, UNDER NODE P1A^GATE
```

```
RESET LINE
SET LINE ATIMER 0
SET LINE BTIMER 0
SET LINE ERANALYSIS 30
SET LINE PORT $BNET
SET LINE TIMEOUT 100
SET LINE LCLLUNAME APCLU1
SET LINE PRTLUNAME PARTLU1
SET LINE UCRATE LU62LINRATE
SET LINE UCSTATE LU62LINSTANCE
```

```
SET LINE UCQUEUE LU62LINQUEUE
SET LINE PROTOCOL LU6.2
SET LINE RETRY 3
ADD LINE L1A^BNET, UNDER SYSNAME \CFS, UNDER NODE P1A^GATE

RESET STATION
SET STATION DESTINATION P1A^BNET
SET STATION DEVICE BNET^DEV
SET STATION LINENAME \SYS1.P1A^NODE.L1A^BNET
ADD STATION S1BNET01, UNDER SYSNAME \CFS, UNDER NODE P1A^GATE
```

NOTE: The device attributes XMITLEN and RECVLEN specify the maximum length (in bytes) of messages transmitted between a device and the XPNET process. To support application messages of 4000 bytes (as required by MasterCard), these attributes must be set to appropriate values.

For more information about configuring the data communications lines used by the Banknet Interface process, see Appendix D.

10. Add the necessary assigns and params to the LCONF for the Banknet interface using the information provided in Section 3 of this document and in the ***BASE24 Logical Network Configuration File Manual***.

In addition to these assigns and params, also add the symbolic process name of the Banknet Interface as one of the "setl-notify" params and as one of the "warm" params if this process is to be warmbooted with a WARMBOOT ALL command.

11. Add an ICF or ICFE record for the Banknet interface process using the information provided later in this document and in the ***BASE24 Base Files Maintenance Manual***.
12. Add a KEYF or KEY6 record for the Banknet interface process using the information provided later in this document and in the ***BASE24 Base Files Maintenance Manual*** and ***BASE24 Transaction Security Manual***.
13. If Cardholder-activated terminals (CAT) are being used with this interface, one record for each terminal must be entered in the Switch Terminal File (STF).
14. Move to the location of the template files used by the network. Create the template files by entering the following commands:

```
FUP /IN $<vol>.SW60BNET.BNETFUP, OUT $S.#BNETFUP /
```

LCONF ASSIGNS AND PARAMS

The Banknet Interface process reads the LCONF whenever it is initialized or reinitialized for the assigns and params required during processing. The assigns and params required by the Banknet Interface process are described below.

Refer to the BASE24 *Logical Network Configuration File (LCONF) Manual* for instructions on adding them to the LCONF.

NOTE: Some of these assigns and params are used throughout BASE24. Therefore, they may already be defined in the LCONF.

Assigns

The following assigns are used by the Banknet Interface process. The assigns marked with asterisks (**) are required if reports are to be generated.

APCFEMT

The physical file name of the Acquirer Processing Code File Extended Memory Table. **Required.**

BNIIF

The fully-qualified file name of the Banknet Institution Identification File. **Required.**

BNRLF

The fully-qualified file name of the Banknet Request Log File. **Required.**

BNRLF assigns also require that a fully-qualified template file name be specified in the TEMPLATE field of the LNCF assign screen. The Banknet Interface uses the specified template file to create the BNRLF.

ICF

The fully-qualified file name of the Interchange Configuration File. **Required if the BASE24-REL param is not present or does not contain the value of 60.**

ICFE

The physical file name of the Enhanced Interchange Configuration File. **Required if the BASE24-REL param contains the value of 60.**

ILF

The fully-qualified file name of the Interchange Log File. **Required.**

Each interface process must have its own ILF assign defined in the LCONF. ILF assigns are tied to interface processes by placing the symbolic name of the interface process in the PROCESS NAME field of the LNCF assign screen.

ILF assigns also require that a fully-qualified template file name be specified in the TEMPLATE field of the LNCF assign screen. BASE24 uses the specified template file to create ILFs.

IPCFEMT

The physical file name of the Issuer Processing Code File Extended Memory Table. **Required.**

KEYF

The physical file name of the Key File. **Required if** the BASE24-REL param is present and contains the value of 50 or greater and LCONF parameter DES-TRIPLE-SINGLE does not exist or does exist with a value of 'N'.

KEY5

The physical file name of the Key File. **Required if** the BASE24-REL param is not present or does not contain the value of 50 or greater. In a Release 3.4/4.0 system this assign is used by the interface to identify the Key File.

KEY6

The physical file name of the Key6 File. **Required if** the BASE24-REL param is present and contains the value of 50 or greater and LCONF parameter DES-TRIPLE-SINGLE has the value 'Y'.

LMCF

The physical file name of the Log Message Configuration File. This assign is **optional** for the Banknet Interface.

RPT **

The physical file name of the Report Perusal File used by BASE24-atm.

RPT assigns also require that a fully-qualified template file name be specified in the TEMPLATE field of the LNCF assign screen. BASE24 uses the specified template file to create Report Perusal Files.

RPT-ERROR-FILE **

The name used when creating the file to receive report error messages.

SAF

The physical file name of the Store-and-Forward File. This assign is **required**.

Each interface process must have its own SAF assign defined in the LCONF. SAF assigns are tied to interface processes by placing the symbolic name of the interface process in the PROCESS NAME field of the LNCF assign screen.

SAF assigns also require that a fully-qualified template file name be specified in the TEMPLATE field of the LNCF assign screen. BASE24 uses the specified template file to create SAFs.

SDF

The physical file name of the Switch Dispute File. This assign is **optional** for the Banknet Interface.

SECURE-DEV1

The name of the primary security module used by the interface process. For security devices other than the Atalla A6000, this assign specifies the Tandem SYSGEN device name. For Atalla A6000 security devices, this assign specifies the PPD name given to the A6000 interface module used with the security device. The device name or interface module name must be compatible with Tandem's Expand product.

If an Atalla Security Module (ASM) or Atalla Network Security Processor (NSP), model A5000 or A6000, is being used, the SECURE-DEV1 assign requires an additional value in the USER FIELD field of the assign screen. For ASMs, the value in the USER FIELD field must indicate the model type (e.g., A1, A2, or A3). For NSPs, the value must be NSP.

If the interface is running in a Release 6.0 network environment using Transaction Security Services, this parameter must contain the PPD name of the primary IAS process and may be process specific. **Required.**

SECURE-DEV2

The name of the secondary security module used by the interface process. This assign is identical the SECURE-DEV1 assign except that it identifies a secondary or backup module for the device defined in SECURE-DEV1. If the interface is running in a Release 6.0 network environment using Transaction Security Services, this parameter must contain the PPD name of the secondary IAS process and may be process specific. **Optional.**

SNF

The physical file name of the Station Name File. This assign is **optional** for the Banknet Interface. If the SNF assign already exists, then the interface will use that assign. A maximum of 16 stations can be defined using the SNF. Stations can be defined in the ICF/E or in the SNF but not both.

STF

The physical file name of the Switch Terminal File. This assign is **optional** for the Banknet Interface. If the STF assign already exists, then the interface will use that assign.

SW-RPT-OBEY-FILE **

The physical file name of the obey file used to start the interchange reports. The interface process creates its report obey files using this name.

Each interface process must have its own SW-RPT-OBEY-FILE assign defined in the LCONF. SW-RPT-OBEY-FILE assigns are tied to interface processes by placing the symbolic name of the interface process in the PROCESS NAME field of the LNCF assign screen.

SW-RPT01-PROG-NAME **

The physical file name of the executable object code that generates the interface's SETL01 report program.

SW-RPT09-PROG-NAME **

The physical file name of the executable object code that generates the interface's STAT09 report program.

The Banknet Interface has its own RPT09 report program and does not use the standard BASE24 interface report. Therefore, each interface process must have its own SW-RPT09-PROG-NAME assign defined in the LCONF. SW-RPT09-PROG-NAME assigns are tied to interface processes by placing the symbolic name of the interface process in the PROCESS NAME field of the LNCF assign screen.

TKN

The physical file name of the Token File. This assign is **optional** and is only used if the BASE24-REL param is present and contains the value of 50 or greater. This assign is only used in a Release 5.0 or above system.

Params

The following params are used by the Banknet Interface process. The params marked with asterisks (**) are required if reports are to be generated, and the params marked with one asterisk (*) are optional if reports are to be generated.

ACCT-NUM-TYP *

A value (B or D) indicating whether account numbers are stored in binary (B) or decimal (D) format. Currently, only decimal format is supported, allowing for account numbers of up to 19 digits in length. This parameter is used for report processing.

ACTVT-INTERVAL-TIMER

The amount of time, in hundredths of seconds, used to maintain a timer for the purpose of sending session activation messages. When the timer expires, an activate message will be sent to each station the interface has configured and is marked "UP". This param is used only if enhanced session management is configured. Valid values are 1 through 8640000 (default 360000). Example: 360000 equals 1 hour.

ATM-LN-CUTOVER

The time (HH:MM) at which all BASE24-atm processes in the logical network are cut over. All institutions must be cut over by this time. Valid values, in military time, are from 00:00 to 23:59 (no default value). This param is **required** if processing BASE24-atm transactions.

BASE24-REL

This parameter indicates the current release of the BASE24 system. Valid values are:

40 = BASE24-atm Release 4.0 and/or BASE24-pos Release 3.4

50 = BASE24 Release 5.0/5.1.2/5.3

60 = BASE24 Release 6.0 or above

BNET-DEBIT-CARD-TYPES

This parameter specifies up to 35 card types, identifying debit card products, e.g. Maestro (but NOT Debit MasterCard), on which acquirer processing specific to debit cards will be performed for POS transactions.

Card types are validated to be two characters in length. A numeric or space character will only be allowed for the second byte of any card type. The card types must be specified as a string of up to 70 characters; for example, card types PM, P, P1 would be configured as "PMP P1".

The default value is a single entry of 'MD' (MasterCard Debit).

BNET-SPPT-NON-EMV-PIN-CHANGE

This parameter allows Latin America and Caribbean (LAC) Region issuers to specify that PIN Change processing is supported for non-EMV transactions. Valid values are:

Y = Allows PIN Change processing for non-EMV transactions

N = Does not allow PIN Change processing for non-EMV transactions (default)

BNET-STAND-IN-RCS

This parameter specifies up to 35 Banknet response codes, identifying response codes for which alternate routing will be invoked for POS transactions.

Response codes are two numeric characters. The response codes must be specified as a string of up to 70 characters; for example, response codes 31, 91, 96 would be configured as "319196". Typically, the param will contain response codes indicating that the issuer could not process the message.

The default value is spaces, indicating that no alternate routing will be performed when a response message is received from Banknet.

BNRLF-NOTIFY-INTERVAL

The number of writes made to the Banknet Request Log File (BNRLF) between calculating the percentage of file full. Each time a write is made to the BNRLF, a counter is incremented. When the counter is equal to the value in this param, a calculation is performed to determine how full the BNRLF is. The counter is then reset to zero. When the file reaches the percent full specified in the BNRLF-NOTIFY-PERCENTAGE param, a message is logged indicating the file is becoming full. Until more file space is made available, the interface process continues to log messages that the file is becoming full at the interval specified in this param. Valid values are 10 through 32766 (default 50).

BNRLF-NOTIFY-PERCENTAGE

The percentage of used file space at which the interface process logs a message indicating the BNRLF is becoming full. The interface process continues to log messages at the interval indicated in the BNRLF-NOTIFY-INTERVAL param until more file space is made available. Valid values are 10 through 99 (default 70).

DATE-DISPLAY-TYPE *

This parameter indicates the format of the month and day portion of the dates on reports. When dates include the year, this param does not affect the year portion of the date. Valid values are:

1 = DDMM

2 = MMDD (default)

DENY-FAILED-ON-BEHALF

This param allows issuers to decline inbound requests when DE 48 sub-element 71 indicates that the transaction failed pre-validation performed by MasterCard. If this param is set to 'Y', then the params ON-BEHALF-DENY-RSLT-CDESxx (where xx are number) specify the exact conditions under which the interface will deny external request messages. Valid values are:

Y = Allow inbound requests to be declined when DE 48 SE 71 is present (default).

N = Do not decline.

DENY-IF-ATC-DISCRP

This param allows issuers to decline inbound requests when DE 48 sub-element 34 indicates that the ATC is outside the values allowed by the issuer. Valid values are:

Y = Decline inbound requests when DE 48 sub-element 34 indicates that the ATC is outside the values allowed by the issuer.

N = Do not decline (default).

DENY-MC-E-COMM-TXN

This parameter indicates whether or not the interface should deny inbound Authorization Request/0100 transactions that do not participate in MasterCard Electronic. Valid values are:

Y = Deny inbound Authorization Request/0100 transactions from acquirers that do not participate in MasterCard Electronic. Do not forward these transactions to BASE24-pos Router Authorization (default)

N = Do not deny inbound Authorization Request/0100 transactions from acquirers that do not participate in MasterCard Electronic. Forward these transactions to BASE24-pos Router Authorization.

DES-TRIPLE-SINGLE

This parameter indicates if Triple to Single DES encryption is **required**. Valid values are:

Y = Triple to Single DES encryption is required (KEY6 assign must exist)

N = Triple to Single DES encryption is not required (KEYF assign must exist)

EKM-ACQ-PEK-INDEX

The enhanced key management acquirer PIN encryption key index. A value indicating the specific PIN key to be used when more than one key is available in a PIN key set. This value will be used in subfield 4 of DE 53 (Security Related Control Information) in authorization request messages containing a PIN block.

Data Representation: n-4

Default value: 0003

EKM-ISS-PEK-INDEX

The enhanced key management issuer PIN encryption key index. A value indicating the specific PIN key to be used when more than one key is available in a PIN key set. This value will be used in subfield 4 of DE 53 (Security Related Control Information) in the Issuer's network management logon message.

Data Representation: n-4

Default value: 0003

EKM-PIN-BLOCK-FORMAT

The enhanced key management PIN block format. A value indicating the type of PIN block format used. This value will be used in subfield 3 of DE 53 (Security Related Control Information) in an Issuer's network management logon message and in an Acquirer's authorization request messages containing a PIN block.

Data Representation: n-2

Default value: 10

ENHNC-KEY-MGMT

A value indicating whether or not the interface is to include DE 53 (Security Related Control Information) in network management request messages and in authorization request messages containing a PIN block. Valid values are:

Y = Send DE 53

N = Do not send DE 53 (default)

ENHNC-SESSION-MGMT

A value indicating whether or not the interface is to support enhanced session management processing which includes network control commands ACTVT STA <station name> and DEACTVT STA <station name>. Valid values are:

Y = Support Enhanced Session Management messages

N = Do not support Enhanced Session Management messages (default)

FALLBACK-ALWD

A value indicating whether an EMV card is permitted to fallback to magnetic stripe for ATM transactions, cash withdrawals from bank branch terminals, and transactions initiated at CAT Level 1 and CAT Level 2 devices. Valid values are:

Y = Banknet processing permits Fallback transactions (default)

N = Banknet processing does not permit Fallback transactions

FIRST-BNET-TRACE-NUM

This value specifies the initial numeric value to be used for the System Trace Audit Number (STAN) by the Banknet Interface process. Valid values are 1 through 999999 (default 1).

FRWD-DCLN-ADVC

This parameter indicates whether the interface is to format an internal declined advice message from an external advice message, and forward the message to BASE24. This occurs either when a declined external advice is received, or when an external request that is declined by the interface for any reason other than a formatting error. Valid values are:

Y = Forward the declined advice to BASE24.

N = Log the declined advice to the ILF and drop the message (default).

FRWD-UNMATCHED-RVSL

This parameter indicates whether the interface is to format an internal reversal message from an unmatched external reversal message, and forward the message to BASE24. This occurs when the original transaction cannot be found on the ILF. Valid values are:

Y = Forward the unmatched external reversal message to BASE24.

N = Log the unmatched external reversal message to the ILF and drop the message (default).

GREATER-AMT-APPROVAL-ALWD

The parameter indicates whether the amount authorized may be more than the value requested for partial approval. It applies to BASE24-pos issuer processing only. Valid values are:

Y = The BASE24-pos system permits greater amount approvals

N = The BASE24-pos system does not permit greater amount approvals (default)

ICHG-COMPLIANCE-ACQ-DATA-LOC

This parameter indicates the offset into the PSTM User Data which the Banknet Interface will use to store the Interchange Compliance Data for acquirer traffic (i.e., outbound transactions to Banknet). Valid values are 0 - 185. Default value is 0.

Note: This param is **required** only in a BASE24-pos Release 3.4/4.0 system. It is not required in a Release 5.0 or above system

ICHG-COMPLIANCE-ISS-DATA-LOC

This parameter indicates the offset into the PSTM User Data which the Banknet Interface will use to store the Interchange Compliance Data for issuer traffic (i.e., inbound transactions from Banknet). Valid values are 0 - 185. Default value is 0.

Note: This param is **required** only in a BASE24-pos Release 3.4/4.0 system. It is not required in a Release 5.0 or above system

ILF-NOTIFY-INTERVAL

The number of writes made to the Interchange Log File (ILF) between calculating the percentage of file full. Each time a write is made to the ILF, a counter is incremented. When the counter is equal to the value in this param, a calculation is performed to determine how full the ILF is. The counter is then reset to zero. When the file reaches the percent full specified in the ILF-NOTIFY-PERCENTAGE param, a message is logged indicating the file is becoming full. Until more file space is made available, the interface process continues to log messages that the file is becoming full at the interval specified in this param. Valid values are 10 through 32766 (default 50).

ILF-NOTIFY-PERCENTAGE

The percentage of used file space at which the interface process logs a message indicating the ILF is becoming full. The interface process continues to log messages at the interval indicated in the ILF-NOTIFY-INTERVAL param until more file space is made available. Valid values are 10 through 99 (default 70).

INCR-BNET-TRACE-NUM-BY

The value used to increment the STAN within this Banknet Interface process. It must be equivalent to the number of Banknet Interface processes within the BASE24 system, and must not be less than the value of the parameter FIRST-BNET-TRACE-NUM. Valid values are 1 through 999999 (default 1).

MAX-GROUP-LGTH

This parameter indicates the maximum length of the MasterCard Customer Id. Valid values are 5 and 6 (default 5).

MAX-STF-RECORDS

This parameter indicates the maximum number of records belonging to the Banknet Interface in the Switch Terminal File (STF). This parameter is used to determine the amount of extended memory required by the interface. If this parameter is set to 0, enough extended memory for all records in the STF is allocated. Valid values are 0 through 32766 (default 0).

MAESTRO-NETWORK-IDxx

MasterCard Maestro Network IDs (where xx are numbers) to allow issuers to define up to 50 MasterCard assigned Maestro network ids. These network IDs are used to identify Maestro fuel preauthorization requests and can be obtained from the MasterCard Customer Interface Specifications descriptions of valid values for DE-63, Network Data, subfield 1, Financial Network Code. MasterCard reserves the right to add network codes, without notifying members. By allowing issuers to define these params, MAESTRO-NETWORK-ID01 through MAESTRO-NETWORK-ID50, issuers can implement changes to these values with a warmboot.

NCD-DFLT-HOPR-CONTENTS

This parameter is used to specify the default setting for the ATM hopper contents in a Non-Currency Dispense (NCD) transaction. Valid values are 0 through 99 (default 10).

NTWK-ASSIGN-TCC

The Banknet ISO Interface can optionally provide a space in position 1 of DE 48, based on the value entered in a new LCONF param. LCONF parameter NTWK-ASSIGN-TCC allows users to configure whether the TRAN-CAT-CDE in DE 48 is assigned by Banknet or assigned by the interface. Valid values are:

Y = The TRAN-CAT-CDE in DE 48 is assigned by the Banknet system.

N = The TRAN-CAT-CDE in DE 48 is assigned by the interface (default).

ON-BEHALF-DENY-RSLT-CDExx

A set of parameters (where xx are numbers) to allow issuers to define up to 50 MasterCard “on-behalf” services and associated result codes. The value of each parameter will indicate one “on-behalf” service identifier (a 2-digit numeric value) followed by up to 20 result codes (each one alphabetic character). If the param DENY-FAILED-ON-BEHALF is set to ‘Y’, then the Banknet Interface will decline authorization requests which indicate that any of these “on-behalf” services have been performed, and have been assigned one of the associated result codes. By allowing issuers to define these params, ON-BEHALF-DENY-RSLT-CDE01 through ON-BEHALF-DENY-RSLT-CDE50, issuers can control which “on-behalf” services should result in a declined transaction when the check fails.

PARTIAL-APPROVAL-ALWD

A value indicating whether the amount authorized may be less than the value requested. This param applies to BASE24-pos issuer processing only. Valid values are:

Y = The BASE24-pos system permits partial approvals (default)

N = The BASE24-pos system does not permit partial approvals

POS-LN-CUTOVER

The time (HH:MM) at which all BASE24-pos processes in the logical network are cut over. All institutions must be cut over by this time. Valid values, in military time, are from 00:00 to 23:59 (no default value). This param is **required** if processing BASE24-pos transactions.

PREPAID-NETWORK-IDxx

MasterCard Prepaid Network IDs (where xx are numbers) to allow issuers to define up to 50 MasterCard assigned Prepaid network ids. These network IDs are used to identify transactions initiated using Prepaid cards and can be obtained from the MasterCard Customer Interface Specifications descriptions of valid values for DE-63, Network Data, subfield 1, Financial Network Code. MasterCard reserves the right to add network codes, without notifying members. By allowing issuers to define these params, PREPAID-NETWORK-ID01 through PREPAID-NETWORK-ID50, issuers can implement changes to these values with a warmboot.

PROCESS-MONEYSEND-TXN

This parameter indicates whether MasterCard *MoneySend* transactions are allowed by the issuer. The interface has not been modified to support *MoneySend* transactions in its acquirer processing. Valid values are:

Y = The interface will forward the request to the BASE24-pos system.

N = The interface will deny the *MoneySend* request back to the network (default).

PRVT-LABEL-USES-DE48SE32

A value indicating whether acquirers want to submit a MasterCard assigned ID in DE 48 SE 32. When the parameter is set to “Y”, the interface will attempt to populate DE 48 SE 32 from the STF SWITCH MERCHANT INFORMATION ID using the retailer id from the PSTM. The interface always populates DE 42 from either the retailer id or the alternate merchant id as configured in the PRDF. Issuers using the Private Label Merchant Verification Service register participating merchants. MasterCard supports two methods to recognize these merchants:

- ☐ Merchants identified by acquirer-assigned Card Acceptor ID Code (DE 42)
- ☐ Merchants identified by a MasterCard Assigned ID (DE 48, subelement 32)

Valid values are:

Y = The interface will add DE 48 SE 32 if a valid STF record is found.

N = The interface will not add DE 48 SE 32 (default).

RACAL-ANSI-METHOD

This parameter allows the Banknet Interface to determine which encryption method should be used by a Thales (formerly Racal) HSM. Valid values are:

Y = Use the ANSI X9.17 Encryption Method.

N = Use the Variant Encryption Method (default).

SAF-0120-MESSAGES

This parameter allows users to configure whether 0120 advice messages are added to the SAF. If this parameter is set to “Y”, the interface will add the 0120 messages to the SAF and will forward them to the network during normal SAF processing. Valid values are:

Y = The interface will add 0120 messages to the SAF.

N = The interface will not add 0120 messages to the SAF (default).

SAF-NOTIFY-INTERVAL

The number of writes made to the Store-and-Forward File (SAF) between calculating the percentage of file full. Each time a write is made to the SAF, a counter is incremented. When the counter is equal to the value in this param, a calculation is performed to determine how full the SAF is. The counter is then reset to zero. When the file reaches the percent full specified in the SAF-NOTIFY-PERCENTAGE param, a message is logged indicating the file is becoming full. Until more file space is made available, the interface process continues to log messages that the file is becoming full at the interval specified in this param. Valid values are 10 through 32766 (default 50).

SAF-NOTIFY-PERCENTAGE

The percentage of used file space at which the interface process logs a message indicating the SAF is becoming full. The interface process continues to log messages at the interval indicated in the SAF-NOTIFY-INTERVAL param until more file space is made available. Valid values are 10 through 99 (default 70).

SECURE-DEV-CONF-CMD100

The command text the interface process is to send to an Atalla Network Security Processor (NSP), model A5000 or A6000, in a 100 command during initialization or warmboot. The 100 command text enables optional features of the security device. The text contained in the 100 command must be obtained directly from Atalla. This param is **required** only if the user wants BASE24 to issue these commands to the Atalla NSP. (NOTE: features enabled for a security device via 100 commands are not process-specific; they affect all processes interacting with the device.)

SECURE-DEV-CONF-TXT

The configuration text the interface is to send to an Atalla Network Security Processor (NSP), model A5000 or A6000, in a 101 command during initialization or warmboot. The configuration text can consist of one or more of the four-character option words described below. If more than one option word is defined in the configuration text, the words must be separated by commas. Each option word contains a three-digit option ID followed by a one-character action flag. An option ID (i.e., 023, 024, 040) can be used only once in each SECURE-DEV-CONF-TXT param.

023E - Enables the NSP's function to remove carriage return and line feed characters from all response messages returned to BASE24.

024E - Enables the NSP's function to identify PIN length errors by returning a unique code in response messages to BASE24. This causes the NSP to return L error responses for PIN length errors and S error responses for sanity check errors.

024D - Disables the NSP's function to identify PIN length errors by returning a unique code in response messages to BASE24. This causes the NSP to return S error responses for both PIN length and sanity check errors.

040x - Specifies that the NSP should check PIN lengths and also sets the minimum valid PIN length for performing the check. The "x" must be a value of 0-9 or A-C to indicate the minimum valid PIN length (A = 10; B = 11; C = 12). A value of 0 effectively disables this check because any PIN would pass.

This param is **required** only if the user wants BASE24 to issue these 101 commands to the Atalla NSP. (NOTE: features enabled or disabled for a security device via 101 commands are not process-specific; they affect all processes interacting with the device.)

SECURE-DEV-FAIL-NOTIFY

The interval at which messages are logged for denied transactions due to security device failures. For example, when the SECURE-DEV-FAIL-NOTIFY param is set to 20, every twentieth transaction denied because of a security device failure causes a message to be logged. The SEC-DEV-FAIL-NOTIFY param sets the interval after the first security device failure. A message is always logged immediately following the first security device failure. Valid values are 10 through 1000 (default 20).

SECURE-DEV-MONITOR-THRESHOLD

The interval, in number of transactions performed by the security device or Transaction Security Services (TSS) process defined in the SECURE-DEV1 assign, at which the security device or TSS process defined in the SECURE-DEV2 assign is checked. The security device is checked by sending it a transaction. Valid values are 0 through 32000 (default 0). A value of 0 indicates the security device defined in the SECURE-DEV2 assign is not checked.

SECURE-DEV-TIM-LMT

The maximum amount of time, in hundredths of a second, that BASE24 waits for a response from the security device before timing out the message. Valid values are 1 through 6000 (default 500). Example: 6000 equals 60 seconds.

SECURE-DEV-TYP

A code identifying the type of security module specified in the SECURE-DEV1 and SECURE-DEV2 assigns. Valid values are:

ASM = Atalla Security Module or Atalla Network Security Processor

RSM = Racal-Guardata Host Security Module (same as VSM)

VSM = Racal-Guardata Host Security Module (same as RSM)

TSS = Transaction Security Services

N/A = A security module is not used (default)

SPAN-FLAG *

This parameter indicates whether totals are to be written to the Span Clearings File by the interface process. This parameter is used for report processing. Valid values are:

Y = Yes, write totals to the Span Clearings File.

N = No, do not write totals to the Span Clearings File (default).

SPPT-MULTI-CRNCY

A value indicating whether or not the interface will support multi-currency by sending BASE24-pos Router Authorization the cardholder billing amount, regardless of whether the cardholder billing currency code in the external message is equal to the currency code specified in the ICF/ICFE. Valid values are:

Y = The Banknet Interface supports Multiple Currency

N = The Banknet Interface does not support Multiple Currency (default)

Note: Only customers who have licensed the Multiple Currency product may configure LCONF parameter SPPT-MULTI-CRNCY to value 'Y'.

SWI-ACQ-AVS-DATA-SUPPRS

This parameter indicates whether Address Verification (AVS) data will be suppressed before logging to the ILF. Valid values are:

Y = Suppress AVS data before logging

N = Do not suppress AVS data (default)

SWI-ACQ-CVD2-DATA-SUPPRS

This parameter indicates whether Card Verification (CVD2) data will be suppressed before logging to the ILF. Valid values are:

Y = Suppress CVD2 data before logging

N = Do not suppress CVD2 data (default)

SWI-ACQ-DISC-DATA-SUPPRS

This parameter indicates whether discretionary data from track 1 and/or track 2 will be suppressed before logging to the ILF. Valid values are:

Y = Suppress discretionary data before logging

N = Do not suppress discretionary data (default)

SWI-ISS-AVS-DATA-SUPPRS

This parameter indicates whether Address Verification (AVS) data will be suppressed before logging to the ILF. Valid values are:

Y = Suppress AVS data before logging

N = Do not suppress AVS data (default)

SWI-ISS-CVD2-DATA-SUPPRS

This parameter indicates whether Card Verification (CVD2) data will be suppressed before logging to the ILF. Valid values are:

Y = Suppress CVD2 data before logging

N = Do not suppress CVD2 data (default)

SWI-ISS-DISC-DATA-SUPPRS

This parameter indicates whether discretionary data from track 1 and/or track 2 will be suppressed before logging to the ILF. Valid values are:

Y = Suppress discretionary data before logging

N = Do not suppress discretionary data (default)

SW-ILF-FILE-RETENTION

This parameter is used to specify the number of days an ILF remains on the system, after which the Banknet Interface process purges the ILF from the system. Valid values are 3 through 99 (default 7).

SW-RPT-STARTUP **

The time (HH:MM) at which the interface process is to run its report obey file. Valid values, in military time, are from 00:00 to 23:59 (no default value).

SW-BNET-CREATE-ILF

The time (HH:MM) at which the interface process should create the next business day's Interchange Log File (ILF). Valid values, in military time, are from 00:00 to 23:59 (no default value).

SW-BNET-LOG-ONLY-COMPL-ADVICE

This parameter indicates whether a "9922" log-only message should be sent to the POS Router/Auth process when a response to an Automated Fuel Dispenser (AFD) completion advice is received from Banknet. Valid values are:

Y = Send log-only message

N = Do not send log-only message (default)

SW-BNET-TRAN-TIME-SET

This parameter specifies the time zone used when setting the transaction date and time in the (P)STM, and how the time offset is calculated. The Customer Interface Specification (CIS) states that the value of DE 7 **MUST** be specified in Coordinated Universal Time (UTC), which is effectively equivalent to Greenwich Mean Time (GMT). The Transaction Time in the (P)STM should contain the time provided in the incoming message, and the Time Offset in the (P)STM should indicate the difference between the transaction time and the local time.

Valid values are:

0 = Use the GMT OFFSET field in the ICF(E) when processing DE 7 on incoming messages. The Time Offset in the (P)STM will be the negative value of the GMT OFFSET from the ICF(E). (default)

1 = Use DE 7 without using the GMT OFFSET from the ICF(E). The Time Offset in the (P)STM will be the negative value of the GMT OFFSET from the ICF(E).

2 = Use DE 7 without using the GMT OFFSET from the ICF(E). Using NSK timestamps, the Time Offset in the (P)STM will be the difference between GMT and LCT. If GMT is greater than LCT, the offset will be positive. If GMT is less than LCT, the offset will be negative.

TIME-ZONE-DIFF

This parameter indicates the number of minutes between the interface and the interchange. During a warmboot or stop/start, the interface will compare the settlement time on the ICF/ICFE, which should be set to the local time of the interchange cutover, to the new param. This will determine if the interchange could be a day behind. This param is **required** if the interchange and interface can be a day or more apart.

USE-OWNER-NAME-FOR-DE43

In POS processing, this parameter is used to specify how DE 43 (Card Acceptor Name/Location) is populated on outbound request messages sent to MasterCard. If this parameter is set to 'Y', then the interface uses PSTM.TERM-OWNER-NAME; otherwise, the interface uses PSTM.TERM-NAME-LOC. Both PSTM.TERM-OWNER-NAME and PSTM.TERM-NAME-LOC are populated by BASE24-pos Device Handlers from PTDF/PTD fields with the same name.

Y = Use PSTM.TERM-OWNER-NAME to populate DE 43.

N = Use PSTM.TERM-NAME-LOC to populated DE 43 (default).

In ATM processing, this parameter is used to specify how DE 42 (Card Acceptor ID Code) and DE 43 is populated on outbound ATM request messages sent to MasterCard. If this parameter is set to 'Y' then the interface uses STM.TERM-OWNER-FIID for DE 42 and STM.RQST.TERM-OWNER-NAME for DE-43. If this parameter is set to 'N' or not present, then the interface uses STM.RQST.TERM-OWNER-NAME for DE-42 and STM.RQST.TERM-NAM-LOC for DE-43.

Y = Use STM.TERM-OWNER-FIID for DE-42 and STM.RQST.TERM-OWNER-NAME for DE-43.

N = Use STM.RQST.TERM-OWNER-NAME for DE-42 and STM.RQST.TERM-NAM-LOC for DE-43 (default).

ICFE SETTINGS

The Enhanced Interchange Configuration File (ICFE) contains parameters relevant to interchange transaction processing. It is used by the financial institution to define institution and terminal interchange sharing, settlement handling, timer limits, and processing options. The options controlled via the ICFE are described below and on the following pages. For more information and screen formats, refer to the ***BASE24 Files Maintenance Manual***. The ICFE fields are described in the order they appear on the screens.

ICFE Screen 1 Settings

The following fields must be set on ICFE screen 1 as indicated:

INTERCHANGE FIID

The FIID uniquely identifying the Banknet interface. This field must match the logical network ID given to the Banknet interface in the BASE24 system. This field and the PROCESS field combine to form the primary key to the ICFE.

FIELD LENGTH: 1-4 alphanumeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: None

PROCESS

The name of the Banknet Interface process as defined in the Network Environment Source File. This is the process that receives and sends transaction messages from and to Banknet. This field and the INTERCHANGE FIID field combine to form the primary key to the ICFE.

FIELD LENGTH: 1-16 alphanumeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: None

SWITCH TYPE

A code identifying the interchange. For the Banknet Interface process, this value must be "BNET".

FIELD LENGTH: 1-4 alphanumeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: None

INTERCHANGE LOGICAL NET

The logical network identifier given to the interchange. This value must match the value in the INTERCHANGE FIID field.

FIELD LENGTH: 1-4 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

REPORTING NAME

The name of the interchange as it appears on BASE24 reports.

FIELD LENGTH: 1-20 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

INSTITUTION ID

The Banknet-assigned identifier of the institution running the Banknet Interface. On outbound messages, this value will be placed in the forwarding institution ID (DE 33) if the acquiring institution is different than the institution running the Banknet Interface. If the CUST ID IND field on Screen 1 of the BNIIF record is set to '3', this value will be placed in the acquiring institution ID (DE 32) in outbound messages.

FIELD LENGTH: 6 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

SWITCH ID

The identification of the interchange center. For customers supporting network management processing using a Group Sign-on Identifier (GSI), this value must be the key to the BNIF record that holds the network management parameters. When used for this purpose, this field must contain 1-4 alphanumeric characters.

FIELD LENGTH: 1-15 alphanumeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: None

STATION 1

The symbolic name of a Banknet station that communicates with the switch. This name must match the name of a station defined in your network environment for the Banknet Interface process.

FIELD LENGTH: 1-16 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

STATION 2

The symbolic name of a second Banknet station that communicates with the switch. This name must match the name of a station defined in your network environment for the Banknet Interface process. If only one station is defined, this field must either be blank or match the station name in the STATION 1 field.

FIELD LENGTH: 1-16 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

SIC CODE

The Standard Industrial Classification Code (SIC) applicable to the institution operating the BASE24-pos network. Used to fill the merchant type field (DE 18) in outbound requests if no SIC Code was specified in the PSTM.

FIELD LENGTH: 1-4 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 0

CURRENCY CODE

Identifies the currency code for transactions that the interchange receives. Valid values are listed in the following publication: ***Codes for the Representation of Currencies and Funds, ISO 4217-2001***. Not used by the Banknet Interface process.

FIELD LENGTH: 3 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 840 (US dollars)

DEFAULT TERM NUM

A default terminal number used to fill the terminal id field in the internal message if an incoming transaction from the interchange does not contain a terminal number. Not used by the Banknet Interface process.

FIELD LENGTH: 1-16 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

DEFAULT ACQUIRER ID NUM

A default acquirer ID number used if an incoming transaction from the interchange does not contain an acquirer's institution identification number. Not used by the Banknet interface process.

FIELD LENGTH: 1-11 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 000000000000

CUSTOMER BALANCE DISPLAY

A value indicating the method in which Banknet acquirers wish balances to be presented to customers at BASE24 ATMs. Valid values are:

0 = Do not display or print
1 = Display only
2 = Print only
3 = Display and print

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 0 (DON'T DISPLAY OR PRINT)

ICFE Screen 2 Settings

The following fields must be set on ICFE screen 2 as indicated:

SETTLEMENT INFORMATION

The following fields contain interchange settlement information.

SETTLEMENT HOUR

The current hour of settlement (local time) for this interchange. For most members, Banknet settles at 8:00 a.m., CST. Valid values include 00 through 23.

FIELD LENGTH: 2 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 00

SETTLEMENT MINUTE

The current minute of settlement (local time) for this interchange. For most members, Banknet settles at 8:00 a.m., CST. Valid values include 00 through 59. The value for Banknet should be 00 since Banknet settles on the hour.

FIELD LENGTH: 2 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 00

SETTLEMENT DAYS

A value indicating the processing schedule for an interchange. Banknet processes 7 days a week. Valid values are:

0 = Processes 5 days a week
1 = Processes 7 days a week

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 1 (PROCESS 7 DAYS).

If replaced with blanks, the default value is 0 (PROCESS 5 DAYS).

SWITCH POSTING DATE (YYMMDD)

The posting date (YYMMDD) for the interchange. Although this field allows entry of one through six alphabetic or numeric characters, the posting dates cannot be processed unless six numeric characters representing valid dates are entered. The interface updates these dates at daily settlement time and upon initialization.

FIELD LENGTH: 6 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

HOLIDAY DATES (YYMMDD)

The interchange's defined holiday dates (YYMMDD). Although these fields allows entry of one through six alphabetic or numeric characters, the holiday dates cannot be processed unless six numeric characters representing valid dates are entered. This interchange does not settle on these holidays and the Interface process does not create an Interchange Log File (ILF) for these dates. This field is not used by this interface.

FIELD LENGTH: 16 fields of 6 numeric characters each
REQUIRED FIELD: No
DEFAULT VALUE: None

REPORT DATA MASKING PARAMETERS

The values in the following fields specify the report data masking parameters. These parameters support the masking of sensitive information in accordance with PCI data security standards.

DATA MASK FLAG

A code indicating whether sensitive data should be masked or unmasked. Valid values are Y (Mask sensitive data) and N (Do not mask sensitive data).

FIELD LENGTH: 1 alphabetic character
REQUIRED FIELD: No
DEFAULT VALUE: Y

RIGHT UNMASKED DIGITS

A code defining the number of rightmost PAN digits to be displayed unmasked. Valid values are 0–9 (Number of rightmost digits to be displayed unmasked).

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: No
DEFAULT VALUE: 4

MIN MASKED DIGITS

A code defining the minimum number of PAN digits to be masked. Valid values are 0–9 (Minimum number of digits to be masked).

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: No
DEFAULT VALUE: 9

MAX LEFT UNMASKED DIGITS

A code defining the maximum number of leftmost PAN digits to be displayed unmasked. Valid values are 0–9 (Maximum number of leftmost digits to be displayed unmasked).

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: No
DEFAULT VALUE: 0

ICFE Screen 3 Settings

The following fields must be set on ICFE screen 3 as indicated:

TIMER LIMITS

The following timer value fields are used to control specific transaction message activity between the BASE24-pos network and Banknet. The values entered in these fields must be checked against Banknet timer values to ensure the integrity of message traffic.

NETWORK MANAGEMENT

The number of seconds BASE24 waits before setting an EXTENDED NETWORK timer for an outstanding network management message (e.g., logon, logoff). An outstanding message is one to which Banknet has not yet responded.

FIELD LENGTH: 4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 30 (SEC)

EXTENDED NETWORK

The number of seconds BASE24 waits before attempting to retry a network management message. Valid values are 0 - 9999.

FIELD LENGTH: 4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 60 (SEC)

WAIT FOR TRAFFIC

The number of seconds BASE24 waits for traffic on the line from Banknet before initiating a network management message. Valid values are 0 - 9999.

FIELD LENGTH: 4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 60 (SEC)

PERFORMANCE PERIOD

The number of minutes the Banknet Interface process uses as an interval for tracking performance (gathering performance statistics) of the interface process. Valid values are 0 - 9999.

FIELD LENGTH: 4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 20 (MIN)

MAXIMUM OUTSTANDING TRANSACTIONS

The following fields specify the maximum number of outstanding inbound and outbound transactions allowed by the interface. The combined value will be utilized to determine the number of extended memory pages that will be required for the transactions in process queue.

OUTBOUND

The maximum number of outstanding outbound transactions.

FIELD LENGTH: 4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 15

INBOUND

The maximum number of outstanding inbound transactions.

FIELD LENGTH: 4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 15

PROCESSING OPTIONS

The following fields define the processing options for Banknet.

ACQUIRER

A value indicating whether the institution acts as an acquirer of Banknet transactions. Valid values are:

Y = Yes, send transactions to Banknet.

N = No, do not send transactions to Banknet.

FIELD LENGTH: 1 alphanumeric character

REQUIRED FIELD: Yes

DEFAULT VALUE: None

ISSUER

A value indicating whether the institution acts as an issuer of Banknet transactions. Valid values are:

Y = Yes, accept transactions from Banknet.

N = No, do not accept transactions from Banknet.

FIELD LENGTH: 1 alphanumeric characters

REQUIRED FIELD: Yes

DEFAULT VALUE: None

PROCESSING MODE

This field is Interchange Interface dependant and is used by the Banknet Interface to indicate whether the process supports UK Domestic Maestro transactions. If this field is set to 'D', then the Interface processes all transactions as UK Domestic Maestro transactions; If this field is set to 'P' (partial alignment), then the Interface supports the alignment of functions with an effective date of March 1, 2011, but not the alignment of functions with an effective date of May 13, 2011; otherwise, no UK Domestic Maestro processing takes place, which is equivalent to all functions being fully aligned. When this field is set to 'D' or 'P', only UK Domestic Maestro transaction should be routed through this Banknet Interface process..

Valid values are:

D = Process transactions as UK Domestic Maestro transactions.

P = Partial alignment.

blank = Normal processing/Full alignment (default).

FIELD LENGTH: 1 alphanumeric characters

REQUIRED FIELD: No

DEFAULT VALUE: None

AUTO SIGNON START

A value indicating whether the Banknet Interface process is set to programmatically log on to Banknet at startup without operator intervention. Valid values are:

Y = Yes, programmatically log on to Banknet at startup

N = No, do not programmatically log on to Banknet at startup

If this field is set to "N", the interface must be manually logged on after initialization.

FIELD LENGTH: 1 alphanumeric characters

REQUIRED FIELD: Yes

DEFAULT VALUE: None

MAXIMUM TIMEOUTS

The maximum number of consecutive timeouts allowed before network management measures are taken to determine the status of the link. Valid values are 0 - 9999.

FIELD LENGTH: 1-4 numeric characters

REQUIRED FIELD: Yes

DEFAULT VALUE: 1

MAX SAF RETRY

The maximum number of times a transaction from the Store-and-Forward File (SAF) is transmitted to Banknet before the record is dumped to the hard-copy log and deleted from the file. A value of zero indicates that the SAF should be retransmitted indefinitely or until satisfactory completion of the message occurs. Valid values are 0 - 9999.

FIELD LENGTH: 4 numeric characters

REQUIRED FIELD: Yes

DEFAULT VALUE: 0

ACK TO SWITCH

A value indicating whether an acknowledgement is required to Banknet. Used to determine whether 0180 acknowledgements will be sent back to Banknet after receiving an 0110 message from Banknet. Valid values are:

Y = Yes

N = No

FIELD LENGTH: 1 alphanumeric characters

REQUIRED FIELD: Yes

DEFAULT VALUE: No

ACK FROM SWITCH

A value indicating whether an acknowledgement is required from Banknet. Not used by the Banknet Interface process.

FIELD LENGTH: 1 alphanumeric characters

REQUIRED FIELD: Yes

DEFAULT VALUE: No

MAXIMUM PIN TRIES

The maximum number of consecutive invalid PINs allowed before an automatic key change occurs. Not used by the Banknet Interface process.

FIELD LENGTH: 1-3 numeric characters

REQUIRED FIELD: No

DEFAULT VALUE: 1 (NO RETRIES)

NETWORK MANAGEMENT MESSAGE ENABLED

A value indicating whether network management messages are enabled or disabled. Valid values are:

Y = Yes, send network management messages to Banknet.

N = No, do not send network management messages to Banknet.

FIELD LENGTH: 1 alphanumeric character

REQUIRED FIELD: Yes

DEFAULT VALUE: No

TYPE OF INTERCHANGE REPORTS

A value indicating the types of reports the interchange requires.
Not used by the Banknet Interface process.

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: No
DEFAULT VALUE: 0

ILF EXTRACT NUMBER

The number of ILF files to be extracted for Host reporting. This field is used in conjunction with the RPT-EXTRACT field in the Extract Configuration File (ECF). For a description of the ECF, refer to the ***BASE24 Base Files Maintenance Manual***.

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: No
DEFAULT VALUE: 3

ICFE Screen 8 Settings

The following fields must be set on ICFE screen 8 as indicated, if ATM processing is supported:

ATM ICFE DATA

The following fields contain information relevant to BASE24-atm.

AUTH PROCESS

The symbolic name of the Authorization process to which the Banknet Interface process routes ATM transaction requests.

FIELD LENGTH: 1-16 alphanumeric characters

REQUIRED FIELD: No

DEFAULT VALUE: None

DEFAULT ROUTING GROUP

The terminal routing group used as a default for transactions from Banknet.

FIELD LENGTH: 1-11 numeric characters

REQUIRED FIELD: Yes

DEFAULT VALUE: 000000000000

ACQUIRER TXN PROFILE

A code identifying a group of BASE24-atm transaction processing codes allowed from Banknet. The value of this field is part of the key used to read the Acquirer Processing Code File (APCF).

FIELD LENGTH: 16 alphanumeric characters

REQUIRED FIELD: No

DEFAULT VALUE: ATM

ISSUER TXN PROFILE

A code identifying a group of BASE24-atm transaction processing codes allowed to be sent to Banknet. The value of this field is part of the key used to read the Issuer Processing Code File (IPCF).

FIELD LENGTH: 16 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: ATM

DEFAULT MERCHANT TYPE

The default merchant type that is sent in a Non-Currency Dispense (NCD) transaction.

FIELD LENGTH: 4 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

TIMER LIMITS

The following fields set the time limits, in seconds, for BASE24-atm transactions.

STORE AND FORWARD

The time limit associated with the store-and-forward response from Banknet for a BASE24-atm transaction. Valid values are 0 - 9999.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 30 (SEC)

OUTBOUND

The time limit associated with transaction responses from Banknet for a BASE24-atm transaction. Valid values are 0 - 9999.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 15 (SEC)

INBOUND

The time limit associated with a transaction response from BASE24-atm. Valid values are 0 - 9999.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 15 (SEC)

COMPLETION

The time limit associated with completion messages for BASE24-atm transactions. Not used by the Banknet Interface process.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 60 (SEC)

COMPLETION ACK

The time limit associated with completion acknowledgement messages for BASE24-atm transactions. Not used by the Banknet Interface process.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 30 (SEC)

SHARING GROUPS

The codes assigned to identify interchange sharing groups for both incoming and outgoing transactions. Zero is not a valid entry. Codes cannot be separated by blank spaces, but unused portions to the right can remain blank.

FIELD LENGTH: 24 fields of 1 numeric character each
REQUIRED FIELD: No
DEFAULT VALUE: None

ICFE Screen 10 Settings

The following fields must be set on ICFE screen 10 as indicated:

POS ICFE DATA

The following fields contain information relevant to BASE24-pos.

AUTH PROCESS

The symbolic name of the Router/Authorization process to which the Banknet Interface process routes POS transaction requests.

FIELD LENGTH: 1-16 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

REFERRAL PHONE NUMBER

The phone number that the Banknet Interface process places in all internal responses. This phone number can be called for an authorization.

FIELD LENGTH: 1-18 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

RETAILER ID DEFAULT

The default retailer ID that the Banknet Interface process places in all internal requests when the interchange does not send a value.

FIELD LENGTH: 1-19 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

TIMEOUT ACTION

A value indicating the action the Banknet Interface process takes when an outgoing request times out at the interchange. Valid values are:

- 0 = Return a decline response
- 1 = Default action
- 2 = Alternate route

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 0 (Return a decline response)

SETTLE ENTITY

Used to determine the value of the draft capture field in the PSTM. Valid values are:

- 0 = No draft capture
- 1 = Draft capture

FIELD LENGTH: 1 alphanumeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 0

ACQUIRER TXN PROFILE

A code identifying a group of BASE24-pos transaction processing codes allowed from Banknet. The value of this field is part of the key used to read the Acquirer Processing Code File (APCF).

FIELD LENGTH: 16 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: ATM

ISSUER TXN PROFILE

A code identifying a group of BASE24-pos transaction processing codes allowed to be sent to Banknet. The value of this field is part of the key used to read the Issuer Processing Code File (IPCF).

FIELD LENGTH: 16 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: ATM

ADJUSTMENT FLAG

A flag indicating whether the interchange allows an adjustment transaction when the new transaction amount (amount 2) is greater than the original transaction amount (amount 1). The Banknet Interface process checks the value in this field to determine whether processing of an adjustment transaction can continue when the new transaction amount (amount 2) is greater than the original transaction amount (amount 1). Valid values are as follows:

Y = Yes, allow adjustments where the new amount is greater than the original amount.

N = No, do not allow adjustments where the new amount is greater than the original amount.

FIELD LENGTH: 1 alphabetic character

REQUIRED FIELD: Yes

DEFAULT VALUE: N

CHARGEBACK FLAG

A flag indicating whether the interchange allows chargeback transactions. The Banknet Interface process checks the value in this field to determine whether processing of a chargeback transaction can continue. Valid values are as follows:

Y = Yes, allow chargeback transactions.

N = No, do not allow chargeback transactions.

FIELD LENGTH: 1 alphabetic character

REQUIRED FIELD: Yes

DEFAULT VALUE: N

TIMER LIMITS

The following fields set the time limits, in seconds, for BASE24-pos transactions.

STORE AND FORWARD

The time limit associated with the store-and-forward response from Banknet for a BASE24-pos transaction. Valid values are 0 - 9999.

FIELD LENGTH: 1-4 numeric characters

REQUIRED FIELD: Yes

DEFAULT VALUE: 30 (SEC)

OUTBOUND

The time limit associated with transaction responses from Banknet for a BASE24-pos transaction. Valid values are 0 - 9999.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 15 (SEC)

INBOUND

The time limit associated with a transaction response from BASE24-pos. Valid values are 0 - 9999.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 15 (SEC)

COMPLETION

The time limit associated with completion messages for BASE24-pos transactions. Not used by the Banknet Interface process.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 60 (SEC)

COMPLETION ACK

The time limit associated with completion acknowledgement messages for BASE24-pos transactions. Not used by the Banknet Interface process.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 30 (SEC)

ICFE Screen 11 Settings

The following fields must be set on ICFE screen 11 as indicated:

POS PRODUCT DATA

The following fields contain information relevant to BASE24-pos.

DEFAULT PRE-AUTH AMOUNT

The default preauthorization amount used in an internal request when the external message does not contain an amount. Not used by the Banknet Interface process.

FIELD LENGTH: 1-5 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0

APPROVAL CODE LENGTH

The number of digits that Banknet requires in an approval code. Valid values are 2 through 6. If the Banknet Interface process supports UK Domestic Maestro transactions, then this field must contain a 4; otherwise this field must contain a 6.

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 6

PRE-AUTH HOLD INCREMENT

A value indicating the time increment to hold preauthorized funds. This field is used in conjunction with the PRE-AUTH HOLD TIME field. Valid values are:

0 = Minutes
1 = Hours
2 = Days

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 0 (Minutes)

PRE-AUTH HOLD TIME

The length of time to hold the preauthorized funds. This field is used in conjunction with the PRE-AUTH HOLD INCREMENT field. Valid values are 0 through 99.

FIELD LENGTH: 1-2 numeric characters

REQUIRED FIELD: Yes

DEFAULT VALUE: 0

ALLOWED SERVICES

Codes that identify the types of cards the Banknet Interchange allows. This field is used for both inbound and outbound BASE24-pos messages. The BASE24 defined codes are:

AD = Administrative	AX = American Express
BD = Business deposit-only	C = Private label credit
CB = Carte Blanche	D = Demonstration
DC = Diners Club	M = MasterCard credit
MD = MasterCard debit	MM = MasterCard dual
P = Proprietary debit	V = VISA credit
VD = VISA debit	VV = VISA dual
** = all	

Institutions can also enter additional codes not listed above, as long as the first position of the code is an alphabetic character other than M or V (M and V are reserved for MasterCard and Visa).

ICFE Screen 12 Settings

The ICFE screen 12 (Banknet specific information) is shown below followed by a description of its fields.

```

BASE24-BASE   ICFE FILE MAINTENANCE   LNET       YY/MM/DD  HH:MM  12 OF 12
INTERCHANGE FIID: _____ PROCESS: _____

                BANKNET SWITCH SPECIFIC DATA
AUTHORIZATION ONLY: _ (Y/N)           LOGON TYPE: _ (G/P)

STATION NAMES: _____

STATION TYPES: _

COUNTRY : _____ COUNTRY CODE: _____ POSTAL CODE: _____

LOCAL CURRENCY CODE: _____ BUY RATE: _____ SELL RATE: _____

ILF HEADER MAX TRAN UPDATE: _____ ALTERNATE HOST: _ (Y/N)

PROTOCOL: _ PROBE PAN: _____ GMT OFFSET _____

ATM PROCESSING: _ (Y/N)   NCD PROCESSING: _ (Y/N)
RECORD LAST CHANGED: 09/07/08 10:10 BY USER: 0255 , 00000255 CHANGE
***** BASE24 *****
NEW PAGE:           FILE DESTINATION:           NEW LOGICAL NETWORK ID:
                  F12-HELP
  
```

The following fields on ICFE screen 12 must be set as indicated.

AUTHORIZATION ONLY

A value indicating whether the Banknet Interface process should send only authorization requests for all transactions, or send both authorization requests and financial requests. The Banknet Interchange currently supports authorization requests only, so this field must be set to 'Y'. Valid values are:

Y = Yes - Send only authorization requests (default value)

N = No - Send both authorization and financial requests

FIELD LENGTH: 1 alphabetic character

REQUIRED FIELD: Yes

DEFAULT VALUE: Y

LOGON TYPE

A value indicating the type of signon request that will be sent to Banknet if the AUTO SIGNON START flag is set to Y or the NCPCOM manual LOGON command doesn't specify a type. Valid values are:

G = Group (default value)

P = Prefix

FIELD LENGTH: 1 alphabetic character

REQUIRED FIELD: Yes

DEFAULT VALUE: G

STATION NAMES

The symbolic names of the third, fourth and fifth Banknet stations that send and receive messages to and from the Banknet Interface process. If used, these names must match the names of stations added in the Network Environment Source File for the Banknet Interface process.

FIELD LENGTH: 16 alphanumeric characters

REQUIRED FIELD: No

DEFAULT VALUE: None

STATION TYPES

The type of up to five Banknet stations that send and receive messages to and from the Banknet Interface process. Banknet no longer allows issuers to differentiate between credit and debit transactions on a particular station, so the default value of 0 is the recommended setting for this field. Valid values are:

0 = Credit

1 = Debit

2 = Both

FIELD LENGTH: 1 numeric character

REQUIRED FIELD: No

DEFAULT VALUE: 0

COUNTRY

A code indicating the country in which the terminal is located. This field is alphabetic. Valid values are listed in the following publication: ***Codes for the Representation of Names of Countries and their Subdivisions, ISO 3166-2006.***

FIELD LENGTH: 3 alphabetic characters
REQUIRED FIELD: No
DEFAULT VALUE: USA

COUNTRY CODE

The Banknet country code of the acquirer. This field is numeric. Valid values are listed in the following publication: ***Codes for the Representation of Names of Countries and their Subdivisions, ISO 3166-2006.***

FIELD LENGTH: 3 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 840

POSTAL CODE

The postal ZIP code of the location of the terminal.

FIELD LENGTH: 9 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 000000000

LOCAL CURRENCY CODE

The default currency code for the Banknet Interface process. Valid values are listed in the following publication: ***Codes for the Representation of Currencies and Funds, ISO 4217-2001.***

FIELD LENGTH: 3 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 840

BUY RATE

The factor used to convert the transaction amount to the settlement amount. The leftmost digit denotes the number of positions the decimal is moved from the right (e.g., 69972522 = 9.972522). Not used by the Banknet Interface process.

FIELD LENGTH: 8 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 6100000000

SELL RATE

The factor used to convert the settlement amount to the cardholder amount. The leftmost digit denotes the number of positions the decimal is moved from the right (e.g., 69972522 = 9.972522). Not used by the Banknet Interface process.

FIELD LENGTH: 8 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 61000000

ILF HEADER MAX TRAN UPDATE

The number of BASE24-initiated transactions logged to the ILF before the trace number in the ILF header record is updated

FIELD LENGTH: 4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0200

ALTERNATE HOST

A value indicating whether the Banknet Interface process supports network management commands as an alternate issuer host. MasterCard allows issuers in the Europe region to route to an alternate issuer host rather than to the Stand-In System. When this value is 'Y' the interface will use the Network Management Information Codes 063, 064, 067, 068 in place of 061, 062, 065, and 066 respectively. Valid values are:

Y = Yes – Use alternate host Network Management Information Codes.
N = No – Use primary host Network Management Information Codes.

FIELD LENGTH: 1 alphabetic character
REQUIRED FIELD: Yes
DEFAULT VALUE: N

PROTOCOL

A value indicating the protocol supported by the Banknet Interface process. For the TCP/IP protocol, this field should be set to a 3 (SNA LU0). Valid values are:

0 = Multipoint Supervisor
1 = Multipoint Tributary
2 = Point-to-Point
3 = SNA LU0
4 = X.25

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 1

PROBE PAN

The PAN in an authorization request message is compared to the value in this field. If the values match, then the transaction is declined and a record is not written to the ILF.

FIELD LENGTH: 19 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: None

GMT OFFSET

The number of minutes that local time varies from Greenwich Mean Time. The characters ' ' and '-' are used in the first byte to indicate whether local time is later than or earlier than GMT.

FIELD LENGTH: 3 numeric characters (values 0-9), and one sign character (' ' or '-')
REQUIRED FIELD: Yes
DEFAULT VALUE: 000

ATM PROCESSING

A value indicating whether the Banknet Interface process provides support for BASE24-atm transactions. Valid values are:

Y = Yes - ATM support provided. Requires the Banknet ATM extension module.
N = No – No ATM support (default value)

FIELD LENGTH: 1 alphabetic character
REQUIRED FIELD: Yes
DEFAULT VALUE: N

NCD PROCESSING

A value indicating whether the Banknet Interface process supports BASE24-atm Non-Currency Dispense (NCD) transactions. Valid values are:

Y = Yes - NCD support provided. Requires the Banknet ATM extension module.
N = No – No NCD support (default value)

FIELD LENGTH: 1 alphabetic character
REQUIRED FIELD: Yes
DEFAULT VALUE: N

ICF SETTINGS

Each interface process must have its own Interchange Configuration File (ICF) record if the BASE24 system is pre-Release 6.0. Otherwise, the Enhanced Interchange Configuration File (ICFE) is used. The following describes how the ICF must be configured for the Banknet Interface.

The screens of the ICF are almost identical to those on the ICFE. Therefore, only fields on ICF screens that differ from the corresponding ICFE screens are described below.

ICF Screen 6 Settings

The following fields must be set on ICF screen 6 as indicated, if the ATM extension module and Non-Currency Dispense (NCD) transactions are supported:

NCD ICF DATA

The following fields contain information relevant to BASE24-atm Non-Currency Dispense (NCD) transactions.

NOT ON US

Codes defining the circumstances in which incoming transactions from Banknet are allowed (that is, where BASE24-atm cardholders use ATMs connected to and controlled by institutions serviced by an interchange). Valid values are:

- 0 = Not allowed
- 1 = Allowed within the county
- 2 = Allowed within the state
- 3 = Allowed nationally
- 4 = Allowed internationally

FIELD LENGTH: 2 fields of 1 numeric character each

REQUIRED FIELD: Yes

DEFAULT VALUE: 0

TRANSACTIONS ALLOWED TO SWITCH

These fields represent the transaction types that are supported by BASE24-atm and can be sent to Banknet. A value of Y is entered against these transactions; all others are set to N.

FIELD LENGTH: 2 fields of 1 alphabetic character each

REQUIRED FIELD: Yes

DEFAULT VALUE: N

DEFAULT MERCHANT TYPE

The default merchant type that is sent in a Non-Currency Dispense (NCD) transaction.

FIELD LENGTH: 4 alphanumeric characters

REQUIRED FIELD: No

DEFAULT VALUE: None

ICF Screen 7 Settings

The following fields must be set on ICF screen 7 as indicated, if the ATM extension module is supported:

ATM ICF DATA

The following fields contain information relevant to BASE24-atm.

NOT ON US

Codes defining the circumstances in which incoming transactions from Banknet are allowed (that is, where BASE24-atm cardholders use ATMs connected to and controlled by institutions serviced by an interchange). Valid values are:

- 0 = Not allowed
- 1 = Allowed within the county
- 2 = Allowed within the state
- 3 = Allowed nationally
- 4 = Allowed internationally

FIELD LENGTH: 12 fields of 1 numeric character each
REQUIRED FIELD: Yes
DEFAULT VALUE: 0

TRANSACTIONS ALLOWED TO SWITCH

These fields represent the transaction types that are supported by BASE24-atm and can be sent to Banknet. A value of Y is entered against these transactions; all others are set to N.

FIELD LENGTH: 12 fields of 1 alphabetic character each
REQUIRED FIELD: Yes
DEFAULT VALUE: N

ICF Screen 8 Settings

The following fields must be set on ICF screen 8 as indicated, if the ATM extension module is supported:

ATM ICF DATA

The following fields contain information relevant to BASE24-atm.

AUTH PROCESS

The symbolic name of the Authorization process to which the Banknet Interface process routes ATM transaction requests.

FIELD LENGTH: 1-16 alphanumeric characters

REQUIRED FIELD: No

DEFAULT VALUE: None

DEFAULT ROUTING GROUP

The terminal routing group used as a default for transactions from Banknet.

FIELD LENGTH: 1-11 numeric characters

REQUIRED FIELD: Yes

DEFAULT VALUE: 00000000000

TIMER LIMITS

The following fields set the time limits, in seconds, for BASE24-atm transactions.

STORE AND FORWARD

The time limit associated with the store-and-forward response from Banknet for a BASE24-atm transaction. Valid values are 0 - 9999.

FIELD LENGTH: 1-4 numeric characters

REQUIRED FIELD: Yes

DEFAULT VALUE: 30 (SEC)

OUTBOUND

The time limit associated with transaction responses from Banknet for a BASE24-atm transaction. Valid values are 0 - 9999.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 15 (SEC)

INBOUND

The time limit associated with a transaction response from BASE24-atm. Valid values are 0 - 9999.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 15 (SEC)

COMPLETION

The time limit associated with completion messages for BASE24-atm transactions. Not used by the Banknet Interface process.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 60 (SEC)

COMPLETION ACK

The time limit associated with completion acknowledgement messages for BASE24-atm transactions. Not used by the Banknet Interface process.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 30 (SEC)

SHARING GROUPS

The codes assigned to identify interchange sharing groups for both incoming and outgoing transactions. Zero is not a valid entry. Codes cannot be separated by blank spaces, but unused portions to the right can remain blank.

FIELD LENGTH: 24 fields of 1 numeric character each
REQUIRED FIELD: No
DEFAULT VALUE: None

ICF Screen 9 Settings

The following fields must be set on ICF screen 9 as indicated:

POS ICF DATA

The following fields contain information relevant to BASE24-pos.

TRANSACTIONS ALLOWED TO SWITCH

These fields represent the transaction types that are supported by BASE24-pos and can be sent to Banknet. A value of Y is entered against these transactions; all others are set to N.

FIELD LENGTH: 20 fields of 1 alphabetic character each

REQUIRED FIELD: Yes

DEFAULT VALUE: N

ICF Screen 10 Settings

The following fields must be set on ICF screen 10 as indicated:

POS ICF DATA

The following fields contain information relevant to BASE24-pos.

AUTH PROCESS

The symbolic name of the Router/Authorization process to which the Banknet Interface process routes POS transaction requests.

FIELD LENGTH: 1-16 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

REFERRAL PHONE NUMBER

The phone number that the Banknet Interface process places in all internal responses. This phone number can be called for an authorization.

FIELD LENGTH: 1-18 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

RETAILER ID DEFAULT

The default retailer ID that the Banknet Interface process places in all internal requests when the interchange does not send a value.

FIELD LENGTH: 1-19 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

TIMEOUT ACTION

A value indicating the action the Banknet Interface process takes when an outgoing request times out at the interchange. Valid values are:

- 0 = Return a decline response
- 1 = Default action
- 2 = Alternate route

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 0 (Return a decline response)

SETTLE ENTITY

Used to determine the value of the draft capture field in the PSTM. Valid values are:

- 0 = No draft capture
- 1 = Draft capture

FIELD LENGTH: 1 alphanumeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 0

TIMER LIMITS

The following fields set the time limits, in seconds, for BASE24-pos transactions.

STORE AND FORWARD

The time limit associated with the store-and-forward response from Banknet for a BASE24-pos transaction. Valid values are 0 - 9999.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 30 (SEC)

OUTBOUND

The time limit associated with transaction responses from Banknet for a BASE24-pos transaction. Valid values are 0 - 9999.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 15 (SEC)

INBOUND

The time limit associated with a transaction response from BASE24-pos. Valid values are 0 - 9999.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 15 (SEC)

COMPLETION

The time limit associated with completion messages for BASE24-pos transactions. Not used by the Banknet Interface process.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 60 (SEC)

COMPLETION ACK

The time limit associated with completion acknowledgement messages for BASE24-pos transactions. Not used by the Banknet Interface process.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 30 (SEC)

KEYF SETTINGS

Each interface process must have its own Key File (KEYF) record. However, if LCONF parameter DES-TRIPLE-SINGLE has the value 'Y', the Key6 File (KEY6) is used instead of the KEYF. The following describes how the KEYF must be configured for the Banknet Interface.

The KEYF screens are not depicted; however, the fields are described as they actually appear on the Release 5.0 and above KEYF screens.

KEYF Screen 1 Settings

The following fields on KEYF screen 1 must be set as indicated.

DPC/FIID

The FIID of the interchange using this KEYF record. The value in this field must match the value in the INTERCHANGE FIID field on ICF/ICFE screen 1.

FIELD LENGTH: 1-4 alphanumeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: None

INTERFACE PROCESS

The name of the interface process associated with the interchange identified in the DPC/FIID field. The value in this field must match the value in the PROCESS field on ICF/ICFE screen 1.

FIELD LENGTH: 1-16 alphanumeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: None

ENCRYPT TYPE

Specifies the type of PIN management used by the interchange. Valid values are:

0 = Clear PINs
1 = Security module PIN management
2 = Software DES PIN management

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: Yes

DEFAULT VALUE: 0

BASE24 ENCRYPT TYPE

Specifies the type of PIN management used by BASE24. Valid values are:

- 0 = Clear PINs
- 1 = Security module PIN management

FIELD LENGTH: 1 numeric character

REQUIRED FIELD: Yes

DEFAULT VALUE: 0

PIN BLOCK FORMAT

Specifies the PIN block format of the PIN in the inbound and outbound external messages. If the Banknet Interface process supports PINs, then this field should be set to a 1 (ANSI PIN block). Valid values are:

- 0 = Clear PINs
- 1 = ANSI (PIN/PAN) PIN block
- 3 = PIN/PAD PIN block

FIELD LENGTH: 1 numeric character

REQUIRED FIELD: Yes

DEFAULT VALUE: 0

ANSI PAN FORMAT

If the value in the PIN BLOCK FORMAT field is 1 (ANSI), the value in this field specifies which PAN digits are used in the formation of the external message's PIN/PAN PIN block. Valid values are:

- 0 = 12 right-most digits of the PAN, excluding the check digit (ANSI standard)
- 1 = 12 right-most digits of the PAN, including the check digit
- 2 = 12 left-most digits of the PAN

FIELD LENGTH: 1 numeric character

REQUIRED FIELD: Yes, this field is required regardless of the value in the PIN BLOCK FORMAT field.

DEFAULT VALUE: 0

MAC ENCRYPT TYPE

A code specifying how message authentication will be performed for messages between the interchange and the interface. Not used by the Banknet Interface.

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: No
DEFAULT VALUE: 0

PIN PAD CHARACTER

If the value in the PIN BLOCK FORMAT field is 3 (PIN/PAD), the value in this field specifies the PAD character used in the formation of the external message's PIN block. Valid values are A through F.

FIELD LENGTH: 1 alphabetic character
REQUIRED FIELD: Yes, this field is required regardless of the value in the PIN BLOCK FORMAT field.
DEFAULT VALUE: F

MAC DATA TYPE

A code specifying the character set being used for the data being authenticated. Not used by the Banknet Interface.

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 0

NUMBER OF KEYS

A code identifying whether the inbound and outbound keys are combined or separate. The value in this field applies to the PIN and MAC keys defined on KEYF screen 2. Not used by the Banknet Interface. Valid values are:

- 1 = Combined--The inbound and outbound PIN keys are the same and the inbound and outbound MAC keys are the same.
- 2 = Separate--The inbound and outbound PIN keys are different and the inbound and outbound MAC keys are different.

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 1

KEY LENGTH

A code identifying whether single- or double-length Key Exchange Keys (KEKs) are being used with this interface. Not used by the Banknet Interface. Valid values are:

- 1 = Single-length KEKs
- 2 = Double-length KEKs

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 1

FULL MESSAGE MAC

A code specifying whether selected data elements or the entire message will be considered when computing the MAC. Not used by the Banknet Interface.

FIELD LENGTH: 1 alphabetic character
REQUIRED FIELD: Yes
DEFAULT VALUE: N

INTERMEDIATE KEYS

Intermediate keys are used whenever a PIN which is encrypted under a secure key (security module PIN management) must be decrypted into the clear.

CLEAR

The clear version of the intermediate key. A PIN that is encrypted under the intermediate key at the security module will be decrypted in software using the value in this field. Valid values for each character in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

FIELD LENGTH: 16 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000000000000000

CHECK DIGITS

The check digits corresponding to the value in the ENCRYPTED INTERMEDIATE KEY field. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters.

FIELD LENGTH: 4 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000

ENCRYPTED

Security module encrypted intermediate key. A PIN which has been encrypted under a secure key will be translated to be encrypted under the intermediate key at the security module. Valid values for each character in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

FIELD LENGTH: 1-16 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000000000000000

EXCHANGE KEYS

When PIN or MAC keys are automatically exchanged between the Interchange Interface and the interchange, they are to be encrypted under this key for the interchange. The exchange key must be manually exchanged between the two entities.

PIN KEY

The security module encrypted form of the key used to exchange PIN keys. There are two sets of 16 hexadecimal characters. The first set is used for single-length keys and both sets are used for double-length keys. Valid values for each character in these fields are 0 through 9 and A through F. Both fields must contain 16 valid characters. If the value in the KEY LENGTH field is 1, the second set of characters must be zeros.

FIELD LENGTH: 2 sets of 16 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000000000000000

CHECK DIGITS

The check digits corresponding to the value in the PIN KEY field. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters.

FIELD LENGTH: 4 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000

MAC KEY

The security module encrypted form of the key used to exchange MAC keys. Not used by the Banknet Interface.

FIELD LENGTH: 2 sets of 16 hexadecimal characters
REQUIRED FIELD: No
DEFAULT VALUE: 0000000000000000

CHECK DIGITS

The check digits corresponding to the value in the MAC KEY field.
Not used by the Banknet Interface.

FIELD LENGTH: 4 hexadecimal characters
REQUIRED FIELD: No
DEFAULT VALUE: 0000

KEYF Screen 2 Settings

The following fields on KEYF screen 2 must be set as indicated.

PIN KEY INFORMATION

OUTBOUND KEYS

The following fields contain the PIN keys used for transactions sent from the interface to the interchange. If the external PIN management is security module PIN management, the keys should be security module encrypted.

PIN KEY1

Used for the encryption or translation of PINs in transactions sent to the interchange. The interface process uses the key in this field when the CURRENT INDEX field contains a value of 1. Valid values for each character in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

FIELD LENGTH: 16 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000000000000000

CHECK DIGITS1

The check digits corresponding to the value in the PIN KEY1 field. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters.

FIELD LENGTH: 4 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000

CURRENT INDEX

Indicates which outbound PIN encryption key is currently being used by the interface. Valid values are:

1 = Value in the PIN KEY1 field
2 = Value in the PIN KEY2 field

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 1

PIN KEY2

Used for the encryption or translation of PINs in transactions sent to the interchange. The interface process uses the key in this field when the CURRENT INDEX field contains a value of 2. Valid values for each character in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

FIELD LENGTH: 16 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000000000000000

CHECK DIGITS2

The check digits corresponding to the value in the PIN KEY2 field. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters.

FIELD LENGTH: 4 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000

KEY COUNTER

The number of times the outbound PIN key has been changed by dynamic key management processing.

This counter is incremented whenever the outbound PIN key is changed by BASE24 or the co-processor. This value should require adjustment via file maintenance only if the co-processor expects a value other than zero when dynamic key management is established. Otherwise, the BASE24 Interchange (BIC) Interface process increments it automatically. Valid values for each character in this field are 0 through 9 and A through F. This field must contain 6 valid characters. Not used by the Banknet Interface.

FIELD LENGTH: 6 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 000000

INBOUND KEYS

The following fields contain the PIN keys used in the transactions sent by the interchange to the interface. If the external PIN management is security module PIN management, the keys should be security module encrypted.

PIN KEY1

Key used by the interchange to encrypt PINs in the transactions sent to BASE24. The interface process uses the key in this field when the CURRENT INDEX field contains a value of 1. Valid values for each character in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

FIELD LENGTH: 16 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000000000000000

CHECK DIGITS1

The check digits corresponding to the value in the PIN KEY1 field. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters.

FIELD LENGTH: 4 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000

CURRENT INDEX

Indicates which inbound PIN encryption key is currently being used by the interface. Valid values are:

1 = Value in the PIN KEY1 field
2 = Value in the PIN KEY2 field

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 1

PIN KEY2

Key used by the interchange to encrypt PINs in the transactions sent to BASE24. The interface process uses the key in this field when the CURRENT INDEX field contains a value of 2. Valid values for each character in this field are 0 through 9 and A through F. This field must contain 16 valid characters.

FIELD LENGTH: 16 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000000000000000

CHECK DIGITS2

The check digits corresponding to the value in the PIN KEY2 field. Valid values for each character in this field are 0 through 9 and A through F. This field must contain four valid characters.

FIELD LENGTH: 4 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000

KEY COUNTER

The number of times the inbound PIN key has been changed by dynamic key management processing.

This counter is incremented whenever the inbound PIN key is changed by BASE24 or the co-processor. This value should require adjustment via file maintenance only if the co-processor expects a value other than zero when dynamic key management is established. Otherwise, the BASE24 Interchange (BIC) Interface process increments it automatically. Valid values for each character in this field are 0 through 9 and A through F. This field must contain six valid characters. Not used by the Banknet Interface.

FIELD LENGTH: 6 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 000000

MAC KEY INFORMATION**OUTBOUND KEYS**

The following fields contain the MAC keys utilized for transactions sent from the interface to the co-network. These fields are not used by the Banknet Interface.

MAC KEY1

Key used by the BASE24 interface to generate MACs in messages from BASE24 to the co-processor. Not used by the Banknet Interface.

FIELD LENGTH: 16 hexadecimal characters
REQUIRED FIELD: No
DEFAULT VALUE: 0000000000000000

CHECK DIGITS1

The check digits corresponding to the value in the MAC KEY1 field. Not used by the Banknet Interface.

FIELD LENGTH: 4 hexadecimal characters
REQUIRED FIELD: No
DEFAULT VALUE: 0000

CURRENT INDEX

Indicates which outbound MAC key is currently being used by the interface. Not used by the Banknet Interface.

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: No
DEFAULT VALUE: 1

MAC KEY2

Key used by the BASE24 interface to generate MACs in messages from BASE24 to the co-processor. Not used by the Banknet Interface.

FIELD LENGTH: 16 hexadecimal characters
REQUIRED FIELD: No
DEFAULT VALUE: 0000000000000000

CHECK DIGITS2

The check digits corresponding to the value in the MAC KEY2 field. Not used by the Banknet Interface.

FIELD LENGTH: 4 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000

KEY COUNTER

The number of times the outbound MAC key has been changed by dynamic key management processing. Not used by the Banknet Interface.

FIELD LENGTH: 6 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 000000

INBOUND KEYS

The following fields contain the MAC keys used in the transactions sent by the co-processor to the interface. These fields are not used by the Banknet Interface.

MAC KEY1

Key used by the BASE24 interface to verify MACs generated by the interchange. Not used by the Banknet Interface.

FIELD LENGTH: 16 hexadecimal characters
REQUIRED FIELD: No
DEFAULT VALUE: 0000000000000000

CHECK DIGITS1

The check digits corresponding to the value in the MAC KEY1 field. Not used by the Banknet Interface.

FIELD LENGTH: 4 hexadecimal characters
REQUIRED FIELD: No
DEFAULT VALUE: 0000

CURRENT INDEX

Indicates which inbound MAC key is currently being used by the interface. Not used by the Banknet Interface.

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: No
DEFAULT VALUE: 1

MAC KEY2

Key used by the BASE24 interface to verify MACs generated by the interchange. Not used by the Banknet Interface.

FIELD LENGTH: 16 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000000000000000

CHECK DIGITS2

The check digits corresponding to the value in the MAC KEY2 field. Not used by the Banknet Interface.

FIELD LENGTH: 4 hexadecimal characters

REQUIRED FIELD: No

DEFAULT VALUE: 0000

KEY COUNTER

The number of times the inbound MAC key has been changed by dynamic key management processing. Not used by the Banknet Interface.

FIELD LENGTH: 6 hexadecimal characters

REQUIRED FIELD: No

DEFAULT VALUE: 000000

KEYF Screen 3 Settings

KEYF screen 3 contains dynamic key management parameters. The following fields on KEYF screen 3 must contain the values indicated below.

PIN KEY VARIANT

A code that specifies the key variant to be applied to the PIN Key Exchange Key (KEK) when translating PIN keys. Valid values are:

- 0 = Variant 0 - Translate the PIN key directly from the PIN KEK parts. The value "0" is used when both sides of the key exchange are using Thales (formerly Racal).
- 1 = Variant 1 - Include the variant 1 constant with the PIN KEK parts when translating the PIN key. The Atalla variant 1 constant is hexadecimal 0800 0000 0000 0000. If either or both sides of the key exchange are using Atalla, the value "1" should be used.

FIELD LENGTH: 1 numeric character

REQUIRED FIELD: Yes

DEFAULT VALUE: 0

MAC KEY VARIANT

A code that specifies the key variant to be applied to the MAC key exchange key when translating MAC keys. Not used by the Banknet Interface.

FIELD LENGTH: 1 numeric character

REQUIRED FIELD: Yes

DEFAULT VALUE: 0

LIMITS

The following fields are used to set the parameters for Dynamic Key Management. Only the PIN key parameters are used by the Banknet Interface. PIN keys can be changed based on time intervals or transaction activity. The counters and timers for the key are reset when the key is changed. If multiple limits are used, the interface will initiate a key exchange message and reset all of the counters and timers whenever any limit is reached.

The PIN key parameters on screen 3 must be set to 0 unless the ENCRYPT TYPE field on KEYF screen 1 is set to 1 (hardware PIN encryption).

PIN KEY TIMER VALUE

The maximum length of time that a PIN key should be used.

The time period can be expressed in minutes, hours, or days, depending on the value in the PIN KEY TIMER INTERVAL field. The shortest length of time is five minutes. Valid values are:

- ☐ 0, 5-1500 if PIN KEY TIMER INTERVAL is M (minutes)
- ☐ 0-1000 if PIN KEY TIMER INTERVAL is H (hours)
- ☐ 0-180 if PIN KEY TIMER INTERVAL is D (days)

BASE24 checks the values in the PIN KEY TIMER VALUE and PIN KEY TIMER INTERVAL fields when the KEYF record is added or updated. If the time is evenly divisible by a longer interval, BASE24 changes the value in the PIN KEY TIMER INTERVAL field to the longer interval and redisplay the value in the PIN KEY TIMER VALUE field expressed in the longer interval. For example, an entry of 60 minutes is automatically changed to an entry of 1 hour and an entry of 48 hours is automatically changed to an entry of 2 days. However, an entry of 61 minutes, even though it is greater than one hour, is not changed because it is not evenly divisible by 60.

This parameter is not used if the value in this field is set to 0. This parameter must be set to 0 unless the ENCRYPT TYPE field on KEYF screen 1 is set to 1 (hardware PIN encryption).

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 0

MAC KEY TIMER VALUE

The maximum length of time that a MAC key should be used. Not used by the Banknet Interface.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 0

PIN KEY TIMER INTERVAL

A code specifying the unit of time used with the value in the PIN KEY TIMER VALUE field to set the maximum length of time that a PIN key should be used. BASE24 changes the value in the PIN KEY TIMER INTERVAL field to the largest possible unit of time when the KEYF record is added or updated. Refer to the PIN KEY TIMER VALUE field for additional information. Valid values are:

D = Days
H = Hours
M = Minutes

FIELD LENGTH: 1 alphabetic character
REQUIRED FIELD: No
DEFAULT VALUE: M

MAC KEY TIMER INTERVAL

A code specifying the unit of time used with the value in the MAC KEY TIMER VALUE field to set the maximum length of time that a MAC key should be used. Not used by the Banknet Interface:

FIELD LENGTH: 1 alphabetic character
REQUIRED FIELD: No
DEFAULT VALUE: M

PIN KEY TRAN

The maximum number of transactions that can be performed before changing the PIN key. Valid values are:

0 = This parameter is not used.
50-100000 = The maximum number of transactions.

This parameter must be set to 0 unless the ENCRYPT TYPE field on KEYF screen 1 is set to 1 (hardware PIN encryption).

FIELD LENGTH: 1-6 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 0

MAC KEY TRAN

The maximum number of transactions that can be performed before changing the PIN key. Not used by the Banknet Interface.

FIELD LENGTH: 1-6 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 0

PIN KEY ERROR

The maximum number of PIN translation errors that can occur with the present PIN key before changing the key. Not used by the Banknet Interface.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 0

MAC KEY ERROR

The maximum number of MAC translation errors that can occur with the present MAC key before changing the key. Not used by the Banknet Interface.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 0

CONSECUTIVE PIN KEY ERROR

The maximum number of consecutive PIN translation errors that can occur with the present PIN key before changing the key. Not used by the Banknet Interface.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 0

CONSECUTIVE MAC KEY ERROR

The maximum number of consecutive MAC translation errors that can occur with the present MAC key before changing the key. Not used by the Banknet Interface.

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 0

KMAC SYNCHRONIZATION ERROR

The maximum number of key synchronization errors that can occur with the present MAC key before changing the key. Not used by the Banknet Interface.

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: No
DEFAULT VALUE: 0

CLEAR OLD KEY TIMER VALUE

The maximum length of time, in seconds, that an old key can be used before it is automatically cleared by the interface process.

The INBOUND PIN KEY and OUTBOUND PIN KEY fields on KEYF screen 2 can contain information for two keys, identified as KEY1 and KEY2. The value in the CURRENT INDEX field for each type of key identifies which key (1 or 2) is currently in use. At the time a new key is placed in service, the other key becomes the old key. For example, if Inbound PIN Key number 1 is currently in use when BASE24 receives a new key, the new key is stored as Inbound PIN Key number 2. Inbound PIN Key number 2 becomes the current key and Inbound PIN Key number 1 becomes the old key. (If Inbound PIN Key number 2 is the current key, it becomes the old key and the new key is stored as Inbound PIN Key number 1.)

When processing a transaction, BASE24 uses the new key for PIN processing or message authentication. If the security device used by BASE24 for PIN processing or message authentication detects an error because the interchange is still using the old key, it can repeat the PIN processing or message authentication using the old key before BASE24 declines the transaction. The value in this field controls how long the old key is available. Valid values are:

0 = The old key is not cleared.
1-9999 = The maximum length of time, in seconds, that an old key can be used before being cleared.

This parameter must be set to 0 unless the ENCRYPT TYPE field on KEYF screen 1 is set to 1 (hardware PIN encryption) or the MAC ENCRYPT TYPE field on KEYF screen 1 is set to 1 (hardware message authentication).

FIELD LENGTH: 1-4 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: 0

ORIGINATING ID

Identifies the sender of dynamic key management messages. This field is not used by the Banknet Interface.

FIELD LENGTH: 4-16 alphanumeric characters

REQUIRED FIELD: No

DEFAULT VALUE: None

RECEIVING ID

Identifies the receiver of dynamic key management messages. This field is not used by the Banknet Interface.

FIELD LENGTH: 4-16 alphanumeric characters

REQUIRED FIELD: No

DEFAULT VALUE: None

KEY PROCESSING TYPE

A code identifying the type of dynamic key management processing this interface can perform. This field is not used by the Banknet Interface.

FIELD LENGTH: 1 alphabetic character

REQUIRED FIELD: No

DEFAULT VALUE: N

NOTARIZATION SUPPORTED

A code identifying whether this interface supports key notarization. This field is not used by the Banknet Interface.

FIELD LENGTH: 1 numeric character

REQUIRED FIELD: No

DEFAULT VALUE: 0

KEY6 SETTINGS

Each interface process must have its own Key6 File (KEY6) record if Triple to Single DES encryption is required. Otherwise, the Key File (KEYF) is used. The following describes how the KEY6 must be configured for the Banknet Interface.

The fields on the screens of the KEY6 are almost identical to those on the KEYF. Therefore, only the screen fields on the KEY6 that differ from those on the KEYF are described below.

KEY6 Screen 1 Settings

MAC KEY LENGTH

A code identifying whether single- or double-length Key MAC keys are being used with this interface. Not used by the Banknet Interface. Valid values are:

- 1 = Single-length MAC keys
- 2 = Double-length MAC keys

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 1

KEY6 Screen 2 Settings

PIN KEY INFORMATION

OUTBOUND KEYS

The following fields contain the PIN keys used for outbound transactions (that is, transactions sent by the interface to the network).

PIN KEY1

A double-length key used for the translation of PINs in transactions sent to the DPC or interchange. The interface process uses the key in these fields when the CURRENT INDEX field contains a value of 1. Valid values for each position in these fields are 0 through 9 and A through F. These fields must contain 32 valid characters, with 16 characters per field.

OCCURS: 2 times
FIELD LENGTH: 16 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000000000000000

PIN KEY2

A double-length key used for the translation of PINs in transactions sent to the DPC or interchange. The interface process uses the key in these fields when the CURRENT INDEX field contains a value of 2. Valid values for each position in these fields are 0 through 9 and A through F. These fields must contain 32 valid characters, with 16 characters per field.

OCCURS: 2 times
FIELD LENGTH: 16 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000000000000000

INBOUND KEYS

The following fields contain the PIN keys used for inbound transactions (that is, transactions sent by the network to the interface).

PIN KEY1

A double-length key used by the DPC or interchange to translate PINs in the transactions sent to the BASE24 transaction processing system. The interface process uses the key in these fields when the CURRENT INDEX field contains a value of 1. Valid values for each position in these fields are 0 through 9 and A through F. These fields must contain 32 valid characters, with 16 characters per field.

OCCURS: 2 times
FIELD LENGTH: 6 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000000000000000

PIN KEY2

A double-length key used by the DPC or interchange to translate PINs in the transactions sent to the BASE24 transaction processing system. The interface process uses the key in these fields when the CURRENT INDEX field contains a value of 2. Valid values for each position in these fields are 0 through 9 and A through F. These fields must contain 32 valid characters, with 16 characters per field.

OCCURS: 2 times
FIELD LENGTH: 16 hexadecimal characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0000000000000000

KEY6 Screen 3 Settings

MAC KEY INFORMATION

OUTBOUND KEYS

The following fields contain the MAC keys used for outbound transactions (that is, transactions sent by the interface to the network).

MAC KEY1

A key used by the BASE24 interface to generate MACs in messages from the BASE24 transaction processing system to the DPC or co- network. The interface process uses the key in these fields when the CURRENT INDEX field contains a value of 1. Valid values for each position in these fields are 0 through 9 and A through F. These fields must contain 32 valid characters, with 16 characters per field. Not used by the Banknet Interface.

OCCURS: 2 times
FIELD LENGTH: 16 hexadecimal characters
REQUIRED FIELD: No
DEFAULT VALUE: 0000000000000000

MAC KEY2

A key used by the BASE24 interface to generate MACs in messages from the BASE24 transaction processing system to the DPC or co-network. The interface process uses the key in these fields when the CURRENT INDEX field contains a value of 2. Valid values for each position in these fields are 0 through 9 and A through F. These fields must contain 32 valid characters, with 16 characters per field. Not used by the Banknet Interface.

OCCURS: 2 times
FIELD LENGTH: 16 hexadecimal characters
REQUIRED FIELD: No
DEFAULT VALUE: 0000000000000000

INBOUND KEYS

The following fields contain the MAC keys used in inbound transactions (that is, transactions sent by the network to the interface).

MAC KEY1

A key used by the BASE24 interface to verify MACs generated by the DPC or interchange. The interface process uses the key in these fields when the CURRENT INDEX field contains a value of 1. Valid values for each position in these fields are 0 through 9 and A through F. These fields must contain 32 valid characters, with 16 characters per field. Not used by the Banknet Interface.

OCCURS: 2 times
FIELD LENGTH: 16 hexadecimal characters
REQUIRED FIELD: No
DEFAULT VALUE: 0000000000000000

MAC KEY2

A key used by the BASE24 interface to verify MACs generated by the DPC or interchange. The interface process uses the key in these fields when the CURRENT INDEX field contains a value of 2. Valid values for each position in these fields are 0 through 9 and A through F. These fields must contain 32 valid characters, with 16 characters per field. Not used by the Banknet Interface.

OCCURS: 2 times
FIELD LENGTH: 16 hexadecimal characters
REQUIRED FIELD: No
DEFAULT VALUE: 0000000000000000

STF SETTINGS

The Switch Terminal File (STF) may be used to provide additional information to the Banknet Interface, such as the level of a Cardholder Activated Terminal (CAT), and the card input capabilities of a terminal.

The STF can be used by acquirers to provide MasterCard assigned IDs for Private Label Processing, for IIAS real-time substantiated transactions, and for e-commerce transactions that are processed under the Maestro® Advance Registration Program™.

The STF may also be used to identify whether a specific terminal has been certified for EMV processing by MasterCard.

For customers with some terminals certified for EMV processing by MasterCard and others not certified for EMV processing by MasterCard, STF records will be required. The “default” certification status will be specified by the EMV CERTIFIED FLAG field value in a “wild card” STF record (i.e. where the TERMINAL ID is set to asterisks). Individual terminals with the opposite certification status must be configured individually in the STF (i.e. where the TERMINAL ID is not set to asterisks), with the EMV CERTIFIED FLAG field set to the opposite value to that in the “wild card” record.

Acquirers running both a BASE24-atm system and a BASE24-pos system must configure two “wild card” STF records, one containing a PROD ID of ‘01’ (ATM), and the other containing a PROD ID of ‘02’ (POS).

For acquirers running a BASE24-atm system, the MERCHANT ID must be set to asterisks in both the “wild card” STF record and individual terminal STF records. For acquirers running a BASE24-pos system, the MERCHANT ID must be set to asterisks in the “wild card” STF record, and may be set to asterisks in individual terminal STF records.

Accessing The STF Screen

There are two ways to access the STF screen:

1. During the initial logon into the BASE24 AFT system, after the user id, password, logical network have been filled in, type STF in the FILE DESTINATION field on the bottom of the BASE24 LOGON Screen, then hit F1.
2. Logon to the BASE24 AFT system, entering the VIRTUAL MENU screen. Choose SWITCH FUNCTIONS and hit F1. Choose STF and hit F1.

STF Screen Settings

The STF screen is shown below followed by a description of its fields.

BASE24-SWI	SWITCH	TERMINAL	FILE	LNET	YY/MM/DD	HH:MM	01 OF 01
SWITCH FIID: _____		MERCHANT ID: _____					
TERMINAL FIID: _____		TERMINAL ID: _____				PROD ID: __ ____	

SWITCH MERCHANT INFORMATION ID: _____ OFFSET: ____ LENGTH: ____ ALT ID: _____	SWITCH TERMINAL INFORMATION ID: _____ OFFSET: ____ LENGTH: ____
--	---

ON PREMISE FLAG: _ (Y/N)	
EMV CERTIFIED FLAG: _ (Y/N)	
CARDHOLDER ACTIVATED TERMINAL: _	_____
TERMINAL INPUT CAPABILITIES: _	_____
MERCHANT CUSTOMER SERVICE PHONE NUM: _____	

RECORD LAST CHANGED: 09/07/08 10:10 BY USER: 0255 , 00000255 CHANGE

***** BASE24 *****

NEW PAGE:	FILE DESTINATION:	NEW LOGICAL NETWORK ID:
F9-NEXT 20 PREFIXES	F11-PREVIOUS 20 PREFIXES	F12-HELP

The following fields on STF screen 1 must be set as indicated.

SWITCH FIID

An identifier for the Banknet financial institution. The value entered in this field must be identical to the value entered in the INTERCHANGE FIID field on ICF/ICFE screen 1.

FIELD LENGTH: 1-4 alphanumeric characters

REQUIRED FIELD: Yes

DEFAULT VALUE: None

MERCHANT ID

An identifier for the merchant. When BASE24-pos is used, this field is required. When BASE24-pos is not used, this field should be filled with asterisks.

FIELD LENGTH: 1-4 alphanumeric characters

REQUIRED FIELD: No

DEFAULT VALUE: None

TERMINAL FIID

The FIID associated with the terminal.

FIELD LENGTH: 1-4 alphanumeric characters

REQUIRED FIELD: No

DEFAULT VALUE: None

TERMINAL ID

The BASE24 terminal identifier. The value entered in this field must be left-justified and blank-filled.

FIELD LENGTH: 1-16 alphanumeric characters

REQUIRED FIELD: No

DEFAULT VALUE: None

PROD ID

An identifier for the product with which the terminal is associated. Valid values are:

01 = ATM

02 = POS

FIELD LENGTH: 2 alphanumeric characters

REQUIRED FIELD: Yes

DEFAULT VALUE: None

SWITCH MERCHANT INFORMATION

The following fields contain information on the interchange merchant.

ID

The MasterCard Assigned ID to which the BASE24 merchant identifier should be translated. The value entered in this field should be left-justified and blank-filled. This field is used by the Banknet Interface to provide a value in DE 48 SE 32 for acquirers that support Private Label Processing or IIAS real-time substantiated transactions or e-commerce transactions that are processed under the Maestro® Advance Registration Program™ (MARF).

FIELD LENGTH: 1-19 alphanumeric characters

REQUIRED FIELD: No

DEFAULT VALUE: None

OFFSET

The position in the BASE24 merchant identifier that must be used as the first digit to append to the interchange merchant identifier to format the merchant identifier in the external message.

FIELD LENGTH: 1-2 numeric characters

REQUIRED FIELD: Yes

DEFAULT VALUE: 0

LENGTH

The number of digits of the BASE24 merchant identifier that must be used as the interchange merchant identifier. The field should be allowed to default to 00 if this field and the OFFSET field are not used for translation purposes.

FIELD LENGTH: 1-2 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0

ALT ID

The Visa Merchant Verification Value to which the BASE24 merchant identifier should be translated. This field is used by the Banknet Interface to provide the MVV for MasterCard-acquired Visa healthcare transactions. When the service is a Visa card the interface will use this value to populate DE 48 SE 32 for IIAS real-time substantiated transactions.

FIELD LENGTH: 1-15 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

SWITCH TERMINAL INFORMATION

The following fields contain information about the interchange's terminal.

ID

The interchange terminal identifier to which the BASE24 terminal identifier should be translated. The value entered in this field should be left-justified and blank-filled. This field is not used by the Banknet Interface.

FIELD LENGTH: 1-19 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

OFFSET

The position in the BASE24 terminal identifier that must be used as the first digit to append to the interchange terminal identifier to format the terminal identifier in the external message. Should be set to zero for the Banknet Interface.

FIELD LENGTH: 1-2 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0

LENGTH

The number of digits of the BASE24 terminal identifier that must be used as the interchange terminal identifier. This field should be allowed to default to 00 if this field and the OFFSET field are not to be used for translation purposes. Should be set to zero for the Banknet Interface.

FIELD LENGTH: 1-2 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: 0

ON PREMISE FLAG

A value indicating whether the terminal is at the same location as the institution that owns the terminal. Valid values are:

Y = On premise
N = Off premise

FIELD LENGTH: 1 alphabetic character
REQUIRED FIELD: No
DEFAULT VALUE: None

EMV CERTIFIED FLAG

A value indicating whether the terminal is certified for EMV processing by the interchange. Valid values are:

Y = Certified
N = Not certified

FIELD LENGTH: 1 alphabetic character
REQUIRED FIELD: No
DEFAULT VALUE: None

CARDHOLDER ACTIVATED TERMINAL

The Banknet Interface uses this field to indicate the Cardholder Activated Terminal (CAT) level of the corresponding terminal, if the information is not provided in the BASE24-pos internal message. Valid values are:

- 0 = Not a CAT
- 1 = Automated Dispensing Machine with PIN/Level 1
- 2 = Self-Service Terminal without PIN/Level 2
- 3 = Limited Amount Terminal/Level 3
- 4 = In-flight Commerce/Level 4
- 5 = Reserved
- 6 = Electronic Commerce/Level 6
- 7 = Transponder Transaction/Level 7

FIELD LENGTH: 1 numeric character

REQUIRED FIELD: No

DEFAULT VALUE: None

TERMINAL INPUT CAPABILITIES

The Banknet Interface uses this field to indicate the card data input capabilities of the corresponding terminal, if the information is not provided in the BASE24-pos internal message. Valid values are:

- 0 = Unknown or unspecified
- 1 = No terminal used
- 2 = Magnetic stripe reader only
- 3 = Contactless M/Chip (proximity chip)
- 4 = Contactless magnetic stripe (proximity chip) only
- 5 = EMV (compatible chip) reader and magnetic stripe reader
- 6 = Key entry only
- 7 = Magnetic stripe reader and key entry
- 8 = EMV (compatible chip) reader, magnetic stripe reader and key entry
- 9 = EMV (compatible chip) reader only

FIELD LENGTH: 1 numeric character

REQUIRED FIELD: No

DEFAULT VALUE: None

MERCHANT CUSTOMER SERVICE PHONE NUM

This field is not used by the Banknet Interface.

FIELD LENGTH: 1-13 numeric characters

REQUIRED FIELD: No

DEFAULT VALUE: None

SNF SETTINGS

The Station Name File (SNF) may contain one record for each Banknet Interface process. Each record contains up to 16 station pairs on which messages are sent to and from Banknet. The key to the SNF record is the Banknet Interface process name.

The SNF is read at initialization and on warmboots to obtain the names of stations to which the Banknet Interface process is connected. If an SNF record does not exist for an interface process, then the process uses the stations configured on ICF/ICFE screen 1 and ICF/ICFE screen 12 instead. You can not use stations from both the ICF/E and the SNF. They are mutually exclusive.

Accessing The SNF Screen

The Banknet Menu screen is accessed from screen 12 of the Interchange Configuration File (ICF/ICFE). From the Banknet menu screen, the SNF screen can be accessed. Provided an operator is assigned the necessary security clearance for the ICF/ICFE, the following procedures can be followed to access the SNF screen. Please refer to the ***BASE24 User Security Manual*** for an explanation of the BASE24 CRT Access system and a complete set of procedures for accessing it.

1. Complete a successful logon to BASE24 via the BASE24 Logon screen by completing the fields displayed on the screen and pressing F1. The Virtual Menu screen is then displayed.
2. Select the SWITCH FUNCTIONS option from the Virtual Menu screen and press F1.
3. Select ICF or ICFE from the Switch Functions Menu screen and press F1. Interchange Configuration File (ICF) screen 1 or Enhanced Interchange Configuration File (ICFE) screen 1 is then displayed. Access to the ICF/ICFE maintenance screens can also be gained by using the FILE DESTINATION field located at the bottom of all BASE24 screens.
4. Enter Banknet in the SWITCH TYPE field on ICF/ICFE screen 1 or read an existing record that has Banknet as its SWITCH TYPE.
5. Go to ICF/ICFE screen 12 by pressing F11. Access to ICF/ICFE screen 12 can also be gained by entering 12 in the NEW PAGE field located at the bottom of the BASE24 screen and pressing F7.
6. Press F8 to access the Banknet Menu.
7. Position the cursor next to SNF entry using the arrow keys, the return key, or the tab key. Press F1 to select this option.

SNF Screen Settings

The SNF screen is shown below followed by a description of its fields.

BASE24-SWI	STATION NAME	FILE	LNET	YY/MM/DD	HH:MM	01 OF 01
INTERCHANGE FIID: _____		PROCESS: _____				
SEND STATION	RECEIVE STATION	TYPE	SWITCH DATA	SWITCH INFO		
_____	_____	-	_____	_____		
_____	_____	-	_____	_____		
_____	_____	-	_____	_____		
_____	_____	-	_____	_____		
_____	_____	-	_____	_____		
_____	_____	-	_____	_____		
_____	_____	-	_____	_____		
_____	_____	-	_____	_____		
_____	_____	-	_____	_____		
_____	_____	-	_____	_____		
_____	_____	-	_____	_____		
_____	_____	-	_____	_____		
_____	_____	-	_____	_____		
_____	_____	-	_____	_____		
_____	_____	-	_____	_____		
_____	_____	-	_____	_____		
RECORD LAST CHANGED: 09/07/08 10:10 BY USER: 0255 , 00000255 CHANGE						
***** BASE24 *****						
NEW PAGE:	FILE DESTINATION:	NEW LOGICAL NETWORK ID:				
	F12-HELP					

The following fields on SNF screen 1 must be set as indicated.

INTERCHANGE FIID

This field is system-protected, and displays the INTERCHANGE FIID from ICF/ICFE screen 1.

FIELD LENGTH: 1-4 alphanumeric characters

REQUIRED FIELD: No

DEFAULT VALUE: None

PROCESS

This field is system-protected, and displays the PROCESS from ICF/ICFE screen 1.

FIELD LENGTH: 1-16 alphanumeric characters

REQUIRED FIELD: No

DEFAULT VALUE: None

SEND STATION

The symbolic names of stations used to send messages to Banknet. If used, these names must match the names of stations added in the Network Environment Source File (NEFS) for the Banknet Interface process, and must be paired with a receive station.

FIELD LENGTH: 16 fields of 16 alphanumeric characters each

REQUIRED FIELD: No

DEFAULT VALUE: None

RECEIVE STATION

The symbolic names of stations used to receive messages from Banknet. If used, these names must match the names of stations added in the Network Environment Source File (NEFS) for the Banknet Interface process, and must be paired with a send station. The name may be identical to the send station on the same line but must be unique between lines.

FIELD LENGTH: 16 fields of 16 alphanumeric characters each

REQUIRED FIELD: No

DEFAULT VALUE: None

TYPE

This field indicates the type of traffic supported by the station pair. Valid values are:

A = Acquirer

I = Issuer

B = Both

FIELD LENGTH: 16 fields of 1 alphabetic character each

REQUIRED FIELD: No

DEFAULT VALUE: B

SWITCH DATA

The first character contains a value indicating the type of service supported by the station pair. Valid values are:

0 = Credit

1 = Debit

2 = Both

FIELD LENGTH: 16 fields of 16 alphanumeric characters each

REQUIRED FIELD: No

DEFAULT VALUE: None

SWITCH INFO

The Banknet Interface process does not use these fields.

FIELD LENGTH: 16 fields of 16 alphanumeric characters each

REQUIRED FIELD: No

DEFAULT VALUE: None

BNIIF SETTINGS

The Banknet Institution Identification File contains one record for every FIID processed through the Banknet Interface process. The purpose of this file is to allow acquirers to configure different Customer IDs (the value sent in DE 32 in authorization request messages) for different acquiring FIIDs. The BNIIF also maintains information about issuer options sent to Banknet in network management messages, e.g. the SECURITY CODE associated with the FIID.

For members supporting enhanced session management, the BNIIF also contains a record for the Group Sign-on Identifier (GSI) used by the Banknet Interface process. The record contains the GSI itself, and other information used in network management messages.

Accessing The BNIIF Screen

The Banknet Menu screen is accessed from screen 12 of the Interchange Configuration File (ICF/ICFE). From the Banknet menu screen, the BNIIF screen can be accessed. Provided an operator is assigned the necessary security clearance for the ICF/ICFE, the following procedures can be followed to access the BNIIF screen. Please refer to the ***BASE24 User Security Manual*** for an explanation of the BASE24 CRT Access system and a complete set of procedures for accessing it.

1. Complete a successful logon to BASE24 via the BASE24 Logon screen by completing the fields displayed on the screen and pressing F1. The Virtual Menu screen is then displayed.
2. Select the SWITCH FUNCTIONS option from the Virtual Menu screen and press F1.
3. Select ICF or ICFE from the Switch Functions Menu screen and press F1. Interchange Configuration File (ICF) screen 1 or Enhanced Interchange Configuration File (ICFE) screen 1 is then displayed. Access to the ICF and ICFE maintenance screens can also be gained by using the FILE DESTINATION field located at the bottom of all BASE24 screens.
4. Enter Banknet in the SWITCH TYPE field on ICF/ICFE screen 1 or read an existing record that has Banknet as its SWITCH TYPE.
5. Go to ICF/ICFE screen 12 by pressing F11. Access to ICF/ICFE screen 12 can also be gained by entering 12 in the NEW PAGE field located at the bottom of the BASE24 screen and pressing F7.
6. Press F8 to access the Banknet Menu.

7. Position the cursor next to BNIIF entry using the arrow keys, the return key, or the tab key. Press F1 to select this option.

BNIIF Screen Settings

BNIIF screen 1 contains the FIID of the financial institution and its associated MCI Institution number (Customer ID), Customer ID Indicator, and Customer IDs associated with specific Currency Codes. It also contains the associated Security Code, Service Type, Traffic Type, AVS Type, Banknet Issuer Prefixes and Prefix Length used in network management messages.

Additional Banknet Issuer Prefix data may be displayed by pressing the F9 function key for the next 20 prefixes, or the F11 function key for the previous 20 prefixes.

The screen is shown below followed by a description of its fields.

BASE24-SWI	BNIIF	FILE	MAINTENANCE	LN	NET	YY/MM/DD	HH:MM	01	OF	01
FINANCIAL ID: _____		INTERCHANGE FIID: _____								
BANKNET MCI ID: _____		SECURITY CODE: _____								
SERVICE TYPE: _ (D/C/B)		TRAFFIC TYPE: _ (A/I/B)								
CUST ID IND: _		AVS TYPE _ (0/1/2/3/4)								
BANKNET ISSUER:										
PREFIX	LGTH	PREFIX	LGTH	PREFIX	LGTH	PREFIX	LGTH			
_____	__	_____	__	_____	__	_____	__			
_____	__	_____	__	_____	__	_____	__			
_____	__	_____	__	_____	__	_____	__			
_____	__	_____	__	_____	__	_____	__			
_____	__	_____	__	_____	__	_____	__			
MC CUST ID	CRNCY CDE	MC CUST ID	CRNCY CDE							
_____	__	_____	__							
_____	__	_____	__							
_____	__	_____	__							
RECORD LAST CHANGED: 09/07/08 10:10 BY USER: 0255 , 00000255 CHANGE										
***** BASE24 *****										
NEW PAGE: FILE DESTINATION: NEW LOGICAL NETWORK ID:										
F9-NEXT 20 PREFIXES F11-PREVIOUS 20 PREFIXES F12-HELP										

The following fields on BNIIF screen 1 must be set as indicated.

FINANCIAL ID (FIID)

The FIID for the financial institution.

If the record relates to the Group Sign-on Identifier (GSI) used by the Banknet Interface process, then the value in this field must match the value in the SWITCH ID field on screen 1 of the ICF/ICFE.

FIELD LENGTH: 1-4 alphanumeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: None

INTERCHANGE FIID

The FIID from screen 1 of the ICF/ICFE. This field can be blank filled if only one copy of the Banknet Interface is being run or this record is valid for all copies.

FIELD LENGTH: 1-4 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

BANKNET MCI ID

The Banknet-assigned MCI identification number for this institution. The number in this field must be the Customer ID that MasterCard assigned to the transaction acquirer.

If the record relates to the Group Sign-on Identifier (GSI) used by the Banknet Interface process, then this field must contain the GSI itself.

FIELD LENGTH: 5-6 numeric characters, depending upon the MAX-GROUP-LGTH LCONF param or its default. This field should be right justified and zero filled to the left.
REQUIRED FIELD: No
DEFAULT VALUE: None

SECURITY CODE

The Banknet-assigned security code required for online requests. This value is included in DE 96 in network management messages.

FIELD LENGTH: 1-5 alphanumeric characters

REQUIRED FIELD: No

DEFAULT VALUE: None

SERVICE TYPE

A value indicating the services type used. Banknet no longer allows this value to be included in subfield 1 of DE 94 in network management messages. This field should be set to C for consistency with the value configured in the STATION TYPES fields on ICFE/ICF screen 12. Valid values are:

D = Debit only

C = Credit only

B = Both debit and credit

FIELD LENGTH: 1 alphanumeric character

REQUIRED FIELD: Yes

DEFAULT VALUE: None

TRAFFIC TYPE

A value indicating the acceptable traffic type associated with this MCI-ID (Acquirer, Issuer or Both). This value is included in subfield 2 of DE 94 in network management messages. Valid values are:

A = Acquirer traffic only
I = Issuer traffic only
B = Both acquirer and issuer

FIELD LENGTH: 1 alphanumeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: None

CUST ID IND

A value indicating how the MasterCard assigned Customer ID, to be mapped to DE 32, is configured. Valid values are:

0 = Terminal FIID (in the BNIIF)
1 = Terminal (Institution ID from Terminal file)
2 = Currency Code per Terminal FIID (in the BNIIF)
3 = Banknet Interface process (Institution ID from ICF/ICFE)

FIELD LENGTH: 1 alphanumeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 0

AVS TYPE

A value indicating the acceptable address verification logon types associated with this MCI-ID. This value is included in subfield 3 of DE 94 in network management messages. Valid values are:

0 = Not supported
1 = Issuer receives complete address data
2 = Issuer receives condensed address data (first five numeric digits)
3 = Issuer receives condensed address data (up to first five numeric digits)
4 = Issuer receives numeric data (address and postal code)

FIELD LENGTH: 1 numeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: 0

PREFIX

The prefixes issued under the above FIID. There are two reasons for including prefixes in BNIF records:

- ☐ Issuers may want to sign-on to Banknet by prefix (using a value of '001' in DE 70) or by prefix within GSI (using a value of '065' in DE 70). This allows issuers to split the routing of their issuer traffic. For example, an issuer may sign-on all debit card prefixes through one interface process, and sign-on credit card prefixes through another interface process. This will result in Banknet routing all debit card transactions to one interface process, and all credit card transactions to another interface process.
- ☐ If an issuer sends File Update messages to Banknet, then configuring prefixes in the BNIF allows the issuer to send different Forwarding Institution Ids (DE 33) to Banknet in the File Update messages. The value of DE 33 is taken from the MCI ID in the BNIF record in which the appropriate prefix is configured. This function may be useful if the BASE24 issuer system is operating on behalf of a number of issuers.

Any prefixes configured should only be 'on-us' prefixes, i.e. prefixes "belonging" to the issuer, so acquirers should not configure prefixes in a BNIF record.

FIELD LENGTH: 1-11 numeric character
REQUIRED FIELD: No
DEFAULT VALUE: None

LENGTH

The length of the associated prefix issued under the above FIID.

FIELD LENGTH: 2 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

MC CUST ID

The MasterCard assigned Customer ID associated with the specified currency code.

FIELD LENGTH: 1-6 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

CRNCY CDE

The currency code associated with the MasterCard assigned Customer ID.

FIELD LENGTH: 1-3 numeric characters

REQUIRED FIELD: No

DEFAULT VALUE: None

TOKEN FILE

The Release 5.0 or above Banknet Interface user may wish to update the token records in the Token File for the following reasons:

1. Log the chosen tokens to the Interchange Log File (ILF) and/or the POS Transaction Log File (PTLF) and/or the ATM Transaction Log File (TLF).
2. Extract the chosen tokens from the Interchange Log File (ILF) and/or the POS Transaction Log File (PTLF) and/or the ATM Transaction Log File (TLF).
3. Send the chosen tokens to be sent to the Host.

An explanation of the Token File screens and screen fields can be found in the ***BASE24 Tokens Manual***.

OTHER BASE24 FILES

APCF/IPCF

Release 6.0 users must configure records for the Banknet Interface process in the Acquirer Processing Code File (APCF) and in the Issuer Processing Code File (IPCF).

The APCF contains information on transaction types that are supported by BASE24 and are allowed to be received from Banknet.

The IPCF contains information on transaction types that are supported by BASE24 and are allowed to be sent to Banknet.

PTD/PTDF/PRDF

MasterCard has a requirement for acquirers to provide a non-zero Merchant Category Code (MCC) in an authorization request message, because the acquirer must identify the merchant type by providing a valid MCC.

To ensure compliance, the SIC Code fields in the POS Terminal Data files (PTD/PTDF) and the POS Retailer Definition File (PRDF) must be set to non-zero values by MasterCard BASE24-pos acquirers.

IPF REFRESH CONFIGURATION

The Banknet IPF Refresh process uses the following TACL assigns and parameters. These assigns and parameters are valid for the current TACL session, and can be entered at a TACL prompt or placed in an obey file.

IPF Refresh Assigns

These assigns are set using the standard TACL ASSIGN command.

Assign	Description
IPFX	This assign identifies the location of the Optimized Current Parameters File used as input for the IPF Refresh Process. Valid values are a HP NonStop (formerly Tandem) file name.
IPF	This assign indicates the location of the disk file used as output from the IPF Refresh Process. Valid values are a HP NonStop (formerly Tandem) file name.
TPLT	This assign indicates the template file that is used as template for the IPF. Valid values are a HP NonStop (formerly Tandem) file name.

IPF Refresh Parameters

These parameters are set using the standard TACL PARAM command.

Parameter	Description
NUM-OF-PROGRAMS	This param identifies the number of PROGRAM-NNN parameters that should follow this param. Valid values are 1–65535. Zero is not valid because at least one acceptance program is required. See PROGRAM-NNN param.
PROGRAM-NNN	Each of these params represents an acceptance program that is required for routing purposes. Replace NNN with a three-character identifier for the acceptance program. The format of the parameter value is XXXZ:YY, where XXX is the MasterCard acceptance program and Z (optional) is the product type id. The last two characters of each param (YY) represent the card type that will be associated in the IPF with all prefixes under that acceptance program. Should a card type not be required for a program, these two characters should be kept blank. The separator ":" should remain after XXX or XXXZ. Valid values are 3 ASCII alphanumeric characters (XXX) or 4 ASCII alphanumeric characters (XXXZ), ':' as a separator, and 2 ASCII alphanumeric characters (YY), i.e. XXX:YY or XXXZ:YY. The number of acceptance programs required is dependent on the value of the NUM-OF-PROGRAMS param. In the RUN command, the NUM-OF-PROGRAMS param must precede all PROGRAM-NNN params.
SUPPRESS-DUP-MESS	This param controls whether duplicate key error messages are included in the error report. As POS and ATM records can both have the same key, duplicate records can be encountered if the TERMINAL-CATEGORY param is set to "BOTH". Valid values are: Y = Suppress error messages N = Do not suppress error messages
TERMINAL-CATEGORY	This param identifies the terminal category that BINs must support if they are to be added to the IPF. Valid values are: ATM = ATM only POS = POS only BOTH = ATM or POS

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Section 4

STARTUP AND CONTROL

INTRODUCTION

This section provides information on the configuration of the Banknet Interface process. It also describes the processing involved for initialization, network control commands, and timers.

HP NONSTOP CONFIGURATION

There are areas in its processing where BASE24 must use timestamps in Greenwich Mean Time (GMT). In the Banknet Interface, for example, the value of the Transmission Date and Time included in field 7 of the message sent to Banknet must be represented in GMT. This means that BASE24 must be able to convert timestamps between GMT and Local Civil Time (LCT), which is the local time of the time zone in which the HP NonStop (formerly Tandem) system is running. Note that the LCT is the same as the Local Standard Time (LST), which is a fixed offset from GMT for the time zone, except that the LCT is also adjusted for any Daylight Savings Time (DST) in the time zone.

The Banknet Interface could use the GMT OFFSET field on Screen 12 of the ICF(E) to calculate the timestamp sent to Banknet in field 7 of the message. However, this approach would require customers to adjust the value of this field every time that DST begins and ends in the time zone. To avoid the need for manual intervention, the Banknet Interface uses the system clock to calculate the Transmission Date and Time.

It is therefore crucial that the configuration of the HP NonStop (formerly Tandem) system allows BASE24 to calculate GMT from the system clock. This means that the `TIME_ZONE_OFFSET` parameter must correctly identify the LST for the time zone. Similarly, if the time zone uses DST, then the `DAYLIGHT_SAVING_TIME` parameter must be set to either `USA66` (for U.S. customers) or `TABLE` (for other customers). If the parameter is set to `TABLE`, then an accurate DST table for the time zone must also be configured. Please refer to Chapter 18 in the *Guardian Programmer's Guide* for further information on managing the system time.

PROCESS CONFIGURATION

The following paragraphs contain an explanation on how the Banknet Interface process is configured. They provide information about the startup logic of the Banknet Interface process and other configuration information about the Banknet Interface.

Startup

Because of the length of time required for initialization, the Banknet Interface process should be assigned a startup logic AUTOMATIC in the Network Environment Source File. AUTOMATIC means the Banknet Interface process is started automatically when the XPNET node is initiated. This ensures that the Banknet Interface process is ready when it receives its first transaction.

Event Messages

The Banknet Interface process sends all of its log messages to the logging facility using a standard routing code of zero.

Format 2 File Support

The Banknet Interface supports format 2 ILFs for D46 and higher Compaq NSK operating systems. Format 2 file support allows BASE24 customers the ability to format Enscribe disk file (ILF) partitions to accommodate format 2 files. Files with format 2 partitions can contain much more data than the existing format 1 files. Files, whether format 1 or format 2, can have up to 16 partitions. Format 1 files are limited to 4 gigabytes (each partition can have up to 2 gigabytes). Format 2 files have new block and label formats and can hold as much as 1024 gigabytes of data per partition and are currently limited by disk volume size. Large format 1 files, limited by the capacity of a single disk volume, have used partitioning as a means to increase file capacity.

Customers wishing to use format 2 ILFs should start at cutover. Create a new ILF format 2 template file and its altfile. To do so, create an obey file named FRMT2ILF with the following text:

```
-RESET
-SET TYPE E
-SET FORMAT 2
-SET EXT ( 4 PAGES, 32 PAGES )
-SET REC 4048
-SET BLOCK 4096
-SET ALTKEY ( "PR", FILE 0, KEYOFF 0, KEYLEN 55 )
-SET ALTFILE ( 0, $<VOL>.<SUBVOL>.xxYYMMDD )
-SET MAXEXTENTS <NUMBER OF MAXEXTENTS>
-CREATE $<VOL>.<SUBVOL>.xxYYMMDD
```

The Format 2 ILF template file in the create command will need to be set to the value specified in the LCONF assign for the ILF for the interface process. The existing ILF template and altfile should be renamed, as the new files created should use the same locations.

Create the format 2 ILF and altfile with the following command:

```
FUP/IN FRMT2ILF/
```

When the interface is cutover, it will create and begin using the new format 2 ILF and new format 2 altfile. The altfile can remain format 1 or change to format 2 along with the new format 2 ILF. Previous ILF files can be format 1 and current ILF files can be format 2.

Tokens

Additional disk space may be required on the Tandem for the 5.0 or above BASE24 database depending on the number of tokens being logged to the files.

The Release 5.0 or above BASE24 user can indicate what tokens should be logged by the Banknet Interface to its Interface Log File (ILF). The user can also indicate to the Release 5.0 or above BASE24-pos Authorization module which tokens should be logged to the POS Transaction Log File (PTLF). The user can also indicate to the Release 5.0 or above BASE24-atm Authorization module which tokens should be logged to the ATM Transaction Log File (TLF).

When deciding whether or not to log a token to any Release 5.0 or above log file the user should consider what the length of the log record will be with the new token added. If logging a new token to a log file increases the length of a log file record so that fewer records can fit in the maximum Tandem data block, more data blocks (i.e. more disk space) will be required to hold the same number of records. This will increase the user's disk space requirements. The maximum Tandem data block is 4096 bytes, which is recommended for all BASE24 log files. For Format 1 files, 22 bytes per block, and 2 bytes per record are reserved for Tandem file system usage in the file type used for BASE24 log files. For Format 2 files, 46 bytes per block, and 2 bytes per record are reserved for Tandem file system usage in the file type used for BASE24 log files.

The Banknet February 1998 Mandates caused the maximum length of the ILF record to exceed 2035 bytes of information. It is now not possible to log more than one record per block.

One ILF record consists of the primary key data, the internal message, the external message, and any token data. If the user decides to log the tokens to the ILF, then additional data will be included in the ILF record. Refer to the BASE24 Tokens Manual for detailed information on individual tokens. If tokens are to be logged to the ILF an additional 6 bytes due to a Header token will also be added to the total length of the ILF plus any token data.

The Switch Tokens should not be logged to the ILF due to blocking considerations. This data is already present in the external message portion of the ILF.

The user should do the same analysis on their 5.0 or above BASE24-pos Transaction Log File (PTLF) as done for the Banknet Interface ILF record size. If the extra bytes added to the PTLF for the chosen tokens increase the size of the PTLF records enough to decrease the number of records that will fit in Tandem's maximum data block, the user may need more disk space.

Users who support the Banknet ATM extension module should do the same analysis on their 5.0 or above BASE24-atm Transaction Log File (TLF) as done for the Banknet Interface ILF record size. If the extra bytes added to the TLF for the chosen tokens increase the size of the PTLF records enough to decrease the number of records that will fit in Tandem's maximum data block, the user may need more disk space.

Additional disk space is not required on the Tandem for the Release 3.4/4.0 BASE24 database because 200 bytes of user data is always logged to the ILF and the PTLF.

Transactional Interface Monitoring (T.I.M.)

The Banknet Interface supports the Transactional Interface Monitoring (T.I.M.) enhancement. This enhancement provides the ability to monitor, in real-time, the actual transaction health between BASE24 and the interchange. T.I.M. provides tracking of more than a dozen transaction conditions/counters at the process level. This enhancement supports issuer and acquirer traffic for both BASE24-atm and BASE24-pos transactions.

SESSION MANAGEMENT

Banknet supports three levels of session management:

- ☐ Socket Connection
- ☐ Session Activation
- ☐ Authorization System Sign-on

The following paragraphs contain an explanation on how the Banknet Interface process supports these levels of session management.

Socket Connection

Socket connection represents the establishment of TCP/IP connections between the member host and Banknet. The host must manage TCP/IP connections by opening and closing sockets, and providing error detection. On a BASE24 system, all requirements for socket connection are managed at the XPNET level.

The Banknet Interface supports the use of “Zero Length Probe” messages to detect broken TCP/IP sockets. The zero length probe functionality allows the network or the member to verify the session status and to close inactive sessions.

XPNET can receive zero length probe messages from Banknet. XPNET discards the zero length probe message, and does not attempt to detect the “broken socket” condition by utilizing an idle timer on the socket.

XPNET can be configured to send zero length probe messages to Banknet by including a parameter in the STATION USERDATA within the declaration of a TCP/IP station in the NEFS:

- ☐ A setting of –B+ will result in XPNET sending a zero length probe message when there has been no traffic detected on the station for the default time interval of 10 seconds.
- ☐ A setting of –BTnn (where nn is a time interval specified in seconds) will result in XPNET sending a zero length probe message when there has been no traffic detected on the station for a time interval of nn seconds.

XPNET does not attempt to perform any error recovery processing if it does not receive an acknowledgement to the zero length probe message. Instead, the standard TCP retry mechanism will ensure that any session problem is detected in a timely manner.

Session Activation

Session activation allows members to control the activation or deactivation of a session. Activation opens the connection, which means that messages will be distributed to that connection until it is deactivated. Deactivation closes the connection, which means that messages will not be distributed to that connection until it is re-activated.

A session activation or deactivation request consists of a Network Management Request/0800 Host Session Activation/Deactivation message which is used to activate or deactivate a socket connection. One activation message must be sent for each socket connection, before receiving authorization traffic.

Session activation must be supported by acquirers and issuers in the MasterCard Europe region, but is not a requirement for members outside Europe.

Session activation is performed if the member has configured the Banknet Interface to support enhanced session management. If the LCONF parameter ENHNC-SESSION-MGMT is set to 'Y', then the interface process supports session activation and deactivation messages.

In a Network Management Request/0800 Host Session Activation/Deactivation message, the PAN field (DE 2) must contain the Group Sign-on Identifier (GSI) assigned to the member by MasterCard. The GSI, together with the associated network management parameters, is configured in a special BNIIF record. To allow the Banknet Interface to identify this BNIIF record, the following configuration parameters must be set appropriately:

- ☐ FINANCIAL ID field on the BNIIF screen. This can be any 4-digit value (it's effectively only an identifier for the BNIIF record), except that it must not match any Terminal FIIDs configured in the system (to avoid confusion with BNIIF records used in acquirer processing).
- ☐ SWITCH ID field on ICF/ICFE screen 1. This field must match the value configured in the FINANCIAL ID field on the BNIIF screen, as the Banknet Interface uses this value as a pointer to the BNIIF record.
- ☐ BANKNET MCI ID field on the BNIIF screen. This must contain the GSI itself (i.e. the value sent in DE 2 in 08xx messages).

If ENHNC-SESSION-MGMT is set to 'Y', then session activation requests are used repeatedly as echo test messages, to validate the connectivity between the member Host and Banknet.

Authorization System Sign-on

Authorization System Sign-on allows members to notify Banknet that they are ready to process authorization messages.

A sign-on or sign-off request consists of a Network Management Request/0800 Sign-on/Sign-off message which is used to logon or logoff a member.

For members outside Europe, i.e. members not supporting enhanced session management, the sign-on/sign-off messages perform a similar function to session activation/deactivation messages for members in the MasterCard Europe region. Members may sign-on either by prefix (with a network management information code of 001) or by group (with a network management information code of 061).

For issuers inside Europe, the sign-on message notifies Banknet that they are ready to process their authorization traffic. Only one sign-on request sent through one connection is sufficient to trigger the reception of authorization traffic through all active connections (i.e. stations) defined for the interface process. Members may sign-on either by GSI (with a network management information code of 061) or by prefix within GSI (with a network management information code of 065).

For acquirers inside Europe, the sign-on message is not relevant and is not required. Acquirers may optionally send sign-on messages to Banknet, but these messages will not affect the processing of the Banknet Interface. For example, a sign-on message will not start the interface sending acquirer traffic to Banknet, as this processing is controlled by session activation messages. Similarly, a sign-off message will not stop the interface sending acquirer traffic to Banknet, as this processing is controlled by session deactivation messages.

If ENHNC-SESSION-MGMT is not set to 'Y', then Network Management Request/0800 Network Connection Status requests are used as echo test messages, to validate the connectivity between the member Host and Banknet.

INITIALIZATION

The following steps are taken by the Banknet Interface process when it is started. To initialize, the Banknet Interface process:

1. Initializes interprocess communication with the network.
2. Reads the name of its Logical Network Configuration File (LCONF). The startup message from XPNET contains the name of the LCONF. This name allows the Banknet Interface process to access the LCONF for the assigns and params it requires for processing.
3. Opens the LCONF, reads the required assigns and params, and closes the LCONF. The process will abend if there is a problem reading the LCONF assigns or params or if a required assign or param is missing. For a list of the required LCONF assigns and params and their descriptions, refer to Section 3.
4. Reads the Log Message Configuration File (LMCF) for its modified log messages. The LMCF contains a record for each modified log message in the system. The Banknet Interface process opens the LMCF, retrieves its modified log messages, and closes the LMCF. Whenever the Banknet Interface process generates a log message, it checks the LMCF records in memory to determine whether the message requires modifications.
5. Opens the Interchange Configuration File (ICF) or the Enhanced Interchange Configuration File (ICFE). If an error occurs, the Banknet Interface process logs an error message and abends.
6. Using its process name, the Banknet Interface process reads its own ICF/ICFE record.
7. Builds the Process Control Table (PCT). The PCT contains configuration and control information for the process. This information is read from the ICF/ICFE, and includes all of the controls required to process transactions.
8. Opens the Station Name File (SNF), and reads station information for the process into the PCT, if an appropriate SNF record is found.
9. Determines the posting date for Banknet by using data on screen 2 of the ICF/ICFE and the LCONF parameter TIME-ZONE-DIFF. Determines the posting date for BASE24-pos by comparing the current time to the BASE24-pos time (POS-LN-CUTOVER) taken from the LCONF. If ATM processing is supported, also determines the posting date for BASE24-atm by comparing the current time to the BASE24-atm time (ATM-LN-CUTOVER) taken from the LCONF. If the optional LCONF parameter SW-BNET-CREATE-ILF is being used, calculate the time for creating the next ILF, calculate the next switch business date and add the cutover timer. If the cutover timer will expire after interchange cutover then create and close the next day's ILF.

10. Opens the Store-and-Forward File (SAF). If the SAF is not present, the interface creates it.
11. Opens the current and previous day's Interchange Log File (ILF). If an error occurs in opening the current ILF, the Banknet Interface process logs an error message and abends.
12. Initializes the extended memory required for this process.
13. Reads the Banknet Institution Identification File (BNIIF) into memory, if the BNIIF exists.
14. Initializes and positions to the end of the Banknet Request Log File (BNRLF).
15. Opens the Key6 File (KEY6) if Triple to Single DES encryption is required. Otherwise, opens the Key File (KEYF). If an error occurs, the Banknet Interface process logs an error message and abends.
16. Using its process name, reads its own KEYF or KEY6 record, assigns Process Control Table (PCT) values, and then closes the KEYF or KEY6.
17. Initializes the security devices.
18. Opens the Switch Terminal File (STF), if the STF exists. The Banknet Interface process loads STF records into extended memory, up to a maximum specified by the LCONF parameter MAX-STF-RECORDS.
19. Initializes performance counters for T.I.M., if required.
20. Opens the Switch Dispute File (SDF), if the SDF exists.
21. Closes the ICF/ICFE.
22. If reports are required, starts a timer set to expire when the reports are to be run.
23. If the automatic logon option was selected, sends a logon message to Banknet, preceded by station activation messages if the LCONF parameter ENHNC-SESSION-MGMT is set to 'Y'.
24. Waits for a message from Banknet or a BASE24 process.

NETWORK CONTROL COMMANDS

Network control commands can be issued from the Network Control Supervisor's (NCS) screen or from the NCPcom prompt.

The commands are entered using the following syntax:

DELIVER PROCESS <process name>, "<Message Format>"

ACTVT STA

Message Format: ACTVT STA
ACTVT STA <station name>

Description: Initiates an activate station sequence for interface processes where the LCONF parameter ENHNC-SESSION-MGMT is set to 'Y'. This command will cause the Banknet Interface process to format and send an activate message to the network for the station specified in the command. If no station is specified in the command, then all stations configured for the process are activated.

ALTER

Message Format: ALTER <emt name>
ALTERATTRIBUTE <emt name>
ALTERATTR <emt name>

Description: Warmboots the APCFEMT or the IPCFEMT.

DEACTVT STA

Message Format: DEACTVT STA
DEACTVT STA <station name>

Description: Initiates a deactivate station sequence for interface processes where the LCONF parameter ENHNC-SESSION-MGMT is set to 'Y'. This command will cause the Banknet Interface process to format and send a deactivate message to the network for the station specified in the command. If no station is specified in the command, then all stations configured for the process are deactivated.

DEBUG

Message Format: DEBUG

Description: Places the interface process in DEBUG. This command should only be used in a TEST environment due to negative performance impacts.

DUMP

Message Format: DUMP

Description: Initiates a programmatic dump of the interface process. This command should only be used in a TEST environment or upon the request of ACI.

LOGOFF

Message Format: LOGOFF

Description: Initiates a logoff sequence. This command causes the Banknet Interface process to send a logoff message to the station or stations.

LOGOFFG

Message Format: LOGOFFG
LOGOFFG <service type> <traffic type> <group id>

Description: Initiates a group logoff sequence. This command causes the Banknet Interface process to send a logoff message to the station or stations matching the information specified in the message.

If the service type is specified in the message, it must be "CREDIT" or "DEBIT". If a service type is not specified, the service type from the BNIIF is used.

If the traffic type is specified in the message, it must be "ACQ" or "ISS". If a traffic type is not specified, the traffic type from the BNIIF is used.

Numeric data on the command line is assumed to be the group id (or GSI). The length of this data must conform to the length specified by the MAX-GROUP-LGTH LCONF param or its default. The group id must be right justified and zero filled to the left. If the group id is not specified in the message, the MCI ID (which may be the GSI) from the BNIIF is used.

LOGOFFP

Message Format: LOGOFFP
LOGOFFP <service type> <traffic type> <prefix>

Description: Initiates a prefix logoff sequence. This command causes the Banknet Interface process to send a logoff message to the station or stations matching the information specified in the message.

If the service type is specified in the message, it must be "CREDIT" or "DEBIT". If a service type is not specified, the service type from the BNIIF is used.

If the traffic type is specified in the message, it must be "ACQ" or "ISS". If a traffic type is not specified, the traffic type from the BNIIF is used.

Numeric data on the command line is assumed to be the prefix. If the prefix is not specified in the message, the prefix from the BNIIF is used.

LOGON

Message Format: LOGON

Description: Initiates a logon sequence. This command causes the Banknet Interface process to send a logon message to the station or stations. The LOGON TYPE on screen 12 of the ICFE will be used to determine group versus prefix logons.

LOGONG

Message Format: LOGONG
LOGONG <service type> <traffic type> <avs type> <group id>

Description: Initiates a group logon sequence. This command causes the Banknet Interface process to send a logon message to the station or stations matching the information specified in the message.

If the service type is specified in the message, it must be "CREDIT" or "DEBIT". If a service type is not specified, the service type from the BNIIF is used.

If the traffic type is specified in the message, it must be "ACQ" or "ISS". If a traffic type is not specified, the traffic type from the BNIIF is used.

The first set of numeric data on the command line is assumed to be the AVS type. If the AVS type is not specified in the message, the AVS type from the BNIIF is used.

The second set of numeric data on the command line is assumed to be the group id (or GSI). The length of this data must conform to the length specified by the MAX-GROUP-LGTH LCONF param or its default. The group id must be right justified and zero filled to the left. If the group id is not specified in the message, the MCI ID (which may be the GSI) from the BNIIF is used.

LOGONP

Message Format: LOGONP
LOGONP <service type> <traffic type> <avs type> <prefix>

Description: Initiates a prefix logon sequence. This command causes the Banknet Interface process to send a logon message to the station or stations matching the information specified in the message.

If the service type is specified in the message, it must be "CREDIT" or "DEBIT". If a service type is not specified, the service type from the BNIIF is used.

If the traffic type is specified in the message, it must be "ACQ" or "ISS". If a traffic type is not specified, the traffic type from the BNIIF is used.

The first set of numeric data on the command line is assumed to be the AVS type. If the AVS type is not specified in the message, the AVS type from the BNIIF is used.

The second set of numeric data on the command line is assumed to be the prefix. If the prefix is not specified in the message, the prefix from the BNIIF is used.

NETWK CONN STAT

Message Format: NETWK CONN STAT

Description: Initiates an echo test sequence. This command causes the Banknet Interface process to send a network connection status (echo test) message to the network.

NEWKEY

Message Format: NEWKEY

Description: Initiates a dynamic key exchange sequence. This command causes the Banknet Interface process to send a key exchange message to the network, requesting a new key from the switch.

PERFORMANCE

Message Format: PERFORMANCE

Description: Requests a display of the performance statistics kept in memory by the interface process. These performance statistics are written to the log via a series of log messages generated by the process. The following is an example of a performance display:

```
*** <xxx> Performance Statistics Summary ***
Minutes in period: <#> <Outbound/Inbound> rqst limit: <#>
      REQUESTS - TIMEOUTS - PERCENT - AVERAGE
PERIOD <#>      <#####>      <#####>      <####>%      <##.##>
PERIOD <#>      <#####>      <#####>      <####>%      <##.##>
PERIOD <#>      <#####>      <#####>      <####>%      <##.##>
PERIOD <#>      <#####>      <#####>      <####>%      <##.##>
PERIOD <#>      <#####>      <#####>      <####>%      <##.##>
```

REFRESH

Message Format: REFRESH
9509

Description: Notifies the interface process that there are records in the Switch Dispute File (SDF). The interface process performs SDF initialization logic.

Message Format: STATUS

```
BASE24-atm Date:      <YYMMDD>
BASE24-pos Date:      <YYMMDD>
Banknet Date:         <YYMMDD>
<#> Outbound Requests Pending of <#> Maximum
<#> Inbound   Requests Pending of <#> Maximum
<#> Timers Active of <#> Maximum
File: <ILF fname>, <#> Percent Full
File: <SAF fname>, <#> Records, <#> Percent Full
File: <BNRLF fname>, <#> Percent Full
SAF Cycle in progress
SDF Cycle in progress
<DEBIT/CREDIT> Station <station> is <UP/DOWN>
        with <#> Timeouts of <#> Maximum
TRACE (%B \\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\\)
```

Message Format: STATUSG

Fiid <fiid>, Logon Group <mci id>, Acq <Y/N>, Iss <Y/N>,
Cr <Y/N>, Db <Y/N>

Message Format: STATUSP

Fiid <fiid>, Group <mci id>, Logon Prefix <prefix>,
Acq <Y/N>, Iss <Y/N>, Cr <Y/N>, Db <Y/N>

SWITCH LOG

Message Format: SWL <log destination>

Description: Switches the interface's primary log destination to the one specified in the command. This command should only be used in a TEST environment.

TRACE

Message Format: TRACE ON
TRACE OFF

Description: The interface process contains trace log messages that are optionally logged. The trace log messages should be requested only in a TEST environment due to negative performance impacts.

If "ON" is specified in the message, the interface starts the trace function and sends log messages to the current logging device.

If "OFF" is specified in the message, the interface stops the trace function.

WARMBOOT

Message Format: WARMBOOT
9501

Description: Warmboots the interface process. A message is logged to indicate whether the warmboot completes or aborts.

TIMERS

This section describes each timer used by the Banknet Interface process. Each timer description includes the timer's function, and how and when the timer is set.

Outbound Requests Timer

An outbound request timer is set each time a 0100 message or 0400 message is sent to Banknet. The number of seconds used for this timer is taken from the OUTBOUND field on ICF/ICFE screen 10 for BASE24-pos transactions, and from the OUTBOUND field on ICF/ICFE screen 8 for BASE24-atm transactions. The outbound request timer is used to indicate whether a transaction sent to Banknet has timed out.

If the timer expires before a response is received from Banknet for the 0100 message, the request is denied and the appropriate performance and timeout counters are incremented. If the expiration of this timer causes the maximum allowed timeouts to be reached, the link to Banknet is marked down.

Inbound Requests Timer

An inbound request timer is set each time a 0200 message is sent to a BASE24 process for authorization. The number of seconds used for this timer is taken from the INBOUND field on ICF/ICFE screen 10 for BASE24-pos transactions, and from the INBOUND field on ICF/ICFE screen 8 for BASE24-atm transactions. The inbound request timer is used to determine that an external transaction has timed out.

If the timer expires before a response is received from the authorizing process, the request is denied. The appropriate performance and timeout counters are incremented.

Store-And-Forward Timer

A store-and-forward timer is set each time a store-and-forward message is sent to Banknet. The number of seconds used for this timer is taken from the STORE AND FORWARD field on ICF/ICFE screen 10 for BASE24-pos transactions, and from the STORE AND FORWARD field on ICF/ICFE screen 8 for BASE24-atm transactions. The store and forward timer is used to determine that a Store-and-Forward File (SAF) record sent to Banknet has timed out.

If the timer expires before an acknowledgement is received from Banknet, the SAF timeout counter is incremented. If the SAF timeout counter is equal to the maximum number of retries set in the MAX SAF RETRY field on ICF/ICFE screen 3, the record is written to the log and deleted from the SAF. If the MAX SAF RETRY field is set to 0, the SAF message will be sent indefinitely until it is successfully processed.

Network Management Timer

A network management timer is set each time an 0800 message is sent to a station. The number of seconds used for this timer is taken from the TIMER LIMITS NETWORK MANAGEMENT field on ICF/ICFE screen 3. The network management timer is used to monitor the communications between BASE24 and Banknet.

If the network management timer expires before a response is received from Banknet, an extended network management timer is set.

Extended Network Management Timer

An extended network management timer is set each time a network management timer expires without a response being received from Banknet. The number of seconds used for this timer is taken from the TIMER LIMITS EXTENDED NETWORK field on ICF/ICFE screen 3.

The network management message is repeated at the interval specified by the extended network management timer until an acknowledgement is received from Banknet or the communications line is marked up.

Wait For Traffic Timer

A wait for traffic timer is set each time a station is marked up and each time a previously set wait for traffic timer expires. The number of seconds used for this timer is taken from the TIMER LIMITS WAIT FOR TRAFFIC field on ICF/ICFE screen 3.

If the timer expires and message traffic has occurred on the line, a new wait for traffic timer is started.

If the timer expires and no message traffic has occurred on the line, a network management logon message is sent. The line is marked up or down, depending on the results of the logon.

Session Activation Timer

A session activation timer is set each time a session activation response message is received from Banknet. The number of hundredths of seconds used for this timer is taken from the ACTVT-INTERVAL-TIMER parameter in the LCONF.

The session activation message is repeated at the interval specified by this timer to validate the connectivity between the member Host and Banknet. The session activation timer is used only if enhanced session management is configured.

Cutover Timer

The optional cutover timer is used to determine when the interface should create the next business day's Interchange Log File (ILF). It is set at initialization or warmboot and again when the previous timer expires. The value for this timer is taken from the SW-BNET-CREATE-ILF parameter in the LCONF. When the timer is set, the next switch business date is determined and added to the timer.

When the cutover timer expires, the Banknet Interface process:

1. Retrieves the next switch business date from the timer and compare the value to the current switch business date;
2. If the next switch business date is greater than the current switch business date, then the interface may not have cutover. Calculate what the switch business date should be. If the date from the timer equals the calculated date, then cutover the interface to the next switch business date;
3. Calculates the new next switch business date and start a new timer;
4. Creates and close the ILF for the new next switch business date;

Report Startup Timer

The report startup timer is used to start the SETL01 and STAT09 report programs. It is set at initialization and at interchange cutover. The value for this timer is taken from the SW-RPT-STARTUP param in the LCONF.

When the report startup timer expires, the obey file containing the commands to run the report programs are executed.

Performance Timer

A performance timer is set at the start of each new performance interval. The number of minutes for this timer is taken from the TIMER LIMITS PERFORMANCE PERIOD field on ICF/ICFE screen 3. The performance timer triggers the end of one performance period and the beginning of the next.

When the performance timer expires, the Banknet Interface process:

1. Moves the interval four performance statistics to interval five.
2. Moves the interval three performance statistics to interval four.
3. Moves the interval two performance statistics to interval three.
4. Moves the interval one performance statistics to interval two.
5. Initializes the interval one performance statistics to zero.
6. Starts a new performance timer.

New Key Timer

The new key timer is used to ensure that a key exchange is initiated after a predetermined time limit has expired. The time limit is set using the PIN KEY TIMER VALUE and PIN KEY TIMER INTERVAL fields on screen 3 of the KEYF or screen 4 of the KEY6 record associated with the interface. If the PIN KEY TIMER VALUE field is left at the default value "0", the key exchange timer will not be activated.

When this timer expires, the key exchange sequence is initiated.

Clear Old Key Timer

The clear old key timer is used to specify how long the old key is available to BASE24 after a key exchange. The clear old keys timer is set using the CLEAR OLDKEY TIMER VALUE field on screen 3 of the KEYF or screen 4 of the KEY6 record associated with the interface. If the CLEAR OLD KEY TIMER VALUE is left at the default value "0", the old working keys will not be cleared from the PCT, KEYF or KEY6.

After a new working key has been activated, BASE24 will use the new key for PIN processing. If the security device detects an error because the interchange is still using the old key, it can repeat the PIN processing using the old key before declining the transaction. When this timer expires, the old key value is cleared from the PCT and the KEYF or KEY6.

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Section 5

MESSAGE HANDLING

INTRODUCTION

This section describes how the Banknet Interface process handles various types of message. It describes the basic message exchange and provides transaction flow diagrams illustrating how the interface operates between BASE24 and Banknet. It also describes how transaction files are used within the message flows.

The following transaction flows are described in this section. In most cases, these flows are described twice: once when the message flow is initiated by BASE24 and once when the message flow is initiated by Banknet.

- ☐ Message Flows from BASE24 to Banknet
- ☐ Message Flows from Banknet to BASE24
- ☐ Network Management Message Flows

The Network Management Message Flows are described only once, because most of the message exchanges are initiated by BASE24.

NOTE: References in this section to transactions sent to Banknet for authorization implies that Banknet is sending the transaction to the appropriate authorizing entity (i.e., the Banknet Interface does not perform authorization).

TRANSACTION FILE USAGE

The following transaction files are described below:

- ☐ Interchange Log File (ILF)
- ☐ Banknet File Request Log File (BNRLF)
- ☐ Store-and-Forward File (SAF)

Logging To The ILF

Requests, advices and reversals handled by the Banknet Interface process are logged to the Interchange Log File (ILF).

A request message received from Banknet is logged to the ILF when the response message is returned to Banknet. A subsequent reversal message will result in the ILF record being updated. An advice message received from Banknet causes a new ILF record to be logged, after the interface has checked the ILF to ensure that the message is not a duplicate transmission.

The Banknet Interface also handles more complex message flows with up to 4 messages (excluding duplicates) per transaction (as described in section 2 in the ***Customer Interface Specification***). Specifically, the interface is able to cater for the following scenarios:

- ☐ An authorization advice and reversal advice received after a request message has been authorized (e.g. because the response message does not reach Banknet, so Banknet reverses the request and authorizes the transaction in stand-in).
- ☐ An authorization advice and two reversal messages received after a request message has been authorized (e.g. because the response message does not reach Banknet, so Banknet reverses the request and authorizes the transaction in stand-in, and then the cardholder cancels the transaction at the terminal).
- ☐ An authorization advice and reversal advice received without a request message being authorized (e.g. because the request message does not reach the interface, so Banknet reverses the request and authorizes the transaction in stand-in).
- ☐ An authorization advice and two reversal messages received without a request message being authorized (e.g. because the request message does not reach the interface, so Banknet reverses the request and authorizes the transaction in stand-in, and then the cardholder cancels the transaction at the terminal).

The interface can also process authorization advices and reversal messages received in different orders (e.g. because the Banknet SAF processing may not send messages in chronological order), and ensures that the interface can detect duplicate messages, regardless of the order in which they are received. The interface can process correctly the combinations of messages shown in the table below.

Number	1 st Message	2 nd Message	3 rd Message	4 th Message
1	0100	0120	0420	
2	0100	0420	0120	
3	0100	0120	0400/0420	0400/0420
4	0100	0400/0420	0120	0400/0420
5	0100	0400/0420	0400/0420	0120
6	0120	0420		
7	0420	0120		
8	0120	0400/0420	0400/0420	
9	0400/0420	0120	0400/0420	
10	0400/0420	0400/0420	0120	

The strategy for managing these message flows is as follows:

- ☐ An authorization advice will be logged to the ILF with a different PRIKEY value, if an authorization request for the same transaction has already been processed.
- ☐ A reversal message will be logged to the ILF with a different PRIKEY value, if a different reversal message for the same transaction has already been processed.
- ☐ An authorization advice will cause a record to be added to the ILF, unless an “unmatched” reversal request for the same transaction has already been processed, in which case the interface will update the record for the “unmatched” reversal request.
- ☐ A reversal message will cause a record to be updated in the ILF, unless any ILF records for the same transaction have already been updated by a reversal message, in which case the interface will add the record as an “unmatched” reversal.
- ☐ If a reversal request or advice has been processed, then the SEM structure in the ILF will contain data from the reversal message, so duplicate checking will be performed on this data if another reversal message for the same transaction is subsequently received.

- ❑ If an authorization request or advice message has been processed, then the ILF.SEM structure in the ILF will contain data from the authorization message, so duplicate checking will be performed on this data if another authorization advice message for the same transaction is subsequently received.
- ❑ If an authorization advice is processed after an associated reversal message, and the reversal message was logged to the ILF and dropped, e.g. because it was “unmatched”, then the authorization advice will also be dropped, after the ILF record has been updated.

Special consideration is required for authorization requests and completion advices from Automated Fuel Dispensers (AFDs), as these messages are processed by BASE24-pos as pre-authorization requests and pre-authorization completions.

The Banknet Interface provides users with the capability of identifying when the ILF is becoming full. The Banknet Interface maintains a counter of records written to the ILF. Each time the counter exceeds the value of the LCONF parameter ILF-NOTIFY-INTERVAL, the interface resets the counter and calculates how full the file has become. If the percentage of the file used exceeds the value of the LCONF parameter ILF-NOTIFY-PERCENTAGE, then a warning message is logged.

Logging To The BNRLF

Online file additions, deletions, updates and inquiries to Banknet exception and verification files are handled through perusal of the Banknet Request Log File (BNRLF).

The Banknet Interface provides users with the capability of identifying when the BNRLF is becoming full. The Banknet Interface maintains a counter of records written to the BNRLF. Each time the counter exceeds the value of the LCONF parameter BNRLF-NOTIFY-INTERVAL, the interface resets the counter and calculates how full the file has become. If the percentage of the file used exceeds the value of the LCONF parameter BNRLF-NOTIFY-PERCENTAGE, then a warning message is logged.

Store-And-Forward Message To Banknet

Non-interactive messages intended for Banknet are stored in the Store-and-Forward File (SAF) and sent to Banknet during the SAF processing cycle. If the link to Banknet is up, the SAF contains records, and a SAF record is not in the process of being sent to Banknet, the Banknet Interface process retrieves the first record from the SAF and sends it to Banknet. Only one message from the SAF is sent at a time. The general processing flow for SAF messages is the same as for normal messages. The Banknet Interface process uses different timers that identify the messages as SAF records. The following message types can be written to the SAF:

- ☐ Acquirer Authorization Advice Messages
- ☐ Acquirer Reversal Request Messages
- ☐ Administrative Messages to Banknet
- ☐ File Maintenance Request Messages

Acquirers can use the LCONF parameter SAF-0120-MESSAGES to configure whether Acquirer Authorization Advice messages are added to the SAF.

Banknet sends acknowledgements of these messages to the Banknet Interface process. When the Banknet Interface process receives the acknowledgement, the SAF timer is deleted and the SAF record is deleted from the SAF. The acknowledgement message from Banknet is dropped.

The Banknet Interface provides users with the capability of identifying when the SAF is becoming full. The Banknet Interface maintains a counter of records written to the SAF. Each time the counter exceeds the value of the LCONF parameter SAF-NOTIFY-INTERVAL, the interface resets the counter and calculates how full the file has become. If the percentage of the file used exceeds the value of the LCONF parameter SAF-NOTIFY-PERCENTAGE, then a warning message is logged.

MESSAGE FLOWS FROM BASE24 TO BANKNET

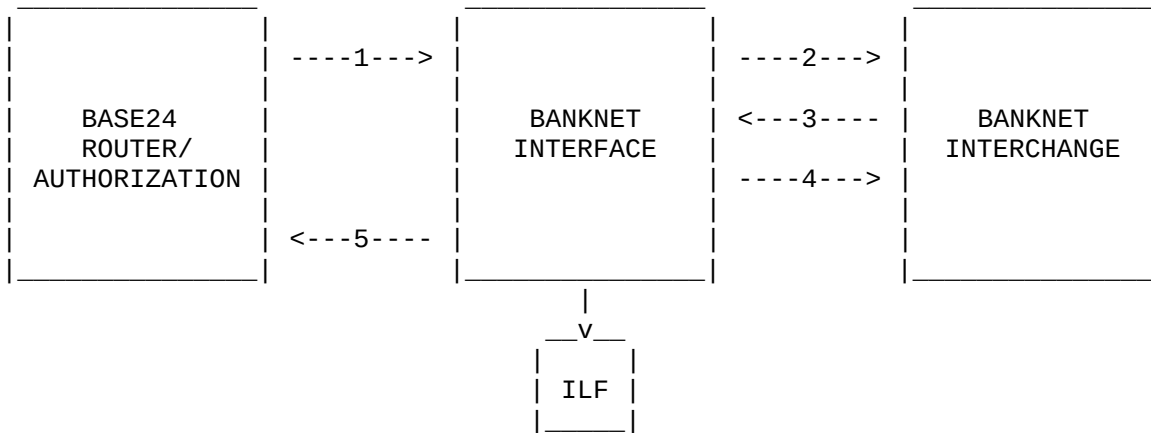
This subsection describes the message flows for the following transactions outbound from BASE24 to Banknet.

- ☐ Authorization Requests
- ☐ Authorization Advices
- ☐ Authorization Reversals
- ☐ Timeouts
- ☐ Administrative Messages
- ☐ File Maintenance Requests
- ☐ Automated Hot Card Updates

If acquirers send authorization transactions through one or more processors, the processors must include the acquirer's ICA number in DE 32 of the Authorization Request/0100 message. The ICA number of the processor that passes the transaction to Banknet must be in the Forwarding Institution ID Code (DE 33). This interface currently handles this logic.

Request Message To Banknet

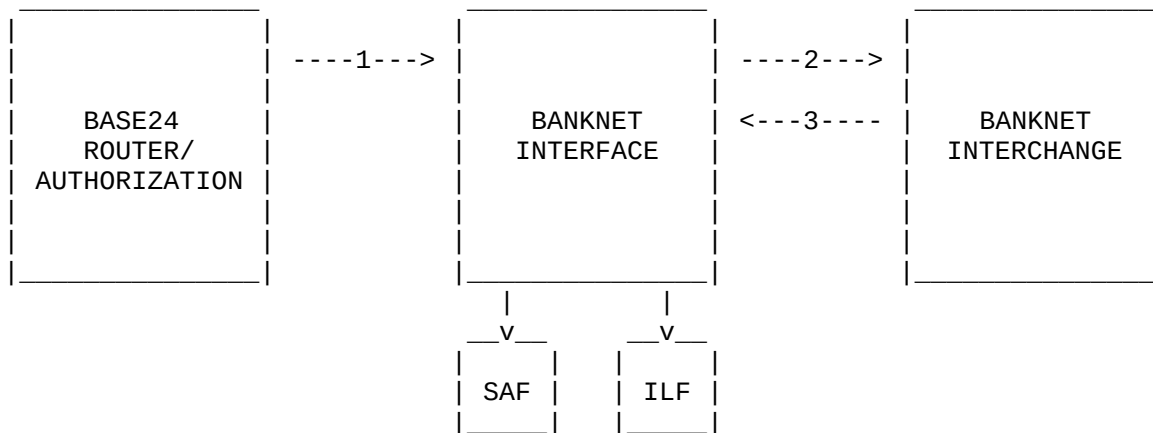
The diagram below illustrates the message processing that takes place when a transaction is initiated by the BASE24 network and must be sent to Banknet for authorization. The steps in the diagram are described below.



1. A BASE24 Authorization process sends the request initiated at a POS device to the Banknet Interface process in the 0200 internal message format.
NOTE: Authorization requests (message type 0100) are not initiated on a BASE24 network.
2. The Banknet Interface process reformats the message into the 0100 external message format and sends the external message to Banknet for authorization. An outbound request timer is set to wait for a response from Banknet.
3. Banknet routes the transaction to the appropriate authorizer and receives the subsequent response. The response is formatted into the 0110 external message format and sent to the Banknet Interface process.
4. The Banknet Interface sends a 0180 positive acknowledgement to Banknet to indicate that the previous 0110 Authorization Request Response was received. The 0180 is optional for Acquirer Processing Systems (determined by the ACK TO SWI field on ICF/ICFE screen 3).
5. The Banknet Interface process deletes the outbound request timer and formats a 0210 internal response message, which includes translating the response code (see Appendix C). The response is returned to the original BASE24 Authorization process and completed at the POS device. The Banknet Interface also logs the transaction to the Interchange Log File (ILF).

Advice Message To Banknet

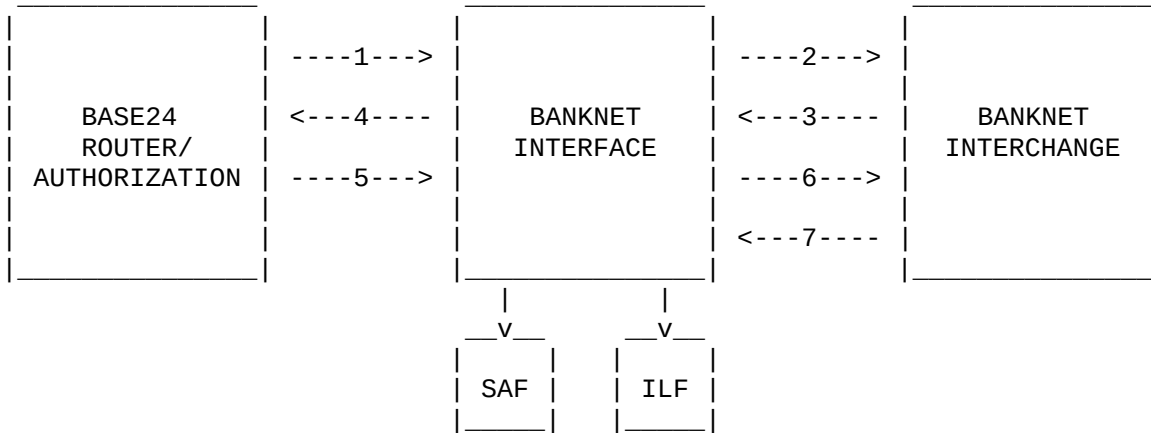
The diagram below illustrates the message processing that takes place when a transaction is initiated by the BASE24 network and authorized before it can be sent to Banknet, e.g. below a floor limit. The steps in the diagram are described below.



1. A BASE24 Authorization process sends an authorized transaction initiated at a POS device to the Banknet Interface process in the 0220 internal message format.
2. The Banknet Interface process reformats the message into an external 0120 Acquirer Authorization Advice, adds it to the SAF and adds an ILF record so it now consists of an internal 0220 and an external 0120. The interface sends the advice message to Banknet during SAF processing.
3. The network sends an external 0130 Acquirer Authorization Advice Response to the Banknet Interface. The interface process deletes the SAF timer and deletes the record from the SAF, and updates the ILF record.

Acquirer Reversal To Banknet

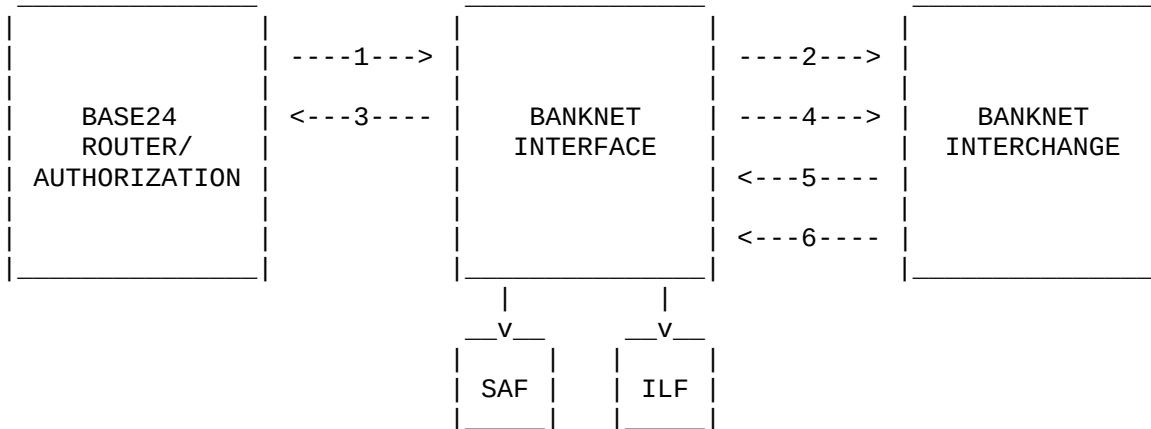
The diagram below illustrates message processing that takes place when BASE24 is unable to deliver an issuer's Authorization Response/0110 to a merchant or when a merchant cancels the full amount of the original authorization amount. The steps in the diagram are described below.



1. BASE24-pos Router Authorization sends an internal 0200 PSTM Request to the Banknet Interface.
2. The interface formats and sends an external 0100 Authorization Request to the network.
3. The network sends an external 0110 Authorization Response to the Banknet Interface.
4. The interface formats and sends an internal 0210 PSTM Response to BASE24-pos Router Authorization and logs a record to the ILF consisting of an internal 0210 and an external 0110.
5. BASE24-pos Router Authorization sends an internal 0420 PSTM Reversal to the Banknet Interface.
6. The interface retrieves the original transaction from the ILF, formats an external 0400 Acquirer Reversal Request, adds it to the SAF and updates the ILF record so it now consists of an internal 0420 and an external 0110. The interface sends the reversal message to Banknet during SAF processing.
7. The network sends an external 0410 Acquirer Reversal Response to the Banknet Interface. The interface process deletes the SAF timer, deletes the record from the SAF and updates the ILF record so it now consists of an internal 0420 and an external 0410.

Timeout Of Request To Banknet

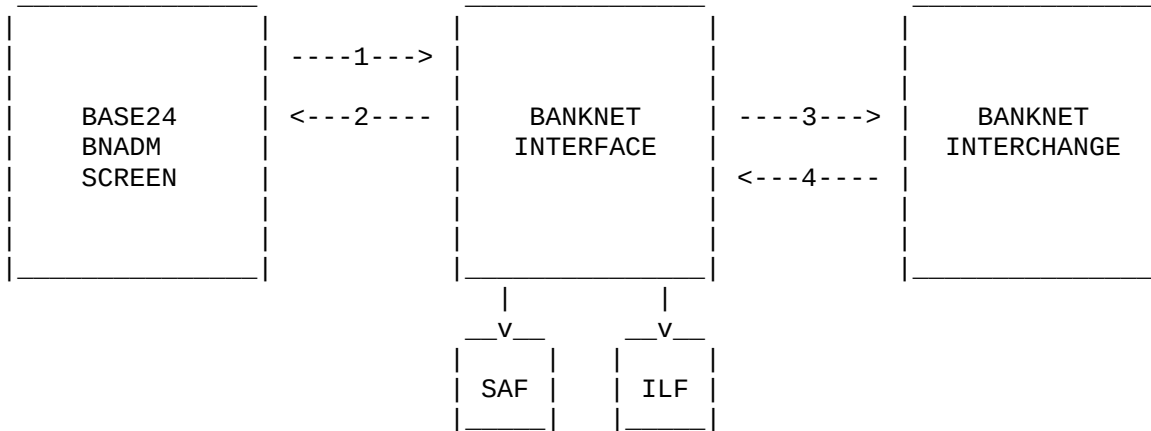
Each time the Banknet Interface process sends a 0100 financial transaction request to Banknet for authorization, the Banknet Interface process sets an outbound request timer. If the outbound request timer expires before a response is received from Banknet, the transaction times out. The diagram below illustrates message processing that takes place when a financial transaction times out. The steps in the diagram are described below.



1. A BASE24 Authorization process sends the request initiated at a POS device to the Banknet Interface process in the 0200 internal message format.
2. The Banknet Interface process reformats the message into the 0100 external message format and sends the external message to Banknet for authorization. An outbound request timer is set to wait for a response from Banknet.
3. When the timer expires before a response is received from Banknet, the Banknet Interface process formats a 0210 internal response message denying the transaction. The response is returned to the BASE24 Authorization process. The transaction is logged to Interchange Log File (ILF).
4. The interface uses the original transaction data in the 0100 external message to format an external 0400 Acquirer Reversal Request, adds it to the SAF and updates the ILF record so it now consists of an internal 0210 and an external 0400. The interface sends the reversal message to Banknet during SAF processing.
5. Banknet sends a 0110 late response to the Banknet Interface process, which is dropped by the Banknet Interface process.
6. The network sends an external 0410 Acquirer Reversal Response to the Banknet Interface. The interface process deletes the SAF timer, deletes the record from the SAF and updates the ILF record so it now consists of an internal 0210 and an external 0410.

Administrative Message To Banknet

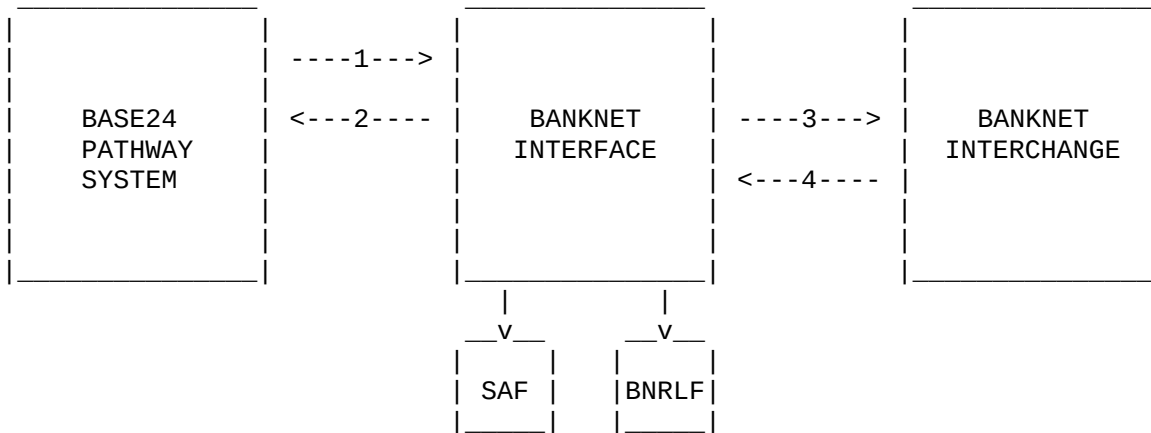
Administrative messages are entered from a BASE24 BNADM screen and sent to Banknet by the Banknet Interface process. The administrative message may be processed by Banknet, or Banknet may forward the message to another entity. The diagram below illustrates the message processing that takes place for an administrative message. The steps in the diagram are described below.



1. The Banknet Interface process receives an administrative message entered at the BNADM screen.
2. The Banknet Interface process responds back to the BNADM screen to acknowledge receipt of the request.
3. The Banknet Interface process reformats the message into an external 0620 Administrative Advice, adds it to the SAF and adds an ILF record. The interface sends the administrative message to Banknet during SAF processing.
4. The network sends an external 0630 Administrative Advice Response to the Banknet Interface, indicating Banknet received the message. The interface process deletes the SAF timer, deletes the record from the SAF and updates the ILF record with information from the response.

File Maintenance Request

The diagram below illustrates the message processing that takes place when a file maintenance request is initiated by the BASE24 network and must be sent to the Banknet Interface process from the Pathway system. The steps in the diagram are described below.



1. The Banknet Interface process receives a file maintenance request from the Pathway system.
2. The Banknet Interface process responds back to the Pathway system to acknowledge receipt of the request.
3. The Banknet Interface process reformats the message into an external 0302 Issuer File Update Request, and adds it to the SAF. The interface sends the file maintenance message to Banknet during SAF processing.
4. The network sends an external 0312 Issuer File Update Response to the Banknet Interface, indicating Banknet received the message. The interface process deletes the SAF timer, deletes the record from the SAF and adds a record to the Banknet File Request Log File (BNRLF), where it can be perused through the Banknet File Request Log File requester.

Automated Hot Card Update

The BASE24 Automated Hot Card Update function allows a Host system to initiate a Card Status Update message to as many as three interchanges, with or without updating a cardholder file on BASE24.

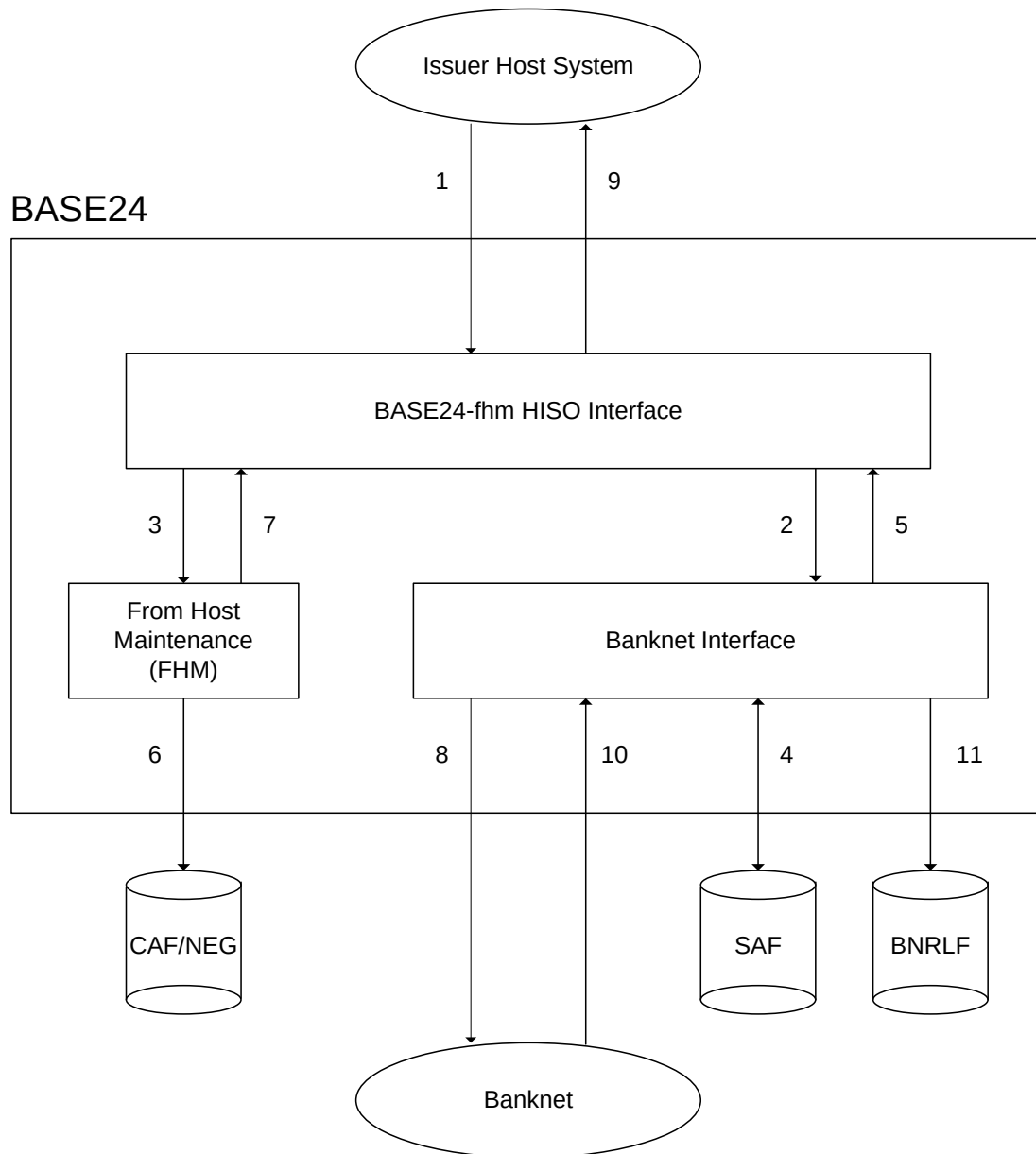
The Host sends the message to the BASE24 From Host Maintenance (FHM) ISO Host Interface (HISO) process. When a Card Status Update Request message is received from the host, the From Host Maintenance ISO Host Interface performs either or both of the following functions:

- ☐ The FHM HISO module builds a request message and sends it to the BASE24 From Host Maintenance process, which updates the Cardholder Authorization File (CAF) or Negative Card File (NEG) on BASE24 with information from the Card Status Update Request message.
- ☐ The FHM HISO module builds a request message and sends it to the specified interchange interface process, so that the interchange databases can be updated. A single Card Status Update Request message can be forwarded to as many as three interchange interface processes simultaneously.

Issuers can utilize the Automated Hot Card Update facility in order to send file maintenance request messages to Banknet.

The ***BASE24 External Message Manual*** includes the format of FHM messages sent to the HISO Interface. It also specifies the ISO field format for File Action request data, which is forwarded to the Banknet Interface as a PATH Request message.

The diagram below illustrates the message processing that takes place for an automated hot card update. The steps in the diagram are described below.



1. The BASE24 HISO Interface receives an 0300 request message from the Issuer Host System, in the format specified in the ***BASE24 External Messages Manual***.
2. The HISO Interface extracts the data required by the Banknet Interface, and sends a PATH request message to the Banknet Interface.
3. The HISO Interface forwards the 0300 request message to the FHM process, if the BASE24 database must also be updated.
4. The Banknet Interface processes the PATH request message, and adds a file maintenance request message to the Store and Forward (SAF) File.
5. The Banknet Interface returns a PATH response message to the HISO Interface.
6. The FHM process updates the CAF or the NEG with the information in the 0300 request message.
7. The FHM process returns an 0310 response message to the HISO Interface.
8. The Banknet Interface retrieves the file maintenance request message from the SAF, and sends the message to Banknet.
9. The HISO Interface returns an 0310 response message to the Issuer Host System, if the BASE24 database was updated, but drops the PATH response message received from the Banknet Interface.
10. The Banknet Interface receives a file maintenance response message from Banknet.
11. The Banknet Interface logs a record to the Banknet File Request Log File (BNRLF).

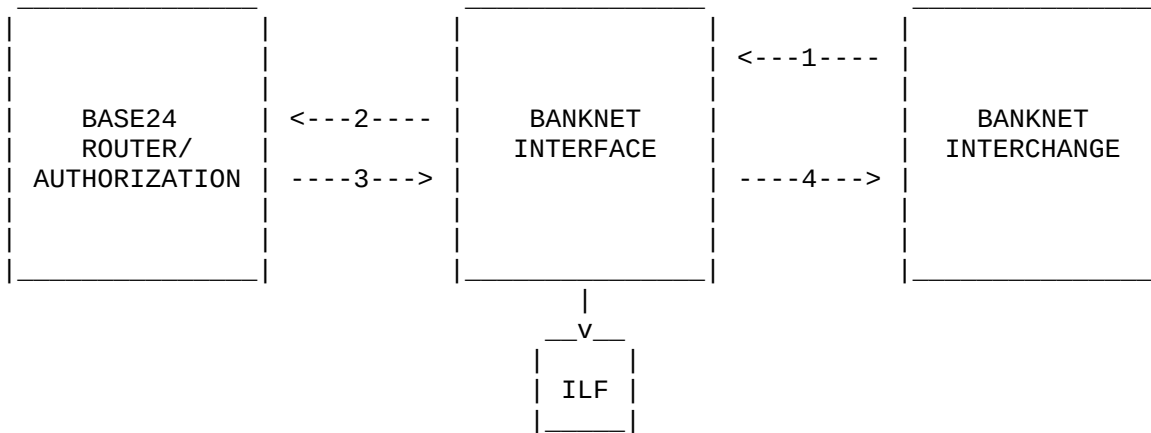
MESSAGE FLOWS FROM BANKNET TO BASE24

This subsection describes the message flows for the following transactions inbound from Banknet to BASE24.

- ☐ Authorization Requests
- ☐ Authorization Advices
- ☐ Reversal Requests
- ☐ Reversal Advices
- ☐ Timeouts (in BASE24)
- ☐ Timeouts (at Banknet)
- ☐ Administrative Messages

Request Message From Banknet

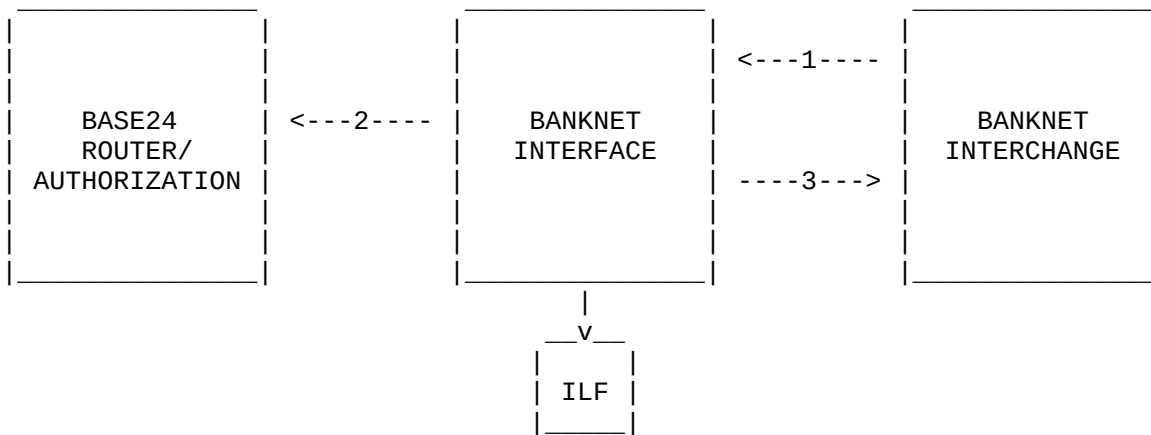
The diagram below illustrates message processing that takes place when a transaction is sent from Banknet to BASE24 for authorization. The steps in the diagram are described below.



1. Banknet sends the request to the Banknet Interface process in the 0100 external message format.
2. The Banknet Interface process reformats the message into the 0200 internal message format and sends the internal message to a BASE24-pos Router/Authorization process for authorization. An inbound request timer is started to wait for the response.
3. The BASE24-pos Router/Authorization process sends the response to the Banknet Interface process in the 0210 internal message format.
4. The Banknet Interface process deletes the inbound request timer and formats a 0110 external response message, which includes translating the response code (see Appendix C). The response is returned to Banknet for completion. The transaction is logged to the Interchange Log File (ILF).

Advice Message From Banknet

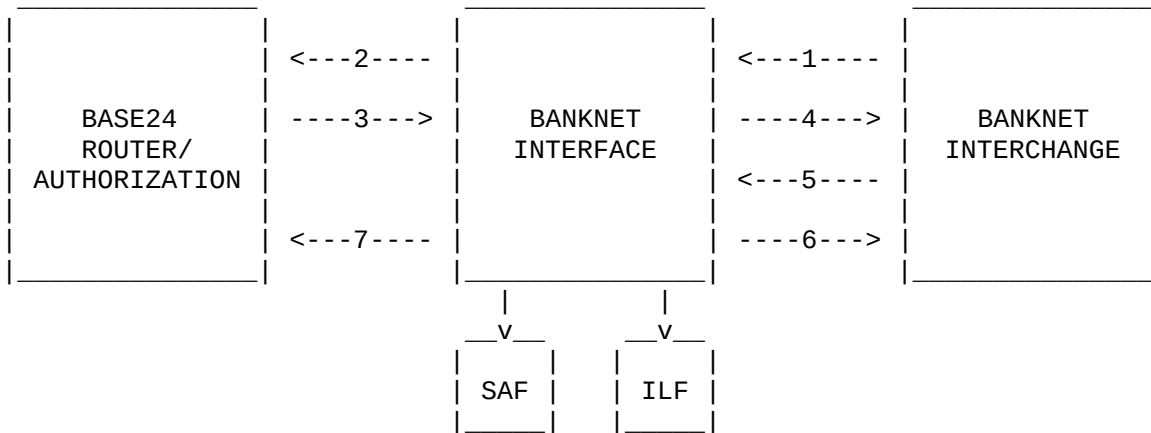
The diagram below illustrates message processing that takes place when an authorized transaction is sent from Banknet to BASE24, e.g. because Banknet has performed “on-behalf” processing. The steps in the diagram are described below.



1. Banknet sends the advice to the Banknet Interface process in the 0120 Authorization Advice external message format.
2. The Banknet Interface process reformats the message into the 0220 internal message format and sends the internal message to a BASE24-pos Router/Authorization process.
3. The Banknet Interface process formats a 0130 Authorization Advice Response message, which is returned to Banknet. The transaction is logged to the Interchange Log File (ILF).

Acquirer Reversal From Banknet

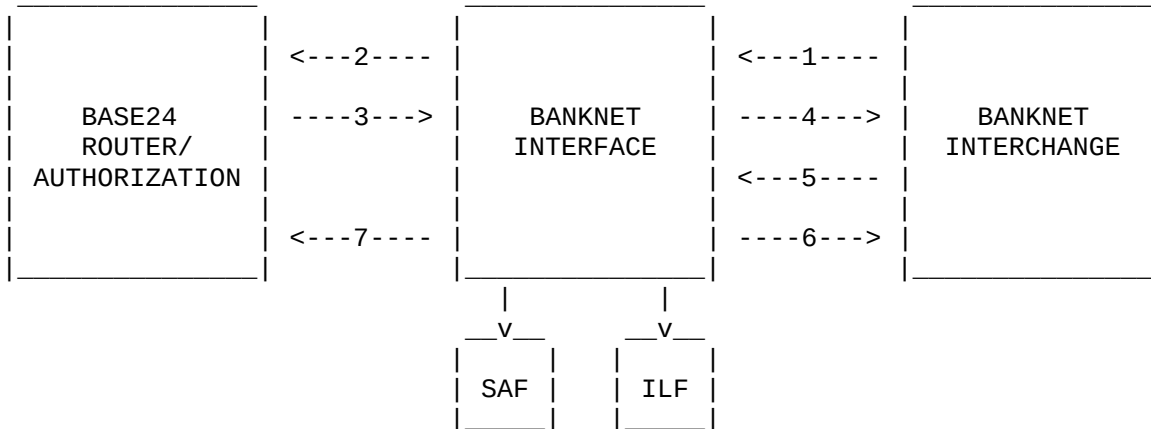
When a POS device connected to the transaction acquirer fails to complete a transaction as authorized, a reversal message must be sent to BASE24. The diagram below illustrates message processing that takes place when a reversal request is sent to BASE24. The steps in the diagram are described below.



1. The network sends an external 0100 Authorization Request into the Banknet Interface.
2. The interface formats and sends an internal 0200 PSTM Request into BASE24-pos Router/Authorization.
3. The interface receives an internal 0210 PSTM Response from BASE24-pos Router/Authorization.
4. The interface formats and sends an external 0110 Authorization Response message to the network and logs a record to the ILF, consisting of an internal 0210 and an external 0110.
5. The network sends an external 0400 Acquirer Reversal Request message to the Banknet Interface.
6. The interface retrieves the original transaction from the ILF and formats and sends an external 0410 Acquirer Reversal Response message to the network.
7. The interface formats and sends an internal 0420 PSTM reversal to BASE24-pos Router/Authorization, and updates the ILF record so that it now consists of an internal 0420 and an external 0400.

Reversal Advice From Banknet

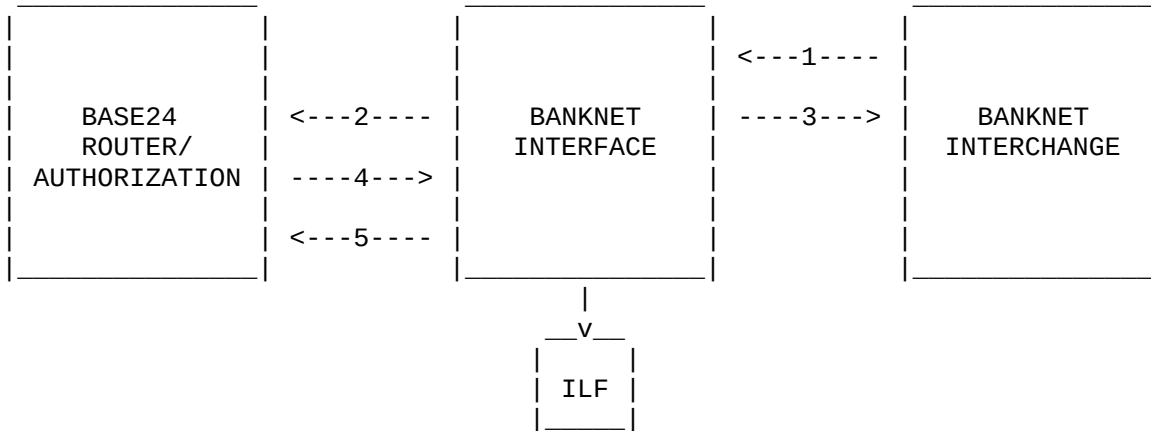
When the Banknet interchange fails to complete a transaction as authorized, e.g. because an approved authorization response message cannot be returned to the acquirer, a reversal message must be sent to BASE24. The diagram below illustrates message processing that takes place when a reversal advice is sent to BASE24. The steps in the diagram are described below.



1. The network sends an external 0100 Authorization Request into the Banknet Interface.
2. The interface formats and sends an internal 0200 PSTM Request into BASE24-pos Router/Authorization.
3. The interface receives an internal 0210 PSTM Response from BASE24-pos Router/Authorization.
4. The interface formats and sends an external 0110 Authorization Response message to the network and logs a record to the ILF, consisting of an internal 0210 and an external 0110.
5. The network sends an external 0420 Reversal Advice message to the Banknet Interface.
6. The interface retrieves the original transaction from the ILF and formats and sends an external 0430 Reversal Advice Response message to the network.
7. The interface formats and sends an internal 0420 PSTM reversal to BASE24-pos Router/Authorization, and updates the ILF record so that it now consists of an internal 0420 and an external 0420.

Timeout Of Request Message From Banknet

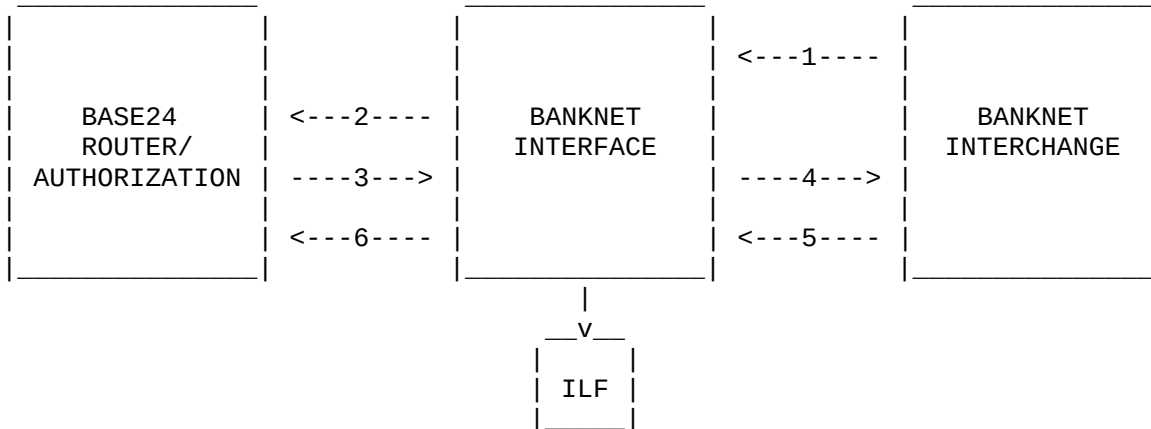
Each time the Banknet Interface process sends a financial transaction to a BASE24 Authorization process for authorization, the Banknet Interface process sets an inbound request timer. If the inbound request timer expires before a response is received, the transaction times out. The diagram below illustrates message processing that takes place when a financial transaction times out. The steps in the diagram are described below.



1. Banknet sends the request to the Banknet Interface process in the 0100 external message format.
2. The Banknet Interface process reformats the message into the 0200 internal message format and sends the internal message to a BASE24-pos Router/Authorization process for authorization. An inbound request timer is set to wait for a response from the BASE24-pos Router/Authorization process.
3. When the inbound request timer expires, the Banknet Interface process formats a 0110 external response message denying the transaction, and returns the response message to Banknet. The transaction is logged to the ILF.
4. The BASE24-pos Router/Authorization process sends a late 0210 response to the Banknet Interface process. If the transaction has been denied by the BASE24-pos Router/Authorization process, the Banknet Interface process simply drops the late response, because denied transactions are not reversed.
5. If the transaction has been approved, the Banknet Interface process formats a 0420 internal reversal message and sends the reversal to the BASE24-pos Router/Authorization process.

Timeout At Banknet

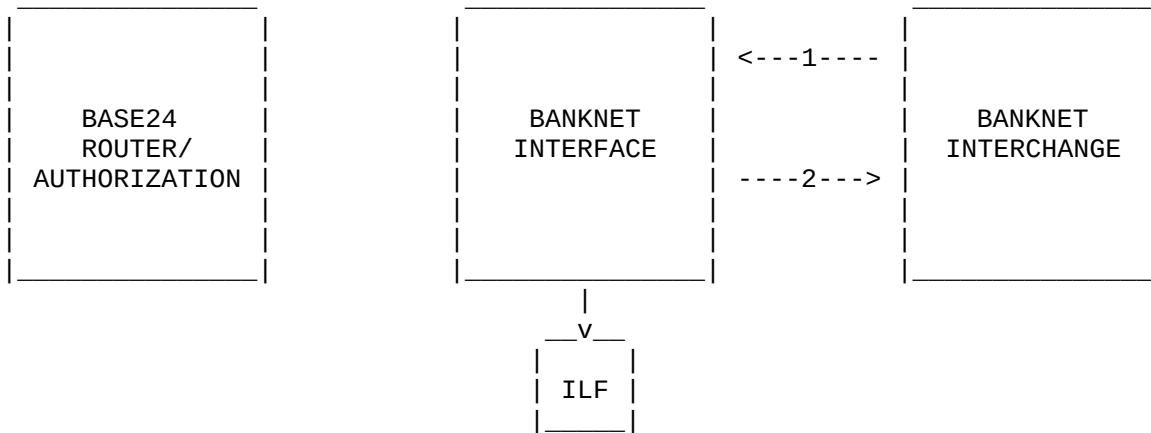
If the Banknet Interface process sends a financial transaction to a BASE24-pos Router/Authorization process for authorization, it is possible for Banknet to time out before receiving a response from BASE24. The diagram below illustrates the message processing that takes place when a financial transaction times out at Banknet. The steps in the diagram are described below.



1. Banknet sends the request to the Banknet Interface process in the 0100 external message format.
2. The Banknet Interface process reformats the message into the 0200 internal message format and sends the internal message to a BASE24 Authorization process for authorization. An inbound request timer is set to wait for a response from the BASE24-pos Router/Authorization process.
3. The BASE24-pos Router/Authorization process sends the response to the Banknet Interface process in the 0210 internal message format.
4. The Banknet Interface process deletes the inbound request timer and formats a 0110 external response message. The response is returned to Banknet for completion. The transaction is logged to the Interchange Log File (ILF).
5. Since Banknet has timed the transaction out it will now format a 0190 Authorization Negative Acknowledgement message and send the message to the Banknet Interface.
6. If the original transaction was approved, the Banknet Interface formats and sends an internal 0420 PSTM reversal to BASE24-pos Router/Authorization, and updates the ILF record.

Administrative Message From Banknet

Administrative messages can be sent from Banknet to BASE24. The diagram below illustrates message processing that takes place when an administrative message is sent to BASE24. The steps in the diagram are described below.



1. Banknet sends an 0620 Administrative Advice to the Banknet Interface process.
2. The Banknet Interface process sends an external 0630 Administrative Advice Response to Banknet and logs the message to the ILF.

NETWORK MANAGEMENT MESSAGE FLOWS

Network management messages are used to manage the operational status of the communications lines between BASE24 and Banknet. The Banknet Interface process supports the following types of network management messages.

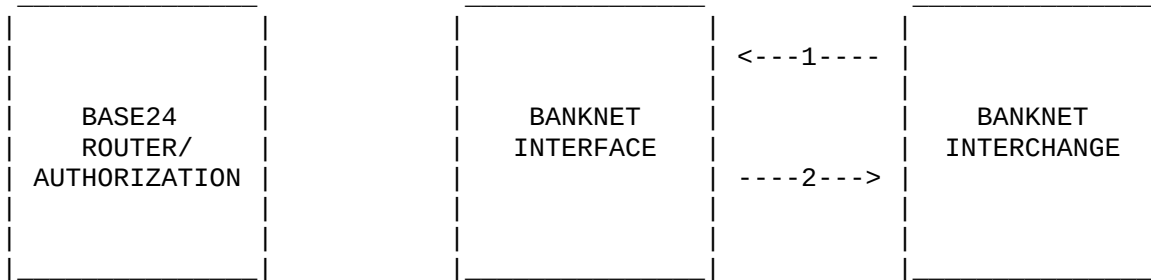
- ☐ Session Activation messages
- ☐ Session Deactivation messages
- ☐ Sign-On messages
- ☐ Sign-Off messages
- ☐ Network Connection Status messages
- ☐ Dynamic Key Management messages

These messages are sent as 0800 messages and require 0810 messages in response. Generally, the 0800 message is sent by BASE24 and the 0810 response message is sent by Banknet. However, Banknet may send Network Connection Status request messages and Dynamic Key Management messages.

This section describes the message flows that occur when network management messages are sent.

Session Activation Initiated By BASE24

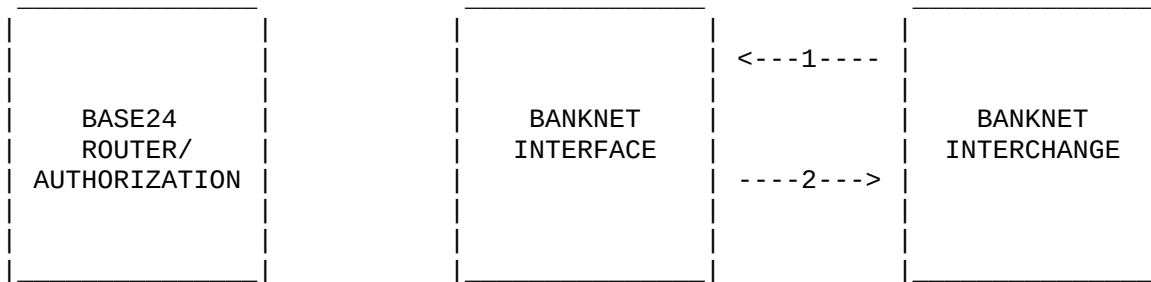
The diagram below illustrates the message processing that takes place between the Banknet Interface process and Banknet when BASE24 activates a session. Session activation messages are used only if the Banknet Interface is configured to support enhanced session management. The steps in the diagram are described below.



1. To activate a session, the Banknet Interface process sends an 0800 Network Management Request message with a network management information code (DE 70) of 081 (host session activation) to Banknet.
2. Banknet responds with an 0810 Network Management Response message. If the response code is set to 00 (approve), the response indicates the activation has been accepted. If the response code is set to 30 (format error), 63 (security violation), 91 (switch down) or 96 (system error) the response indicates the activation was rejected.

Session Deactivation Initiated By BASE24

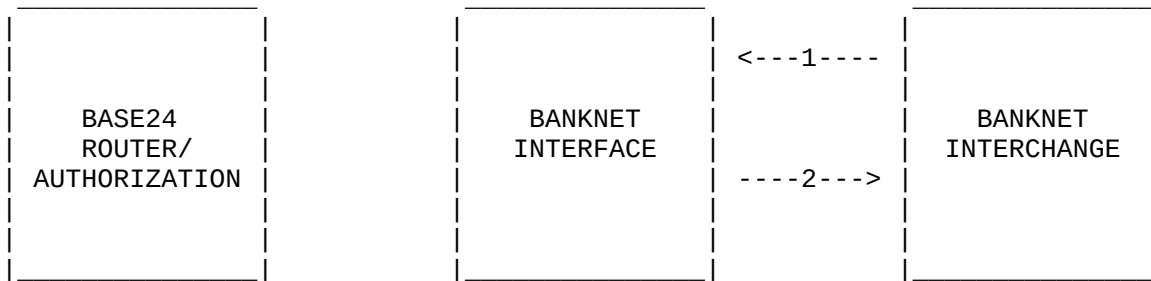
The diagram below illustrates the message processing that takes place between the Banknet Interface process and Banknet when deactivates a session. Session deactivation messages are used only if the Banknet Interface is configured to support enhanced session management. The steps in the diagram are described below.



1. To initiate a sign-off, the Banknet Interface process sends an 0800 Network Management Request message with a network management information code (DE 70) of 082 (host session deactivation) to Banknet.
2. Banknet responds with an 0810 Network Management Response message. If the response code is set to 00 (approve), the response indicates the deactivation has been accepted. If the response code is set to 30 (format error), 63 (security violation), 91 (switch down) or 96 (system error) the response indicates the deactivation was rejected.

Sign-On Initiated By BASE24

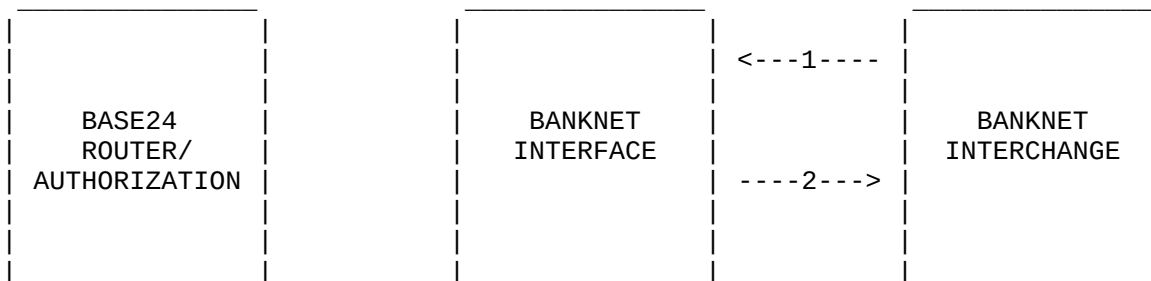
The diagram below illustrates the message processing that takes place between the Banknet Interface process and Banknet when BASE24 initiates a sign-on. The steps in the diagram are described below.



1. To initiate a sign-on, the Banknet Interface process sends an 0800 Network Management Request message with a network management information code (DE 70) of 001 (sign-on by prefix), 061 (sign-on by group or GSI) or 065 (prefix sign-on by GSI) to Banknet.
2. Banknet responds with an 0810 Network Management Response message. If the response code is set to 00 (approve), the response indicates the sign-on has been accepted. If the response code is set to 30 (format error), 63 (security violation), 91 (switch down) or 96 (system error) the response indicates the sign-on was rejected.

Sign-Off Initiated By BASE24

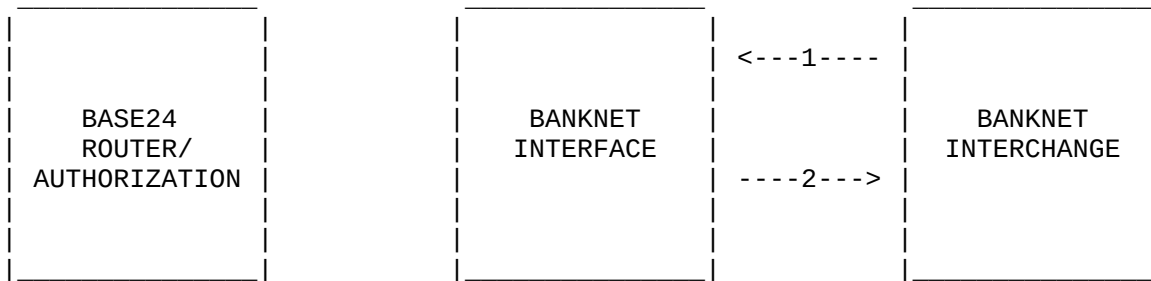
The diagram below illustrates the message processing that takes place between the Banknet Interface process and Banknet when BASE24 initiates a sign-off message. The steps in the diagram are described below.



1. To initiate a sign-off, the Banknet Interface process sends an 0800 Network Management Request message with a network management information code (DE 70) of 002 (sign-off by prefix), 062 (sign-off by group or GSI) or 066 (prefix sign-off by GSI) to Banknet.
2. Banknet responds with an 0810 Network Management Response message. If the response code is set to 00 (approve), the response indicates the sign-off has been accepted. If the response code is set to 30 (format error), 63 (security violation), 91 (switch down) or 96 (system error) the response indicates the sign-off was rejected.

Network Connection Status Initiated By BASE24

The diagram below illustrates the message processing that takes place between the Banknet Interface process and Banknet when BASE24 initiates a network connection status message. The steps in the diagram are described below.



1. To initiate a network connection status (echo test), the Banknet Interface process sends an 0800 Network Management Request message with a network management information code (DE 70) of 270 (echo test) to Banknet.
2. Banknet responds with an 0810 Network Management Response message. If received, the Banknet Interface process can assume that the network connection is functioning correctly.

It is also possible for Banknet to initiate a network connection status message. When the Banknet Interface process receives an 0800 Network Management Request message with a network management information code (DE 70) of 270 (echo test), the interface responds with an approved 0810 Network Management Response message.

Dynamic Key Exchange

The Banknet Interface supports dynamically changing the working keys online. Dynamic key exchanges may be initiated by MasterCard or by BASE24. BASE24 will use the same key for inbound and outbound transactions. In order to participate in dynamic key exchanges, the Key Exchange Key (KEK) on screen 1 of the KEYF or KEY6 record associated with the interface is required to be set up as described in the **BASE24 Transaction Security Manual**.

When MasterCard initiates the key exchange, an 0800 message is received by the Banknet Interface with a NMM information code of 161. Processing is invoked that translates the new working key from data received in DE 48 of the 0800 message. If a Thales (formerly Racal) HSM is being used, then the value of the LCONF parameter RACAL-ANSI-METHOD indicates how the new working key is encrypted. If the translation is successful, the new working key is placed in the PCT pending later activation. The interface responds with an 0810 message indicating the new key is ready to be activated. When MasterCard is ready to start processing with the new key, the MasterCard Authorization System will send the interface an 0820 message. The interface will then write the new key from the PCT to the KEYF or KEY6 file, and subsequent transactions will be processed using the new working key.

BASE24 may initiate a key exchange in the following ways:

- ☐ The operator may begin the key exchange process by issuing the interface a "NEWKEY" command.
- ☐ A key exchange may be initiated after a predetermined number of transactions. The transaction limit is set using the PIN KEY TRAN field on screen 3 of the KEYF or KEY6 record associated with the interface. Entering a value in this field will activate a transaction counter. When the number of inbound or outbound transactions with PINs have been processed that match the transaction limit, the key exchange sequence is initiated. If the PIN KEY TRAN field is left at the default value "0", the transaction counter will not be activated.
- ☐ A key exchange may be initiated after a predetermined time limit has expired. The time limit is set using the PIN KEY TIMER VALUE and PIN KEY TIMER INTERVAL fields on screen 3 of the KEYF or screen 4 of the KEY6 record associated with the interface. When this timer expires, the key exchange sequence is initiated. If the PIN KEY TIMER VALUE field is left at the default value "0", the "new key" timer will not be activated.

The interface will initiate the key exchange by sending an 0800 message to the switch with a NMM information code of 162. After the switch has responded with an 0810 message, the new key is exchanged and activated using a process identical to that when MasterCard initiates the key exchange (described above).

Any key exchange will reset the transaction counters and the key exchange timer, if they have been activated. After a new working key has been activated, BASE24 will use the new key for PIN processing. If the security device detects an error because the interchange is still using the old key, it can repeat the PIN processing using the old key before declining the transaction. A “clear old keys” timer may be set to control how long the old key is available after a key exchange. When this timer expires, the old key values are cleared from the PCT and the KEYF or KEY6.

ACI Worldwide, Inc.

Section 6

INTERFACE FUNCTIONS

INTRODUCTION

This section describes some of the specific transaction processing functions supported by the Banknet Interface.

TRACE NUMBERS

When the Banknet Interface process receives an outbound transaction request, it assigns a trace number to the transaction. The trace number is sent to Banknet in Field 11 of the Standard External Message (SEM). Trace numbers are used to uniquely identify a transaction, and a new trace number is assigned to each new request or advice. Reversal transactions contain the trace number of the original transaction in Field 90 (Original Data Elements) of the message.

Trace numbers must be unique, and the Banknet Interface process keeps track of the range of trace numbers it is assigning by writing trace numbers periodically to a special record in the ILF (the first record in the file). The Banknet Interface process updates the trace number in the ILF each time it processes the number of transactions set in the ILF HEADER MAX TRAN UPDATE field on ICF/ICFE screen 12. This way, it can keep track of the range of trace numbers it is assigning even if the process is stopped and restarted. At initialization, the Banknet Interface process always reads the trace number from the ILF and increases that number by the number in the ILF MAX TRAN UPDATES field in order to arrive at the first new trace number to assign. Using this technique, the Banknet Interface process can avoid creating duplicate trace numbers even in situations where the process is stopped and restarted.

When a single Banknet Interface process exists in a BASE24 system, the trace number begins at 000001 and is incremented by 1 until 999999 is reached. At this point, the trace number begins again at 000001.

To ensure a unique Systems Trace Audit Number (STAN) is used when multiple Banknet Interface processes exist in a single BASE24 system, two LCONF parameters need to be configured. These are FIRST-BNET-TRACE-NUM and INCR-BNET-TRACE-NUM-BY. These parameters are optional, as they have no value for a BASE24 system that operates with a single Banknet Interface process. If these parameters are not present, then the Banknet Interface process will assume that they have a value of 1.

The FIRST-BNET-TRACE-NUM needs to be defined per Banknet Interface process. It is used to define the starting Trace Number for each process; therefore this parameter should be set to a different value for each Banknet Interface process operating within a BASE24 system. The INCR-BNET-TRACE NUM-BY is the value by which the Trace Number is to be incremented, and should be set to the number of Banknet Interface processes that are configured.

When the Trace Number is incremented in the processing, it is incremented by the value contained within the INCR-BNET-TRACE NUM-BY parameter. If the value exceeds 999999, then the number is reset to the value in the FIRST-BNET-TRACE-NUM parameter.

CURRENCY CONVERSION

The Banknet Interface does not support currency conversion, because the switch performs all required currency conversion processing.

In authorization messages sent to the issuer, Banknet converts all amounts from the transaction currency to the cardholder billing currency. If the issuer has licensed the Multiple Currencies product, then multiple cardholder billing currencies can be supported (the BASE24 internal message is populated with amounts in the cardholder billing currency for the transaction). Otherwise, the issuer must configure the cardholder billing currency in the LOCAL CURRENCY CODE field on ICFE/ICF Screen 12. The LCONF parameter SPPT-MULTI-CRNCY indicates whether the issuer supports multiple cardholder billing currencies.

BANKNET CURRENCY EXPONENT TABLE (BNETTBL)

The Banknet Currency Exponent Table (BNETTBL) gives users the ability to control the adjustment of internal and external amounts when the exponents defined by MasterCard differ from the exponents defined by BASE24. BASE24 exponents are based on ISO4217 and defined in BAUTILS. For example, the Indonesian Rupiah (IDR) is defined by ISO4217 with an exponent of 2 while MasterCard defines the exponent as 0. If the MasterCard definition of the Rupiah's exponent is added to the BNETTBL the Banknet interface will adjust between internal and external amounts based on the difference between the exponents. External amounts from MasterCard (in Rupiahs) are converted to internal BASE24 amounts (in Rupiahs) by multiplying the amount by 100. Internal BASE24 amounts (in Rupiahs) are converted to external MasterCard amounts (in Rupiahs) by dividing the amount by 100. ACI ships the BNETTBL empty so that customers have complete control over the adjustment of amounts. After modifying the BNETTBL, the customer must recompile and restart the interface to invoke any changes. The currency must exist in the BNETTBL and the exponent must differ from the exponent in BASE24 for any amount adjustment to take place. The following is an example of three currencies in the BNETTBL. The table is searched sequentially and thus need not be in any order. The num^crncy^expnt^l must match the number of currencies in the table for it to compile correctly. Only add currencies that differ in their exponent and that you process.

```
literal num^crncy^expnt^l = 3;

string .crncy^expnt^tbl^s[0:num^crncy^expnt^l * 4] := [
! |----- Currency Code
! |
! |--- Currency Exponent
! |
! |
! v v
"108", 2, ! Burundi Franc BIF
"152", 2, ! Chilean Peso CLP
"360", 0, ! Indonesian Rupiah IDR

0 ]; ! End of Table - required to terminate a search
```

PIN BLOCK ENCRYPTION

The Banknet Interface process supports three PIN encryption options:

☐ **Clear PIN processing**

PINs are handled in the clear. They are not encrypted. This option should not be used.

☐ **Hardware encrypted PINs**

A security module is accessed for PIN processing. PINs are encrypted under the secure keys of a security module. A PIN will be encrypted at the point of entry (the ATM) and will remain encrypted until it is received by the issuer (the final destination) for verification. The PIN never appears in clear text, except at the point of entry and at the final destination. The security modules that can be used include the Atalla Security Module, the Atalla High Performance Security Module, and the Thales (formerly Racal) Security Module.

☐ **Software encrypted PINs**

PINs are encrypted via software code, using a DES (Data Encryption Standard) algorithm. No security module is in use. The PINs are in clear text within the processor itself. The Banknet Interface process can be set up to use software encryption/decryption even when VisaNet is set up to use hardware on their end.

Triple To Single Des Encryption

The Banknet Interface supports Triple to Single DES encryption where PINs may be translated from encryption under a double length key (3DEA) to encryption under a single length key (DEA) or translated from DEA to 3DEA. Triple to Single DES encryption is supported when translating PINs in the following situations:

- ☐ Internal clear PIN to external PIN encrypted in hardware
- ☐ Internal PIN encrypted in hardware to external PIN encrypted in hardware
- ☐ External PIN encrypted in hardware to internal PIN encrypted in hardware

If LCONF parameter DES-TRIPLE-SINGLE is 'Y', this indicates that the interface will translate PINs from being single DES encrypted inside BASE24 to triple DES encrypted when sent to the interchange and will translate PINs from being triple DES encrypted when received from an interchange to being single DES encrypted inside BASE24. The Key6 File (KEY6) is used instead of the Key File (KEYF) in order to store and process the double length PIN encryption keys. The KEY6 supports double length PIN encryption and key encryption keys. The intermediate keys remain single length within the KEY6.

Triple Des Encryption Via Security Services

The Banknet Interface supports Triple DES encryption using Transaction Security Services only in a Release 6.0 network environment. PINs may be translated from encryption under a single or double length key to encryption under a single or double length key. Triple DES encryption using Transaction Security Services is supported when translating PINs in the following situations:

- ☐ Internal clear PIN to external PIN encrypted in hardware
- ☐ Internal PIN encrypted in hardware to external PIN encrypted in hardware
- ☐ External PIN encrypted in hardware to internal PIN encrypted in hardware

The interface will translate PINs from being single or triple DES encrypted inside BASE24 to single or triple DES encrypted when sent to the interchange and will translate PINs from being single or triple DES encrypted when received from an interchange to being single or triple encrypted inside BASE24.

LCONF parameter DES-TRIPLE-SINGLE must not exist or may exist with the value 'N' if the Banknet Interface is to support Triple DES encryption using Transaction Security Services. The Key File (KEYF) is utilized. LCONF parameter SECURE-DEV-TYP must have the value 'TSS'. LCONF assign SECURE-DEV-1 must specify the name of the primary IAS process and may be process specific. LCONF assign SECURE-DEV-2 may specify the name of the secondary IAS process and may be process specific.

Single Des Encryption

If LCONF parameter DES-TRIPLE-SINGLE does not exist or has the value 'N' and LCONF parameter SECURE-DEV-TYP does not have the value 'TSS', single DES encryption is supported by the Banknet Interface. Note that MasterCard no longer permits the encryption of PIN blocks using single DES encryption.

Enhanced Key Management

Acquirers using the Banknet Authorization System to perform PIN translation must provide DE 53 (Security-Related Control Information) in authorization request messages.

Acquirers provide DE 53 to identify the PIN Block Format (configured using the LCONF parameter EKM-PIN-BLOCK-FORMAT) and the key used for PIN Block encryption (configured using the LCONF parameter EKM-ACQ-PEK-INDEX). The Banknet Interface includes DE 53 in authorization request messages if the LCONF parameter ENHNC-KEY-MGMT is set to 'Y'.

Issuers performing their own PIN validation or using the Banknet Authorization System to perform stand-in PIN verification must provide DE 53 in sign-on network management request messages. Issuers provide DE 53 to identify the PIN Block Format (configured using the LCONF parameter EKM-PIN-BLOCK-FORMAT) and the key used for PIN Block encryption (configured using the LCONF parameter EKM-ISS-PEK-INDEX) in subsequent authorization request messages. The Banknet Interface includes DE 53 in issuer sign-on network management request messages if the LCONF parameter ENHNC-KEY-MGMT is set to 'Y'.

ADDRESS VERIFICATION SUPPORT

The Address Verification Service (AVS) provides greater security to merchants and cardholders. It protects against fraudulent use of cards, in non face-to-face transactions, by verifying the cardholder's billing address. This verification is accomplished by matching the cardholder billing address that the acquirer sends with address information either the issuer or Banknet maintains.

For additional information on Address Verification refer to the ***BASE24-pos Address Verification Manual***. The request for AVS is sent in data element 48 of the external message (SEM), the response data will also be found in this location. The AVS sub-elements in DE 48 are documented in Appendix B.

International Address Verification Support

MasterCard also supports IAVS for MasterCard's Visa gateway. This is accomplished by the increased size of the address field in DE 120 from 20 bytes to 40 bytes.

CARDHOLDER-ACTIVATED TERMINALS

Cardholder-activated terminals (CAT) are unattended terminals that accept bankcards, debit, charge, and proprietary cards. Because these terminals are unattended, the traditional point-of-sale (POS) acceptance procedures do not apply (e.g., examination of card for abnormalities, signature comparisons, etc.). Each CAT is identified by an indicator (1 - 7) in POS Data (field 61, position 10) of the Authorization Request/0100 message.

Cardholder-Activated Terminals (CAT) may be supported via the Switch Terminal File (STF). In general, the STF is used to map BASE24 terminal IDs to switch-assigned terminal IDs. The Banknet Interface uses the STF to assign a CAT value to a terminal, so every CAT needs to have an entry in the STF. Customers who are using the interface as issuer only or acquirers who do not have CATs, do not need to add any STF records. If an STF already exists for another one of the customer's interfaces, then the customer just needs to add their Banknet records to that file.

On an inbound request, the CAT level (POS Data - DE 61, position 10) will be moved to position 8 of the interchange response in the PSTM and the Point of Service Data Token.

On an outbound request, the STF will be searched. If a record is found, the value of the Cardholder Activated Terminal Level will be moved into the POS Data (DE 61) position 10, which is the CAT level.

INTERCHANGE COMPLIANCE

The MasterCard Interchange Compliance program is designed to improve data integrity in the authorization process and clearing process as it applies to interchange and other MasterCard rules. This in turn will reduce the number of transactions submitted by acquirers for an inappropriate interchange rate.

The MasterCard Interchange Compliance program attempts to successfully match each day's clearing transactions with the appropriate authorization transactions received through the MasterCard International Banknet telecommunications network (Banknet network). The information is then analyzed to determine if they were submitted for the appropriate interchange rate.

The fields used to match the transactions are the cardholder number, financial network code, Banknet reference number, and Banknet date. Banknet assigns the financial network code, Banknet reference number and Banknet date before forwarding a transaction from the acquirer to the issuer. The assigned data will remain with the transaction on the response side as well. The Banknet Interface supports Interchange Compliance by storing the data in the BASE24-pos Interchange Compliance Token (Id = '20') or the BASE24-atm Interchange Compliance Token (Id = 'A6'), and adding it to the internal message. If the token already exists, then it will be updated. In both tokens, the TRACE-ID field holds the 3-character financial network code (from DE 63.1), the first 6 characters of the Banknet reference number (from DE 63.2), and the 4-character Banknet date (from DE 15). For more information on the fields in the Interchange Compliance Tokens, refer to the ***BASE24 Tokens Manual***.

The Banknet acquirers and issuers must be able to retrieve these fields to format their 635 Interchange Compliance addendum records. Acquirers that submit transactions for the Merit III, Merit II, and Supermarket rates without a valid 635 Interchange Compliance addendum record should anticipate an adjustment to their interchange rate.

TRANSACTION CATEGORY CODE (TCC)

The TCC is sent to Banknet by acquirers in DE 48 in Authorization Request/0100 messages. The Banknet Interface uses the transaction type and a table that maps TCC values from Merchant Category Code (MCC) values to determine the TCC.

Acquirer authorizations initiated with a swiped card should not contain the transaction category code (TCC) "T". The "T" designation represents a mail order/telephone order and should be used only if the cardholder and card are not present. The merchant category code (MCC) for Telecommunications Services ("4814") is designated as TCC "T". The Banknet Interface modifies the TCC to contain "R" (retail purchase) when an authorization using MCC 4814 is initiated through the use of a magnetic stripe reading telephone. The Banknet Interface also modifies a TCC value of 'T' to 'R' for other MCCs, if the POS Data (DE 61) indicates card and cardholder presence. Conversely, the Banknet Interface modifies a TCC value of 'R' to 'T', if the POS Data (DE 61) indicates that the card and cardholder are not present.

Note that the Banknet Interface leaves the TCC set to a space (allowing the Banknet Authorization System to set the value appropriately) if the LCONF parameter NTWK-ASSIGN-TCC is set to 'Y'.

MAGNETIC STRIPE READ/CVC SUPPORT

MasterCard incorporated enhanced security features on all new and reissued MasterCards beginning 1 January 1994. As a part of the security features, issuers must encode the Card Validation Code (CVC 1) on both track 1 and track 2 providing a mechanism to validate the card and respond with reasonable assurance that the track data is present and accurate. To support this feature, there are three sub-elements (87, 88 and 89) in DE 48 in the message between BASE24 and the interchange. A description of these fields is included in Appendix B. With this feature the issuers will be able to provide an indicator (DE 48, sub-element 87) in their authorization response if an invalid CVC is identified. BASE24 Release 3.4/4.0 does not support the CVC verification.

MasterCard recommends issuers provide invalid Card Validation Code (CVC) error codes in sub-element 87 of the Additional Data (DE 48) in Authorization Request/0100 messages. For Release 5.0 or above of BASE24-pos, this is currently done. For Release 3.4/4.0 of BASE24-pos, the Host can perform CVC verification but there is no method to inform the Banknet Interface process of the result.

BASE24 acquirers provide a magnetic stripe read indicator of "90" in the POS entry mode field of an electronically card read authorization request. A value of "90" indicates that the full track data has been read from the data encoded on the card and recorded within the authorization request without alteration or truncation. A POS entry mode field indicator of "02" indicates that full magnetic stripe was not read and transmitted. A value of "02" is also used by MasterCard to displace a "90" when the acquirer is not certified.

If a customer is running BASE24 release 3.4/4.0 as an acquirer and has terminals that are not in compliance, then it will be necessary for them to modify the BASE24 software to send "02" in the POS entry mode to Banknet.

CVC 2 SUPPORT

Card Validation Code 2 (CVC 2) support is designed to provide added security for the card authentication process for authorization requests primarily in non-face-to-face environments. CVC 2 support allows an acquirer to receive online verification of the CVC 2 value from registered issuers under the following conditions:

- ☐ The merchant provides the CVC 2 value.
- ☐ The acquirer is registered with MasterCard indicating their ability to provide online CVC 2 verification data.
- ☐ The acquirer transmits the CVC 2 value in Data Element (DE) 48, sub-element 92 of an authorization request.

- The issuer is capable of performing online verification of the CVC 2 value in sub-element 87 of DE 48. Issuers capable of online CVC 2 verification must register with MasterCard Member Services.

CVC 2 request data is supported by BASE24 in the PS51 token CVD-FLD, and by Banknet in sub-element 92 of DE 48. CVC 2 response data is supported by BASE24 using the PS50 token CRD-VRFY-FLG, and by Banknet in sub-element 87 of DE 48.

MasterCard will allow CVC 1 and CVC 2 to be transmitted in the same transaction. BASE24-pos Release 6.0 Router/Auth now supports verification of both values in the same transaction. If a customer is not running BASE24-pos Release 6.0, only the CVC 1 value will be verified. If CVC 1 passes verification, the value "P" will be returned in sub-element 87 of DE 48 indicating that CVC 2 was not processed. If CVC 1 fails validation, the value "Y" will be returned in this field as in current CVC 1 processing.

PAYMENT SERVICE 2000 SUPPORT

Payment Services 2000 (PS2000) is a Visa program that increases the quality, value, efficiency, and control of the payment process within the Visa system. Visa will assign PS2000 data to each online transaction request received, when a transaction meets certain requirements. This data will remain with the transaction in all related messages. The end result will be faster clearing of transactions and reduced back office operating costs for Visa members.

To participate, Banknet acquirers must be able to recognize and indicate potential PS2000 eligible transactions when sending requests to Visa, as well as send Visa assigned PS2000 data from original transactions when sending reversals. Data element 48 of the standard external message (SEM) accommodates this via sub-elements 90 and 91. A description of these sub-elements can be found in Appendix B.

To qualify for PS2000, a transaction must be a purchase using a Visa card requiring a single authorization, executed by a U.S. merchant. The merchant's terminal must be able to read the entire track 1 or track 2 off of the magnetic stripe on the back of the card, and the terminal must be uniquely identified, including a zip code for its location.

The Banknet Interface checks for the conditions in Release 3.4/4.0 BASE24-pos acquired transactions and sends the information to Visa through Banknet correctly. The interface can also send the PS2000 data within the Release 3.4/4.0 BASE24-pos PSTM to be logged in the POS Transaction Log File (PTLF), to aid in settlement with Visa's BASE II Settlement system.

The Banknet Interface also supports PS2000 processing in Release 5.0 or above BASE24-pos. A new token in the Release 5.0 or above PSTM can be used to check for PS2000 eligibility, and to store PS2000 data for logging to the Release 5.0 or above PTLF. If the new PS2000 token is not present in the PSTM when the interface is checking for PS2000 eligibility, the interface will make checks as is done on the Release 3.4/4.0 PSTM. If the PSTM request is found to be PS2000 eligible, the interface will add the PS2000 token to the PSTM for any further processing.

In order to implement the Banknet Interface PS2000 support in a Release 5.0 or above BASE24-pos system, the POS Token record in the Token File must be updated to log the PS2000 token to the Interchange Log File (ILF) and POS Transaction Log File (PTLF), and extract the token from these files. Refer to the ***BASE24 Tokens Manual*** if assistance is required.

SWITCH TOKEN SUPPORT

The Banknet Interface supports the generic switch token which facilitates the passing and logging of interchange specific data in BASE24. Using this token, the Interface can have data sent to the Host or co-networks and may be logged to the POS Transaction Log File (PTLF) or ATM Transaction Log File (TLF) to be extracted by a Host. It gives members the ability to log switch-specific data to the PTLF and TLF that is not currently being logged.

Two token IDs are available for the switch token:

- ☐ B0 (Acquirer Switch token)
- ☐ B1 (Issuer Switch token)

The generic switch token includes data taken from the external message.

For requests and advices received from Banknet, the interface adds the Acquirer Switch token (B0). For reversal messages received from Banknet, the interface checks for the presence of the Acquirer Switch token (B0) in the ILF. If the token exists, the interface updates the token; otherwise the interface will add the token.

For responses received from Banknet, the interface adds the Issuer Switch token (B1).

For information on the Banknet definition of the generic switch token, refer to Appendix A.

CHIP-BASED DEBIT AND CREDIT SUPPORT

The Banknet EMV module is an additional licensable module, built on top of the Switch 6.0 version of the Banknet Interface.

The source files of the standard Banknet Interface include a number of EMV-specific 'hooks' or user exits. These provide an interface between the standard Banknet Interface and the Banknet EMV module. The Banknet EMV modules is bound together with the standard Banknet Interface code to provide support for Banknet EMV transactions, in addition to non-EMV transactions.

For details of the Banknet EMV processing, refer to the ***BASE24-atm EMV Support Manual*** and the ***BASE24-pos EMV Support Manual***.

EMV Transactions

An EMV transaction sent to Banknet must contain EMV data in DE 55. To identify an EMV transaction, a value of '05' (Integrated Circuit Card) is included in digits 1 and 2 of the Point of Service Entry Mode (DE 22).

MasterCard also supports a means for acquirers to provide the Card Sequence Number (DE 23) in Authorization Request/0100 messages for chip transactions. Chip card issuers can use Data Element 23 to distinguish between cards with the same primary account number, which will allow two cards with the same primary account number (PAN) to have unique identifiers.

Fallback Transactions

Chip fallback occurs when both the card and the terminal support EMV technology, and a failure of either the card or the terminal prevents the use of this technology. If appropriate, a secondary, or "fallback", technology, e.g. reading the data on the magnetic stripe, may then be used in order to complete the transaction. When fallback occurs, the first two digits of the POS Entry Mode (data element 22) are set to '80' (fallback to magnetic stripe) or '79' (fallback to PAN key entry).

MasterCard Europe does not permit fallback for ATM transactions (both credit and debit), PIN based in-branch transactions (i.e. cash withdrawals from a bank branch terminal), and CAT Level 1 and CAT Level 2 transactions. If fallback occurs, the acquirer should decline these types of transaction.

However, other MasterCard regions do not have this restriction. Therefore, the Banknet Interface supports an LCONF parameter that indicates whether fallback is permitted for ATM transactions (both credit and debit), PIN based in-branch transactions (i.e. cash withdrawals from a bank branch terminal), and CAT Level 1 and CAT Level 2 transactions. Acquirer processing checks the FALLBACK-ALWD parameter, and uses the value to determine whether the authorization request should be declined or forwarded to the issuer.

A MasterCard transaction can be identified as “fallback”, only if the terminal has been certified for EMV processing by MasterCard. The Switch Terminal File (STF) allows the Banknet Interface to identify whether a specific terminal has been certified for EMV processing by MasterCard.

When processing a BASE24-acquired non-EMV POS transaction containing the EMV Status Token (indicating that the terminal is EMV-capable), the Banknet Interface will use the STF memory table to determine whether the transaction is “fallback”. If an STF record is located, and the record contents indicate that the terminal has not been certified for EMV processing by MasterCard, then the transaction will not be identified as “fallback”, i.e. the POS Entry Mode (DE 22) will not be set to a value of ‘80x’ (card read) or ‘79x’ (key entered) in the message sent to Banknet.

When processing a BASE24-acquired non-EMV ATM transaction containing the EMV Status Token (indicating that the terminal is EMV-capable), the Banknet Interface will use the STF memory table to determine whether the transaction is “fallback”. If an STF record is located, and the record contents indicate that the terminal has not been certified for EMV processing by MasterCard, then the transaction will not be identified as “fallback”.

Chip Card Authorization Support For Visa

Members that use the MasterCard Authorization System are able to forward full grade chip data for Visa chip card transactions through MasterCard's Visa gateway.

When acquirers send full grade chip data for Visa chip cards, MasterCard uses Integrated Circuit Card (ICC) System-Related Data (DE 55) values to forward the data through the Visa gateway to Visa issuers. If DE 55 is present for a transaction conducted using a Visa card, the Authorization System will forward the transaction through the Visa gateway.

Members that use the MasterCard Authorization System can also receive full grade chip data originating at a Visa acquirer through MasterCard's Visa gateway.

MasterCard supports a value of "95" in POS Entry Mode (DE 22) to indicate a Visa transaction initiated by a chip card with unreliable Card Verification Value (CVV) data.

On inbound processing, the "95" in the POS Entry Mode is mapped to an "05" (Integrated Circuit Card) in the PSTM Point of Service Entry Mode. The original POS Entry Mode from the SEM is available in the switch token.

The POS Entry Mode value "95" is not supported in outbound transactions at this time.

ON-BEHALF PROCESSING BY BANKNET

An "on-behalf" service provides issuers with the option of requesting Banknet to validate information in the message from the acquirer, before the authorization request is forwarded to the issuer. If Banknet performs processing on behalf of the issuer, then the authorization request message will contain DE 48, sub-element 71, specifying the result of up to ten "on-behalf" services performed on the transaction.

The processing of DE 48 SE 71 in the Banknet Interface supports the following functions:

- ☐ The Banknet Interface can check the results of all (up to 10) "on behalf" services performed on the transaction, instead of only the first two.
- ☐ It is possible for an issuer to configure the interface to decline a transaction in which some "on behalf" checks have failed, but to accept others.
- ☐ It is possible for an issuer to configure the interface to decline a transaction in which an "on behalf" result code for a service contains a particular result code, but to accept the same "on behalf" result code for another service.
- ☐ The Banknet Interface code adds the results of all ten "on behalf" services to the generic switch token.

The LCONF parameter DENY-FAILED-ON-BEHALF allows the issuer to decline authorization requests where Banknet "on behalf" processing has failed.

The specific conditions under which the issuer declines the transaction are configured in the LCONF parameters ON-BEHALF-DENY-RSLT-CDESxx (where xx are numbers). Each parameter contains the identifier of an "on-behalf" service (a 2-digit numeric value), followed by a set of up to 20 result codes for the service (each one alphabetic character) that will cause the transaction to be declined.

Example LCONF parameter values are:

Param: ON-BEHALF-DENY-RSLT-CDES01
Text: 02FGIU

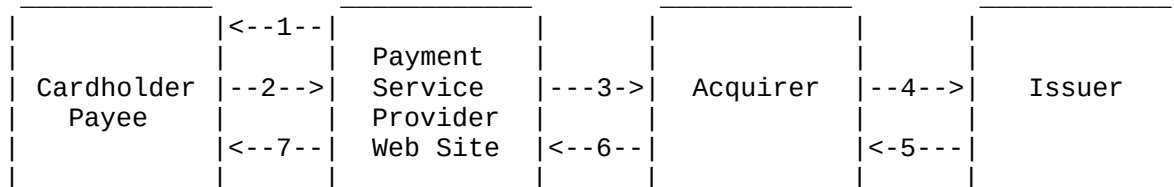
Param: ON-BEHALF-DENY-RSLT-CDES02
Text: 15AEINU

Refer to the definition of DE 48 SE 71 in MasterCard's **Customer Interface Specification** for a full description of "on-behalf" service identifiers and result codes.

PAYMENT TRANSACTION SUPPORT

MasterCard's Authorization System supports authorizations for Payment Transactions. These transactions are used when a consumer chooses to receive a payment using their MasterCard.

The figure below illustrates how the Payment Transaction pays the cardholder payee when using Authorization System enhancements:



1. Once the payer has requested a transfer of funds using his or her account to the payee, the MasterCard Payment Service Provider sends an email message to the payee to advise that funds are currently payable to the payee.
2. The payee enters MasterCard account information via the Payment Service Provider Web site.
3. The Payment Service Provider Web site forwards authorization information to an acquirer.
4. The acquirer transmits an Authorization Request (Payment Transaction)/0100 message to the payee's issuer.
5. The payee's issuer responds with an Authorization Request Response/0110 message to the acquirer.
6. The acquirer forwards authorization information to the Payment Service Web site.

7. The Payment Service Web site confirms that funds have been transferred to the payee's MasterCard account, usually providing a confirmation number and transaction details.

The Banknet Interface supports Payment Transaction as merchandise return transactions.

On inbound Payment Transactions, the external message contains a Processing Code (DE 3) value of "28" and a Transaction Category Code (DE 48) value of "P". The "from" account type is "00", and the "to" account type can be "00" or "30". These transactions are mapped to a BASE24-pos transaction code of "14", merchandise return. In addition, the Transaction Subtype Token is added, with the TXN-SUBTYP field set to indicate payment from third party "B000", and field ISS-PROC-CDE field populated with the value in DE 3.

The Banknet Interface also supports outbound Payment Transactions. On merchandise return requests received from BASE24, the Banknet Interface checks for the presence of the Transaction Subtype Token. If the token is present, and the TXN-SUBTYP field in this token is set to indicate payment from third party "B000", then DE 3 (Processing Code) is set to "28" (payment transaction) in the authorization request message sent to Banknet.

The Banknet Authorization system edits for valid values in Processing Code and Transaction Category Code for Payment Transactions.

E-COMMERCE TRANSACTIONS

The Banknet Interface supports e-commerce transactions, identified by a DE 22 (Point-of-Service Entry Mode) subfield 1 (POS Terminal PAN Entry Mode) value of '81' or '82', in both its acquirer and issuer processing.

DE 48 SE 42 (Electronic Commerce Indicators) and DE 48 SE 43 (Universal Card Authentication Field) are used in e-commerce transactions to identify the cardholder authentication performed. DE 48 SE 42 and DE 48 SE 43 are sent in authorization request, authorization advice and reversal request messages. DE 48 SE 42 includes a Universal Card Authentication Field (UCAF) collection indicator field that is used to indicate the presence/type of UCAF data in DE 48 SE 43.

Electronic Commerce Indicator (ECI) Support

MasterCard requires the Electronic Commerce Security Level Indicator (DE 48, sub-element 42) on all electronic commerce authorization requests. Sub-element 42 identifies the following attributes of the transaction:

- ☐ the security protocol (channel encryption or no encryption);

- ☐ the cardholder authentication (chip cryptogram used, electronic commerce service indicator or cardholder certificate not used);
- ☐ the UCAF collection indicator (UCAF data collection is not supported, UCAF data collection is supported by the merchant but UCAF data was not populated, UCAF data collection is supported by the merchant and UCAF data was populated, UCAF data collection is supported by the merchant and a Static authentication value was provided).

Various combinations of values for these attributes are valid, and depend on the security related initiative used for the transaction.

Universal Card Authentication Field (UCAF) Support

MasterCard provides members with the Universal Cardholder Authentication Field (UCAF) for standardizing the transport of cardholder authentication data for electronic commerce transactions, including mobile commerce, within authorization messages.

UCAF (DE 48, sub-element 43) is a universal, multi-purpose data transport infrastructure that MasterCard will use to communicate authentication information between the cardholder, merchant, issuer, and acquirer communities. UCAF is a variable length field up to 32 bytes, with a flexible data structure that MasterCard can tailor to support the needs of various issuer security and authentication approaches, including Secure Payment Application (SPA) and other future security solutions.

In its generic form, the UCAF data structure consists of a control byte assigned by MasterCard to represent a particular use of UCAF. The control byte is then followed by issuer discretionary data that may be standardized for specific security applications using UCAF to transport authentication data to the issuer for validation. UCAF provides the basic infrastructure that SPA and other emerging solutions will use to allow merchants to collect authentication data from cardholders for each online transaction and then standardize the communication of this authentication data to acquirers, MasterCard, and issuers.

Merchants implement UCAF by using hidden fields on their Web merchant's payment page. These hidden fields are transparent to cardholders but can be detected by issuer and cardholder security solutions, and are designed to collect and pass authentication data generated by issuer and cardholder security schemes.

Security Related Initiatives

MasterCard has several e-commerce and online security related initiatives in the marketplace, including:

- ☐ MasterCard® SecureCode™
- ☐ Maestro® Advance Registration Program™ (MARF)
- ☐ MasterCard Online Checkout Service

Each of these programs uses various common infrastructure components to provide program-related features such as cardholder authentication. In authorization messages, information about cardholder authentication at the point of interaction is provided in DE 48 SE 42 and DE 48 SE 43.

For MasterCard® SecureCode™, DE 48 SE 42 position 1 (security protocol) will indicate channel encryption, DE 48 SE 42 position 2 (cardholder authentication) will be “cardholder certificate not used” (preferred) or “chip cryptogram used”, and DE 48 SE 42 position 3 (UCAF collection indicator) will specify that UCAF data was populated. DE 48 SE 43 contains the encoded MasterCard® SecureCode™.

For the Maestro® Advance Registration Program™, DE 48 SE 42 position 1 (security protocol) will indicate channel encryption, DE 48 SE 42 position 2 (cardholder authentication) will be “cardholder certificate not used” or “chip cryptogram used”, and DE 48 SE 42 position 3 (UCAF collection indicator) will specify that a static account holder authentication value (AAV) was provided. DE 48 SE 43 contains the static AAV assigned to the merchant.

For the MasterCard Online Checkout Service, DE 48 SE 42 position 1 (security protocol) will indicate channel encryption or no encryption, DE 48 SE 42 position 2 (cardholder authentication) will be “electronic commerce service indicator”, and DE 48 SE 42 position 3 (UCAF collection indicator) will specify either that UCAF data is supported by the merchant but UCAF data but was not populated (Real Card Number applies) or that UCAF data was populated (Virtual Card Number applies). If present, DE 48 SE 43 contains the SecureCode provided by the merchant.

For a non-authenticated e-commerce transaction, DE 48 SE 42 position 1 (security protocol) will indicate channel encryption or no encryption, DE 48 SE 42 position 2 (cardholder authentication) will be “cardholder certificate not used” or “chip cryptogram used”, and DE 48 SE 42 position 3 (UCAF collection indicator) will specify that UCAF data collection is not supported or that UCAF data collection is supported by the merchant but UCAF data was not populated. DE 48 SE 43 is not present.

VISA AUTHORIZATION CHARACTERISTIC INDICATOR VALUES SUPPORT

MasterCard supports the data requirements for transactions that qualify for the Visa Custom Payment Service (CPS).

MasterCard members that acquire Visa point-of-interaction (POI) transactions can receive Visa Authorization Characteristic Indicator (ACI) values in the Authorization Request Response/0110 message. MasterCard DE 48, sub-element 90 in the Authorization Request Response/0110 message may contain any valid value from Visa.

MASTERCARD ELECTRONIC PROGRAM

MasterCard Electronic is a MasterCard brand designed to meet the needs of issuers targeting higher-risk cardholders. MasterCard Electronic requires 100% online authorization to allow close risk management.

The value of DE 48 sub-element 76 (MasterCard Electronic Acceptance Indicator) in authorization request messages indicates whether the acquirer participates in the MasterCard Electronic program. The LCONF parameter DENY-MC-E-COMM-TXN allows the issuer to decline MasterCard Electronic transactions if the acquirer (or the merchant) does not participate in the program.

If DE 48 sub-element 76 (MasterCard Electronic Acceptance Indicator) contains “C” or “M” and the LCONF parameter DENY-MC-E-COMM-TXN is not set to “N”, then the transaction will be declined with a response code of ‘57’ (transaction not supported). The message will then be dropped, unless the interface has been configured to forward declined advices to the POS Router/Authorization process.

The Banknet Interface does not populate DE 48 sub-element 76 in authorization request messages sent to Banknet.

DATA SUPPRESSION

MasterCard does not allow members to store sensitive transaction data on their systems. This includes the discretionary data from the card's magnetic stripe and/or chip (Track-1 and Track-2), manual card verification digits (CVC2), and AVS data.

Acquirers can configure the Banknet Interface to ensure that this data is suppressed, i.e. not logged to the ILF, using the following LCONF parameters:

- ☐ SWI-ACQ-DISCR-DATA-SUPPRS
- ☐ SWI-ACQ-CVD2-DATA-SUPPRS
- ☐ SWI-ACQ-AVS-DATA-SUPPRS

Issuers can configure the Banknet Interface to ensure that this data is suppressed, i.e. not logged to the ILF, using the following LCONF parameters:

- ☐ SWI-ISS-DISCR-DATA-SUPPRS
- ☐ SWI-ISS-CVD2-DATA-SUPPRS
- ☐ SWI-ISS-AVS-DATA-SUPPRS

Note that issuers should suppress the Track-1/Track-2 discretionary data only if the Banknet Interface is running in a BASE24 Release 6.0 system. Earlier releases of BASE24-pos and BASE24-atm use the discretionary data in reversal message processing, and the track data is retrieved from the ILF when the Banknet Interface receives an external reversal request or advice.

DECLINED ADVICES

A Banknet Interface process on an issuer system accepts 0120 advice messages from Banknet, and converts them to the BASE24 internal message format before forwarding them to the POS Router/Authorization process or the ATM Authorization process.

When processing an advice message received from Banknet, the interface may forward the advice to the appropriate authorization process, even if the transaction is not approved. The Banknet Interface can also forward an advice to BASE24 if a request message is declined by the interface itself, e.g. if the acquirer (or the merchant) does not participate in the MasterCard Electronic program.

The LCONF parameter FRWD-DCLN-ADVC allows the issuer to configure whether declined advices should be forwarded to BASE24 or logged to the ILF and dropped. Transaction messages rejected because of formatting errors will not be forwarded to the BASE24-pos Router/Authorization process or the BASE24-atm Authorization process, regardless of the setting of the LCONF parameter.

UNMATCHED REVERSALS

A Banknet Interface process on an issuer system accepts 0400 reversal requests and 0420 reversal advice messages from Banknet, and locates the original transaction message in the ILF. The Banknet Interface converts the reversal to the BASE24 internal message format before forwarding the message to the POS Router/Authorization process or the ATM Authorization process.

When processing a reversal message received from Banknet, the interface may forward the reversal to the appropriate authorization process, even if a matching request/response message cannot be found.

The LCONF parameter FRWD-UNMATCHED-RVSLS allows the issuer to configure whether unmatched reversals should be forwarded to BASE24 or logged to the ILF and dropped. This functionality is not available for PIN Change or PIN Unblock reversal messages.

CONTACTLESS TRANSACTION SUPPORT

Contactless (proximity) transactions are identified by specific values in sub-field 1 (Card Data Entry Mode) of DE 22 (Point of Service Entry Mode), and terminals with a contactless reader are identified by specific values in sub-field 11 (POS Card Data Terminal Input Capability Indicator) of DE 61 (POS Data).

For DE 22, sub-field 1, the values are as follows:

- ☐ 07 (PAN auto-entry via contactless M/Chip)
- ☐ 91 (PAN auto-entry via contactless magnetic stripe)

For DE 61, sub-field 11, the values are as follows:

- ☐ 3 (PAN auto-entry via contactless M/Chip)
- ☐ 4 (PAN auto-entry via contactless magnetic stripe only)

Contactless transactions where the PAN is entered via contactless M/Chip are authorized by BASE24-pos using EMV functionality. Contactless transactions where the PAN is entered via contactless magnetic stripe are authorized by BASE24-pos using dynamic card verification (DCV) functionality.

On-Behalf Processing By Banknet

An “on-behalf” service provides issuers with the option of requesting Banknet to validate the Application Transaction Counter (ATC), before a contactless magnetic stripe authorization request is forwarded to the issuer. If Banknet validates the ATC on behalf of the issuer, then the authorization request message will contain DE 48, sub-element 34, specifying the result of the validation.

The LCONF parameter DENY-IF-ATC-DISCRP allows the issuer to decline contactless magnetic stripe authorization requests where the Banknet Authorization system finds that the ATC is outside the range of values permitted by the issuer.

CARD TYPE VALIDATION

Certain transactions are valid only if performed on particular card types. When sending transaction messages to Banknet, the Banknet Interface validates the transaction type against the card type.

The BNET-DEBIT-CARD-TYPES parameter is used to specify debit card types explicitly, by containing a list of BASE24 card types representing debit card products.

Gambling Transactions

For credit card transactions, CVC2 data is mandatory in non-face-to-face gambling transactions (MCC = 7995), where the Banknet processing code is 00 (Purchase). Therefore, when the card type does not appear in the BNET-DEBIT-CARD-TYPES parameter list, this type of transaction is allowed only when CVC2 data is present. If the card type does appear in the list, then CVC2 data is not required.

Cash Withdrawals From Bank Branch Terminals

These transactions are only allowed on debit cards. Therefore, the transaction is allowed only when the card type is contained in the BNET-DEBIT-CARD-TYPES parameter list. If it is not contained in the list, then the transaction is processed as a credit card cash advance.

E-Commerce Transactions

These transactions must contain DE 35 (Track-2 data) when performed on debit cards. If the card type is contained in the BNET-DEBIT-CARD-TYPES parameter list, then the Banknet Interface “constructs” the Track-2 data if DE 35 is not present in the outgoing authorization request message.

ALTERNATE ROUTING

If a BASE24-pos authorization request cannot be sent to Banknet, e.g. because of a communications error, then it is possible for “alternate” routing to occur. The transaction message is routed either to another interface process or back to the POS Router/Authorization process in an attempt to obtain authorization for the transaction.

It is also possible to invoke alternate routing when the Banknet Interface receives a response message containing a response code indicating some type of system malfunction, e.g. 31, 91 or 96. This processing may be configured to apply to any response code returned by Banknet, by specifying the relevant response code value(s) in the LCONF parameter BNET-STAND-IN-RCS.

PARTIAL APPROVALS

The BASE24-pos Router/Authorization process supports the concept of an authorization for a lesser amount, if the cardholder has insufficient funds to complete the transaction. For Banknet, an authorization for a lesser amount is permitted, as long as the acquirer indicates that partial approvals are supported and that the transaction is not a purchase with cash back.

The LCONF parameter PARTIAL-APPROVAL-ALWD allows a POS issuer to specify whether authorizations for a lesser amount are supported. When an acquirer supports partial approvals, as indicated in an authorization request (using a sub-field in DE 48 sub-element 61), the Banknet Interface checks the value of the LCONF parameter. If the PARTIAL-APPROVAL-ALWD parameter contains a value other than ‘N’ (and the transaction is not a purchase with cash back), then the Banknet Interface sets a field in the PSTM to allow the BASE24-pos Router/Authorization process to approve the transaction for a lesser amount.

From Release 6.0 version 10, the BASE24-pos Router/Authorization process supports the concept of an authorization for a greater amount, where the request message contains a notional transaction amount. For Banknet, an authorization for a greater amount is permitted, as long as the acquirer indicates that partial approvals are supported and that the transaction originates from an automated fuel dispenser (AFD), which is identified by a Merchant Category Code (MCC) of “5542”.

The LCONF parameter GREATER-AMT-APPROVAL-ALWD allows a POS issuer to specify whether authorizations for a greater amount are supported. When an acquirer supports partial approvals, as indicated in an authorization request (using a sub-field in DE 48 sub-element 61), the Banknet Interface checks the value of the LCONF parameter. If the GREATER-AMT-APPROVAL-ALWD parameter contains a value of 'Y', the MCC (DE 18) in the message is "5542", and the transaction amount (DE 4) contains one major unit of currency, then the Banknet Interface sets a field in the PSTM to allow the BASE24-pos Router/Authorization process to approve the transaction for a greater amount.

NON-CURRENCY DISPENSE TRANSACTION SUPPORT

Non-currency dispense (NCD) transactions are supported in BASE24-atm, using the Non-Currency Dispense Token (which provides information about the NCD transaction).

On requests sent to Banknet, a BASE24-atm transaction code of '10' (withdrawal) is mapped to a Banknet processing code of '00' (purchase), if the internal message contains the NCD token and the NCD segment is present in the ICF/ICFE record.

On requests received from Banknet, a processing code of '00' (purchase) is mapped to a BASE24-atm transaction code of '10' (withdrawal), if DE 18 (Merchant Category Code) is set to '6012' (indicating a NCD transaction), and the NCD segment is present in the ICF/ICFE record. The NCD Token is also added, with the HOPR-CONTENTS field set to the value of the LCONF parameter NCD-DFLT-HOPR-CONTENTS.

MASTERCARD inCONTROL COMMERCIALIZATION

The first application to be launched from the MasterCard inControl platform allows for the generation of a unique MasterCard inControl virtual card number for every transaction. : The interface maps the DE 48 SE 33 account number to the virtual PAN field in the Banknet Switch Token, when subfield 1 of DE 48 SE 33 contains the value "V".

MASTERCARD *MONEYSEND* PAYMENT TRANSACTION

The interface has not been modified to support *MoneySend* transactions in its acquirer processing. The LCONF parameter, PROCESS-MONEYSEND-TXN indicates whether *MoneySend* transactions are allowed by the issuer. If the LCONF parameter does not contain the value "Y", the interface will deny *MoneySend* transactions back to the network.

The token, Person to Person Transaction 2 Token (Token ID = "S7") is used to support the data in DE 124 (Member-defined Data) for both Sender and Recipient data.

The MasterCard *MoneySend* service uses the Payment Transaction to facilitate person-to-person payments. The Banknet Interface supports Payment Transaction as merchandise return transactions. For Payment Transaction, the Transaction Subtype Token is added, with the TXN-SUBTYP field set to indicate payment from third party "B000".

The MasterCard *MoneySend* On-Us Funding Transaction is identified by a Merchant Category Code (MCC) of "6538". The Transaction Subtype Token is added, with the TXN-SUBTYP field set to indicate payment to third party "B001".

PRIVATE LABEL PROCESSING

MasterCard is supporting enhancements to private label processing to offer merchant verification, co-brand proprietary program support, new private label product codes, and more. To make sure that only participating private label merchants are allowed to accept participating private label cards, transaction verification will be enabled so that non-participating merchants' transactions are not approved. MasterCard allows participating merchants to be registered and identified by acquirer-assigned Card Acceptor ID Code (DE 42) or by a MasterCard Assigned ID (DE 48, subelement 32).

Acquirers may use the Alternate Merchant ID Token to vary the value in DE42 either by supplying the token from the host or by configuring an alternate merchant id by card type on screens 3 and 4 of the PRDF.

Acquirers that wish to use the MasterCard Assigned ID (DE 48, subelement 32) must add records to the STF by retailer ID along with the MasterCard Assigned ID in the STF SWITCH MERCHANT ID. To activate the search of the STF the LCONF parameter PRVT-LABEL-USES-DE48SE32 must exist and contain the value "Y". The value in DE48 SE32 will be logged to a new field in the Banknet Switch Token.

Card activation transactions (with or without an initial load) are not supported by the interface. Any Card activation transactions received by the interface will be declined.

POS ACCOUNT BALANCE RESPONSES

An issuer may elect to return a balance for approved and non-approved purchase transactions. Set the return balances flag on page 8 of the CPF to enable Router Auth to return the POS Balances Token. When the interface finds the token the amount will be added to DE 54 (Additional Amounts) in the response message.

An acquirer may add the POS Balances Token in a transaction. If DE 54 (Additional Amounts) is present in the response from Banknet and the transaction is not a balance inquiry, then the Banknet Interface adds the POS Balances Token.

ALTERNATE ISSUER TRANSACTION ROUTING

Issuers in the Europe region may use an alternate issuer host instead of the Stand-In System as their alternate authorization route. The BASE24 Banknet interface can be configured to run as the alternate issuer host. The alternate host is identified by the value Y in the ALTERNATE HOST field on ICF(E) Screen 12.

When operating as the alternate host, the interface will use the alternate network management information codes in DE 70:

- ☐ 063 = Group sign-on alternate issuer route
- ☐ 064 = Group sign-off alternate issuer route
- ☐ 065 = Prefix sign-on (by Group Sign-on ID for primary route)
- ☐ 066 = Prefix sign-off (by Group Sign-on ID for primary route)
- ☐ 067 = Prefix sign-on (by Group Sign-on ID for alternate issuer route)
- ☐ 068 = Prefix sign-off (by Group Sign-on ID for alternate issuer route)

DE 48, subelement 12 (Routing Indicator), value A (Alternate issuer host routing) will be present in the Authorization Request/0100 message to indicate that alternate issuer routing has been invoked. This value will be logged in the switch token. DE 121 (Authorizing Agent ID Code) with value 000002 will be logged in the switch token for issuers and acquirers.

UK DOMESTIC MAESTRO PROCESSING

When there are functional differences in the BASE24 Banknet Interface processing between UK Domestic Maestro transactions and other transactions, separate Banknet Interface processes must be configured to support UKDM and non-UKDM transactions. UK Domestic Maestro transactions will need to be routed to a Banknet Interface process that strictly supports UK Domestic Maestro transactions. The Banknet UK Domestic Maestro Technical Alignment is a phased approach to eliminate these functional differences.

The Processing Mode field on Screen 3 of the ICF(E) record is checked by the Banknet Interface to determine the UKDM processing to be performed. A Banknet Interface process configured to support non-aligned UKDM transactions is identified by a value of 'D' (domestic). A Banknet Interface process configured to support the alignment of UKDM functions with an effective date of March 1, 2011, but not the alignment of functions with an effective date of May 13, 2011 is identified by a value of 'P' (partial alignment). From May 13, 2011, all UKDM functions must be fully aligned. Because the processing for UKDM and non-UKDM transactions will be identical a value other than 'D' or 'P' indicates a standard Banknet Interface. It will then no longer be necessary for customers to configure separate Banknet Interface processes for UKDM and non-UKDM transactions.

HEALTHCARE TRANSACTION PROCESSING

MasterCard supports the issuance of MasterCard Assigned IDs for IIAS certified merchants. The Auto-Substantiation Data token carries the IIAS indicator and the Master Assigned ID. For Visa healthcare transactions routed through the Banknet interface, the token could be carrying a Visa Merchant Verification Value.

The SPDH device handler is being updated to provide the data in this token. The interface will use the data in the token to format DE 48 SE 61 Subfield 3—Real-time Substantiation Indicator and DE 48 SE 32 MasterCard Assign ID. If the Auto-Substantiation Data token does not exist but the amount types in the Healthcare/Transit token indicate real-time substantiation, the interface will attempt to find an STF record that matches the retailer id. If a record is found and the card service is Visa the alternate switch merchant ID will be used to populate DE48 SE32, otherwise the switch merchant ID will be used to populate DE 48 SE 32. The IIAS indicator will be set to “1”. For Visa transactions, when the response is received, DE 48 SE 96 will be logged to the switch token.

When the interface receives a request with DE 48 SE 61 Subfield 3—Real-time Substantiation Indicator equal to a “1” or “4”, the Auto-Substantiation Data token will be added using DE 48 SE 61 Subfield 3 and DE48 SE32. DE 48 SE 32 is always added to the switch token when it is provided.

POS ADJUSTMENT PROCESSING

Banknet does not support adjustment processing codes. Therefore, upon receipt of an internal BASE24-pos adjustment transaction, the Banknet Interface formats and send an external 0400 Acquirer Reversal Request message outbound to the network.

When the interface receives a PSTM adjustment transaction, the interface formats and sends an external 0400 reversal message to the interchange. In order for the interface to correctly format the reversal, it retrieves the original request or advice from the ILF. This ensures that the data in the request/advice and reversal are consistent. The ILF is searched for the original request/advice as described below:

- ☐ The ILF alternate key (PRIKEY) is formatted in the same way as for all other acquired transactions, in an attempt to locate the original request/advice.
- ☐ If the ILF record for the original request/advice cannot be found, it is possible a PSTM 0200/0220 adjustment has a different transaction time and sequence number than the original request. Therefore, the transaction date/time and sequence number within the original data fields of the PSTM are used in the PRIKEY to attempt to locate a matching ILF record.
- ☐ If the ILF record for the original request still cannot be found (possibly because the original data elements in the PSTM have not been passed), then a new read will be attempted with a reduced PRIKEY which does not include transaction date/time and sequence number. If a record is found, then additional information from the current PSTM will be checked (Transaction Amount, Retailer ID, Terminal ID, Approval Code) to ensure that the correct record has been located.

Internal adjustments containing a non-zero completed amount (i.e. PSTM.TRAN.AMT-2 contains a non-zero value) are supported as external partial reversal requests. The value in PSTM.TRAN.AMT-2 is used to format DE 95 (Replacement Amounts) in the external reversal request message.

If the ILF record for the original request is not found in these searches, then an 0400 reversal request is not sent to Banknet. Instead, an adjustment request is approved “blind”, and an adjustment advice is logged to the ILF.

PRE-AUTHORIZATION PROCESSING

Banknet allows acquirers to request preauthorization on transactions for which the amount is not yet determined. Once the issuer approves the preauthorization request and after the transaction has taken place and the final amount is determined, the acquirer can send an authorization advice message to the issuer upon transaction completion.

At present the service is available only for Petrol Transactions (DE 18 = "5542").

Acquirer Processing

Upon receipt of an internal 0200 Petrol (POS Retailer SIC Code contains value "5542") preauthorization transaction, an external 0100 Authorization Request message is formatted with:

- ☐ subfield 7 of DE 61 (POS Data) containing the value "4" (preauthorization)
- ☐ DE 18 containing the value 5542
- ☐ subfield 10 of DE 61 containing value 1 or 2
- ☐ subfield 1 of DE 22 containing value 02, 05, 07, 80, 90 or 91

Upon receipt of an external 0110 Petrol (POS Retailer SIC Code contains value "5542") Authorization Response message, the ILF record is written with an alternate key (PRIKEY) formatted as follows:

- ☐ DE 2 (PAN)
- ☐ DE 32 (Acquiring Institution Id Code)
- ☐ PSTM.PRE-AUTH-SEQ-NUM
- ☐ PSTM.TERM-FIID
- ☐ PSTM.RETL-SIC-CDE
- ☐ PSTM.TRAN.TRAN-CDE.TC

Upon receipt on an internal 0220 preauthorization completion transaction, an external 0120 Authorization Advice message is formatted with the following data from the original request:

- ☐ DE 2 (PAN)
- ☐ DE 7 (Transmission Date and Time)
- ☐ DE 32 (Acquiring Institution ID Code)
- ☐ DE 42 (Card Acceptor ID Code)

The 0120 Authorization Advice message contains the final transaction amount in DE 4. DE 60 (Advice Reason Code), subfield 1 (Advice Reason Code) will contain new value 191 (Acquirer Processing System [APS] Completed Authorization Transaction).

The interface retrieves the original request from the ILF, to ensure that the data in the request and advice are consistent. The ILF is searched for the original request with a PRIKEY which does not include transaction date/time, as the original transaction date/time value will not be present in the PSTM. The sequence number in the PRIKEY is set to PSTM.PRE-AUTH-SEQ-NUM, which will contain either the sequence number or the approval code of the original preauthorization request. The transaction date/time field is replaced by the PSTM fields TERM-FIID and RETL-SIC-CDE, as MasterCard has specified that a preauthorization completion must be initiated from the same merchant as the original preauthorization request. The last 2 digits of the transaction date/time field are replaced by the value "11", which is the internal transaction code for a preauthorization request.

If a record is found, then additional information from the current PSTM is checked (Retailer ID, Currency Code) to ensure that the correct record has been located. If a record is not found, then the preauthorization completion message is processed like any other offline-approved advice message.

A new record is written to the ILF for the preauthorization completion to allow information about the preauthorization request to be retained.

Note that the usual PRIKEY format is used when the preauthorization request message is added to the suspense queue. If no response to a preauthorization request is received from Banknet, then the original ILF record is also written with the usual PRIKEY format. This means that the Banknet Interface cannot retrieve the preauthorization request/response record from the ILF, so a subsequent preauthorization completion message is processed like any other offline-approved advice message.

Issuer Processing

Automated Fuel Dispenser (AFD) CAT Level 2 Petrol transactions are identified differently from Maestro CAT Level 1 Petrol transactions.

Upon receipt of an external 0100 Authorization Request message, the following data is used to identify a Petrol transaction:

- ☐ Maestro Pre-Authorization Petrol transactions
 - o DE 18 containing the value 5542
 - o DE-63, Network Data, subfield 1, Financial Network Code has been defined by the issuer as a Maestro network id using LCONF parameters MAESTRO-NETWORK-ID01 through MAESTRO-NETWORK-ID50.
 - o subfield 10 of DE 61 (CAT Level) containing value 1
 - o subfield 7 of DE 61 (POS Data) containing the value “4” (preauthorization)
- ☐ AFD CAT Level 2 Petrol transactions
 - o DE 18 containing the value 5542
 - o DE-63, Network Data, subfield 1, Financial Network Code has **not** been defined by the issuer as a Maestro network id using LCONF parameters MAESTRO-NETWORK-ID01 through MAESTRO-NETWORK-ID50.
 - o subfield 10 of DE 61 (CAT Level) containing value 2

An internal 0200 PSTM Request message is formatted with the pre-authorization sequence number is set to the last 6 bytes of DE 32 concatenated with the last 6 bytes of DE 7. The pre-authorization hold time is taken from the values configured in the PRE-AUTH HOLD INCREMENT and PRE-AUTH HOLD TIME on ICFE/ICF Screen 11.

Upon receipt of an external 0400 Acquirer Reversal Request or 0420 Reversal Advice message, Petrol transaction are identified in the same manner as the 0100 Request message.

DE 18 containing the pre-authorization sequence number is set to the last 6 bytes of DE 32 concatenated with the last 6 bytes of DE 7. Logic in the POS Router/Authorization process attempts to remove the pre-authorization “hold” applied by the original pre-authorization request.

Upon receipt of an external 0120 Authorization Advice message, Petrol transaction are identified in the same manner as the 0100 Request message with the additional requirement that subfield 1 of DE 60 contains value 191.

An internal 0220 PSTM Request message is formatted with a BASE24-pos transaction code of '12' (pre-authorization completion). The pre-authorization sequence number is set to the last 6 bytes of DE 32 concatenated with the last 6 bytes of DE 7. Logic in the POS Router/Authorization process attempts to find the pre-authorization "hold" applied by the original pre-authorization request.

INTERCHANGE COMPLIANCE FOR AFD COMPLETION ADVICES

MasterCard allows acquirers of Automated Fuel Dispenser (AFD) transactions to notify issuers of the final amount of the transaction by sending an Authorization Advice/0120 message. Banknet responds to this AFD completion advice message with an Authorization Advice Response/0130 message. Acquirers globally must include data from the Authorization Advice Response/0130 message in DE 63 subfield 2 (Trace Id) in the First Presentment/1240 (clearing) message.

Authorization requests and completion advices from Automated Fuel Dispensers (AFDs), are processed by BASE24-pos as pre-authorization requests and pre-authorization completions respectively. Within BASE24-pos, the information required to construct the Trace Id (sent in DE 63 subfield 2 in the First Presentment/1240 message) is carried in the Interchange Compliance token (Id = '20'). The Interchange Compliance token also specifies the type of message to which the Trace Id relates. For a pre-authorization request transaction, the Banknet Interface sets the first character of the VALID-CDE field to '1', to indicate that the data within the token has been taken from the Authorization Request Response/0110 message. For a pre-authorization completion transaction completion, the Banknet Interface sets the first character of the VALID-CDE field to '2', to indicate that the data within the token has been taken from the Authorization Advice Response/0130 message. For all other authorization messages, the Banknet Interface sets the VALID-CDE field to spaces.

Typically, acquirers (or acquirer processors) obtain the data from the Interchange Compliance token using one of the following methods:

- ☐ Extract the Banknet Interface's Interchange Log File (ILF), and retrieve the Interchange Compliance token from the data in each authorization record;
- ☐ Extract the BASE24 Logical Network's POS Transaction Log File (PTLF), and retrieve the Interchange Compliance token from the data in each authorization record;
- ☐ Return the Interchange Compliance token from each authorization response message to a front-end acquiring system, so that the front-end system can retrieve the data.

For each of these options, the Interchange Compliance token in an AFD transaction (created using data in the Authorization Advice Response/0130 message) must be available to acquirer systems for creating MasterCard clearing files.

Logging to the ILF

When processing an Authorization Advice Response/0130 message, the Banknet Interface updates the ILF record added for the associated Authorization Advice/0120 message with information from the advice response message. This includes the contents of the Interchange Compliance token, so the required information is available to acquirer systems that create MasterCard clearing files from ILF Extract data. Note that the Trace Id (sent in DE 63 subfield 2 in the First Presentment/1240 message) is also present in SEM.USER-FLD within the ILF record.

Logging to the PTLF

When processing an Authorization Advice Response/0130 message, the Banknet Interface must send an internal message to a POS Router/Authorization process to ensure that the Interchange Compliance token (containing data from the advice response message) is included in a PTLF record. The Banknet Interface can be configured (using the LCONF parameter SW-BNET-LOG-ONLY-COMPL-ADVICE) to send a '9922' log-only message to a POS Router/Authorization process. The log-only message includes the updated Interchange Compliance token, and is written to the PTLF by the POS Router/Authorization process. Acquirer systems that create MasterCard clearing files from PTLF Extract data must process this record (either instead of or in addition to the PTLF record for the original advice message) in order to retrieve the required information.

Returning a Message to a Front-End Acquiring System

BASE24-pos supports pre-authorization completion transactions using both request messages and advice messages. To return data in the Interchange Compliance token to a front-end acquiring system, pre-authorization completions for AFD transactions must be sent to the Banknet Interface as request messages.

If the pre-authorization completion request originates from an AFD and the corresponding pre-authorization request message is found on the ILF, then the Banknet Interface maps the internal request message to an external Authorization Advice/0120 (completion advice) message. The Banknet Interface then maps the external Authorization Advice Response/0130 message to an internal response message. The internal response message includes the Interchange Compliance token, containing data from the advice response message.

The Interchange Compliance token is written to the PTLF with other data from the internal response message, and is also returned to the acquiring endpoint (e.g. a POS Device Handler or an acquiring Host Interface). Therefore, the data in the Interchange Compliance token can be returned to a front-end acquiring system, so that the front-end system can create MasterCard clearing files.

If the original pre-authorization request message cannot be found in the ILF, e.g. because the request message was routed to an alternate destination, then the interface attempts to route the pre-authorization completion request message to an alternate destination. A pre-authorization completion request message where all alternate routing destinations have been exhausted is approved 'blind', and is simply logged to the ILF. The Interchange Compliance token is not present in this transaction.

If the AFD completion advice message times out or cannot be sent to Banknet, then the pre-authorization completion request message is approved 'blind', and a response sent immediately to the POS Router/Authorization process. The Interchange Compliance token in this response does not contain data from the Authorization Advice Response/0130 message, but instead contains the equivalent data from the original Authorization Request Response/0110 message.

FUNDING TRANSACTION AT AN ATM

MasterCard allows cardholders to submit bill payment requests at an ATM when the service is supported. The authorization received by the issuer is a POS purchase transaction with DE 18 containing the value 6539 (Funding Transaction [Excluding *MoneySend*]).

An internal 0200 PSTM Request message is formatted with the TXN-SUBTYP field in the Transaction Subtype token (Id = 'BM') set to 'B001' (Payment to Third Party).

Section 7

CUTOVER AND REPORTING

INTRODUCTION

This section describes the Banknet Interface process's cutover processing and report generation.

CUTOVER PROCESSING

Within the BASE24 system, interchange cutover processing is a transaction management and reporting function that produces reports for each business day. These reports are provided to aid in reconciling the BASE24 network with Banknet.

The business day for a particular transaction is defined by the date (MMDD) sent to BASE24 by Banknet in DE 15 (Date, Settlement) of authorization messages. Banknet initiates cutover seven days a week at 08:00 CST (8 a.m.).

When Banknet changes the date in DE 15, the SWITCH POSTING DATE on ICF/ICFE screen 2 is updated to the next business date. A new ILF is created (if required) and opened, the old ILF is closed, and the LCONF parameter SW-ILF-FILE-RETENTION is checked to determine whether any old ILFs should be deleted.

The optional LCONF parameter SW-BNET-CREATE-ILF can specify the time at which the interface process should create the next business day's ILF. Typically, a new ILF is created when a new settlement date is provided by Banknet in DE 15. For customers that run a "backup" BASE24 system, a new ILF may only be created on one system. Such customers may configure the ILF creation to occur BEFORE switch cutover so the ILFs are "ready and waiting" when Banknet increments the switch posting date. For customers that receive little traffic it is possible for the ILF not to be created at all. Such customers may configure the ILF creation to occur AFTER switch cutover so the interface can always be cutover, even when no messages are received from Banknet.

REPORTING

BASE24's interchange interface processes each produce two reports of interchange transaction activity: the SETL01 Clearings Statement Report and the STAT09 Transaction Detail Report. These reports can be used in conjunction with BASE24's other reports to reconcile the differences between the BASE24 processing day and the interchange processing day.

A brief description is provided below for both the SETL01 and STAT09 reports. For additional information on these reports, refer to the ***BASE24-pos Settlement and Reporting Manual*** and/or the ***BASE24-atm Settlement and Reporting Manual***.

SETL01 Clearings Statement Report

The SETL01 Clearings Statement Report reflects the debit, credit, and net position of the interchange to BASE24. This report contains the totals of all approved transactions logged to the Interchange Log File (ILF).

The first section of the report provides transaction totals for the BASE24 processing day. The second section of the report provides totals for transactions that occurred during the BASE24 processing day, but not during the interchange's processing day. The third section of the report provides totals for transactions that occurred during the interchange's processing day but not during the BASE24 processing day. Totals from each of these sections are used to compute adjusted BASE24 processing day totals that can be reconciled with the interchange's processing day totals.

STAT09 Transaction Detail Report

The STAT09 Transaction Detail Report provides a complete listing of all interchange activity processed through BASE24. The report is based on the transactions logged to the current day's ILF.

This report contains one detail item for each transaction processed by the Interchange Interface process during its processing day. The information provided includes the cardholder number, transaction type, transaction time, posting date, and transaction amount.

To comply with the requirements of the PCI data security standards, some or all of the cardholder number in the report may be displayed masked with asterisks (*). The setting of the DATA MASK FLAG on screen 2 of the ICF/ICFE record determines whether the data in this field is masked. If masking is enabled for the interface process, then the values of the other three report data masking parameters on ICF/ICFE screen 2 determine which digits are masked.

Configuration

The support of Interchange reports for the Banknet Interface process is not mandatory. If it is required to produce the standard BASE24 interchange reports for the Banknet Interface process, then assigns and parameters must be defined in the Logical Network Configuration File (LCONF).

The LCONF assigns required to specify the settlement report production are:

- ☐ RPT
- ☐ RPT-ERROR-FILE
- ☐ SW-RPT-OBEY-FILE
- ☐ SW-RPT01-PROG-NAME
- ☐ SW-RPT09-PROG-NAME

The LCONF parameter required for settlement report production is:

- ☐ SW-RPT-STARTUP

The LCONF parameters also used in settlement report production are:

- ☐ ACCT-NUM-TYP
- ☐ DATE-DISPLAY-TYPE
- ☐ SPAN-FLAG

For a detailed explanation of the above assigns and parameters, refer to Section 3.

Starting Reports

The interface process is responsible for starting the interchange reports. It does so by creating an obey file to start the report programs and then running the obey file at a specified time.

The interface process gets the information it needs for creating and running the obey file from the LCONF. If the report assigns and params are not included in the LCONF, the interface reports will not be run. These assigns and params are documented in Section 3.

The interface process sets a reports timer to expire based on the SW-RPT-STARTUP param in the LCONF. When the timer expires, the interface process creates and runs an obey file containing the commands necessary to run the report programs from Tandem's TACL.

Under normal circumstances, the report programs start and stop without operator intervention. However, if the interface process does not start the programs, they can be started manually from a TACL prompt by running the report obey file created by the Interface process. If the report programs abend or fail, they can be restarted in the same way.

Section 8

BASE24 CRT OPERATIONS

INTRODUCTION

BASE24 supports several Banknet-specific operations that are performed from BASE24 CRT screens. There are two sets of BASE24 screens used to perform Banknet-specific operations:

- ☐ Banknet Administrative (BNADM) screens - the BNADM screens are used to create and send administrative messages to Banknet.
- ☐ Banknet Request Log File (BNRLF) screens - the BNRLF screens are used to create and send file request messages to Banknet and also to peruse request log messages stored in the Banknet Request Log File (BNRLF).

This section describes how to use the BNADM and BNRLF screens. The information provided is intended for persons familiar with BASE24 CRT access and other BASE24 operations (e.g., files maintenance, network control, device control). Additional information on BASE24 CRT access is available in the ***BASE24 User Security Manual***.

The Banknet Institution Identification File (BNIIF) screen is a file maintenance screen for Banknet and is described in Section 3.

ACCESSING THE BANKNET OPERATION SCREENS

The Banknet Menu screen is accessed from screen 12 of the Interchange Configuration File (ICF/ICFE). From this screen, provided an operator is assigned the necessary security clearance for the ICF/ICFE, the following procedures can be followed to access the BNADM or the BNRLF screens. The same procedure can be used to access the BNIIF or the SNF screens (see Section 3). Please refer to the **BASE24 User Security Manual** for an explanation of the BASE24 CRT Access system and a complete set of procedures for accessing it.

1. Complete a successful logon to BASE24 via the BASE24 Logon screen by completing the fields displayed on the screen and pressing F1. The Virtual Menu screen is then displayed.
2. Select the SWITCH FUNCTIONS option from the Virtual Menu screen and press F1.
3. Select ICF or ICFE from the Switch Functions Menu screen and press F1. Interchange Configuration File (ICF) screen 1 or Enhanced Interchange Configuration File (ICFE) screen 1 is then displayed. Access to the ICF/ICFE maintenance screens can also be gained by using the FILE DESTINATION field located at the bottom of all BASE24 screens.
4. Enter Banknet in the SWITCH TYPE field on ICF/ICFE screen 1 or read an existing record that has Banknet as its SWITCH TYPE.
5. Go to ICF/ICFE screen 12 by pressing F11. Access to ICF/ICFE screen 12 can also be gained by entering 12 in the NEW PAGE field located at the bottom of the BASE24 screen and pressing F7.
6. Press F8 to access the Banknet Menu.
7. Choose the desired screen by placing the cursor to the left of the entry and press F1.

Banknet Menu Screen

The Banknet Menu screen is shown below.

```
BASE24-BASE  ICFE FILE MAINTENANCE  LNET      YY/MM/DD  HH:MM  01 OF 01
INTERCHANGE FIID: _____        PROCESS: _____
```

- _ BANKNET INSTITUTION IDENTIFICATION FILE (BNIIF)
- _ BANKNET ADMINISTRATIVE REQUEST (BNADM)
- _ BANKNET FILE REQUEST LOG FILE (BNRLF)
- _ STATION NAME FILE (SNF)

```
RECORD LAST CHANGED: 09/07/08 10:10 BY USER: 0255 , 00000255 CHANGE
***** BASE24 *****
NEW PAGE:          FILE DESTINATION:          NEW LOGICAL NETWORK ID:
                  F12-HELP
```

Using The Server Suffix

A field, named SERVER SUFFIX, was added to each screen to allow the entry of a "suffix." The suffix entered by the operator is used by the system to determine the interface process that will actually send the transaction to the interface. It is through a series of linkages within the system that the suffixes entered on the screens actually become tied to a particular interface process. The SERVER SUFFIX field allows entries of 0-9 and blank. However, the suffixes used in the system are configuration-specific and controlled by the personnel configuring the system. It is very important that an operator work with other system personnel to identify which suffixes can and should be entered. In a system not using this feature, operators can allow the suffix field to default to blank. However, again, the values are configuration-specific.

BANKNET ADMINISTRATIVE MESSAGE (BNADM) SCREEN

The BNADM screen provides a method of sending administrative messages to Banknet and in addition displays the response.

BNADM Screen

The Banknet Administrative Message screen is shown below.

BASE24-BASE	ICFE	FILE	MAINTENANCE	LNET	YY/MM/DD	HH:MM	01	OF	01
-------------	------	------	-------------	------	----------	-------	----	----	----

INTERCHANGE FIID: _____ PROCESS: _____

DESTINATION MCI ID: _____

NAME/ADDRESS: _____

TEXT: _____

SERVER SUFFIX: _

RECORD LAST CHANGED: 09/07/08 10:10 BY USER: 0255 , 00000255 CHANGE
***** BASE24 *****

NEW PAGE: FILE DESTINATION: NEW LOGICAL NETWORK ID:
F12-HELP

DESTINATION MCI ID

MCI identification number for this institution.

FIELD LENGTH: 6 numeric characters

This field should be right justified and zero filled to the left.

REQUIRED FIELD: Yes

DEFAULT VALUE: None

NAME/ADDRESS

The name and address of the financial institution.

FIELD LENGTH: 50 alphanumeric characters

REQUIRED FIELD: No

DEFAULT VALUE: None

TEXT

The message to be sent to the institution.

FIELD LENGTH: 594 alphanumeric characters

REQUIRED FIELD: No

DEFAULT VALUE: None

SERVER SUFFIX

The server suffix to use for sending the message. For information on how to use the server suffix, please refer to the introduction in this section. Valid values are "1" through "9" or " " depending on which server is to be used.

FIELD LENGTH: 1 alphanumeric character

REQUIRED FIELD: No

DEFAULT VALUE: None

Function Keys

The BNADM screens require the use of a special set of function key assignments. These assignments are described below for Tandem 6520/6530 terminals.

- F1 SEND REQUEST - Sends the displayed information on the BNADM screen to the Banknet Interface process. A data check is performed prior to the message being sent. If the data is not valid, the field in error will be highlighted and an error message will be displayed at the bottom of the screen.
- F7 GO TO NEW PAGE - Retrieves the next page of information for this message.
- F8 CLEAR SCREEN - Clears all field data on the displayed BNADM screen and replaces it with default values.
- F9 NEXT SCREEN - Displays the NEXT message types default screen.
- F10 PRINT SCREEN - Prints the currently displayed screen to the designated spooler location (BASE24 Logon screen default value is \$S.#HOLD). When the F10 function key is pressed, a message including the spooler location will be displayed at the bottom of the screen.
- F11 PREVIOUS SCREEN - Displays the PREVIOUS message types default screen.
- F12 DISPLAY HELP SCREEN - Displays the help screen for the BNADM screens.
- F16 EXIT OR GO TO FILE - Displays the Banknet Administrative Message Menu. The F16 function key will also display a requested file providing a valid file acronym has been entered in the "FILE DESTINATION" field. If an invalid acronym is entered, an error message will be displayed at the bottom of the screen.
- SF16 LOGOFF - Displays the Logon screen. The shifted F16 function key will also log the user off BASE24 from the Logon screen.

BANKNET REQUEST LOG FILE (BNRLF) SCREENS

The Banknet Request Log File (BNRLF) contains one record for every Banknet external file request response received.

Requests are created by entering data on the appropriate file request screen. The file request screens are accessed via the BNRLF menu screen. The desired screen is selected by placing the cursor to the left of the entry and pressing the F1 key.

Responses are logged to the BNRLF where they can be perused via the Banknet Request Log File perusal screens. The perusal screen can be accessed via the F9 key from any of the file request screens. It is intended that the records be deleted upon verification (by using the F4 key).

BNRLF Menu Screen

BASE24-BASE	ICFE FILE MAINTENANCE	LNET	YY/MM/DD	HH:MM	01 OF 01
INTERCHANGE FIID: _____		PROCESS: _____			

- _ MCC102 ACCOUNTS FILE
- _ MCC105 RECURRING PAYMENT CANCELLATION SERVICE FILE
- _ MCC103 ACCOUNTS MANAGEMENT FILE
- _ MCC104 ACCOUNTS MANAGEMENT FILE (AMS REGION, COUNTRY AND SUBCOUNTRY)
- _ MCC106 PAYPASS MAPPING FILE
- _ MCC107 MASTERCARD ENHANCED VALUE PLATFORM ACCOUNT FILE
- _ MCC108 MASTERCARD PRODUCT GRADUATION ACCOUNT FILE
- _ MCC109 PAYPASS APPLICATION TRANSACTION COUNTER (ATC) FILE

RECORD LAST CHANGED: 09/07/08 10:10 BY USER: 0255 , 00000255 CHANGE
***** BASE24 *****
NEW PAGE: FILE DESTINATION: NEW LOGICAL NETWORK ID:
F12-HELP

MCC102 Accounts File Screen

The Banknet MCC102 Accounts File screen (shown below) is used for online update, add and delete requests for the MasterCard Accounts File.

BASE24-BASE	ICFE FILE MAINTENANCE	LNET	YY/MM/DD	HH:MM	01 OF 08
INTERCHANGE FIID: ____		PROCESS: ____			

ACCOUNT NUMBER: ____	SECURITY CODE: ____
ENTRY REASON: _	PIN LENGTH: ____
DATE LAST ACTIVITY: ____ (MMDDYY)	CARD SEQUENCE NUMBER: ____
TIME LAST ACTIVITY: ____ (HHMM)	PURGE DATE: ____ (YYYYMMDD)
VIP TRAN AMOUNT: ____	CARD EXPIRATION DATE: ____ (YYYYMM)
VIP TRAN CURRENCY CODE: ____	
SERVER SUFFIX: _	

RECORD LAST CHANGED: 09/07/08 10:10 BY USER: 0255 , 00000255 CHANGE
***** BASE24 *****
NEW PAGE: FILE DESTINATION: NEW LOGICAL NETWORK ID:
 F12-HELP

ACCOUNT NUMBER

The account number of the cardholder's primary account.

FIELD LENGTH: 1-19 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: None

SECURITY CODE

The Banknet-assigned processor value used when BASE24 is the issuer processor.

FIELD LENGTH: 5 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: None

ENTRY REASON

A code for the entry reason. The codes are as follows:

P = Pick up card	L = Lost	S = Stolen
X = Counterfeit	Other	F = Fraud
V = VIP Status	G = Gold	C = Credit
U = Unauthorized use		

FIELD LENGTH: 1 alphanumeric character

REQUIRED FIELD: No

DEFAULT VALUE: None

PIN LENGTH

The length of the PIN to be encrypted.

FIELD LENGTH: 1-2 numeric characters

REQUIRED FIELD: Yes

DEFAULT VALUE: None

DATE LAST ACTIVITY

The date of the interface's last transaction activity. This field is only used on an inquiry and will be ignored on add, change and delete.

FIELD LENGTH: 6 numeric characters

REQUIRED FIELD: No

DEFAULT VALUE: None

CARD SEQUENCE NUMBER

The card sequence number of the listed card.

FIELD LENGTH: 3 numeric characters

REQUIRED FIELD: No

DEFAULT VALUE: None

TIME LAST ACTIVITY

The time of the interface's last transaction activity. This field is only used on an inquiry and will be ignored on add, change and delete.

FIELD LENGTH: 4 numeric characters

REQUIRED FIELD: No

DEFAULT VALUE: None

PURGE DATE

The purge date for account listings submitted to the Account File in YYYYMMDD format.

FIELD LENGTH: 8 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

VIP TRAN AMOUNT

The amount of a VIP transaction. This field will be zero except for entry reason 'V'.

FIELD LENGTH: 12 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

CARD EXPIRATION DATE

The expiration date of the listed card in YYYYMM format.

FIELD LENGTH: 6 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

VIP TRAN CURRENCY CODE

The VIP currency code. This field will be spaces except for entry reason 'V'.

FIELD LENGTH: 3 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

SERVER SUFFIX

The server suffix to use for sending the message. For information on how to use the server suffix, please refer to the introduction in this section. Valid values are "1" - "9" or " " depending on which server is to be used.

FIELD LENGTH: 1 alphanumeric character
REQUIRED FIELD: No
DEFAULT VALUE: None

MCC105 Recurring Payment Cancellation Service File Screen

The Banknet MCC105 Recurring Payment Cancellation Service File screen (shown below) is used for online add, delete and update requests for the Recurring Payment Cancellation Service File.

BASE24-BASE	ICFE FILE MAINTENANCE	LNET	YY/MM/DD	HH:MM	02 OF 08
INTERCHANGE FIID: _____		PROCESS: _____			

PRIMARY ACCOUNT NUMBER: _____

ACQUIRER ID: _____

CARD ACCEPTOR ID CODE: _____

SINGLE/LOW TRANSACTION AMOUNT: _____ SECURITY CODE: _____

HIGH TRANSACTION AMOUNT: _____ PURGE DATE: _____ (YYYYMMDD)

RECORD LAST CHANGED: 09/07/08 10:10 BY USER: 0255 , 00000255 CHANGE
***** BASE24 *****
NEW PAGE: FILE DESTINATION: NEW LOGICAL NETWORK ID:
F12-HELP

ACCOUNT NUMBER

The account number of the cardholder's primary account.

FIELD LENGTH: 1-19 alphanumeric characters

REQUIRED FIELD: Yes

DEFAULT VALUE: None

ACQUIRER ID

Valid acquirer ICA.

FIELD LENGTH: 6 numeric characters

REQUIRED FIELD: Yes

DEFAULT VALUE: None

CARD ACCEPTOR ID CODE

Must be unique per acquirer ICA.

FIELD LENGTH: 15 alphanumeric characters

REQUIRED FIELD: Yes

DEFAULT VALUE: None

SINGLE/LOW TRANSACTION AMOUNT

Indicates a single transaction amount or the start of a transaction amount range. Transaction amount must be stated in the transaction currency and not the cardholder billing currency.

FIELD LENGTH: 12 numeric characters

REQUIRED FIELD: No

DEFAULT VALUE: Zeroes

SECURITY CODE

The Banknet-assigned processor value used when BASE24 is the issuer processor.

FIELD LENGTH: 5 numeric characters

REQUIRED FIELD: No

DEFAULT VALUE: None

HIGH TRANSACTION AMOUNT

Indicates the end of a transaction amount range. Do not use this field if specifying a single amount. If used, must be greater than Single/Low Transaction Amount. Transaction amount must be stated in the transaction currency and not the cardholder billing currency.

FIELD LENGTH: 12 numeric characters

REQUIRED FIELD: No

DEFAULT VALUE: Zeroes

PURGE DATE

The issuer-defined purge date must be no more than 15 months beyond the account list date in YYYYMMDD format.

FIELD LENGTH: 8 numeric characters

REQUIRED FIELD: Yes

DEFAULT VALUE: None

MCC103 Account Management System File Screen

The Banknet MCC103 Account Management System File Screen (shown below) is used for online add, read (inquiry) and delete requests for the Warning Bulletin File.

BASE24-BASE	ICFE FILE MAINTENANCE	LNET	YY/MM/DD	HH:MM	03 OF 08
INTERCHANGE FIID: _____		PROCESS: _____			

ACCOUNT NUMBER: _____	CUST IDENTIFIER: _____
ENTRY REASON: _____	PRODUCT TYPE: _____
SECURITY CODE: _____	

REGION	PURGE DATE (YYMMDD)
☐ UNITED STATES	_____
☐ CANADA	_____
☐ LATIN AMERICA/CARIBBEAN	_____
☐ ASIA/PACIFIC	_____
☐ EUROPE	_____
☐ SOUTH ASIA/MIDDLE EAST/AFRICA	_____

SERVER SUFFIX: _____

RECORD LAST CHANGED: 09/07/08 10:10 BY USER: 0255 , 00000255 CHANGE

***** BASE24 *****

NEW PAGE: FILE DESTINATION: NEW LOGICAL NETWORK ID:

 F12-HELP

ACCOUNT NUMBER

The account number of the cardholder's primary account.

FIELD LENGTH: 1-19 alphanumeric characters

REQUIRED FIELD: Yes

DEFAULT VALUE: None

CUST IDENTIFIER

The customer identification number.

FIELD LENGTH: 1-6 alphanumeric characters

REQUIRED FIELD: No

DEFAULT VALUE: None

ENTRY REASON

The Banknet information reason code for placing the account in the Warning Bulletin File. Valid values are:

C =Credit
F = Fraud
X = Counterfeit
O = Other

FIELD LENGTH: 1 alphanumeric character
REQUIRED FIELD: No
DEFAULT VALUE: None

PRODUCT TYPE

The Banknet product type for the card number. Valid values are:

MCC = MasterCard Card
MCB = MasterCard BusinessCard Card
MCD = MasterCard Debit
MCG = Gold MasterCard Card
MNS = Non-standard Card
OTH = Other

FIELD LENGTH: 3 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

SECURITY CODE

The Banknet-assigned processor value used when BASE24 is the issuer processor.

FIELD LENGTH: 5 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: None

REGION

A value indicating whether the specific geographic region is to be affected by this Electronic Warning Bulletin. Valid values are:

Y = Yes

N = No

FIELD LENGTH: 1 alphanumeric character

REQUIRED FIELD: No

DEFAULT VALUE: None

PURGE DATE

A value indicating the member requested purge date. The valid values for year position must be 94-25 which will be processed as 1994 through 2025. The month positions must be 1-12, and the day, 1-31.

FIELD LENGTH: 6 alphanumeric characters

REQUIRED FIELD: No

DEFAULT VALUE: None

SERVER SUFFIX

The server suffix to use for sending the message. For information on how to use the server suffix, please refer to the introduction in this section. Valid values are "1" - "9" or " " depending on which server is to be used.

FIELD LENGTH: 1 alphanumeric character

REQUIRED FIELD: No

DEFAULT VALUE: None

MCC104 Accounts Management File Screen

The Banknet MCC104 Accounts Management File screen (shown below) is used for online add, read (inquiry) and delete requests for the Accounts Management File.

BASE24-BASE	ICFE FILE MAINTENANCE	LNET	YY/MM/DD	HH:MM	04 OF 08
INTERCHANGE FIID: _____		PROCESS: _____			

ACCOUNT NUMBER: _____	CUST IDENTIFIER: _____
ENTRY REASON: _____	CARD PROGRAM: _____ SECURITY CODE: _____

REGION	PURGE DATE (YYMMDD)	COUNTRY	SUB-COUNTRY
____ UNITED STATES	_____	_____	_____
____ CANADA	_____	_____	_____
____ LATIN AMERICA/CARIBBEAN	_____	_____	_____
____ ASIA/PACIFIC	_____	_____	_____
____ EUROPE	_____	_____	_____
____ SOUTH ASIA/MIDDLE EAST/AFRICA	_____	_____	_____

RECORD LAST CHANGED: 09/07/08 10:10 BY USER: 0255 , 00000255 CHANGE
***** BASE24 *****

NEW PAGE:	FILE DESTINATION:	NEW LOGICAL NETWORK ID:
	F12-HELP	

ACCOUNT NUMBER

The account number of the cardholder's primary account.

FIELD LENGTH: 1-19 alphanumeric characters

REQUIRED FIELD: Yes

DEFAULT VALUE: None

CUST IDENTIFIER

A Master Card Assign a Customer ID.

FIELD LENGTH: 6 numeric characters

REQUIRED FIELD: Yes

DEFAULT VALUE: None

ENTRY REASON

Reason for listing this card.

FIELD LENGTH: 1 alphanumeric characters

REQUIRED FIELD: Yes

DEFAULT VALUE: None

CARD PROGRAM

Type of card bearing the account holder.

FIELD LENGTH: 3 alphanumeric characters

REQUIRED FIELD: Yes

DEFAULT VALUE: None

SECURITY CODE

The Banknet-assigned processor value used when BASE24 is the issuer processor.

FIELD LENGTH: 5 numeric characters

REQUIRED FIELD: No

DEFAULT VALUE: None

UNITED STATES

The Region associated with account listing.

FIELD LENGTH: 1 alphanumeric character

REQUIRED FIELD: Yes

DEFAULT VALUE: None

PURGE DATE1

Purge Date (Format YYMMDD) associated with this account number.

FIELD LENGTH: 6 numeric characters

REQUIRED FIELD: No

DEFAULT VALUE: None

COUNTRY1

It contains the country associated with this account listing. Must be a participating country within the issuer's region.

FIELD LENGTH: 3 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

SUB COUNTRY1

It contains the sub country associated with this account listing. Must be a participating sub country with in the issuer's country.

FIELD LENGTH: 3 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

CANADA

The Region associated with account listing.

FIELD LENGTH: 1 alphanumeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: None

PURGE DATE2

Purge Date (Format YYMMDD) associated with this account number.

FIELD LENGTH: 6 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

COUNTRY2

It contains the country associated with this account listing. Must be a participating country within the issuer's region.

FIELD LENGTH: 3 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

SUB COUNTRY2

It contains the sub country associated with this account listing. Must be a participating sub country with in the issuer's country.

FIELD LENGTH: 3 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

LATIN AMERICA/CARIBBEAN

The Region associated with account listing.

FIELD LENGTH: 1 alphanumeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: None

PURGE DATE3

Purge Date (Format YYMMDD) associated with this account number.

FIELD LENGTH: 6 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

COUNTRY3

It contains the country associated with this account listing. Must be a participating country within the issuer's region.

FIELD LENGTH: 3 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

SUB COUNTRY3

It contains the sub country associated with this account listing. Must be a participating sub country with in the issuer's country.

FIELD LENGTH: 3 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

ASIA/PACIFIC

The Region associated with account listing.

FIELD LENGTH: 1 alphanumeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: None

PURGE DATE4

Purge Date (Format YYMMDD) associated with this account number.

FIELD LENGTH: 6 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

COUNTRY4

It contains the country associated with this account listing. Must be a participating country within the issuer's region.

FIELD LENGTH: 3 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

SUB COUNTRY4

It contains the sub country associated with this account listing. Must be a participating sub country with in the issuer's country.

FIELD LENGTH: 3 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

EUROPE

The Region associated with account listing.

FIELD LENGTH: 1 alphanumeric character
REQUIRED FIELD: Yes
DEFAULT VALUE: None

PURGE DATE5

Purge Date (Format YYMMDD) associated with this account number.

FIELD LENGTH: 6 numeric characters

REQUIRED FIELD: No

DEFAULT VALUE: None

COUNTRY5

It contains the country associated with this account listing. Must be a participating country within the issuer's region.

FIELD LENGTH: 3 alphanumeric characters

REQUIRED FIELD: No

DEFAULT VALUE: None

SUB COUNTRY5

It contains the sub country associated with this account listing. Must be a participating sub country with in the issuer's country.

FIELD LENGTH: 3 alphanumeric characters

REQUIRED FIELD: No

DEFAULT VALUE: None

SOUTH ASIA/MIDDLE EAST/AFRICA

The Region associated with account listing.

FIELD LENGTH: 1 alphanumeric character

REQUIRED FIELD: Yes

DEFAULT VALUE: None

PURGE DATE6

Purge Date (Format YYMMDD) associated with this account number.

FIELD LENGTH: 6 numeric characters

REQUIRED FIELD: No

DEFAULT VALUE: None

COUNTRY6

It contains the country associated with this account listing. Must be a participating country within the issuer's region.

FIELD LENGTH: 3 alphanumeric characters

REQUIRED FIELD: No

DEFAULT VALUE: None

SUB COUNTRY6

It contains the sub country associated with this account listing. Must be a participating sub country with in the issuer's country.

FIELD LENGTH: 3 alphanumeric characters

REQUIRED FIELD: No

DEFAULT VALUE: None

MCC106 Paypass Mapping File Screen

The Banknet MCC106 PayPass Mapping File screen (shown below) is used for online add, update and delete requests for the PayPass Mapping File.

BASE24-BASE	ICFE FILE MAINTENANCE	LNET	YY/MM/DD	HH:MM	05 OF 08
INTERCHANGE FIID: _____		PROCESS: _____			

PAYPASS PAN: _____
PAN: _____

REPLACEMENT PAYPASS PAN: _____
REPLACEMENT PAN: _____

CARD EXPIRATION DATE: _____ (YYMM)

SECURITY CODE: _____

RECORD LAST CHANGED: 09/07/08 10:10 BY USER: 0255 , 00000255 CHANGE
***** BASE24 *****
NEW PAGE: FILE DESTINATION: NEW LOGICAL NETWORK ID:
F12-HELP

PAYPASS PAN

The account number that is assigned to the PayPass card or device and transmitted from the card or device to the PayPass terminal.

FIELD LENGTH: 1-19 alphanumeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: None

PAN

The account number that is embossed, encoded or both on the card.

FIELD LENGTH: 1-19 alphanumeric characters
REQUIRED FIELD: Yes, on adds and updates
DEFAULT VALUE: None

REPLACEMENT PAYPASS PAN

The replacement number that is assigned to the PayPass card or device and transmitted from the card or device to the PayPass terminal.

FIELD LENGTH: 1-19 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

REPLACEMENT PAN

The account number that is embossed, encoded or both on the card.

FIELD LENGTH: 1-19 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

CARD EXPIRATION DATE

The expiration date that is embossed, encoded or both on the actual card in YYMM format.

FIELD LENGTH: 4 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

SECURITY CODE

The Banknet-assigned processor value used when BASE24 is the issuer processor.

FIELD LENGTH: 5 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

MCC107 MasterCard Enhanced Value Platform Account File

The Banknet MCC107 MasterCard Enhanced Value Platform Account File screen (shown below) is used for online add, update, delete and inquiry requests for the MasterCard Enhanced Value Platform Account File.

BASE24-BASE	ICFE FILE MAINTENANCE	LNET	YY/MM/DD	HH:MM	06 OF 08
INTERCHANGE FIID: _____		PROCESS: _____			

LOW ACCOUNT NUMBER: _____

HIGH ACCOUNT NUMBER: _____

ACCOUNT CATEGORY: _ PURGE DATE: _____ (CCYYMMDD)

PROGRAM ID: _____ SECURITY CODE: _____

RECORD LAST CHANGED: 09/07/08 10:10 BY USER: 0255 , 00000255 CHANGE
***** BASE24 *****
NEW PAGE: FILE DESTINATION: NEW LOGICAL NETWORK ID:
F12-HELP

LOW ACCOUNT NUMBER

The single cardholder account or the first account in the range of accounts.

FIELD LENGTH: 1-19 alphanumeric characters

REQUIRED FIELD: Yes

DEFAULT VALUE: None

HIGH ACCOUNT NUMBER

The last account in the range of accounts.

FIELD LENGTH: 1-19 alphanumeric characters

REQUIRED FIELD: Yes

DEFAULT VALUE: None

ACCOUNT CATEGORY

The category associated with the account number or account number range. Valid values are 'B' or a space.

FIELD LENGTH: 1 alpha character
REQUIRED FIELD: No
DEFAULT VALUE: Space

PURGE DATE

The purge date (format CCYYMMDD) associated with this account number or account number range.

FIELD LENGTH: 8 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: None

PROGRAM ID

The program ID assigned to the issuer when the issuer enrolled in MasterCard Enhanced Value Platform.

FIELD LENGTH: 6 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

SECURITY CODE

The Banknet-assigned processor value used when BASE24 is the issuer processor.

FIELD LENGTH: 5 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

MCC108 MasterCard Product Graduation Account File Screen

The Banknet MCC108 MasterCard Product Graduation Account File screen (shown below) is used for online add, update, delete and inquiry requests for the MasterCard Product Graduation Account File.

BASE24-BASE ICFE FILE MAINTENANCE LNET YY/MM/DD HH:MM 07 OF 08
INTERCHANGE FIID: _____ PROCESS: _____

ACCOUNT NUMBER: _____

ACCOUNT CATEGORY: _ PRODUCT CODE: _____
PURGE DATE: _____ (CCYYMMDD) SECURITY CODE: _____
PROGRAM ID: _____ PREVIOUS PRODUCT CODE: _____
CUSTOMER SPECIFIC INDEX: _____

RECORD LAST CHANGED: 09/07/08 10:10 BY USER: 0255 , 00000255 CHANGE
***** BASE24 *****
NEW PAGE: FILE DESTINATION: NEW LOGICAL NETWORK ID:
F12-HELP

ACCOUNT NUMBER

The single cardholder account.

FIELD LENGTH: 1-19 alphanumeric characters

REQUIRED FIELD: Yes

DEFAULT VALUE: None

ACCOUNT CATEGORY

The category associated with the account number or account number range.

FIELD LENGTH: 1 alphanumeric character

REQUIRED FIELD: Yes

DEFAULT VALUE: None

PRODUCT CODE

The product code to which the issuer wants to migrate the account found in the account number field.

FIELD LENGTH: 3 alphanumeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: None

SECURITY CODE

The Banknet-assigned processor value used when BASE24 is the issuer processor.

FIELD LENGTH: 5 numeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

PURGE DATE

The purge date (format CCYYMMDD) associated with this account number or account number range.

FIELD LENGTH: 8 numeric characters
REQUIRED FIELD: Yes
DEFAULT VALUE: None

PROGRAM ID

The program ID is required if the issuer is registering the account from the MasterCard Enhanced Value Platform.

FIELD LENGTH: 6 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

PREVIOUS PRODUCT CODE

The product code from which the account is migrating.

FIELD LENGTH: 3 alphanumeric characters
REQUIRED FIELD: No
DEFAULT VALUE: None

CUSTOMER SPECIFIC INDEX

The customer specific index is the unique custom number that is assigned by MasterCard.

FIELD LENGTH: 7 numeric characters

REQUIRED FIELD: No

DEFAULT VALUE: None

MCC109 Paypass Application Transaction Counter (ATC) File

The Banknet MCC109 PayPass Application Transaction Counter (ATC) File screen (shown below) is used for online update, add and delete requests for the PayPass Application Transaction Counter (ATC) File.

BASE24-BASE	ICFE FILE MAINTENANCE	LNET	YY/MM/DD	HH:MM	08 OF 08
INTERCHANGE FIID: _____		PROCESS: _____			

PAYPASS ACCOUNT NUMBER: _____

CARD SEQUENCE NUMBER: _

ACCOUNT EXPIRATION DATE (YYMM): _____

PAYPASS ATC VALUE: _____

RECORD LAST CHANGED: 09/07/08 10:10 BY USER: 0255 , 00000255 CHANGE
***** BASE24 *****
NEW PAGE: FILE DESTINATION: NEW LOGICAL NETWORK ID:
F12-HELP

PAYPASS ACCOUNT NUMBER

The number that is assigned to the PayPass card of NCFF and transmitted from the card or NCFF to the PayPass terminal.

FIELD LENGTH: 1-19 alphanumeric characters.

REQUIRED FIELD: Yes, must be left justified with trailing spaces.

DEFAULT VALUE: None.

CARD SEQUENCE NUMBER

Application PAN Sequence Number (May be zero).

FIELD LENGTH: 1 numeric character.

REQUIRED FIELD: Yes.

DEFAULT VALUE: Zero.

ACCOUNT EXPIRATION DATE

The expiration date that is embossed, encoded, or both on the card that represents the cardholder PAN. Format is YYMM.

FIELD LENGTH: 4 numeric characters.

REQUIRED FIELD: Yes.

DEFAULT VALUE: None.

PAYPASS ATC VALUE

PayPass Application Transaction Counter Value. Value should be right-justified, zero-filled.

FIELD LENGTH: 5 numeric characters.

REQUIRED FIELD: Yes, must be right-justified, leading zeroes.

DEFAULT VALUE: None.

Function Keys

The BNRLF screens require the use of a special set of function key assignments. These assignments are described below for Tandem 6520/6530 terminals.

- F1 VALIDATE DATA - Checks the data currently displayed on the BNRLF screens for errors. If the data is error-free, a "DATA O.K." response will appear on the bottom of the screen. If the data is not valid, the field in error will be highlighted and an error message will be displayed on the bottom of the screen. The F1 function key will also display default values for fields which have not been edited.

- F2 READ RECORD - This key will send an online file inquiry to MasterCard.

- F3 ADD RECORD - This key will send an online file add to MasterCard.

- F4 SEND DELETION - Sends a message to MasterCard for deletion of a record from the Warning Bulletin File.

- F5 UPDATE RECORD - This key will send an online file update to MasterCard.

- F7 GO TO PAGE - Retrieves the page number indicated at the bottom of this screen.

- F8 CLEAR SCREEN - Clears all field data on the displayed BNRLF screen and replaces it with default values.

- F9 GO TO PERUSAL SCREEN - Displays the perusal screen.

- F10 PRINT SCREEN - Prints the currently displayed screen to the designated spooler location (BASE24 Logon screen default value is \$S.#HOLD). When the F10 function key is pressed, a message including the spooler location will be displayed at the bottom of the screen.

- F11 PAGE BACK - Displays the previous page.

- F12 DISPLAY HELP SCREEN - Displays the help screen for the BNRLF screens.

- F16 EXIT OR GO TO FILE - Displays the Banknet Administrative Message Menu. The F16 function key will also display a requested file providing a valid file acronym has been entered in the "FILE DESTINATION" field. If an invalid acronym is entered, an error message will be displayed at the bottom of the screen.

- SF16 LOGOFF - Displays the Logon screen. The shifted F16 function key will also log the user off BASE24 from the Logon screen.

Function Keys For BNRLF Perusal Screen

The BNRLF perusal screens require the use of a special set of function key assignments. These assignments are described below for Tandem 6520/6530 terminals.

- F2 READ RECORD - Reads the first record from the BNRLF.
- F4 DELETE RECORD - Deletes a record from the BNRLF. To delete a record, the record must first be read. The F4 function key must then be used to delete the record from the BNRLF.
- F6 RETRIEVE NEXT RECORD - Reads the next record from the BNRLF.
- F9 NEXT SCREEN - Displays the NEXT message types default screen.
- F10 PRINT SCREEN - Prints the currently displayed screen to the designated spooler location (BASE24 Logon screen default value is \$S.#HOLD). When the F10 function key is pressed, a message including the spooler location will be displayed at the bottom of the screen.
- F11 PAGE BACK - Displays the previous page.
- F12 DISPLAY HELP SCREEN - Displays the help screen for the BNRLF perusal screens.
- F16 EXIT OR GO TO FILE - Displays the Banknet Administrative Message Menu. The F16 function key will also display a requested file providing a valid file acronym has been entered in the "FILE DESTINATION" field. If an invalid acronym is entered, an error message will be displayed at the bottom of the screen.
- SF16 LOGOFF - Displays the Logon screen. The shifted F16 function key will also log the user off BASE24 from the Logon screen.

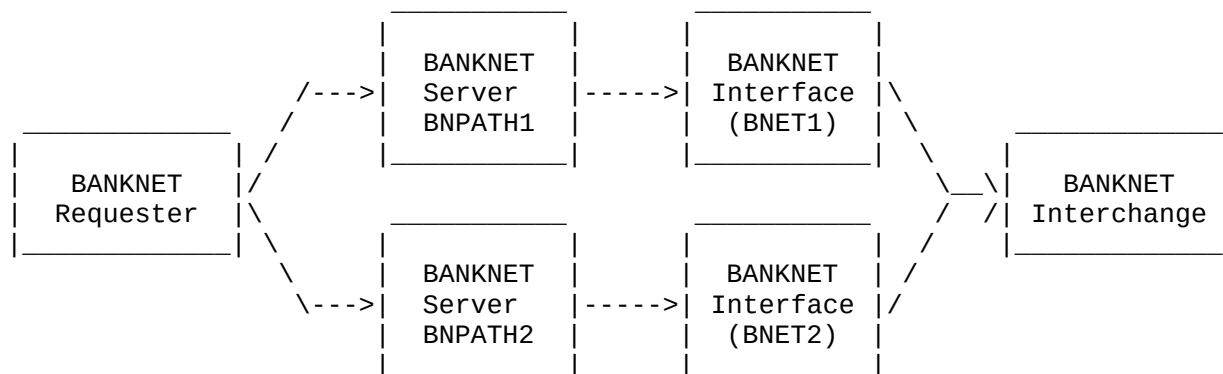
MULTIPLE COPY SUPPORT FOR ADMINISTRATIVE TRANSACTIONS

Multiple Copy Support allows file update and administrative messages to be sent from each copy of the Banknet Interface process when multiple copies of the interface are being used to communicate with MasterCard.

This is accomplished by allowing the customer to set up multiple Banknet Pathway (BNPATH) servers, each corresponding to a separate interface process. Using this new approach, the requester controlling the Banknet administrative screens can communicate with any of the BNPATH servers based on a suffix entered on the administrative screens at the time the transaction is entered. In the past, only one BNPATH server could be set up, which in turn forwarded file update and administrative transactions to the one interface process with which it was associated. Under the new approach, up to 11 BNPATH servers can be set up, each forwarding transactions to a different interface process.

This allows operators to selectively send out file update and administrative transactions to the interchange through different interfaces when necessary (e.g., different financial institutions not sharing common links to the interchange).

An illustration of using two server paths follows:



Section 9

IPF REFRESH

INTRODUCTION

The Banknet Interchange Prefix File (IPF) Refresh Process constructs a full positive list of all BINs that must be sent through Banknet for authorization. The External Interchange Prefix File (IPFX) is the disk file format in which BASE24 receives input records to the IPF. Tape processing is not supported.

The Banknet IPF Refresh module supports full refresh only (i.e. where a new IPF is created). Partial refresh (i.e. where an existing IPF is updated) is not supported.

PROCESSING OVERVIEW

MasterCard provides Acquirers with the Integrated Product Message (IPM) Member Parameter Extract (MPE) file. The IPM MPE file is used as input to MasterCard's IPM MPE Maintenance Process (Pre-Edit Utility), which runs on a PC and updates an existing database of BINs. This program creates optimized parameter files in a directory called "OneDayFile", including a file called CURRENTPARM.TXT. This file must be transferred as a structured (Type E) file to the HP NonStop (formerly Tandem) system, for the Banknet IPF Refresh process to use as input data (IPFX).

The IPFX is a structured, entry-sequenced file containing variable length records, with a maximum record length of 4072 bytes. The first record in the file is an IP0000T1 Keys Table record, and is used to identify a valid IPFX input file. Only IP0040T1 Account Range Table entries within the file will be processed as detail records. MasterCard IP0040T1 Account Range Table records provide the required BIN information.

The Banknet IPF Refresh module validates run parameters, renames the current IPF to "OLDIPF", converts the detail records from the Optimized Current Parameters File record format on disk to the BASE24 IPF record format, and writes the BASE24 IPF records to a new IPF. When the refresh is completed, the new IPF is moved into production.

CONFIGURATION

Configuration information for the Banknet IPF Refresh module must be set using standard TACL ASSIGN and TACL PARAM commands. These commands are valid for the current TACL session, and can be entered at a TACL prompt or placed in an obey file.

The assigns are:

- ☐ IPFX
- ☐ IPF
- ☐ TPLT

The params are:

- ☐ NUM-OF-PROGRAMS
- ☐ PROGRAM-NNN
- ☐ SUPPRESS-DUP-MESS
- ☐ TERMINAL-CATEGORY

When creating params, the NUM-OF-PROGRAMS param must precede all PROGRAM-NNN params, and must match the number of PROGRAM-NNN params.

For more details of assign and param settings, refer to Section 3.

RUNNING IPF REFRESH

The Banknet IPF Refresh module is located on subvolume SW60IPBN.

The RUNIPBNT file provided with the Banknet IPF Refresh module is an OBEY file containing TACL ASSIGN and TACL PARAM commands. The file also contains a command to purge an obsolete IPF, plus a TACL RUN command to execute the Banknet Refresh program (IPBNREF). The RUNIPBNT file can be edited to set the required values of the assigns and params listed above.

The RUNIPBNT obey file contains the following RUN command:

```
RUN IPEUREF / OUT outfile /
```

where *outfile* specifies the device or spooler location used to display informational messages and errors.

IPF REFRESH REPORT

The Banknet IPF Refresh module provides a report that displays informational messages and error messages.

The report layout is shown below.

```
BASE24 Banknet IPF Refresh
Copyrighted by Applied Communications Inc. Limited, 2008

TOTAL RECORDS READ FROM IPFX DISK FILE:          aaaaaaaaaa
TOTAL IP0040T1 RECORDS READ FROM IPFX DISK FILE:   bbbbbbbb
TOTAL IP0040T1 RECORDS PROCESSED FROM IPFX DISK FILE: cccccccc
TOTAL DUPLICATE RECORDS NOT WRITTEN TO IPF DISK FILE: dddddddd
TOTAL PREFIX RECORDS WRITTEN TO IPF DISK FILE:     eeeeeeee

IPF REFRESH IS COMPLETED
```

Where: aaaaaaaaaa is the number of records read;

bbbbbbbb is the number of IP0040T1 records read;

ccccccc is the number of records processed (i.e. this value excludes those records that don't meet the TERMINAL-CATEGORY and/or PROGRAM-NNN requirements);

ddddddd is the number of duplicate records encountered;

eeeeeee is the number of records actually added to the IPF.

PROCESSING CONSIDERATIONS

Acquirers should configure the IPF Refresh parameters carefully, to ensure that the resulting IPF records contain accurate data.

Terminal Categories and Acceptance Programs, the ranges of which are configurable by the Acquirer, identify a subset of the input file's records. The Acquirer can use these parameters to limit the number of records added to the IPF. For example, a BASE24-pos acquirer may not want to include BINs for ATM-only (e.g. CIRRUS) cards.

The Acquirer can also configure Card Types associated with Acceptance Programs that will be included in the specific IPF records. These provide a means of alternate routing.

Acquirers running both a BASE24-atm and a BASE24-pos system may want to create a separate IPF for ATM and POS transactions. This approach can ensure that all transactions routed to Banknet are valid for a specific card program. For example, a BASE24 acquirer may want to reject POS transactions performed on ATM-only (e.g. CIRRUS) cards.

This solution may not be appropriate for institutions using BASE24-pos to acquire debit card cash withdrawals from bank branch terminals. These transactions are equivalent to ATM cash withdrawals, and are therefore valid when performed on ATM-only cards. For these transactions, a separate entry can be configured in the PREGEN file, e.g. by specifying a unique routing group, which would reference the IPF used for ATM transactions. To access the additional entry, the same routing group value must be configured on the PTD/PTDF record for the bank branch terminal.

Appendix A

Token Documentation

The table below identifies the various tokens used by the Banknet Interface along with their token IDs. For more information on these tokens, refer to the ***BASE24 Tokens Manual***.

ID	Token Name	Product
03	BASE24-atm Release 5.0 Token	ATM
04	BASE24-pos Release 5.0 Token	POS
06	PIN Change Token	ATM
16	Alternate Merchant ID Token	POS
17	Visa Payment Service 2000 Token	POS
20	Interchange Compliance Token	POS
23	Track 1 Token	BASE
24	ATM Flag 1 Token	ATM
25	Surcharge Data Token	BASE
A5	Non-Currency Dispense Token	ATM
A6	ATM Interchange Compliance Token	ATM
B2	EMV Request Data Token	BASE
B3	EMV Discretionary Data Token	BASE
B4	EMV Status Token	BASE
B5	EMV Response Data Token	BASE
B6	EMV Script Data Token	BASE
BA	Acquirer Routing Token	BASE
BE	Original Currency Token	BASE
BM	Transaction Subtype Token	BASE
BS	Pre-Pay Switch Token	BASE
BY	Switch Common Data Token	BASE
C0	BASE24-pos Release 5.1 Token	POS
C2	Purchase Card and Fleet Card Token	POS
C4	Point of Service Data Token	POS
C6	Trans Stain XID Token	POS
C7	Cardholder Serial Number Token	POS

ID	Token Name	Product
C8	Merchant Serial Number Token	POS
CE	Authentication Data Token	POS
CF	IAVS Data Token	POS
CH	POS Data1 Token	POS
CI	POS Merchant Token	POS
CT	Transaction Specific Data Token	POS
CV	Healthcare/Transit Token	POS
CW	Healthcare/Service Token	POS
CY	Auto Substantiation Data Token	POS
F1	E-Commerce Additional Data Token	POS
S7	Person-to-Person Transaction 2 Token	POS
S8	Pan Mapping Token	POS

The table below identifies the switch-specific tokens used by the Banknet Interface along with their token IDs. Each of these tokens is described below.

ID	Token Name	Product
B0	Switch Token (Acquirer)	BASE
B1	Switch Token (Issuer)	BASE

Switch Token (B0, B1)

The fields in the Banknet Switch Token, which redefines the Switch Token (swi-tn), are described below.

<u>Position</u>	<u>Level</u>	<u>Field Name and Description</u>	<u>Data Type</u>
1-220		BNET-SWI-TKN-DATA	
1	02	LGTH The length of the token data. The length includes the FIID and the data present in the data buffer.	BINARY 16 SIGNED.
3	02	FIID The FIID for the specific switch. For Banknet, the FIID will be "BNET".	PIC X(4).
7	02	VERSION-ID Indicates the version of the switch token. At this time the version is 14.	PIC X(2).
9	02	LOCAL-TIM Bit-12 (sem.local^tim) The Local Transaction Time specified in HHMMSS.	PIC X(6).
15	02	LOCAL-DAT Bit-13 (sem.local^dat) The Local Transaction Date specified in MMDD.	PIC X(4).
19	02	ADVICE-RSN-CDE Bit-60 Sub-field 1 (sem.advice^cde.reason^cde) Indicates the general category of (or reason for) the advice message.	PIC X(3).
22	02	POS-ENTRY-MDE Bit-22 (sem.pos^entry^mde) The Banknet value for Point of Service Entry Mode.	PIC X(3).

25	02	RESP-CDE	PIC X(2).
		Bit-39 (sem.resp^cde)	
		Contains the response code associated with the transaction.	
		This token field will not contain significant data in Acquirer tokens, B0, when BASE24 is the authorizer.	
27	02	CARD-VRFY-RESULT	PIC X.
		Bit-48 Sub-element 87 (sem.addl^data.info.cvc^sub^fld^87.tag^data)	
		Contains the result of the card verification.	
28	02	VISA-XSTNG-DBT-IND	PIC X.
		Bit-48 Sub-element 85 (sem.addl^data.info.visa^xstng^dbt^ind^sub^fld^85.tag^data)	
		Indicates whether the transaction was a payment on an existing debt.	
29	02	VISA-SVC-DVLPMT-IND	PIC X.
		Bit-48 Sub-element 86 (sem.addl^data.info.visa^svc^dvlpmt^ind^sub^fld^86.tag^data)	
		Indicates whether the transaction contained the VISA Relationship Participant Indicator.	
30	02	VISA-DEF-BILL-IND	PIC X.
		Bit-48 Sub-element 78 (sem.addl^data.info.visa^def^bill^ind^sub^fld^78.tag^data)	
		Identifies transactions for which the billing for merchandise occurred after the merchandise was delivered to the customer.	
31	02	PROC-CDE	PIC X(6).
		Bit-3 (sem.proc^cde)	
		Contains the transaction processing code.	

37	02	ON-BEHALF 04 SVC 04 RSLT-1	OCCURS 10 TIMES. PIC X(2). PIC X.
Bit-48 Sub-element 71 (sem.addl^data.info. on^behalf^svc^sub^fld^71.tag^data)			
Contains information about on-behalf processing performed by Banknet.			
67	02	MCHIP-PRO-IND	PIC X(2).
69	02	MCHIP-PROC-INFO	PIC X.
Bit-48 Sub-element 74 (sem.addl^data.info. adnl^proc^info^sub^fld^74.tag^data)			
Contains information about the M/Chip processing performed by the issuer.			
70	02	TRAN-FEE	PIC X(9).
Bit-28 (sem.tran^fee)			
Contains the fee associated with the transaction.			
79	02	E-COM-SEC-LVL-IND	PIC 9(3).
Bit-48 Sub-element 42 (sem.addl^data.info. ecom-sec-sub-fld-42.lvl-of-sec)			
Contains the electronic commerce security level indicator and UCAF collection indicator.			
82	02	AVS-RESULT	PIC X.
Bit-48 Sub-element 83 (sem.addl^data.info. avs-resp.result)			
Contains the result of the address verification.			
83	02	ACCT-NUM-IND	PIC X.
Bit-48 Sub-element 33 (sem.addl^data.info. paypass^mapping^sub^fld^33.acct^num.tag^data)			
Indicates the type of PAN present in the second sub-field of the sub-element.			
84	02	ADVC-DETL-CDE	PIC 9(4).
Bit-60 Sub-field 2 (sem.advice^cde.detail^cde)			
Provides additional (specific) information as to the exact nature of the advice message.			

88	02	AUTH-AGENT-ID-CDE	PIC X(6).
		Bit-121 (sem.auth^inst.cde)	
		Identifies the specific authorizing agent that actually approved the transaction, if different to the issuer.	
94	02	PMNT-TXN-TYP-IND	PIC X(3).
		Bit-48 Sub-element 77 (sem.addl^data.info. pmnt^txn^typ^ind^sub^fld^77.tag^data)	
		Value that identifies the type of payment transaction taking place.	
97	02	CHIP-BIT-ERR-RSLTS	PIC X(50).
		Bit-48 Sub-element 79 (sem.addl^data.info. chip^bit^err^rsrts^sub^fld^79.tag^data)	
		Provides the TVR and CVR bitmask and expected values registered by the issuer.	
147	02	CRD-LVL-RSLT	PIC X(2).
		Bit-48 Sub-element 46 (sem.addl^data.info. crd^lvl^rslt^sub^fld^46.tag^data)	
		Contains the Visa card-level result.	
149	02	MC-GTWY-TXN-IND	PIC X(8).
		Bit-48 Sub-element 47 (sem.addl^data.info. mc^gtwy^txn^ind^sub^fld^47.tag^data)	
		Contains the MasterCard payment gateway transaction indicator.	
157	02	MC-ASGN-ID-SUB-FLD-32	PIC X(6).
		Bit-48 Sub-element 32 (sem.addl^data.info. mc-asgn-id-sub-fld-32.tag^data)	
		Contains the MasterCard assigned ID for the merchant.	
163	02	VPAN	PIC X(19).
		Bit-48 Sub-element 33 (sem.addl^data.info. paypass^mapping^sub^fld^33.acct^num.tag^data)	
		Virtual PAN provided by the MasterCard inControl platform.	

163	02	PAYPASS-MAPPED-PAN	REDEFINES VPAN PIC X(19).
		Bit-48 Sub-element 33 (sem.addl^data.info. paypass^mapping^sub^fld^33.acct^num.tag^data) Represents the cardholder's PayPass mapped PAN.	
163	02	PAYPASS-PAN	REDEFINES VPAN PIC X(19).
		Bit-48 Sub-element 33 (sem.addl^data.info. paypass^mapping^sub^fld^33.acct^num.tag^data) Represents the cardholder's PayPass PAN.	
182	02	RTE-IND	PIC X.
		Bit-48 Sub-element 12 (sem.addl^data.info. rte-ind-sub-fld-12.tag^data) Routing indicator passed to the issuer to indicate that alternate issuer host routing has been invoked.	
183	02	VISA-MKT-SPCFC-DATA	PIC X.
		Bit-48 Sub-element 96 (sem.addl^data.info. visa-mkt-spcfc-data-sub-fld-96.tag^data) Contains the Visa market-specific data identifier.	
184	02	NTL-POS-DATA. 04 cond-cde	PIC 9(11).
		Bit-61 (sem.ntl-pos-data) Contains Point-of-Service condition codes.	
195	02	PIN-SVC-CDE	PIC 99.
		Bit-48 Sub-element 80 (sem.addl^data.info. pin-svc-cde-sub-fld-80.cde) Indicates the results of PIN processing by the Authorization system.	
197	02	PIN-CAP-CDE	PIC 99.
		Bit-26 (sem.pin-cap-cde) Indicates the max number of pins capable of being entered other than 12. Not actual pin length. 04-12.	

199	02	AUTH-SYS-ADV-DAT-TIM	PIC 9(10).
		Bit-48 Sub-element 15 (sem.addl^data.info. auth-sys-adv-sub-fld-15.tag-data)	
		Contains the Authorization System Advice Date and Time in AFD completion advices.	
209	02	FRAUD-DATA.	
		04 SCORE	PIC X(3).
		04 RSN-CDE	PIC X(2).
		04 RULE-ADJ-SCORE	PIC X(3).
		04 RULE-RSN-CDE-1	PIC X(2).
		04 RULE-RSN-CDE-2	PIC X(2).
		Bit-48 Sub-element 75 (sem.addl^data.info. fraud-scoring-data-sub-fld-75.fraud-score)	
		Contains the transaction risk score, the score reason code, the rule adjusted score and up to two rule reason codes.	
221	02	PMNT-INITIATION-CHAN	PIC X(2).
		Bit-48 Sub-element 23 (sem.addl^data.info. pmnt-initiation-chan)	
		Contains Device Type Indicator describing mobile and other non-card payments.	

Appendix B

Sub-Element Documentation

The following are the sub-elements added to Data Element 48 in the standard external message (SEM) between BASE24 and the interchange to support various functions.

<u>Sub-Element</u>	<u>Description</u>
12	Passed to the issuer to indicate that alternate issuer host routing has been invoked.
13	Contains data to support ATM mobile top-up transactions.
15	Contains the Authorization System Advice Date and Time in AFD completion advices.
20	Used in Authorization Advice/0120 and Reversal Request/0400 messages to indicate whether the request message contained an online PIN.
23	Contains Device Type Indicator describing mobile and other non-card payments.
32	Contains the MasterCard Assigned ID to support Healthcare IAS merchant validation, Private label merchant verification, and is required for e-commerce MARP transactions.
33	Contains data to support the PAN Mapping Services.
34	Contains Dynamic ATC Information.
35	Contains PayPass NCFR Request/Response.
38	Contains the applicable account category based on information provided by the issuer.
40	Used to carry merchant and cardholder certificates for Visa SET transactions.
42	Used to provide an authorization level of security in electronic commerce transactions.
43	Used to communicate the UCAF authentication information among the cardholder, merchant, issuer, and acquirer communities. A redefine of this field is used to carry Visa Cardholder Authentication Verification Value (CAVV) request data.
44	Used to carry the Visa Electronic Commerce Transaction Identifier (XID).
45	Used in to carry the Visa CAVV Result Code.
46	Contains the Visa card-level result.
47	Contains the MasterCard payment gateway transaction indicator.
61	Indicates whether the merchant terminal supports a specific program or service, including partial approvals.
63	Contains network data generated by Banknet for matching purposes.
71	Used to notify the issuer of the on-behalf service performed and the result of the service performed.
72	Used to carry data during cryptogram processing.
74	Used in Authorization Request Response/0110 and Reversal Request Response/0410 messages to provide additional information about chip transaction processing.

75	Contains the Fraud Scoring Data when Expert Monitoring Fraud Scoring Service is performed on the transaction.
76	Used to indicate whether the acquirer participates in MasterCard Electronic.
77	Used to identify the type of payment transaction being performed.
78	Used to identify Visa transactions for which the billing for merchandise occurred after the merchandise was delivered to the customer.
79	Used to provide the TVR and CVR bitmask, and the expected values registered by the issuer.
80	Contains a PIN Service Code identifying whether the MasterCard Authorization System attempted to verify or translate the PIN.
82	Request for Address Verification Service.
83	Contains the results of the Address Verification.
84	Contains additional information about recurring payment transactions.
85	Used to communicate U.S. Existing Debt in transactions forwarded to Visa.
86	Used to communicate Relationship Participant in transactions forwarded to Visa.
87	Used to indicate that the CVC 1 or CVC 3 is incorrect, or to contain the CVC 2 result code. Also used to contain the CVV 2 result code for Visa transactions.
88	Advises both acquirer and issuer if the POS entry mode is changed from "90" to "02".
89	Advises both acquirer and issuer that the track data, POS data, or transaction category code contains an edit error.
90	Used to indicate a custom payment service transaction for Visa.
91	Used to provide authorization response data when a custom payment service is requested for Visa.
92	Used to send the CVC 2 data to the issuer. A redefine of this field is used to send CVV 2 data for Visa transactions.
93	Used to allow Visa Fleet Card transactions to be processed by Banknet.
94	Used to allow Visa Commercial Card inquiries to be processed by Banknet.
95	Used to identify unique use of a MasterCard card for a specific program or service established between issuers and merchants.
96	Contains market-specific data in a Visa transaction.
98	Contains the ID of the user of a Fleet card.
99	Contains the ID of the vehicle used in conjunction with a Fleet card purchase.

Appendix C

Response Code Translation

This Appendix contains gives a description of the response code mappings performed by the Banknet Interface.

BASE24-pos TO BANKNET RESPONSE CODE

BASE24-pos RESPONSE CODE		BANKNET RESPONSE CODE	
000	Approved - with balances	00	Approved
001	Approved - no balances	00	Approved
002	Approved - country club	00	Approved
003	Approved - maybe	00	Approved
004	Approved - with ID	08	Honor with ID
005	Approved - blind	00	Approved
006	Approved - VIP	00	Approved
007	Approved - admin	00	Approved
008	Approved - NNEG Hit OK	00	Approved
009	Approved - commercial	00	Approved
050	Declined	05	Do not honor
051	Expired card	54	Expired card
052	PIN tries exceeded	75	PIN tries exceeded
053	No sharing	57	Transaction not permitted to cardholder
054	No security box	96	System error
055	Invalid transaction	12	Invalid transaction
056	No support	57	Transaction not permitted to cardholder
057	Lost/stolen card	41	Lost card - pick up
058	Declined card status	62	Restricted card
059	Restricted card	62	Restricted card
060	No accounts	14	Invalid card number
061	No PBF	14	Invalid card number
062	PBF update error	96	System error
063	Invalid auth type	05	Do not honor
064	Bad track2	12	Invalid transaction
065	Adjustment not allowed	57	Transaction not permitted to cardholder
066	Invalid cash adv increment	13	Invalid amount
067	Invalid transaction date	05	Do not honor
068	TLF error	96	System error
069	Message edit error	57	Transaction not permitted to cardholder
070	No IDF	96	System error

Response Code Translation

BASE24-pos RESPONSE CODE		BANKNET RESPONSE CODE	
071	Invalid auth destination	92	Unable to route transaction
072	Card on NNEG	62	Restricted card
073	Invalid service destination	92	Unable to route transaction
074	Unable to authorize	05	Do not honor
075	Invalid PAN length	14	Invalid card number
076	Insufficient funds	51	Insufficient funds
077	Pre-auth limit reached	61	Exceeds withdrawal amount limit
078	Duplicate transmission	94	Duplicate transmission
079	Refund max reached	57	Transaction not permitted to cardholder
080	Offline refund max reached	57	Transaction not permitted to cardholder
081	Max credit/refund reached	57	Transaction not permitted to cardholder
082	Usage exceeded	65	Exceeds withdrawal count limit
083	Max refund credit reached	57	Transaction not permitted to cardholder
084	Customer-selected NEG rsn	62	Restricted card
085	Enquiry not allowed	57	Transaction not permitted to cardholder
086	Over floor limit	57	Transaction not permitted to cardholder
087	Max no. of refunds reached	57	Transaction not permitted to cardholder
088	Place Call	01	Refer to Card Issuer
089	Card inactive or closed	62	Restricted card
090	Referral file full	96	System error
091	NEG file problem	96	System error
092	Advance less than minimum	13	Invalid amount
093	Delinquent	51	Insufficient funds
094	Over limit table	51	Insufficient funds
095	Amount over maximum	13	Invalid amount
096	PIN required	55	Invalid PIN
097	Mod 10 check	14	Invalid card number
099	Bad PBF	62	Restricted card
100	Referral; Unable to process	01	Refer to Card Issuer
101	Referral; Issue call	01	Refer to Card Issuer
102	Referral; Call	01	Refer to Card Issuer
103	Referral; NEG file problem	01	Refer to Card Issuer
104	Referral; CAF file problem	01	Refer to Card Issuer
105	Referral; No support	01	Refer to Card Issuer
106	Referral; Amount over max	01	Refer to Card Issuer
107	Referral; Over daily limit	01	Refer to Card Issuer
108	Referral; CAPF not found	01	Refer to Card Issuer
109	Referral; Advance < min	01	Refer to Card Issuer
110	Referral; Number times	01	Refer to Card Issuer
111	Referral; Delinquent	01	Refer to Card Issuer
112	Referral; Over limit	01	Refer to Card Issuer
113	Referral; Timeout	91	Authorization system inoperative
115	Referral; PTLF full	01	Refer to Card Issuer

BASE24-pos RESPONSE CODE		BANKNET RESPONSE CODE	
120	UAF file problem	01	Refer to Card Issuer
121	Admin file problem	01	Refer to Card Issuer
122	Admin file problem	01	Refer to Card Issuer
130	ARQC verification failure	01	Refer to Card Issuer
131	CVR check failure	01	Refer to Card Issuer
132	TVR check failure	01	Refer to Card Issuer
133	Reason online code failure	01	Refer to Card Issuer
134	Fallback transaction	01	Refer to Card Issuer
150	Merchant not on file	14	Invalid card number
200	Error; Invalid account	14	Invalid card number
201	Error; Incorrect PIN	55	Invalid PIN
202	Error; Advance < minimum	13	Invalid amount
203	Error; Need admin card	14	Invalid card number
204	Error; Enter lesser amount	61	Exceeds withdrawal amount limit
205	Error; Invalid advance amt	13	Invalid amount
206	Error; CAF not found	14	Invalid card number
207	Error; Invalid tran date	05	Do not honor
208	Error; Invalid expiry date	05	Do not honor
209	Error; Invalid tran code	05	Do not honor
251	PWCB – insufficient funds	51	Insufficient funds
400	ARQC verification failure	05	Do not honor
401	HSM parameter error	05	Do not honor
402	HSM failure	05	Do not honor
403	KEYI not found	05	Do not honor
404	ATC check failure	05	Do not honor
405	CVR check failure	05	Do not honor
406	TVR check failure	05	Do not honor
407	Reason online code failure	05	Do not honor
408	Fallback transaction	05	Do not honor
900	Capture; PIN tries exceeded	04	Pick up card
901	Capture; Expired card	04	Pick up card
902	Capture; NEG capture	04	Pick up card
903	Capture; CAF stolen	43	Stolen card, pick up
904	Capture; Advance < min	04	Pick up card
905	Capture; Num times used	04	Pick up card
906	Capture; Delinquent	04	Pick up card
907	Capture; Over limit table	04	Pick up card
908	Capture; Amount > max	04	Pick up card
909	Capture	04	Pick up card
910	Capture; ARQC check failure	04	Pick up card
911	Capture; CVR check failure	04	Pick up card
912	Capture; TVR check failure	04	Pick up card

Any BASE24 codes not listed are translated to 05 (do not honor).

If a Balance Inquiry or Card Verification transaction (except a 'test' recurring payment) is approved, then the external response code is set to 85 (not declined).

If a Card Verification transaction (except a 'test' recurring payment) is declined, then the external response code is set to 05 (do not honor).

BANKNET TO BASE24-pos RESPONSE CODE

BANKNET RESPONSE CODE		BASE24-pos RESPONSE CODE	
00	Approved	001	Approved – no balances
01	Refer to card issuer	101	Referral; Issue call
03	Invalid merchant	150	Merchant not on file
04	Capture card	909	Capture
05	Do not honor	050	Declined
08	Honor with identification	004	Approved – with ID
10	Partially approved	000	Approved
12	Invalid transaction	055	Invalid transaction
13	Invalid amount	205	Error; Invalid advance amt
14	Invalid card number	200	Error; Invalid account
15	Invalid card issuer	073	Invalid destination
30	Format error	069	Bad message edit
41	Lost card, pick up	909	Capture
43	Stolen card, pick up	903	Capture; CAF stolen
51	Insufficient funds	076	Low funds
54	Expired card	051	Expired card
55	Invalid PIN	201	Error; Incorrect PIN
57	Transaction not permitted to cardholder	055	Invalid transaction
58	Transaction not permitted to terminal	055	Invalid transaction
61	Exceeds withdrawal amount limit	204	Error; Enter lesser amount
62	Restricted card	059	Restricted
63	Security violation	074	Unable to authorize
65	Exceeds withdrawal count limit	082	Usage exceeded
75	Allowable number of PIN tries exceeded	052	PIN tries exceeded
76	Invalid/nonexistent “to” account	200	Error; Invalid account
77	Invalid/nonexistent “from” account	200	Error; Invalid account
78	Invalid/nonexistent account (general)	200	Error; Invalid account
84	Invalid authorization lifecycle	050	Declined
85	Not declined	001	Approved - no balances
86	PIN validation not possible	054	No security box
88	Cryptographic failure	054	No security box
91	Authorization system inoperative	074	Unable to process
92	Unable to route transaction	073	Invalid destination
94	Duplicate transmission detected	078	Duplicate transmission
96	System error	074	Unable to process
N7	Visa decline – CVV2 failure	074	Unable to process

Any codes not listed are translated to the BASE24 response code 050 (Declined).

BASE24-atm TO BANKNET RESPONSE CODE

BASE24-atm RESPONSE CODE		BANKNET RESPONSE CODE	
00	Approved	00	Approved
01	Approved – no balances	00	Approved
50	Unauthorized usage	62	Restricted card
51	Expired card	54	Expired card, card return
52	Invalid card	14	Invalid card number
53	Incorrect PIN	55	Invalid PIN
54	BASE24 database problem	96	System problem
55	Ineligible transaction	57	Transaction not permitted to cardholder
56	Ineligible account	14	Invalid card number
57	Transaction not supported	57	Transaction not permitted to cardholder
58	NSF – no available balance	51	Not sufficient funds
59	NSF – with available balance	51	Not sufficient funds
60	No further withdrawals	65	Over withdrawal frequency limit
61	Enter lesser amount	61	Over withdrawal limits
62	PIN tries exceeded	75	PIN tries exceeded
63	Daily withdrawal limit reached	61	Exceeds withdrawal limits
64	Invalid cash advance amount	13	Invalid amount
65	No statement print info	57	Transaction not permitted to cardholder
66	No statement print info	57	Transaction not permitted to cardholder
67	Invalid cash back amount	13	Invalid amount
68	External decline	57	Transaction not permitted to cardholder
69	No sharing with issuer	57	Transaction not permitted to cardholder
70	System error	96	System problem
71	Contact bank	57	Transaction not permitted to cardholder
72	Destination not available	91	Authorization system inoperative
73	Routing lookup problem	92	Unable to route transaction
74	Message edit error	57	Transaction not permitted to cardholder
80	Signature check failed	57	Transaction not permitted to cardholder
81	HSM parameter error	57	Transaction not permitted to cardholder
82	HSM failure	57	Transaction not permitted to cardholder
83	KEYI not found	57	Transaction not permitted to cardholder
84	ATC check failure	57	Transaction not permitted to cardholder
85	CVR check failure	57	Transaction not permitted to cardholder
86	TVR check failure	57	Transaction not permitted to cardholder
87	ARQC verification failure	57	Transaction not permitted to cardholder
88	Fallback transaction	57	Transaction not permitted to cardholder

If the BASE24-atm response code is 50 (unauthorized usage), and the ERR-FLG element of the AT-FLG1 token is 2 (lost card), then the external response code is set to 41 (lost card).

If the BASE24-atm response code is 50 (unauthorized usage), and the ERR-FLG element of the AT-FLG1 token is 3 (stolen card), then the external response code is set to 43 (stolen card).

If the first byte of the BASE24-atm response code is 1 (capture), then the external response code is set to 04 (capture card).

Any BASE24 codes not listed are translated to 57 (transaction not permitted to cardholder).

If a PIN Change transaction is declined and the ERR-FLG element of the AT50 token is P (PIN change error), then the external response code is set to 89 (unacceptable PIN – retry).

Otherwise, if a PIN Change transaction is declined, then the external response code is set to 71 (PIN not changed).

If a PIN Unblock transaction is declined, then the external response code is set to 70 (contact card issuer).

If a Balance Inquiry, a PIN Change transaction or a PIN Unblock transaction is approved, then the external response code is set to 85 (not declined).

BANKNET TO BASE24-atm RESPONSE CODE

BANKNET RESPONSE CODE		BASE24-atm RESPONSE CODE	
00	Approved	001	Approved – no balances
01	Refer to card issuer	068	External decline
03	Invalid merchant	068	External decline
04	Capture card	168	External decline
05	Do not honor	068	External decline
08	Honor with identification	068	External decline
12	Invalid transaction	055	Ineligible transaction
13	Invalid amount	064	Invalid cash advance amount
14	Invalid card number	056	Ineligible account
15	Invalid card issuer	052	Invalid card
30	Format error	074	Bad message edit
41	Lost card, pick up	150	Unauthorized usage
43	Stolen card, pick up	150	Unauthorized usage
51	Insufficient funds	058	NSF – no available balance
54	Expired card	051	Expired card
55	Invalid PIN	053	Invalid PIN
57	Transaction not permitted to cardholder	055	Ineligible transaction
58	Transaction not permitted to terminal	055	Ineligible transaction
61	Exceeds withdrawal amount limit	061	Enter lesser amount
62	Restricted card	052	Invalid card
63	Security violation	050	Unauthorized usage
65	Exceeds withdrawal count limit	060	No further withdrawals
70	Contact card issuer	055	Ineligible transaction
71	PIN not changed	068	External decline
75	Allowable number of PIN tries exceeded	062	PIN tries exceeded
76	Invalid/nonexistent “to” account	056	Ineligible account
77	Invalid/nonexistent “from” account	056	Ineligible account
78	Invalid/nonexistent account (general)	056	Ineligible account
84	Invalid authorization lifecycle	068	External decline
85	Not declined	001	Approved – no balances
86	PIN validation not possible	070	System error
88	Cryptographic failure	070	System error
89	Unacceptable PIN – retry	053	Invalid PIN
91	Authorization system inoperative	072	Destination not available
92	Unable to route transaction	073	Routing lookup problem
94	Duplicate transmission detected	068	External decline
96	System error	070	System error

Any codes not listed are translated to the BASE24 response code 068 (external decline).

Appendix D

Data Communications

This Appendix contains configuration information for the following communications protocols:

- ☐ TCP/IP
- ☐ X.25
- ☐ SNA LU-0

TCP/IP PROTOCOL

Network Environment Declarations

The following information gives example Device, Station and Line declarations that can be used when communicating with Banknet utilizing the TCP/IP protocol. These definitions may be declared via a network control facility. For further information, refer to the **NET24-XPNET Network Control Command Reference Manual**.

RESET PROCESS

```
SET PROCESS BCPU -1
SET PROCESS LIBRARY $SYSACI.XPNET.SKELB
SET PROCESS PROGRAM $<vol>.XXXXOBJ.BNET
SET PROCESS PPD $BNET
SET PROCESS PRIORITY 139
SET PROCESS CPU -1
SET PROCESS IAUDIT OFF
SET PROCESS OAUDIT OFF
ADD PROCESS P1A^BNET, UNDER SYSNAME \sys, UNDER NODE P1A^NODE
```

RESET DEVICE

```
SET DEVICE INPUTFORMAT TXT
SET DEVICE OUTPUTFORMAT TXT
SET DEVICE POLLFORMAT NUL
SET DEVICE RECVLEN 4096
SET DEVICE SELECTFORMAT NUL
SET DEVICE TYPE 1
SET DEVICE XMITLEN 4096
ADD DEVICE BNET^CTS^DEV, UNDER SYSNAME \sys, UNDER NODE P1A^NODE
```

RESET LINE

```
SET LINE ERANALYSIS 30
==
== The PORT name must match the PPD name of the CTS/E Server process
==
SET LINE PORT $BNET
SET LINE CMETHOD ACCEPT
SET LINE EXPDATA ROUTE
SET LINE FMM ALL
SET LINE PROTOCOL CTS
SET LINE MODE ASYNC
ADD LINE L1A^BNET, UNDER SYSNAME \sys, UNDER NODE P1A^NODE
```

RESET STATION

```
SET STATION DESTINATION P1A^BNET
SET STATION RETRY 1
SET STATION CLASS SRV
SET STATION DEVICE BNET^CTS^DEV
SET STATION LINENAME \sys.P1A^NODE.L1A^CTS^BNET
SET STATION USERDATA OFFSET 0 "-1 -K+" 0 0 0 0 0 0 0 0
ADD STATION S1ABNET, UNDER SYSNAME \sys, UNDER P1A^NODE
```

X.25 PROTOCOL

Network Environment Declarations

The following information gives example Device, Station and Line declarations that can be used when communicating with Banknet utilizing the X.25 protocol. These definitions may be declared via a network control facility. For further information see the **NET24-XPNET Network Control Command Reference Manual**.

RESET PROCESS

```
SET PROCESS LIBRARY $SYSACI.XPNET.SKELB
SET PROCESS PROGRAM $vol.xxxxOBJ.BNET
SET PROCESS PPD $T1B1
SET PROCESS PRIORITY 118
SET PROCESS CPU 1
SET PROCESS QAT 70
ADD PROCESS P1A^BNET, UNDER SYSNAME \CFS, UNDER NODE P1B^GATE
```

RESET DEVICE

```
SET DEVICE RECVLEN 4000
SET DEVICE XMITLEN 1000
SET DEVICE TYPE 29
ADD DEVICE BNET1^SWITCH^X25, UNDER SYSNAME \CFS, UNDER NODE P1B^GATE
```

RESET LINE

```
SET LINE ETE 9
SET LINE PORT $X25XA1.#PU01
SET LINE WFCALL ON
SET LINE RECALL 2
SET LINE DUPLEX FULL
SET LINE PROTOCOL X.25
ADD LINE L1A^BNET1, UNDER SYSNAME \CFS, UNDER NODE P1B^GATE
```

RESET STATION

```
SET STATION DESTINATION P1B^BNET1
SET STATION DEVICE BNET1^SWITCH^X25
SET STATION LINENAME \CFS.P1B^GATE.L1A^BNET1
SET STATION QAT 10
SET STATION QMT 100
SET STATION RETRY 1
ADD STATION S1ABNET, UNDER SYSNAME \CFS, UNDER NODE P1B^GATE
```

NOTE: For D Series Operating Systems and above, Subsystem Control Facility (SCF) commands will replace the CMI commands used to start the X.25 sessions in the following example. For further information on SCF, see Tandem's **Subsystem Control Facility (SCF) Reference Manual**.

CMI Obey File

The following is a list of Communication Management Interface (CMI) commands to start the X.25 sessions. For further information on CMI, see Tandem's ***Communication Management Interface Operator Reference Manual***.

The customer's system management group should be responsible for correctly configuring the X.25 ports. Any sample configuration we provide may not cover all of the available options the customer may wish. Here is a sample configuration and it may not include how the customer may eventually implement its X.25 configuration.

The user will also have to type the following commands under CMI. (D customers and above will be using SCF commands and should refer to the SCF example following the CMI example.)

CMI Example:

```
assume line $<ldev name>
abort
alter , netaddr <my DTE address>
alter , framemode <dte or dce>
alter , extended off
alter , netid <my netid>
abort su under
delete su under
add su #<port name>, recsize 128, protocol ptp, type(9,0)
add su #<port name>, recsize 128, protocol ptp, type(9,0)
add su #<port name>, recsize 128, protocol ptp, type(9,0)
start
```

SCF Example:

```
assume line $<ldev name>
abort
alter , netaddr <my DTE address>
alter , framemode <dte or dce>
alter , extended off
alter , netid <my netid>
abort sub all
delete sub all
add su #<port name>, recsize 128, protocol ptp, devtype(9,0)
add su #<port name>, recsize 128, protocol ptp, devtype(9,0)
add su #<port name>, recsize 128, protocol ptp, devtype(9,0)
start sub all
start
```

SNA LU-0 PROTOCOL

Network Environment Declarations

The following information gives example Device, Station and Line declarations that can be used when communicating with Banknet utilizing the SNA LU-2 protocol. These definitions should be declared via a network control facility. For further information refer to the ***NET24-XPNET Network Control Command Reference Manual***.

```
RESET DEVICE
  SET DEVICE PROFILE PBNETLU0
  SET DEVICE RECVLEN 4095
  SET DEVICE XMITLEN 1024
  ADD DEVICE BNET^LU^T2, UNDER SYSNAME \CFS,
                        UNDER NODE P1B^GATE
```

```
RESET LINE
  SET LINE PROTOCOL PLU
  ADD LINE L1BPLU0, UNDER SYSNAME \CFS,
                  UNDER NODE P1B^GATE
```

```
RESET LINE
  SET LINE PROTOCOL PLU
  ADD LINE L1BPLU1, UNDER SYSNAME \CFS,
                  UNDER NODE P1B^GATE
```

```
RESET STATION
  SET STATION DESTINATION P1A^BNET
  SET STATION DEVICE BNET^LU^T2
  SET STATION LINENMAE \CFS.P1B^GATE.L1BPLU0
  SET STATION RETRY 3
  ADD STATION S1ABNETLU0, UNDER SYSNAME \CFS,
                        UNDER NODE P1B^GATE
```

```
RESET STATION
  SET STATION DESTINATION P1A^BNET
  SET STATION DEVICE BNET^LU^T2
  SET STATION LINENMAE \CFS.P1B^GATE.L1BPLU1
  SET STATION RETRY 3
  ADD STATION S1ABNETLU1, UNDER SYSNAME \CFS,
                        UNDER NODE P1B^GATE
```

RDT File

This information is an example of the compile commands used by Tandem's SNAX-HLS product. The SNAX/HLS operating environment presumes that a Resource Definition Table (RDT) exists. The RDT accessed by SNAX/HLS is created by compiling an edit file. These command shown below can be used as input to the SNAX-HLS compiler to create the RDTDATA file. For further information regarding the RDTDATA, RDTTEXT and HLSRDT, see the Tandem ***SNAX/HLS Configuration and Control Manual***.

```
reset BIND
! Bind BBNETLU0 TEST BIND FOR BNET SNA LU_0
set bind bind-type          NONNEGOTIABLE
set bind fm-profile          3
set bind ts-profile          3
set bind pri-chain-use       SINGLE-RU
set bind pri-rq-mode          IMMEDIATE
set bind pri-chain-protocol   any
set bind pri-two-phase-commit FALSE
set bind pri-compression-ind  FALSE
set bind pri-eb-send          TRUE
set bind sec-chain-use        SINGLE-RU
set bind sec-rq-mode          IMMEDIATE
set bind sec-chain-protocol   exception
set bind sec-two-phase-commit FALSE
set bind sec-compression-ind  FALSE
set bind sec-eb-send          TRUE
set bind no-segments          FALSE
set bind fm-header-use        FALSE
set bind brkt-reset           betb
set bind brkt-term            CONDITIONAL
set bind alt-code-use          FALSE
set bind seqno-use            FALSE
set bind bind-queuing          FALSE
set bind send-recv            HDUX-FF
set bind recovery-mode         LOSER
set bind contention-winner     SECONDARY
set bind hdx-ff-reset-sender   SECONDARY
set bind sec-send-pacing-size  0
set bind sec-max-ru            1024
set bind sec-recv-pacing-size  0
set bind pri-max-ru            1024
set bind lu-session-type       0
set bind ps-usage              000000000000000000000000
set bind plu-name              NETPNAME
```



```
add BIND          BBNETLU0
!-----
reset PROFILE
! Profile PBNETLU0 TEST PROFILE FOR BNET SNA LU_2 (NETPNAME)
set profile bind-name          BBNETLU0
set profile open-mode          ACQUIRE
set profile hs-polarity        PRIMARY
set profile want-sscp-text     FALSE
set profile response-type      1
set profile pipeline-lu-ind    FALSE
set profile translate          FALSE
set profile lmo                FALSE
set profile max-requests       2
set profile want-bind-data     NONE
set profile user-responses     FALSE
set profile want-flow-control  FALSE
set profile encode-eri         FALSE
set profile single-fmh         FALSE
set profile omit-modem-wait    FALSE
add PROFILE        PBNETLU0
```

Creating An RDT File

An RDTDATA file can be created by the following set of commands.

1. Edit the source file:

```
> EDIT RDTTEXT
(Perform changes)
```

```
*EXIT
```

2. Compile the RDTTEXT file:

```
> RUN HLSRDT / IN RDTTEXT, OUT $$.#LP / RDTDATA
```

For more information see the Product Overview section of Tandem's **SNAX/HLS Configuration and Control Manual**. The three lines of command shown below is another way to create the RDTDATA file.

```
> purge ordtdata
> rename rdtdata,ordtdata
> run $system.system.hlsrdt/in rdttext,out $$.#rdtcomp/rdtdata
```

Operating SNAX/HLS

The following set of commands is used to start the new HLSOBJ. For further details see section Operating SNAX/HLS of Tandem's ***SNAX/HLS Configuration and Control Manual*** for more details.

```
> stop $visx
> hlsobj/in rdtdata, out $sudo,  nowait, name $hlsx, pri 140/
    abend^on^error on, max^lus 32, initial^trace 15
```

NOTE: For G Series Operating Systems and above, the Communication Management Interface (CMI) will no longer be supported. Subsystem Control Facility (SCF) commands will replace the CMI commands used to start the SNA sessions in the following example. For further information on SCF, see Tandem's ***Subsystem Control Facility (SCF) Reference Manual***.

CMI Obey File

The following is a list of Communication Management Interface (CMI) commands to start the SNA sessions. For further information on CMI, see Tandem's ***Communication Management Interface Operator Reference Manual***.

```
allow all errors
assume line $snap3
abort lu under
abort pu under
abort
delete lu under
delete pu under
delete
add line,character set ascii
add pu $snap3.#pu1,type(13,2),recsize 265,maxlus 2,address %hC1,
window 7
add lu $snap3.#lu1,type(14,2),puname #pu1,recsize 265,protocol &
    snalu, address 1, character set ascii
add lu $snap3.#lu0,type(14,2),puname #pu1,recsize 265,protocol &
    snalu, address 2, character set ascii
start
start pu under
start lu under
```

NOTE: The values of addresses mentioned above were set to match the Banknet Test System. These might need to be changed depending upon the configuration.